

CONSERVATION PROPERTIES OF FINITE-DIFFERENCE TIME-DOMAIN METHODS
FOR THE MAXWELL AND SCHRÖDINGER EQUATIONS WITH APPLICATION TO
THE DEVELOPMENT OF NEW SCHEMES WITH GUARANTEED STABILITY

by

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Conservation Properties of Finite-Difference Time-Domain Methods for the Maxwell and Schrödinger Equations With Application to the Development of New Schemes with Guaranteed Stability

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Abstract

The finite-difference time-domain (FDTD) method is a popular technique for solving the Maxwell equations. The method works by updating unknowns in a leap-frog manner, starting with initial conditions. The update equations are explicit, and hence have relatively low computational requirements. However, the explicit nature comes at the cost of conditional stability, which makes it substantially harder to create new FDTD-based methods, compared to implicit or frequency-domain methods. Methodologies for stability analysis exist, but can lead to lengthy derivations when applied to complex problems.

FDTD has also been applied to solving the Schrödinger equation. Research related to this scheme, along with other numerical methods for quantum mechanical problems, is becoming increasingly important. Another type of FDTD methods that were recently inspired by quantum applications are schemes that use potentials as unknowns in order to facilitate modelling of light-matter interaction. Much like in the conventional FDTD, the difficulty of stability analysis poses a significant challenge in advancing these schemes.

This thesis draws inspiration from results in control theory in order to facilitate stability analysis of these FDTD methods based on the concepts of energy conservation and dissipation, as well as probability conservation in the case of FDTD for the Schrödinger equation. For the fields-based FDTD, expressions are proposed to quantify the energy stored in a region and the energy entering the region through the boundary. Based on these expressions, we formulate rigorous conditions under which the region is unable to generate spurious energy. These results provide a way to create new schemes by coupling the FDTD equations for the region to other dissipative or energy conserving models. As an example we develop a subgridding scheme by coupling two FDTD grids of different resolution in an energy-conserving manner. The theory is extended to FDTD for the Maxwell equations written in terms of potentials and a subgridding scheme with potentials is developed as a demonstration of application. Furthermore, this thesis extends the approach to propose a similar framework for FDTD for the Schrödinger equation, based on the concept of probability conservation.

Conditions for the energy conservation of the scheme are also derived, motivated by the possibility of extending the results to the coupled Maxwell-Schrödinger system of equations in the future.

To Mom, Dad, and Jesse

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Notation and Abbreviations

a^*	the complex conjugate of a
$\otimes$	Kronecker product
$\hbar$	reduced Planck's constant
i	square root of negative one ($\sqrt{-1}$): not to be confused with i used for spatial indices
$\mathbf{I}_m$	$m \times m$ identity matrix
$\text{elem}\{m, n\}$	$n \times 1$ vector containing a single 1 in position m , with the other elements equal to zero
$\text{diag}(\mathbf{A}_1, \mathbf{A}_2, \dots)$	block-diagonal matrix with $\mathbf{A}_1, \mathbf{A}_2, \dots$ on its diagonal
$\succ 0$	positive definite matrix
$\succeq 0$	positive semidefinite matrix
$\hat{n}$	outward normal vector to a region
$\hat{n}_{\text{inward}}$	inward normal vector to a region
$\rho(\mathbf{A})$	spectral radius of the matrix $\mathbf{A}$
$\mathbf{T}_m$	$m \times 1$ vector of ones
$\mathcal{F}\{h(t)\}$	Fourier transform of the function h
ADI	alternating direction implicit
CFL	Courant–Friedrichs–Lewy
FDTD	finite-difference time-domain
FEM	finite element method
FIT	finite integration technique
PDE	partial differential equation
PEC	perfect electric conductor
PML	perfectly matched layer

Introduction

Many physical phenomena are described by partial differential equations (PDEs) relating the temporal and spatial derivatives of quantities of interest. Examples include the Maxwell equations in electromagnetics [1], the Schrödinger equation in quantum mechanics [2], the Navier-Stokes equations in fluid dynamics [3], the equation of conduction of heat [4], and many others. Solving these partial differential equations is essential in many areas of science and engineering. Analytical solutions to these equations can be found only for a small set of simple problems. Hence, numerical methods are required to study practical scenarios. Numerical approaches for solving partial differential equations date back to the works of Ritz, Galerkin, Courant, Friedrichs, Lewy, von Neumann, Crank, Nicolson, and others [5–7]. Interestingly, many of the major contributions were made decades before the inception of modern electronic computers. With the subsequent advancement of computing power, simulations based on numerical methods for solving PDEs became an integral part of engineering research and design methodologies. Examples include electromagnetic simulators, which at their core implement numerical solution techniques for the Maxwell equations. Those simulators are essential for antenna design [8, 9], electromagnetic analysis of electronic devices [10–12], and many other applications. Likewise, numerical solutions to the Poisson and Schrödinger equations can provide insights into the behaviour of semiconductor qubits [13–15], MOSFET transistors [16, 17], carbon nanotube field-effect transistors [18, 19], and other devices.

One of the major challenges in simulating practical scenarios is their multi-scale nature. Consider, for example, the electromagnetic simulations in [20], which were designed to assess the effect of reflecting environments on the safety of workers in the vicinity of base station antennas. The resolution selected to model the human tissues was 1–2 mm, whereas the distance between the person and the antenna (30 cm – 10 m) was orders of magnitude larger. In such multi-scale problems, the fine resolution imposed by the small scale features and the small scale spatial variations in the unknowns is inappropriate and often unfeasible for discretizing the free space or other large-scale features. Using the fine resolution throughout the entire simulation domain can impose prohibitive computational requirements. This challenge can be addressed via local resolution refinement [21–26]. Another approach involves solving the multi-scale problem via separate simulations, which employ different resolution or entirely different numerical techniques [20, 27–29]. The latter group of methods rely on the assumption that the problem can be decoupled without significant loss of accuracy [20]. One approach to reduce the error associated with this assumption is to solve the

decoupled problems by using the results of one simulation to excite the other in an iterative manner until convergence is achieved [28].

A related challenge is the large amount of volume that is densely filled by small-scale features. Consider again the scenario in [20]. In that scenario, the height of the human model was three orders of magnitude larger than the resolution of the grid needed to model the tissues. In problems where fine discretization is needed over a large volume, mitigating the high computational requirements is non-trivial.

Due to the aforementioned challenges, the application of high fidelity models is limited by the computational resources and availability of techniques that can reduce their computational cost. Often, approximate models are used instead. Examples of such models include lumped circuit element representations of electromagnetic phenomena [30] or methods for independent analysis of a single electron or a hole in a semiconductor lattice [31, 32].

Numerous methods have been proposed for solving PDEs. One of such methods is the finite-difference time-domain (FDTD) scheme [33–35], which was originally developed for solving the Maxwell equations in electromagnetics and was later applied to other PDEs, such as the Schrödinger equation in quantum mechanics [36, 37], as well as the coupled Maxwell-Schrödinger system of equations [38]. The method uses centered finite differences to discretize temporal and spatial derivatives in the PDEs. Update equations are then obtained, describing the evolution of the unknowns between two consecutive time steps. Starting with some initial conditions, the scheme uses these update equations to compute the unknowns at each time step. The FDTD is a common method of choice in electromagnetics and photonics, owing its popularity to the simplicity of the algorithm, ease of parallelization, and low computational cost associated with each time step.

The inexpensive update procedure is a result of explicit matrix-free update equations. This property comes at a cost of stability limit that sets an upper bound on the time step. If the stability limit is violated, simulations may produce unphysical values of unknowns, which grow indefinitely as the iterations progress. Moreover, modifications to the basic FDTD scheme require careful derivation of stability conditions. The complexity of stability analysis using existing techniques serves as a motivation for the theory developed in this thesis. Facilitation of stability analysis is important for the development of new FDTD-based schemes that address the computational challenges of the method and extend its capabilities.

1.1 The Maxwell Equations

The temporal evolution of electromagnetic fields is described by the Maxwell equations [1]

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad (1.1a)$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} + \vec{J}, \quad (1.1b)$$

$$\nabla \cdot \vec{D} = \rho_v, \quad (1.1c)$$

$$\nabla \cdot \vec{B} = 0, \quad (1.1d)$$

where $\vec{E}(x, y, z, t)$ is the electric field intensity (also known as electric field), $\vec{B}(x, y, z, t)$ is the magnetic flux density, $\vec{H}(x, y, z, t)$ is the magnetic field intensity or magnetic field, $\vec{D}(x, y, z, t)$ is the

electric flux density, $\vec{J}(x, y, z, t)$ is the current density, and $\rho_v(x, y, z, t)$ is the charge density. The latter are also known as, respectively, the volume current and volume charge densities. The current density $\vec{J}$ can include contributions due to conduction current density $\vec{J}_c$ and impressed source current density $\vec{J}_s$ [1, 39]. As will be discussed later, $\vec{J}$ can also include a contribution originating from a quantum mechanical model of a charged object. Vectors $\vec{D}$ and $\vec{H}$ are defined in terms of $\vec{E}$ and $\vec{B}$ as [1]

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P}, \quad (1.2)$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}, \quad (1.3)$$

where ε_0 and μ_0 are, respectively, the permittivity and permeability of free space, $\vec{P}$ is the polarization vector resulting from induced electric dipoles in the medium, and $\vec{M}$ is the magnetization vector that can be present in magnetic materials. The charge density is related to the current density through the continuity equation [1, 39]

$$\frac{\partial \rho_v}{\partial t} = -\nabla \cdot \vec{J}. \quad (1.4)$$

In linear isotropic non-dispersive media, (1.2) and (1.3) take on the following form [1]:

$$\vec{D} = \varepsilon \vec{E}, \quad (1.5)$$

$$\vec{H} = \frac{1}{\mu} \vec{B}, \quad (1.6)$$

where $\varepsilon(x, y, z)$ and $\mu(x, y, z)$ are, respectively, the permittivity and permeability of the medium, which are often expressed in terms of the relative permittivity $\varepsilon_r(x, y, z)$ and relative permeability $\mu_r(x, y, z)$ as $\varepsilon = \varepsilon_r \varepsilon_0$ and $\mu = \mu_r \mu_0$. If the linear isotropic medium is also conductive, the resulting conduction current density is related to the electric field as [1, 39]

$$\vec{J}_c = \sigma \vec{E}, \quad (1.7)$$

where σ is the conductivity of the medium.

1.2 Maxwell Equations Written Using Potentials

The electromagnetic field can be described in terms of scalar and vector potentials $\phi(x, y, z, t)$ and $\vec{A}(x, y, z, t)$ through [39]

$$\vec{E} = -\nabla \phi - \frac{\partial \vec{A}}{\partial t}, \quad (1.8)$$

$$\vec{B} = \nabla \times \vec{A}. \quad (1.9)$$

The choice of $\vec{A}$ and ϕ is not unique. If one replaces them with

$$\phi' = \phi - \frac{\partial \Gamma}{\partial t}, \quad (1.10a)$$

$$\vec{A}' = \vec{A} + \nabla \Gamma, \quad (1.10b)$$

where $\Gamma(x, y, z, t)$ is an arbitrary scalar function of position and time, the resulting fields $\vec{E}$ and $\vec{B}$ remain unchanged [39]. Choosing $\Gamma(x, y, z, t)$ corresponds to choosing a gauge condition, such as, for example, the Lorenz gauge for a homogeneous non-conductive medium [1, 39, 40]

$$\nabla \cdot \vec{A} + \mu \varepsilon \frac{\partial \phi}{\partial t} = 0. \quad (1.11)$$

Using (1.8) and (1.9), the fields in the Maxwell equations (1.1a)–(1.1d) can be replaced with potentials. Equations (1.1a) and (1.1d) are satisfied by construction as a result of writing $\vec{E}$ and $\vec{B}$ in the form (1.8)–(1.9) [39]. Using the constitutive relations (1.5) and (1.6) and assuming uniform μ and ε and zero σ , (1.1b) and (1.1c) reduce to [1, 39]

$$\frac{1}{\mu} \nabla \times \nabla \times \vec{A} = -\varepsilon \frac{\partial}{\partial t} \nabla \phi - \varepsilon \frac{\partial^2 \vec{A}}{\partial t^2} + \vec{J}_s, \quad (1.12a)$$

$$-\varepsilon \nabla^2 \phi - \varepsilon \frac{\partial}{\partial t} \nabla \cdot \vec{A} = \rho_v, \quad (1.12b)$$

where $\vec{J}$ is assumed to only comprise the impressed source current density $\vec{J}_s$. Under the Lorenz gauge condition, the scalar potential can be eliminated from (1.12a) and the vector potential can be eliminated from (1.12b). After simplification, one obtains the nonhomogeneous wave equations for the vector and scalar potentials [1, 39]

$$\nabla^2 \vec{A} - \mu \varepsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J}_s, \quad (1.13a)$$

$$\nabla^2 \phi - \mu \varepsilon \frac{\partial^2 \phi}{\partial t^2} = -\frac{\rho_v}{\varepsilon}. \quad (1.13b)$$

Treatment of heterogeneous media with generalized Lorenz gauge conditions can be found in [41, 42]. In [39] the case of non-zero conductivity for homogeneous media is addressed.

The equation describing the vector potential (1.13a) is commonly used in antenna theory [43]. Electromagnetic potentials play an important role in quantum mechanics [44], although the question remains open whether the potentials are an indispensable part of the theory [45–47]. Nevertheless, as will be discussed in Section 1.3, the vector and scalar potentials appear in the Schrödinger equation describing the interaction of a charged quantum particle with the electromagnetic field. Hence, in the context of numerical methods, there is an interest in developing techniques for solving the Maxwell equations using potentials instead of fields as the unknowns, as was done in [38, 48]. Moreover, the use of potentials has been proposed to address the low frequency breakdown in integral equation methods [49, 50].

Being able to solve the Maxwell equations is essential for the design of antennas [43], safety assessment related to exposure of human tissue to electromagnetic radiation in various environments [51], analysis of signal propagation in wireless communication systems [52], and in numerous other applications.

1.3 The Schrödinger Equation

The quantum state of a charged particle, such as an electron or a proton, is governed by the Schrödinger equation, which for a spinless non-relativistic particle in the electromagnetic field is given by [53, 54]

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi, \quad (1.14)$$

where $i = \sqrt{-1}$, $\hbar$ is the reduced Planck's constant, $\psi(x, y, z, t)$ is the complex-valued wavefunction that allows computing the probability of finding the particle in a given location, as well as other quantities. The operator $\hat{H}$ is the Hamiltonian operator given by [53, 54]

$$\hat{H} = \frac{1}{2m} \left(\frac{\hbar}{i} \nabla - q\vec{A} \right) \cdot \left(\frac{\hbar}{i} \nabla - q\vec{A} \right) + q\phi + U, \quad (1.15)$$

where m is the mass of the particle, $U(x, y, z)$ is the confinement potential measured in the units of energy. The potentials $\phi(x, y, z, t)$ and $\vec{A}(x, y, z, t)$ are associated with the electromagnetic field created by external sources and by the charged particle itself [55]. The gauge condition on the potentials only affects the phase of the wavefunction. In particular, replacing ϕ and $\vec{A}$ with ψ' and $\vec{A}'$ in (1.10a)–(1.10b) corresponds to the following change in ψ [54]

$$\psi' = \psi e^{i \frac{q\Gamma}{\hbar}}. \quad (1.16)$$

Many of the important quantities, such as the probability density $|\psi|^2$ or the expectation values of position or mechanical momentum do not change with the transformation (1.16) and are thus gauge-invariant [56]. For a detailed treatment of the implications associated with the transformations (1.10a)–(1.10b) and (1.16), the Reader is referred to [56].

Here, the Schrödinger equation (1.14) and the corresponding Hamiltonian (1.15) are written for a single charged particle in presence of an electromagnetic field. However, the Schrödinger equation can be used to describe the quantum behavior of other entities, such as a molecule [57, 58], a semiconductor crystal [31], an LC oscillator circuit [59], or the electromagnetic field [60].

Solving the Schrödinger equation is important for understanding the underlying mechanisms of chemical reactions [57, 61], parasitic quantum effects such as leakage current in MOS transistors [62], and operation of various devices, such as quantum dot-based qubits [63], carbon nanotube field-effect transistors [64, 65], and single-electron transistors [66].

The Maxwell equations and the Schrödinger equation are mutually coupled, since the moving charged particle affects the electromagnetic field via the contribution to the charge density ρ_v in (1.1c) and the current density $\vec{J}$ in (1.1b). These contributions are given by [38, 53]

$$\rho_{v,q} = q\psi^*\psi, \quad (1.17)$$

$$\vec{J}_q = \frac{q}{m} \text{Re} \left\{ \psi^* \left(\frac{\hbar}{i} \nabla - q\vec{A} \right) \psi \right\}, \quad (1.18)$$

where the asterisk $*$ denotes the complex conjugate. The values of $\rho_{v,q}$ and $\vec{J}_q$ do not depend on the gauge. The charge density contribution (1.17) and the current density contribution (1.18) obey

the corresponding continuity equation by construction [53]

$$\frac{\partial \rho_{v,q}}{\partial t} = -\nabla \cdot \vec{J}_q. \quad (1.19)$$

Often, quantum mechanical models of charged particles do not take into account the field created by the particle due to $\rho_{v,q}$ and $\vec{J}_q$. In such simplified models [37, 67], the potentials $\vec{A}$ and ϕ in (1.15) describe only the externally applied electromagnetic field. When no external electromagnetic field is present, the simplification results in the well-known form of the Schrödinger equation that can be found in quantum mechanics textbooks [2]

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + U\psi. \quad (1.20)$$

Nevertheless, accounting for the feedback effect of the particle on the electromagnetic field [38, 55, 68, 69] is preferred from the standpoint of accuracy. For example, Takeuchi, Ohnuki, and Sako [55] demonstrated via hybrid Maxwell-Schrödinger FDTD simulations that neglecting the feedback effect can lead to erroneous predictions when designing control pulses to alter the quantum state of an electron in a narrow tube. The inaccuracy introduced by the simplification became more pronounced when the cross section of the tube was made smaller. Equations (1.1a)–(1.1d), (1.8), (1.9), (1.14), (1.15), (1.17), (1.18), along with constitutive relations such as (1.5)–(1.6) and gauge conditions such as (1.11), correspond to a semi-classical model of light-matter interaction, where the field is treated classically and the particle is treated quantum mechanically [54]. A fully quantum mechanical model would involve quantization of the electromagnetic field [54, 60, 70].

The effect of the electromagnetic field on the particle is manifested in the presence of the potentials ϕ and $\vec{A}$ in the Hamiltonian (1.15). Hence, the Maxwell equations written in terms of potentials can be used to describe the electromagnetic field directly in terms of the unknowns that appear in the Schrödinger equation. In [38], this approach was used to simplify an FDTD formulation of the Maxwell-Schrödinger system of equations.

1.4 Numerical Solution Approaches

Numerous techniques have been proposed for computing solutions to PDEs. The variability in the approaches comes from the way one treats the temporal and spatial derivatives. In the following discussion these approaches are discussed in detail.

1.4.1 Treatment of the Time Derivatives and Temporal Discretization

A large class of methods is formed by considering solutions that oscillate in time with a specific angular frequency ω

$$f(x, y, z, t) = g(x, y, z) e^{\pm i\omega t}. \quad (1.21)$$

The assumption of the form (1.21) allows replacing temporal derivatives with multiplication by $\pm i\omega$. In methods for the Maxwell equations this approach gives rise to frequency-domain methods, such as [71–73]. For the Schrödinger equation, assumption (1.21) yields the time-independent Schrödinger equation with the corresponding eigenfunctions as solutions [2]. Different numerical methods have

been proposed for solving the Schrödinger equation in its time-independent form [74–76]. An underlying assumption when restricting one’s attention to solutions of the form (1.21) is that a linear combination of these solutions is also a solution to the PDE, which is true only if the model is linear. Nevertheless, research has been done to extend frequency-domain analysis to nonlinear differential equations where this property does not hold [77, 78].

Time-domain methods [33, 36, 38, 79] typically involve defining a discrete set of temporal samples t_n and using them to approximate the derivatives with respect to time. For example, consider a linear system of m first order differential equations

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \quad (1.22)$$

where $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{R}^m$ is a time-dependent vector of unknowns and $\mathbf{A} \in \mathbb{R}^{m \times m}$ is a matrix that could result, for example, from performing the spatial discretization of a PDE. By partitioning the time into uniform intervals as $t_n = n\Delta t$, the samples of $\mathbf{x}$ at time t_n and t_{n+1} can be used to define a discrete time derivative at t_n [80]

$$\frac{\mathbf{x}^{n+1} - \mathbf{x}^n}{\Delta t} = \mathbf{A}\mathbf{x}^n \quad (1.23)$$

in a method known as the forward Euler method. Alternatively, in the backward Euler method, these two samples can be used to approximate the derivative at t_{n+1} [80]

$$\frac{\mathbf{x}^{n+1} - \mathbf{x}^n}{\Delta t} = \mathbf{A}\mathbf{x}^{n+1}. \quad (1.24)$$

Starting with some initial conditions $\mathbf{x}^0$, (1.23) or (1.24) can be solved recursively to obtain the vector of unknowns $\mathbf{x}^{n+1}$ from the previous value $\mathbf{x}^n$. The forward Euler method is an example of an explicit scheme, where only the discretization of the time derivative in (1.22) involves the unknown sample $\mathbf{x}^{n+1}$, whereas the discretization of the right hand side of (1.22) only involves earlier samples. As a result, no matrix inversion is required to obtain $\mathbf{x}^{n+1}$ from previously computed samples. An important drawback of the forward Euler scheme is the ability of the algorithm to amplify error, known as instability. For systems with decaying solutions the forward Euler method can be made stable by choosing a sufficiently small time step [80]. However, for systems where solutions do not decay with time, the algorithm is unstable for any time step [80]. A known example of this is the forward Euler method applied to discretizing the Schrödinger equation [81]. The backward Euler method is an implicit scheme, requiring a system of m equations to be solved at each time step. In contrast to the forward Euler method, the backward Euler method is stable for any time step, even if the solutions of (1.22) do not decay to zero [80].

Under some continuity assumptions, the local truncation error on the computation of $\mathbf{x}^{n+1}$ using the forward Euler (1.23) and the backward Euler (1.24) methods decays with the square of the time step [80]. Such methods are said to be first-order¹ [80]. Achieving a faster decay of the local truncation error is possible through the use of Crank-Nicolson method [82], which involves

¹The order can also be understood as the rate of decay of the error associated with the approximation of the derivative that is used to construct the scheme [35]. For the case of the first-order accurate forward and backward Euler methods, this error decays linearly with the time step.

discretizing (1.22) using the so-called trapezoidal rule [80]

$$\frac{\mathbf{x}^{n+1} - \mathbf{x}^n}{\Delta t} = \mathbf{A} \frac{\mathbf{x}^{n+1} + \mathbf{x}^n}{2}. \quad (1.25)$$

The method is unconditionally stable [80, 81, 83], similarly to the backward Euler scheme, while the local truncation error associated with the Crank-Nicolson scheme decays with the third power of the time step, which is a property of second-order methods [80]. These qualities make the Crank-Nicolson scheme a popular method of choice in circuit simulators [80]. Nevertheless, this scheme is implicit and hence requires the solution of a system of equations at every time step, which can incur a high computational requirements. Research has been done in alleviating the computational cost of the update procedure in Crank-Nicolson schemes [83, 84].

The forward Euler, the backward Euler, and the Crank-Nicolson methods belong to the class of Runge-Kutta schemes, which only use the sample $\mathbf{x}^n$ to compute $\mathbf{x}^{n+1}$ [80]. In contrast, a more general class of methods, known as linear multistep methods, can involve earlier samples in the update procedure [80]. An example of a linear multistep method is the following scheme [26, 81]

$$\frac{\mathbf{x}^{n+1} - \mathbf{x}^{n-1}}{2\Delta t} = \mathbf{A} \mathbf{x}^n, \quad (1.26)$$

which approximates the derivative in (1.22) using a central difference. This scheme is explicit but, unlike the forward Euler method, it is a second-order method with a local truncation error that decreases with a cube of the time step [80]. The improved accuracy is a result of the central difference used to approximate the time derivative in (1.22). If the eigenvalues of $\mathbf{A}$ are imaginary, this scheme can be made stable by choosing a sufficiently small time step [26, 81, 85].

Another popular class of time-domain methods are the leap-frog update schemes [33, 36] for equations that can be written in the form

$$\frac{d\mathbf{x}_1}{dt} = \mathbf{A}_{12}\mathbf{x}_2, \quad (1.27a)$$

$$\frac{d\mathbf{x}_2}{dt} = \mathbf{A}_{21}\mathbf{x}_1, \quad (1.27b)$$

where the time derivative of $\mathbf{x}_1$ depends on $\mathbf{x}_2$ and vice versa. For example, the form (1.27a)–(1.27b) is applicable as is to the spatially discretized Maxwell curl equations (1.1a) and (1.1b) in the absence of current² [35]. With the leap-frog approach, $\mathbf{x}_1$ is sampled at integer time points $t = n\Delta t$, denoted as $\mathbf{x}_1^n$, and $\mathbf{x}_2$ is sampled at time points shifted by $\Delta t/2$, namely $\mathbf{x}_2^{n-0.5}$. With these samples, (1.27a)–(1.27b) are both discretized using the central differences as

$$\frac{\mathbf{x}_1^{n+1} - \mathbf{x}_1^n}{\Delta t} = \mathbf{A}_{12}\mathbf{x}_2^{n+\frac{1}{2}}, \quad (1.28a)$$

$$\frac{\mathbf{x}_2^{n+\frac{1}{2}} - \mathbf{x}_2^{n-\frac{1}{2}}}{\Delta t} = \mathbf{A}_{21}\mathbf{x}_1^n. \quad (1.28b)$$

As was pointed out in [26] for FDTD-type schemes for the Schrödinger equation, (1.28a)–(1.28b) is equivalent to (1.26) when applied to equations of the form (1.27a)–(1.27b). Leap-frog methods

²The Maxwell equations with nonzero current can also be discretized using the leap-frog approach with some modifications, as described in [35] and Section 1.5.

involve explicit computations and are conditionally stable, similarly to other explicit methods such as forward Euler or the second-order scheme (1.26). The stability analysis methods and results are reviewed in detail later for the case of FDTD-type methods for the Maxwell equations and for the Schrödinger equation.

1.4.2 Spatial Discretization

Different approaches exist for discretizing the unknowns in space and for writing the spatial derivatives in the PDEs. Common approaches include finite element methods (FEMs) [79, 86–90], finite difference methods [33, 36, 91], and spectral methods [92–95].

In FEM and spectral approaches, the discretization is achieved by approximating the unknowns as a linear combination of basis functions. As an example, consider the time-independent Schrödinger equation in one dimension [2]

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U(x)\psi(x) = E\psi(x), \quad (1.29)$$

with suitable boundary conditions. Constant E in (1.29) is the energy, known as eigenenergy, for which there is an eigensolution $\psi(x)$ satisfying (1.29) and the boundary conditions. The eigenenergies are to be determined, along with the corresponding eigensolutions. With the finite element method [86, 87] or spectral method [93], the unknown ψ is written as a linear combination of basis functions

$$\psi(x) = \sum_{m=0}^M \psi_m B_m(x). \quad (1.30)$$

One would then construct $M + 1$ linear equations using (1.29), forming a system [86, 87]

$$\mathbf{H}\boldsymbol{\psi} = E\mathbf{S}\boldsymbol{\psi}, \quad (1.31)$$

where $\boldsymbol{\psi}$ is a vector of coefficients ψ_m in (1.30) and $\mathbf{H}$ and $\mathbf{S}$ are matrices with elements that can be computed from the predefined basis functions $B_m(x)$. One can then solve the generalized eigenvalue problem (1.31) [96] to determine the values of the possible energy levels E and the corresponding eigenfunctions $\psi(x)$. The difference between finite element and spectral methods is in the type of basis functions used in (1.30) [92]. In spectral methods the basis functions typically encompass the entire domain, such as the sinusoidal functions in [93]. The efficiency of the approach depends on how well the basis functions represent solutions of a PDE. In contrast, the FEM approach utilizes local basis functions and does not require making predictions on the global spatial behavior of the solution [86].

In finite-difference techniques, the spatial derivatives are approximated via difference quotients. When used in conjunction with explicit time domain schemes, finite difference discretization avoids any need for matrix inversion. The FDTD method [33, 36] leverages this property to allow for low-cost updates, which will be discussed in detail in Section 1.5. FDTD uses central differences to obtain approximations of the derivatives that are second-order accurate [35], as discussed in Section 1.4.1. Higher order finite difference methods have also been proposed [26, 97].

Another group of spatial discretization techniques that should be mentioned are the boundary element methods [98–100] and surface integral equation formulations [101, 102]. These methods

discretize the surfaces of objects, in contrast to FEM, spectral, finite difference, which perform the discretization over the volume of the entire domain. Developing a surface-based formulation is challenging in scenarios involving general inhomogeneous objects. In such scenarios, volume integral equation methods can be used [103, 104], which similarly avoid discretizing the entire domain but require a volume-based discretization of objects. In electromagnetics, the method of moments [105] is often applied to formulations involving surface and volume integral equations. The name of the method refers to the procedure for constructing a linear system from the integral equations. The aforementioned approaches can lead to significant computational savings in some multi-scale problems. For example, they are known to be advantageous in antenna simulations, as they avoid the need to discretize the large amount of empty space in the domain [20, 106].

1.4.3 Challenges of Common Numerical Methods

Each of the methods outlined in Sections 1.4.1 and 1.4.2 has its limitations. For example, FEM methods, in general, require the solution of systems of equations [86, 90]. Although the matrices involved tend to be sparse, the FEM can still incur long simulation times when applied to complex multi-scale problems. Explicit FEM methods have been proposed in the time domain. For a detailed discussion on explicit and implicit time-domain FEM methods in electromagnetics, the Reader is referred to [79]. Integral equation methods often involve dense matrices, which can lessen the computational savings gained from a surface-based formulation, even when paired with acceleration algorithms. The application of the integral equation methods to broadband simulations is also known to be challenging due to the phenomenon known as the low-frequency breakdown [107]. In FDTD, the efficiency of the method is curtailed by the stability limit on the time step, which becomes particularly restrictive when small geometrical features impose a fine spatial discretization [35, 108]. Moreover, the spatial sampling in FDTD is also less flexible than in methods such as the FEM, which typically use a conformal spatial discretization. This makes the FDTD prone to the so-called stair-casing error when applied to problems with curved boundaries. Research is ongoing in addressing these challenges [50, 109–114].

Commercial simulation tools are available based on different numerical techniques. The preferred technique is largely dependent on the problem, which is why some software tools, such as FEKO [115] and CST [116] provide access to multiple numerical methods. Table 1.1 outlines some of the commercial tools and the underlying numerical methods used for solving the Maxwell equations and the Schrödinger equation. Moreover, the open source package Meep (MIT Electromagnetic Equation Propagation) for solving the Maxwell equations using FDTD [117] has received substantial recognition.

Table 1.1: Some examples of commercial tools for solving the Maxwell equations and the Schrödinger equation.

Software	Main solution method
Maxwell equations	
SEMCAD X [118]	FDTD
Ansys Lumerical [119]	FDTD
Ansys HFSS [120]	FEM ^a
FEKO [115]	Method of moments and others
COMSOL [121]	FEM
CST Studio Suite [116]	Multiple methods, including FEM and FIT ^b
Sonnet Suites [122]	Method of moments
Schrödinger equation	
COMSOL [121]	FEM
QTCAD [123]	FEM
Q-Chem [124] ^c	Spectral method
NanoDCAL [125] ^c	Spectral method
Jaguar [126] ^c	Spectral method

^a HFSS also uses the method of moments.

^b FDTD can be thought of as a special case of the finite-integration technique (FIT) [127].

^c Solve an auxiliary single-electron Schrödinger equation in the context of density functional theory [128, 129] and Hartree-Fock theory [31] for modeling systems of multiple electrons and nuclei.

1.5 The Finite-Difference Time-Domain Approach

The FDTD method for solving the Maxwell equations [33–35] discretizes (1.1a)–(1.1b). The discretization is applied to the following form of the two Maxwell curl equations

$$\epsilon \frac{\partial \vec{E}}{\partial t} + \sigma \vec{E} = \nabla \times \vec{H} - \vec{J}_s, \quad (1.32a)$$

$$\mu \frac{\partial \vec{H}}{\partial t} = -\nabla \times \vec{E}. \quad (1.32b)$$

In order to discretize (1.32a)–(1.32b), the region is divided into cells, known as primary cells, as shown in Figure 1.1. The components of the electric field, namely E_x , E_y , and E_z are sampled at the edges of the primary grid. The centers of the primary cells are connected by the edges of the so-called secondary grid [35]. Those edges, referred to as secondary edges, intersect the primary cell faces at ninety degrees, as shown in Figure 1.1³. This way, the primary edges, where electric field components are sampled, have the four samples of magnetic field needed for the computation of the $\nabla \times \vec{H}$ in (1.32a), as evident from Figure 1.1. Likewise, the secondary grid edges have the four electric field samples needed for computing the $\nabla \times \vec{E}$ in (1.32b). The electric field is sampled at the integer time instances $t = n\Delta t$ and the magnetic field is sampled at the time instances shifted by half of a time step $t = (n - 0.5)\Delta t$. With this temporal and spatial sampling, (1.32a)–(1.32b) for

³The precise definition of the primary and secondary grids, as well as the corresponding cells, edges, and nodes will be given in Chapter 2 and Appendix A.

the E_x and the H_z components in Figure 1.1 are discretized as [35]

$$\varepsilon_x|_{i+\frac{1}{2},j,k} \frac{E_x|_{i+\frac{1}{2},j,k}^{n+1} - E_x|_{i+\frac{1}{2},j,k}^n}{\Delta t} + \sigma_x|_{i+\frac{1}{2},j,k} \frac{E_x|_{i+\frac{1}{2},j,k}^{n+1} + E_x|_{i+\frac{1}{2},j,k}^n}{2} = \frac{H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_z|_{i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_y|_{i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - J_{s,x}|_{i+\frac{1}{2},j,k}^{n+\frac{1}{2}}, \quad (1.33a)$$

$$\mu_z|_{i+\frac{1}{2},j+\frac{1}{2},k} \frac{H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n-\frac{1}{2}}}{\Delta t} = - \left(\frac{E_y|_{i+1,j+\frac{1}{2},k}^n - E_y|_{i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_x|_{i+\frac{1}{2},j+1,k}^n - E_x|_{i+\frac{1}{2},j,k}^n}{\Delta y} \right), \quad (1.33b)$$

where Δx , Δy , and Δz are the dimensions of the primary cells. The subscripts and the superscripts indicate, respectively, the spatial and temporal indices of the corresponding samples. For example, $E_x|_{i+0.5,j,k}^n$ denotes the x component of the electric field at the coordinates $x = (i - 0.5)\Delta x$, $y = (j - 1)\Delta y$, $z = (k - 1)\Delta z$ and time $t = n\Delta t$. The scalars ε_x , σ_x , and μ_z denote the effective values of permittivity, conductivity, and permeability along the corresponding edges. The equations for E_y , E_z , H_x , and H_y are written similarly to (1.33a)–(1.33b). Starting with initial conditions for $\vec{H}$ at $t = -0.5\Delta t$ and $\vec{E}$ at $t = 0$, the magnetic field can be updated explicitly using (1.33b) as [35]

$$H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n-\frac{1}{2}} - \frac{\Delta t}{\mu_z|_{i+\frac{1}{2},j+\frac{1}{2},k}} \left(\frac{E_y|_{i+1,j+\frac{1}{2},k}^n - E_y|_{i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_x|_{i+\frac{1}{2},j+1,k}^n - E_x|_{i+\frac{1}{2},j,k}^n}{\Delta y} \right). \quad (1.34)$$

Values of H_y and H_z components are updated analogously. Likewise, the electric field is updated using (1.33a) as [35]

$$E_x|_{i+\frac{1}{2},j,k}^{n+1} = \frac{\varepsilon_x|_{i+\frac{1}{2},j,k} - \frac{\sigma_x|_{i+\frac{1}{2},j,k}}{\Delta t}}{\frac{\varepsilon_x|_{i+\frac{1}{2},j,k}}{\Delta t} + \frac{\sigma_x|_{i+\frac{1}{2},j,k}}{2}} E_x|_{i+\frac{1}{2},j,k}^n + \frac{1}{\frac{\varepsilon_x|_{i+\frac{1}{2},j,k}}{\Delta t} + \frac{\sigma_x|_{i+\frac{1}{2},j,k}}{2}} \cdot \left(\frac{H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_z|_{i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_y|_{i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - J_{s,x}|_{i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right). \quad (1.35)$$

The values of samples $E_y|^{n+1}$ and $E_z|^{n+1}$ are computed analogously. The update procedure given by (1.34) and (1.35) is then repeated in a leap-frog manner.

Boundary conditions determine the update procedure on the boundary of the region. For example, if the boundary of the region represents a perfect electric conductor (PEC), a homogeneous Dirichlet boundary condition is imposed on the tangential components of the electric field. To implement this boundary condition, the electric field samples on the boundary edges are simply set to zero [35] and the rest of the samples in the region are updated using (1.34), (1.35), and the analogous equations for the remaining field components. Other boundary conditions include perfect magnetic conductor boundary [35], periodic boundary conditions [130], and absorbing boundary conditions

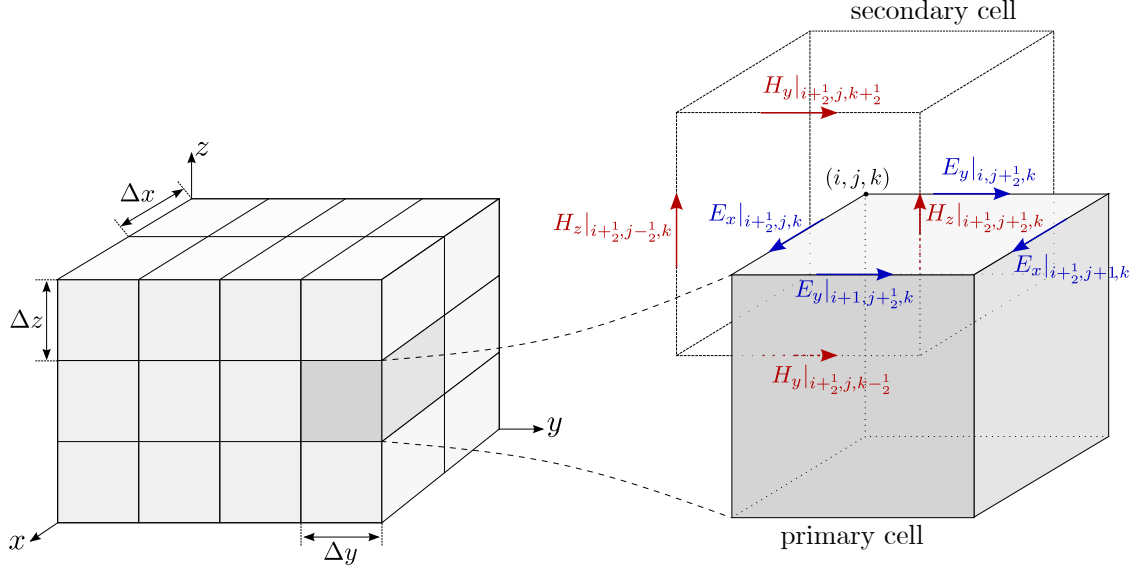


Figure 1.1: Illustration of the spatial discretization in FDTD for the Maxwell equations and the samples involved in (1.33a) and (1.33b).

modeling unbounded domains [131, 132]. One of the most commonly used types of absorbing boundary conditions is the so-called perfectly matched layer (PML), which is a layer of cells that absorbs and attenuates incident waves [133, 134].

In order to ensure that an FDTD simulation is stable, the time step needs to respect the Courant–Friedrichs–Lewy (CFL) stability limit [35]

$$\Delta t < \Delta t_{\text{CFL}} = \frac{\sqrt{\mu_{\min} \varepsilon_{\min}}}{\sqrt{\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2}}}, \quad (1.36)$$

where $\mu_{\min}$ and $\varepsilon_{\min}$ are the smallest values of permeability and permittivity in the region, respectively.

The memory complexity of the FDTD algorithm is $O(n_x n_y n_z)$, where n_x , n_y , and n_z is the number of primary cells in x , y , and z directions, respectively. The time complexity of the algorithm is $O(n_x n_y n_z n_t)$, where n_t is the number of time steps required to complete the simulation. Due to the stability limit (1.36), an improvement in the spatial discretization imposes a reduction of Δt . Hence, refining the grid by a factor of two results in an eight-fold increase in the number of operations required by the algorithm to obtain a solution over a given time interval.

The FDTD approach can also be used to discretize the Schrödinger equation (1.20) [36, 135] by rewriting it in terms of the real and imaginary parts of the wavefunction, ψ_R and ψ_I , respectively

$$\hbar \frac{\partial \psi_R}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi_I + U \psi_I, \quad (1.37a)$$

$$\hbar \frac{\partial \psi_I}{\partial t} = \frac{\hbar^2}{2m} \nabla^2 \psi_R - U \psi_R. \quad (1.37b)$$

In the FDTD method for the Schrödinger equation, the wavefunction is sampled at the primary nodes, as shown in Figure 1.2. The real part ψ_R is sampled at the integer time points and the

imaginary part at the time points shifted by $\Delta t/2$. The temporal derivatives and the Laplacian operators in (1.37a)–(1.37b) are then discretized using the central differences, yielding [108, 135]

$$\hbar \frac{\psi_R|_{i,j,k}^{n+1} - \psi_R|_{i,j,k}^n}{\Delta t} = -\frac{\hbar^2}{2m} \left(\frac{\psi_I|_{i+1,j,k}^{n+\frac{1}{2}} - 2\psi_I|_{i,j,k}^{n+\frac{1}{2}} + \psi_I|_{i-1,j,k}^{n+\frac{1}{2}}}{(\Delta x)^2} + \frac{\psi_I|_{i,j+1,k}^{n+\frac{1}{2}} - 2\psi_I|_{i,j,k}^{n+\frac{1}{2}} + \psi_I|_{i,j-1,k}^{n+\frac{1}{2}}}{(\Delta y)^2} + \frac{\psi_I|_{i,j,k+1}^{n+\frac{1}{2}} - 2\psi_I|_{i,j,k}^{n+\frac{1}{2}} + \psi_I|_{i,j,k-1}^{n+\frac{1}{2}}}{(\Delta z)^2} \right) + U|_{i,j,k} \psi_I|_{i,j,k}^{n+\frac{1}{2}}, \quad (1.38a)$$

$$\hbar \frac{\psi_I|_{i,j,k}^{n+\frac{1}{2}} - \psi_I|_{i,j,k}^{n-\frac{1}{2}}}{\Delta t} = \frac{\hbar^2}{2m} \left(\frac{\psi_R|_{i+1,j,k}^n - 2\psi_R|_{i,j,k}^n + \psi_R|_{i-1,j,k}^n}{(\Delta x)^2} + \frac{\psi_R|_{i,j+1,k}^n - 2\psi_R|_{i,j,k}^n + \psi_R|_{i,j-1,k}^n}{(\Delta y)^2} + \frac{\psi_R|_{i,j,k+1}^n - 2\psi_R|_{i,j,k}^n + \psi_R|_{i,j,k-1}^n}{(\Delta z)^2} \right) - U|_{i,j,k} \psi_R|_{i,j,k}^n. \quad (1.38b)$$

Given the initial conditions on $\psi_R|_0$ and $\psi_I|^{-0.5}$, (1.38a)–(1.38b) are used to update the real and imaginary parts of the wavefunction in a leap-frog manner, analogously to (1.33a)–(1.33b) for the Maxwell equations. In this thesis, the scheme involving the leap-frog updates (1.38a)–(1.38b) will be referred to as “FDTD-Q” (i.e. quantum FDTD). The update procedure for the wavefunction samples on the boundary of the region is determined by the boundary conditions. Dirichlet [108, 135], periodic [136, 137], or absorbing [26, 135, 138, 139] boundary conditions have been used in FDTD and other numerical methods for the Schrödinger equation.

In order to ensure that the updates (1.38a)–(1.38b) are stable, the time step needs to be below the CFL stability limit given by [140]

$$\Delta t < \Delta t_{\text{CFL}} = \frac{2}{\frac{2\hbar}{m} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right) + \frac{|U|_{\text{max}}}{\hbar}}, \quad (1.39)$$

where $|U|_{\text{max}}$ is the largest absolute value of the potential in the region⁴.

The application of the scheme (1.38a)–(1.38b) has been demonstrated in [37] by simulating two electrons in a quantum dot and in [135] it was used to develop a technique for finding the eigenfunctions of arbitrary nanoscale structures. The scheme was also used for modeling anti-reflective coating [143] and electron diffraction through a double slit [144].

The FDTD approach has also been applied to the Maxwell equations written in terms of potentials. In [38, 48, 145], the potentials-based FDTD methods facilitated the formulation of discretized Maxwell-Schrödinger systems of equations, which were used to model the interaction between a quantum particle and electromagnetic field. Potentials-based formulations have also been proposed as means of reducing the computational cost in FDTD [146, 147]. For example, in [146] it was argued that the memory requirements can be reduced by computing each component of $\vec{A}$ from (1.13a) separately and obtaining ϕ , $\vec{H}$, and $\vec{E}$ as a post-processing step. Chapter 3 will provide more detail on one of the existing FDTD discretization methods [38], which is applicable to lossless

⁴In some sources [108, 139, 141], the CFL limit (1.39) is given with an extra factor of 2 in the second term of the denominator, giving a smaller time step. However, references [26, 138, 140, 142] agree that the stability limit (1.39) is sufficient, which is also corroborated by the derivations in Chapter 4 of this thesis.

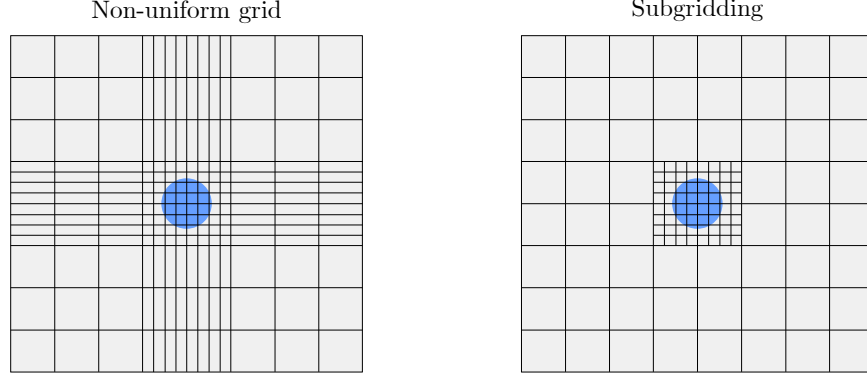


Figure 1.3: Distinction between the nonuniform discretization and subgridding approaches.

When using non-uniform grids or subgridding, the time step needs to be below the CFL limit associated with the finest spatial resolution in the simulation domain [26, 35, 155]. As a result, in problems with fine geometrical features the stability limit on the time step can be much smaller than the temporal resolution needed to sample the waveforms⁵. Different approaches have been proposed to address this issue, such as subgridding schemes that update the coarse and fine grids at different time steps [22, 23, 25, 156, 157]. Other works proposed to use unconditionally stable implicit methods to update unknowns in the refined regions of the simulation domain [112, 158]. The use of implicit schemes unavoidably increases the computational cost of individual updates. However, depending on the problem at hand, this approach can be justified, as it allows using a larger time step. Methods such as alternating direction implicit (ADI) FDTD [159, 160] or related hybrid methods [161] aim at minimizing the computational overhead associated with the use of implicit updates. Other techniques to alleviate restrictive stability limits in multi-scale problems

⁵To illustrate this, consider a simulation scenario involving electromagnetic waves with free-space wavelength λ_0 and sub-wavelength geometrical features requiring spatial resolution of Δ_0 . From (1.36), the stability limit becomes

$$\Delta t < \Delta t_{\text{CFL}} = \Delta_0 \frac{\sqrt{\mu_0 \varepsilon_0}}{\sqrt{3}}, \quad (1.40)$$

where it was assumed that $\mu_{\min}$ and $\varepsilon_{\min}$ are those of free space. The time step required for accurate sampling of the waveform is governed by the period of the waveforms, which for the waves of wavelength λ_0 is given by [1]

$$T = \sqrt{\mu_0 \varepsilon_0} \lambda_0. \quad (1.41)$$

Hence,

$$\frac{\Delta t_{\text{CFL}}}{T} = \frac{1}{\sqrt{3}} \frac{\Delta_0}{\lambda_0}. \quad (1.42)$$

If the scale of the geometrical features requires Δ_0 to be significantly smaller than the wavelength λ_0 , the waveforms would be over-sampled in time, incurring significant computational cost.

Similarly, we can repeat the calculation for FDTD discretization of the Schrödinger equation. Equation (1.39) reduces to

$$\Delta t < \Delta t_{\text{CFL}} = \frac{2}{\frac{2\hbar}{m} \left(\frac{3}{\Delta_0^2} \right) + \frac{|U|_{\max}}{\hbar}} = \frac{m}{3\hbar} \Delta_0^2, \quad (1.43)$$

assuming the potential U is 0 (or smaller if $U \neq 0$). For a plane wave solution with wavelength λ_0 the corresponding temporal period is given by [2]

$$T = \frac{m}{\pi\hbar} \lambda_0^2. \quad (1.44)$$

Hence, the ratio of the time step and the period of characteristic waveforms is

$$\frac{\Delta t_{\text{CFL}}}{T} = \frac{\pi}{3} \left(\frac{\Delta_0}{\lambda_0} \right)^2. \quad (1.45)$$

which has a quadratic dependence on the ratio of spatial step to wavelength.

include spatial filtering [113], eigenvalue perturbation [111, 162], and removal of unstable modes [163, 164].

It should also be noted that in the case of FDTD-Q, the dependence of the CFL limit (1.39) on the spatial step is quadratic. As a result, temporal over-sampling can occur even without the presence of small sub-wavelength features. For instance, based on (1.45), resolving a plane wave solution with 30 cells per wavelength would require 859 temporal samples per period, which is far larger than would be needed to accurately resolve a sinusoidal waveform. This motivates the use of higher order schemes [26, 165], which allow increasing the time step by using a coarser spatial discretization.

An important research direction in FDTD is the creation of hybrid techniques. Examples include hybridizations with the FEM [166], the method of moments [106], ray tracing [29], as well as the implicit schemes mentioned earlier [112, 158, 161]. These methods aim to leverage the advantages of FDTD, while applying a more suitable numerical scheme for specific aspects of the problem at hand, such as the thin-wire antenna in [106] that can be modelled more effectively using the method of moments.

Another active topic of research in FDTD involves the extension of the method’s modeling capabilities. These include support for different types of media [167, 168], simulations of metasurfaces [169], periodic structures [170], and so on. Multi-physics methods have also attracted significant attention in both FDTD and other numerical schemes for Maxwell and Schrödinger equations. For example, in [171] a plasmon-molecule interaction was modeled using the Maxwell equations for the plasmon coupled to the von Neumann equation describing the quantum mechanics of the molecule. In [172], microwave rectifiers with Schottky diodes were simulated using the Maxwell equations coupled to the drift-diffusion model describing the diodes. As was mentioned earlier, in [38, 48, 145], FDTD-based methods for modeling interaction between electromagnetic field and a quantum particle were proposed using the potentials-based Maxwell equations for the field and the Schrödinger equation for the quantum particle.

1.6 Existing Stability Analysis Approaches in FDTD

Despite the substantial progress in the development of the FDTD method, stability remains a challenging aspect in the creation of new schemes. Some of the methods proposed in the literature are known to suffer from instability [173, 174]⁶. In other methods, stability has been achieved using a trial-and-error approach to determine the time step and other parameters of the method for which the instabilities cease to manifest [22, 176, 177]. Unfortunately, the divergent behavior can depend on the scenario being simulated. Moreover, it may become apparent after many time steps: a phenomenon known as *late-time instability*. Hence, the trial-and-error approach can never be used to guarantee stability for a general simulation scenario. Many provably stable FDTD-based methods have been proposed in the past [26, 154, 155, 157, 161, 178–181]. However, as this section argues, a rigorous, convenient, and physically intuitive framework that could be used for creating complex stable FDTD schemes was not available prior to the work proposed in this thesis, apart from

⁶According to the discussion in [175], the type of instability studied in [173] does not constitute a genuine instability due to the temporal growth being linear as opposed to exponential. Nevertheless, the growing behavior is an undesirable numerical artifact that does not represent the true physics of the problem.

the preliminary developments in two dimensions [182]. This section reviews the stability analysis approaches that have been used in the literature in FDTD for the Maxwell and Schrödinger equation.

1.6.1 Von Neumann Analysis

Approaches related to von Neumann analysis [183] (also known as Fourier analysis) are commonly used for deriving stability conditions in numerical methods, including FDTD for Maxwell [34, 113, 184, 185] and Schrödinger [36, 38, 108, 138, 186] equations. These approaches involve assuming a specific form of the solution to the discretized equations, for example a plane wave solution [34, 108]. A complex scalar growth factor is determined, which describes the temporal evolution of the assumed solution. The stability condition is then found by imposing that the magnitude of the growth factor does not exceed one.

This class of methods is meant for simple scenarios involving uniform discretization, homogeneous materials and, for the case of the Schrödinger equation, constant potential profile. Some exceptions exist, such as [187, 188], where von Neumann analysis was used to analyze schemes for non-uniform grids. However, these works do not take into account the fact that the growth factors for nonuniform grids depend on grid size, which in turn depends on position. Often, derivations based on von Neumann-type analysis methods generalize the stability limit to scenarios such as inhomogeneous media and others by taking the worst-case values of the CFL limit in the region [138, 184]. However, to the best of our knowledge, no rigorous argument has been given in the literature that the worst-case time step computed via von Neumann analysis provides a valid stability limit in cases where the underlying assumptions of the analysis are violated.

1.6.2 Iteration Matrix Method

A more general approach [26, 35, 127, 163, 189–191] involves analyzing the eigenvalues of a matrix associated with the update procedure. In particular, when the governing partial differential equations are discretized using one’s method of choice, one can formulate a matrix expression relating the vector of unknowns to its value at the previous time step. Assuming the discretized equations are linear, they can ultimately be written in the form

$$\mathbf{x}^{n+1} = \mathbf{A}\mathbf{x}^n. \quad (1.46)$$

With that, the stability enforcement entails verifying that the iteration matrix $\mathbf{A}$ has all eigenvalues inside or on the unit circle. For the eigenvalues that lie on the unit circle, the algebraic multiplicity must equal the geometric multiplicity in order to avoid polynomially growing solutions [192]. The iteration matrix method can be applied to much more complicated cases than von Neumann analysis. However, there are scenarios that are challenging for this method to handle. For example, consider analyzing a scenario where several different types of equations are coupled, as illustrated in Figure 1.4. Many of the advanced techniques in Section 1.5.1 involve such coupled equations. The simplest example would be a subgridding scheme [22, 23, 25, 154, 155, 157, 193, 194], where the FDTD equations of two grids are coupled via interpolation rules on the subgridding boundary. More advanced examples of multi-component scenarios could include macromodels [111, 176, 195] or circuit elements [152, 196, 197] incorporated in an FDTD grid, or hybridizations with other numerical methods [106, 112, 158, 166]. FDTD-based schemes for solving multi-physics problems can also

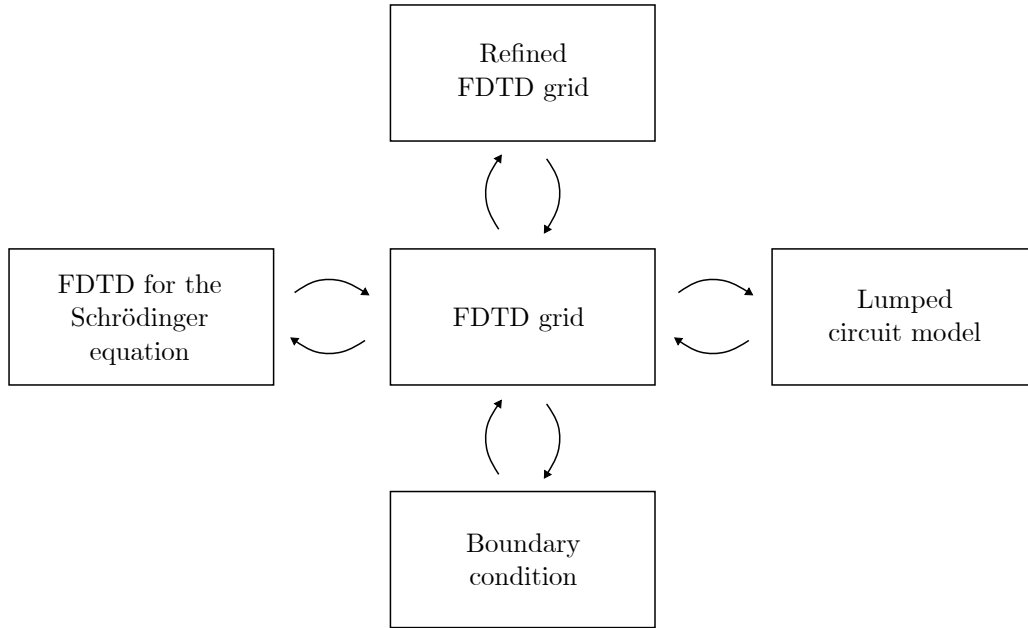


Figure 1.4: Advanced FDTD schemes created by coupling different equations.

comprise of coupled systems of discretized time-domain equations [38, 145, 171]. In such coupled scenarios, the analysis of eigenvalues of the iteration matrix $\mathbf{A}$ becomes more cumbersome, since the matrix would contain blocks corresponding to each of the models and the equations coupling those models. Numerical calculation of eigenvalues of $\mathbf{A}$ is possible [190]. However, this would be computationally taxing for large problems and would provide little insight into which equations cause the instability. In [154] and [157], the iteration matrix method was used to show that the proposed subgridding schemes (spatial and spatiotemporal, respectively) are stable for a sufficiently small time step. However, the value of the stable time step that could be used in practice had to be determined experimentally. It would be therefore advantageous to have stability conditions on individual components of the system that could be verified and enforced independently.

1.6.3 Energy-based Approaches and Conservation Laws

Another approach to study stability is to construct an energy function associated with a time-domain scheme and to study the evolution of this function over the temporal iterations. Even though this approach draws inspiration from the concept of energy conservation, its application [140, 198, 199] is frequently physics-agnostic, with the functions in question defined as norms that do not necessarily have to correspond to the physical energy of the system being modeled. In [198] the approach yielded stability conditions on the linear operators describing second order finite difference equations. Those conditions were later adapted to numerical methods in electromagnetics [200], including FDTD [178, 179]. In [140], the stability limit (1.39) for the FDTD-Q scheme was derived for a general case of spatially varying potential U . Temporally varying U was also considered for the one-dimensional case. The norm function in question, whose time evolution was investigated,

quantified the error between two solutions stemming from errors in the initial conditions⁷. Stability arguments based on physically meaningful definitions of numerical energy in electromagnetics have also been presented [23, 196, 197, 201, 202] for FDTD and the finite-integration technique (FIT), which is a closely related method.

Of the aforementioned works, [23, 178, 179, 196, 197, 201] have analyzed multi-component FDTD-based schemes. In [23, 201], a criterion was proposed to ensure that a subgridding scheme does not violate the principle of energy conservation for the case of discrete-space continuous-time semi-discretization of the Maxwell equations. A subgridding scheme satisfying the aforementioned criterion was proposed [23]. However, it remained unclear how the temporal discretization affects the energy conservation and stability properties of the scheme. As a result, a stable time step had to be determined experimentally. Likewise, in [154] the proposed subgridding method was shown to be energy-conserving prior to the temporal discretization, but the stability of the time marching scheme had to be analyzed using the iteration matrix method, which did not yield a provable expression for the stable time step as per discussion in Section 1.6.2. In [178, 179], a requirement for the interpolation condition similar to [23, 201] was derived based on the results in [200]. In [179], the time step was taken as the smallest CFL limit of the two grids. However, no proof was provided that the stability limit of the subgridding scheme is the same as that of an isolated region discretized by FDTD. In general, stability limit of a scheme composed from coupled time-domain models can differ from that of the uncoupled models. For instance, in the spatiotemporal subgridding scheme in [23], a smaller time step than CFL limit of individual FDTD regions had to be used.

In [197] the role of the time step in the numerical energy conservation was studied and it was found that the CFL limit ensures that the energy associated with the region is non-negative, which is an essential property of the physical energy associated with the electromagnetic field that formed the basis for the stability analysis in [197]. In addition to the basic FDTD scheme with PEC boundaries, the authors of [197] study the case when lumped elements are incorporated into the scheme. By modifying the stability and energy conservation proof, it is found that the stable time step is unchanged. Whereas the modifications required to the proof were fairly simple, the question remained whether the analysis of more complicated schemes, such as subgridding [23, 201], could be conveniently performed with the method in [197]. Moreover, any subsequent modification to the schemes studied in [197] would require revising the stability proof. Ideally, a scheme that is composed of multiple coupled models should only require stability conditions to be enforced once for each model and each set of coupling equations. One step was made in this direction in [197] by arguing that the limit on the time step can be derived by considering the contribution of each primary cell to the total energy. However, a general framework where the energy conservation and stability properties are analyzed separately for each part of the system was still missing. In [196], stability analysis technique for FDTD with linear and nonlinear lumped elements and resistive voltage sources was proposed. With that technique, it was possible to study the stability properties of the lumped elements in a modular fashion by ensuring that they dissipate numerical energy. Reference [196] also contains a brief discussion on how absorbing boundary conditions could be handled using the proposed approach. However, those developments remained to be extended to other complex FDTD methods.

⁷Interestingly, if, instead of the error in the wavefunction, the derivations considered the discretized wavefunction itself, the analyzed norm could be interpreted as the total probability. Hence, as will become more clear in Chapter 4, there are connections between the work in [140], the numerical probability conservation concepts in [36], and the theory proposed in this thesis.

Moreover, the time step derived using the technique in [196] was smaller than the conventional CFL limit (1.36). In [178], some modularity of stability analysis was achieved in deriving a stable time step bound for subgridding schemes with reduced macromodels. The approach was to use a triangle inequality to estimate the norm of a matrix composed of contributions from the uncoupled models (an FDTD region and a reduced macromodel) and the coefficients coupling those models. Unfortunately, a triangle inequality can only provide an approximation of the true norm of a matrix.

In the field of circuit modeling [203, 204] and control theory [205], it is understood that stable systems can be created by properly combining energy-conserving or energy-dissipating sub-systems. Thus, the energy conservation concepts can provide a way to ensure the stability of a coupled system by studying the properties of individual components. As was discussed earlier, this is a much needed capability when analyzing the stability of new FDTD-type methods. Previously, this concept has been applied to develop a modular stability framework for FDTD in two dimensions [182]. However, many practical problems are three-dimensional in nature, calling for a generalized theory that can be applied to 3D FDTD. A modular stability framework such as this would allow FDTD simulation tools to seamlessly incorporate general user-defined models, which can be verified for dissipativity in a stand-alone manner, giving those tools the flexibility of circuit simulators. Moreover, in light of the interest in modeling of quantum systems, the question arises whether these concepts can be extended to PDEs describing quantum phenomena. For instance, in [36], it was argued that the basic FDTD-Q scheme in one dimension conserves probability. Hence, in principle, the probability conservation could be used as a basis for stability framework for FDTD-Q-type schemes.

1.7 Thesis Goals

The goal of this thesis is to answer the following questions:

1. Can the concepts of energy conservation and dissipation from control theory [205, 206] and the prior investigations in 2D FDTD [182] be used to create a modular framework for stability analysis and enforcement in 3D FDTD, where the analysis would be performed individually on each component of the system? In particular,
 - (a) for a region of space discretized via FDTD equations, can we create a self-contained model with clearly defined stored energy and supply rate, which quantifies the rate at which the region exchanges energy with the surrounding space [205, 206];
 - (b) can we find conditions under which the region conserves or dissipates energy, which can be verified and enforced a priori;
 - (c) is it possible to create new FDTD methods by combining different models in an energy-conserving manner?
2. Can the proposed approach be extended to the FDTD method based on potentials [38]?
3. Can the proposed approach be extended to the FDTD method for the Schrödinger equation [36, 135]?

1.8 Thesis Organization

This thesis is organized as follows. Chapter 2 proposes an energy dissipation theory for stability analysis of schemes based on 3D FDTD [33, 35]. Chapter 3 extends the theory to FDTD-P [38]. Chapter 4 proposes an analogous probability conservation theory for FDTD-Q [36, 135] and investigates the energy conservation properties of the scheme. Conclusions are drawn in Chapter 5.

Energy Dissipation Theory for FDTD for the Maxwell Equations

In this chapter, we propose the energy dissipation theory for the conventional FDTD equations [35] discussed in Chapter 1. The theory gives rise to a modular framework for the creation of stable schemes that respect the principle of energy conservation (for the case of lossless media) and energy dissipation (for the case of media with nonzero conductivity). As an application of the theory, we propose a new subgridding method that supports arbitrary odd grid refinement ratios and can handle scenarios where the subgridding interface traverses material boundaries. The content of the chapter is a modified version of [193, 207, 208], which I authored with contributions from Xinyue Zhang, and is an extension of the preliminary work on the topic in two dimensions [182].

As a first step in extending the concept of dissipation from control theory literature [205, 206] to FDTD, we write the FDTD equations for a region in the form of a dynamical system [209], which is a central object of study in control theory. The equations forming the dynamical system are written in a way that facilitates the definition of energy stored in the region and the rate at which the region exchanges the energy through the boundary. An important component in formulating the dynamical system are special equations introduced on the boundary of the region, which allow studying the region in a self-contained manner. These equations will facilitate the coupling of FDTD equations for the region to models for neighbouring regions in order to create new stable schemes. In the case of subgridding studied here, the FDTD region is coupled to another FDTD region of different resolution. In the subsequent discussion the term *dissipative* will be used to describe systems that are unable to generate energy. The term includes energy conserving systems, which are also unable to incur energy losses.

2.1 Dynamical System Formulation of a 3-D FDTD Region

Consider the region shown in Figure 1.1, described by the Maxwell equations discretized using the FDTD equations in Section 1.5. The impressed current density $\vec{J}_s$ is assumed to be zero. The region contains n_x , n_y , and n_z primary cells in the x , y , and z directions, respectively. For simplicity, we assume that the region is discretized uniformly with primary cells of dimensions $\Delta x \times \Delta y \times \Delta z$. For

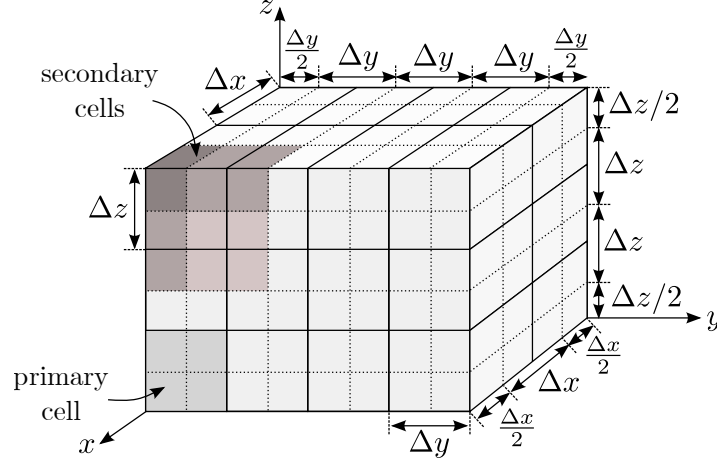


Figure 2.1: The primary and secondary grids.

reasons that will become clear shortly, the secondary grid [35] is defined as shown in Figure 2.1, with the secondary cells adjacent the boundary having the corresponding dimensions halved as shown in the figure.

The conventional FDTD samples of electric and magnetic field are located, respectively, at the *conventional primary edges* and *conventional secondary edges*. These edges can hence be defined as follows. The conventional primary edges are simply the edges of the primary cells. The conventional secondary edges are the edges of the secondary grid that are either strictly inside the region or are normal to the boundary. The remaining edges of the secondary cells are referred to as the *tangential secondary edges*, since they are tangential to the boundary. The latter will be needed to formulate the special update equations for the electric fields on the boundary. Different entities of the primary and secondary grid, along with the corresponding notation are summarized in Appendix A. The faces of the boundary are referred to as *west* ($i = 1$), *east* ($i = n_x + 1$), *south* ($j = 1$), *north* ($j = n_y + 1$), *bottom* ($k = 1$), and *top* ($k = n_z + 1$). Shortened notation w, e, s, n, b, t will also be used throughout the thesis.

The material properties, such as permittivity, conductivity, and permeability, are assumed to be constant within each primary cell. Following the reasoning in [35, 197], we define the permittivity associated with the conventional primary edges as the average of the values in the adjacent primary cells. For example, on a conventional x -directed primary edge located strictly inside the region, the permittivity is given by [35, 197]

$$\varepsilon_x|_{i+\frac{1}{2},j,k} = \frac{\varepsilon|_{i+\frac{1}{2},j-\frac{1}{2},k-\frac{1}{2}} + \varepsilon|_{i+\frac{1}{2},j+\frac{1}{2},k-\frac{1}{2}} + \varepsilon|_{i+\frac{1}{2},j-\frac{1}{2},k+\frac{1}{2}} + \varepsilon|_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}}}{4}. \quad (2.1)$$

For the conventional primary edges that lie on a face of the boundary, the permittivity is the average value obtained from two primary cells. The same approach is used for defining the values of conductivity along the conventional primary edges. The definition of permeability on the conventional secondary edges involves averaging of the values of reluctivity μ^{-1} (reciprocal of permeability)¹. For

¹As will become clear later in this chapter and in Chapter 3, with these definitions of material properties along the primary and secondary edges, the FDTD equations for the region (using fields or potentials) can be obtained by coupling the equations for individual primary cells. This is a similar argument as that used in [35] to justify the averaging procedure for permittivity values.

example, for a z -directed conventional secondary edge shared between two primary cells [197],

$$\mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k} = \frac{\mu^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k-\frac{1}{2}} + \mu^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}}}{2}, \quad (2.2)$$

where

$$\mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k} = \frac{1}{\mu_z|_{i+\frac{1}{2},j+\frac{1}{2},k}}, \quad (2.3)$$

$$\mu^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}} = \frac{1}{\mu|_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}}}. \quad (2.4)$$

Permittivity and permeability are assumed to be positive at every primary cell in the region.

2.1.1 Equations at Each Conventional Edge

The first type of equations that form the dynamical system for the region are equations describing the temporal evolution of the electric field samples. For an electric field sample strictly inside the region, such as the $E_x|_{i+0.5,j,k}$ sample shown in Figure 2.2a, we take the conventional FDTD update equation (1.33a) multiplied on both sides by $\Delta x \Delta y \Delta z$

$$\begin{aligned} \Delta x \Delta y \Delta z \left(\frac{\varepsilon_x|_{i+\frac{1}{2},j,k}}{\Delta t} + \frac{\sigma_x|_{i+\frac{1}{2},j,k}}{2} \right) E_x|_{i+\frac{1}{2},j,k}^{n+1} &= \Delta x \Delta y \Delta z \left(\frac{\varepsilon_x|_{i+\frac{1}{2},j,k}}{\Delta t} - \frac{\sigma_x|_{i+\frac{1}{2},j,k}}{2} \right) E_x|_{i+\frac{1}{2},j,k}^n \\ &+ \Delta x \left(\Delta y H_y|_{i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} - \Delta y H_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \Delta z H_z|_{i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} + \Delta z H_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right). \end{aligned} \quad (2.5)$$

The product $\Delta x \Delta y \Delta z$ is the volume associated with $E_x|_{i+0.5,j,k}$. In particular, Δx is the length of the primary edge where the sample is located and $\Delta y \Delta z$ is the area of the secondary face shaded in pale-red in Figure 2.2a. Equation (2.5) can be thought of as an FDTD-type discretization of the Maxwell-Ampère law in the integral form²

$$\int_{\Delta l} \iint_{\Delta S''} \left(\varepsilon \frac{\partial \vec{E}}{\partial t} + \sigma \vec{E} \right) \cdot d\vec{S} \, dl = \int_{\Delta l} \oint_{\partial \Delta S''} \vec{H} \cdot d\vec{s} \, dl, \quad (2.6)$$

where $\Delta S''$ is the secondary face shaded in Figure 2.2a, $\partial \Delta S''$ is the closed contour directed consistently with $d\vec{S}$ according to the right hand rule, and Δl is the primary edge where the electric field is sampled.

For the electric fields on the region's faces, such as the south $E_x|_{i+0.5,1,k}$ sample in Figure 2.2b, the conventional FDTD equation would involve some magnetic field samples that are beyond the considered region, which we wish to avoid. Instead, we write a modified equation for the electric field samples on the $j = 1$ face using the so-called hanging variables³ [155] introduced on the *tangential*

²The concept of discretizing PDEs via their integral form is related to finite volume methods [210] and the finite integration technique [127].

³The concept of hanging variables is related to the mortar methods [211].

secondary edges

$$\begin{aligned} \Delta x \frac{\Delta y}{2} \Delta z \left(\frac{\varepsilon_x|_{i+\frac{1}{2},1,k}}{\Delta t} + \frac{\sigma_x|_{i+\frac{1}{2},1,k}}{2} \right) E_x|_{i+\frac{1}{2},1,k}^{n+1} &= \Delta x \frac{\Delta y}{2} \Delta z \left(\frac{\varepsilon_x|_{i+\frac{1}{2},1,k}}{\Delta t} - \frac{\sigma_x|_{i+\frac{1}{2},1,k}}{2} \right) E_x|_{i+\frac{1}{2},1,k}^n \\ &+ \Delta x \left(\frac{\Delta y}{2} H_y|_{i+\frac{1}{2},1,k-\frac{1}{2}}^{n+\frac{1}{2}} - \frac{\Delta y}{2} H_y|_{i+\frac{1}{2},1,k+\frac{1}{2}}^{n+\frac{1}{2}} - \Delta z \boxed{H_z|_{i+\frac{1}{2},1,k}^{n+\frac{1}{2}}} + \Delta z H_z|_{i+\frac{1}{2},\frac{3}{2},k}^{n+\frac{1}{2}} \right), \end{aligned} \quad (2.7)$$

where $H_z|_{i+0.5,1,k}$ is a hanging variable, shown in green in Figure 2.2b. Equations such as (2.7) and the hanging variables play serve several purposes:

- they eliminate the need to make any assumptions on the nature of the model representing the space outside the considered region;
- they create a clear distinction between the region and surrounding space, which is important for unambiguously defining the energy stored in the region;
- as will be discussed later, they help in quantifying the power that enters the region through the boundary;
- they will also facilitate the coupling between the model of the region and the model of the surrounding space.

In a similar way, we write a modified FDTD equation for the samples located at the edges where two faces of the boundary come together. Those equations involve two hanging variables for each sample. For instance, in the case of the sample shown in Figure 2.2c, the update equation reads⁴

$$\begin{aligned} \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \left(\frac{\varepsilon_x|_{i+\frac{1}{2},1,n_z+1}}{\Delta t} + \frac{\sigma_x|_{i+\frac{1}{2},1,n_z+1}}{2} \right) E_x|_{i+\frac{1}{2},1,n_z+1}^{n+1} &= \\ \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \left(\frac{\varepsilon_x|_{i+\frac{1}{2},1,n_z+1}}{\Delta t} - \frac{\sigma_x|_{i+\frac{1}{2},1,n_z+1}}{2} \right) E_x|_{i+\frac{1}{2},1,n_z+1}^n &+ \\ + \Delta x \left(\frac{\Delta y}{2} H_y|_{i+\frac{1}{2},1,n_z+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{\Delta y}{2} \boxed{H_y|_{i+\frac{1}{2},1,n_z+1}^{n+\frac{1}{2}}} - \frac{\Delta z}{2} \boxed{H_z|_{i+\frac{1}{2},1,n_z+1}^{n+\frac{1}{2}}} + \frac{\Delta z}{2} H_z|_{i+\frac{1}{2},\frac{3}{2},n_z+1}^{n+\frac{1}{2}} \right). \end{aligned} \quad (2.8)$$

In (2.8), the y - and z -directed hanging variables, $H_y|_{i+0.5,1,n_z+1}^{n+0.5}$ and $H_z|_{i+0.5,1,n_z+1}^{n+0.5}$, respectively, are used to complete the line integral of magnetic field around $E_x|_{i+0.5,1,n_z+1}^n$. The update equations for E_y^n and E_z^n are defined analogously to E_x^n and to (2.5)–(2.8).

The update equation for the z -directed magnetic samples strictly inside the region is the con-

⁴Equations such as (2.8) allow justifying the averaging procedure discussed at the beginning of Section 2.1 by following a similar argument to [35], as was alluded to in footnote 1 on p. 24. If one connects four FDTD regions by adding equations such as (2.8) and imposes the continuity of the tangential magnetic field (valid for finite conductivity [1]), one obtains the same equation as (2.5) for internal edges, with an equivalent permittivity and conductivity values on the primary edge equal to the average of those in the four joined regions. This confirms that the averaged values of material properties defined at the conventional primary edges are compatible with the equations of the dynamical system. Similar arguments can be used to show that averaging the primary cell values of reluctivity and the parameter $\chi = \mu\varepsilon^2$ (defined and used in Chapter 3) is the most natural way to obtain equivalent values of the material properties at secondary edges and primary nodes, respectively.

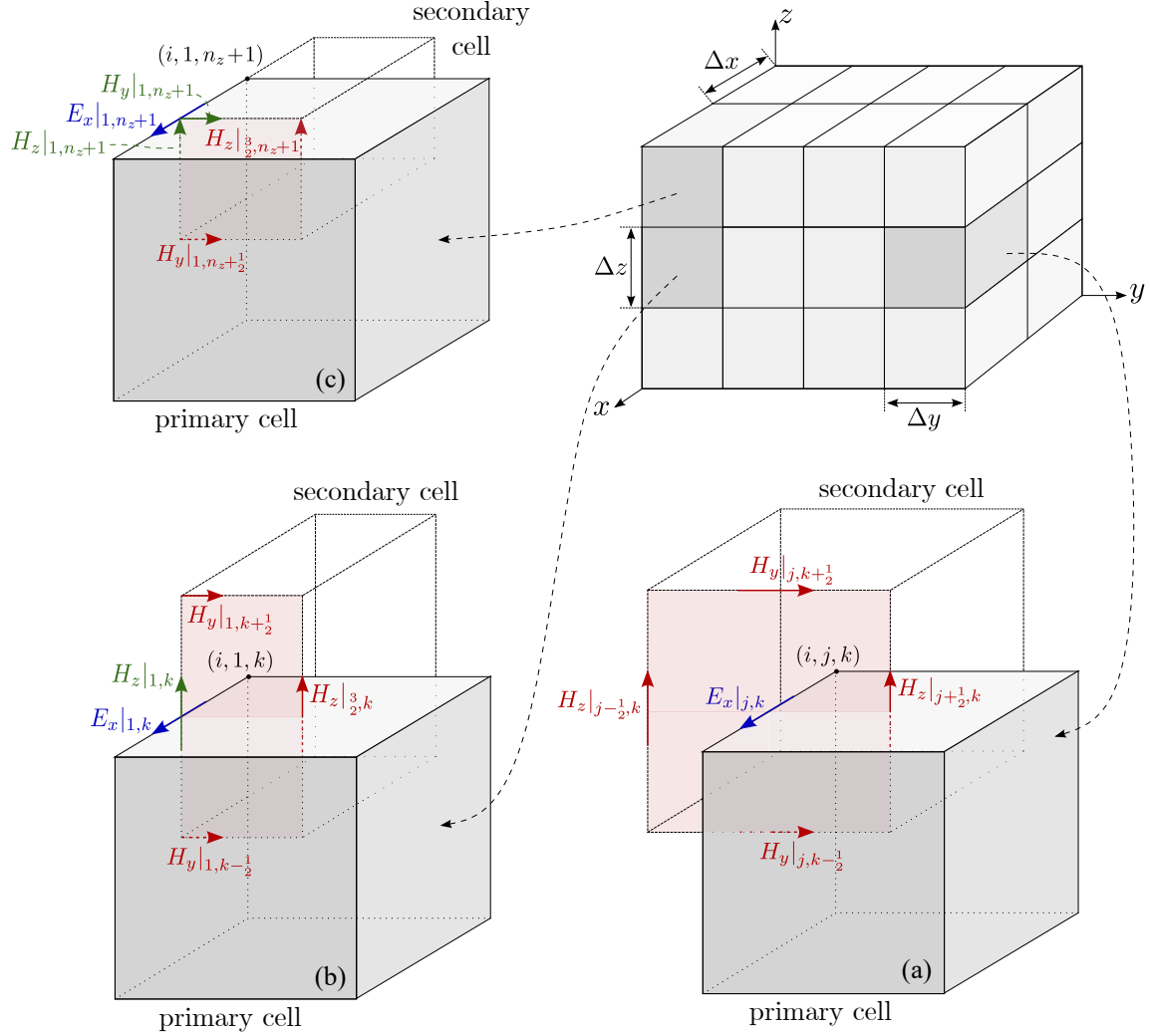


Figure 2.2: Illustration of the variables involved (a) in (2.5) for internal E_x samples (b) in (2.7) for E_x samples on the south face, and (c) in (2.8) for E_x samples on the top-south edge of the boundary. The common subscript “ $i+0.5$ ” is omitted for clarity.

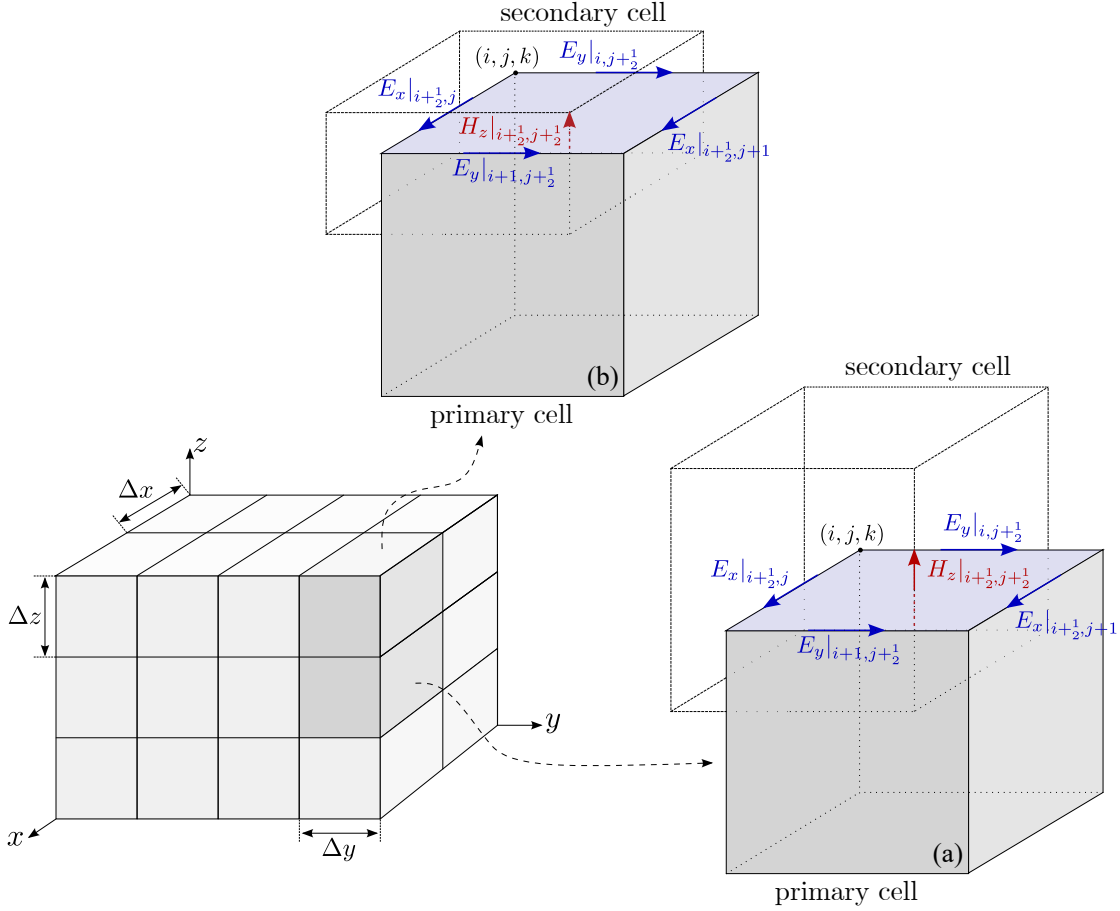


Figure 2.3: Illustration of the variables involved (a) in (2.9) for internal H_z samples and (b) in (2.10) for the normal H_z samples on the top face of the boundary. The common subscript “ k ” is omitted for clarity.

ventional update FDTD equation (1.33b), multiplied by $\Delta x \Delta y \Delta z$

$$\Delta z \Delta x \Delta y \frac{\mu_z|_{i+\frac{1}{2}, j+\frac{1}{2}, k}}{\Delta t} H_z|_{i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} = \Delta z \Delta x \Delta y \frac{\mu_z|_{i+\frac{1}{2}, j+\frac{1}{2}, k}}{\Delta t} H_z|_{i+\frac{1}{2}, j+\frac{1}{2}, k}^{n-\frac{1}{2}} - \Delta z \left(\Delta x E_x|_{i+\frac{1}{2}, j, k}^n - \Delta x E_x|_{i+\frac{1}{2}, j+1, k}^n - \Delta y E_y|_{i, j+\frac{1}{2}, k}^n + \Delta y E_y|_{i+1, j+\frac{1}{2}, k}^n \right), \quad (2.9)$$

where Δz is the length of the conventional secondary edge where H_z sample is located and $\Delta x \Delta y$ is the area of the corresponding primary face. For the boundary magnetic field sample located at the conventional secondary edge of length $\Delta z/2$ normal to the top boundary, the equation involves a factor of $\Delta z/2$, as opposed to Δz

$$\frac{\Delta z}{2} \Delta x \Delta y \frac{\mu_z|_{i+\frac{1}{2}, j+\frac{1}{2}, n_z+1}}{\Delta t} H_z|_{i+\frac{1}{2}, j+\frac{1}{2}, n_z+1}^{n+\frac{1}{2}} = \frac{\Delta z}{2} \Delta x \Delta y \frac{\mu_z|_{i+\frac{1}{2}, j+\frac{1}{2}, n_z+1}}{\Delta t} H_z|_{i+\frac{1}{2}, j+\frac{1}{2}, n_z+1}^{n-\frac{1}{2}} - \frac{\Delta z}{2} \left(\Delta x E_x|_{i+\frac{1}{2}, j, n_z+1}^n - \Delta x E_x|_{i+\frac{1}{2}, j+1, n_z+1}^n - \Delta y E_y|_{i, j+\frac{1}{2}, n_z+1}^n + \Delta y E_y|_{i+1, j+\frac{1}{2}, n_z+1}^n \right) \quad (2.10)$$

and similarly for the samples on the bottom face of the region's boundary. The update equations for H_y and H_z are obtained in a fashion similar to (2.9)–(2.10).

2.1.2 Compact Matrix Form

In order to shorten the notation, we write the equations at each node from Section 2.1.1 in a compact matrix form.

Matrix Equation for the Conventional Magnetic Field Samples

Collecting all update equations for H_z samples, such as (2.9) and (2.10), into matrix form, we obtain

$$\mathbf{\Lambda}_{l,z}'' \mathbf{\Lambda}_{S,z}' \frac{\mathbf{\Lambda}_{\mu,z}}{\Delta t} \mathbf{h}_z^{n+\frac{1}{2}} = \mathbf{\Lambda}_{l,z}'' \mathbf{\Lambda}_{S,z}' \frac{\mathbf{\Lambda}_{\mu,z}}{\Delta t} \mathbf{h}_z^{n-\frac{1}{2}} - \mathbf{\Lambda}_{l,z}'' (\mathbf{C}_{zx} \mathbf{\Lambda}_{l,y}' \mathbf{e}_x^n + \mathbf{C}_{zy} \mathbf{\Lambda}_{l,y}' \mathbf{e}_y^n), \quad (2.11)$$

where vector $\mathbf{h}_z$ contains the conventional H_z samples, $\mathbf{e}_y$ and $\mathbf{e}_z$ contain the E_y^n and E_z^n samples, respectively. Matrix $\mathbf{\Lambda}_{l,x}''$ is a diagonal matrix containing the length of each conventional secondary edge associated with samples in $\mathbf{h}_z$, namely Δz for internal samples and $\Delta z/2$ for the samples normal to the top and bottom boundaries. Diagonal matrix $\mathbf{\Lambda}_{S,z}'$ contains the areas of primary faces where H_z is sampled. For the case of uniform discretization, this matrix reduces to $\Delta y \Delta z \mathbf{I}$, where $\mathbf{I}$ denotes an identity matrix. As a result, the product $\mathbf{\Lambda}_{l,x}'' \mathbf{\Lambda}_{S,x}'$ contains the volumes that can be associated with each sample in $\mathbf{h}_z$. The coefficient matrix $\mathbf{\Lambda}_{\mu,z}$ is a diagonal matrix containing the magnetic permeability values on the edges where the corresponding elements of $\mathbf{h}_z$ are sampled. Matrix $\mathbf{C}_{zx}$, which consists of zeros, +1's and -1's, is a discrete circulation/curl operator for computing the contribution of E_x to the curl around each z -directed conventional secondary edge. Matrix $\mathbf{C}_{zy}$ is defined analogously. Matrices $\mathbf{\Lambda}_{l,y}'$ and $\mathbf{\Lambda}_{l,z}'$ contain the lengths of the primary edges where the elements of $\mathbf{e}_y^n$ and $\mathbf{e}_z^n$ are sampled, respectively. For the uniform discretization, these matrices reduce to $\Delta y \mathbf{I}$ and $\Delta z \mathbf{I}$, respectively. The expressions of the matrices involved in (2.11) depend on how the scalar samples are ordered in $\mathbf{h}_z$, $\mathbf{e}_x$, and $\mathbf{e}_y$. A possible choice of ordering and the resulting expressions for the matrices in (2.11) are described in Appendix B.

In a similar fashion to (2.11), we write in matrix form the update equations for $\mathbf{h}_x$ and $\mathbf{h}_y$. Further combining the resulting equations and (2.11) into a single matrix update equation, we obtain

$$\mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \frac{\mathbf{\Lambda}_\mu}{\Delta t} \mathbf{h}^{n+\frac{1}{2}} = \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \frac{\mathbf{\Lambda}_\mu}{\Delta t} \mathbf{h}^{n-\frac{1}{2}} - \mathbf{\Lambda}_l'' \mathbf{C} \mathbf{\Lambda}_l' \mathbf{e}^n, \quad (2.12)$$

where

$$\mathbf{e}^n = \begin{bmatrix} \mathbf{e}_x^n \\ \mathbf{e}_y^n \\ \mathbf{e}_z^n \end{bmatrix}, \quad \mathbf{h}^{n-\frac{1}{2}} = \begin{bmatrix} \mathbf{h}_x^{n-\frac{1}{2}} \\ \mathbf{h}_y^{n-\frac{1}{2}} \\ \mathbf{h}_z^{n-\frac{1}{2}} \end{bmatrix}, \quad (2.13)$$

$$\mathbf{\Lambda}_l' = \text{diag}(\mathbf{\Lambda}_{l,x}', \mathbf{\Lambda}_{l,y}', \mathbf{\Lambda}_{l,z}'), \quad (2.14)$$

$$\mathbf{\Lambda}_l'' = \text{diag}(\mathbf{\Lambda}_{l,x}'', \mathbf{\Lambda}_{l,y}'', \mathbf{\Lambda}_{l,z}''), \quad (2.15)$$

$$\mathbf{\Lambda}_S' = \text{diag}(\mathbf{\Lambda}_{S,x}', \mathbf{\Lambda}_{S,y}', \mathbf{\Lambda}_{S,z}'), \quad (2.16)$$

$$\mathbf{\Lambda}_\mu = \text{diag}(\mathbf{\Lambda}_{\mu,x}, \mathbf{\Lambda}_{\mu,y}, \mathbf{\Lambda}_{\mu,z}), \quad (2.17)$$

and $\mathbf{C}$ is the discrete circulation/curl operator given by

$$\mathbf{C} = \begin{bmatrix} \mathbf{0} & \mathbf{C}_{xy} & \mathbf{C}_{xz} \\ \mathbf{C}_{yx} & \mathbf{0} & \mathbf{C}_{yz} \\ \mathbf{C}_{zx} & \mathbf{C}_{zy} & \mathbf{0} \end{bmatrix}. \quad (2.18)$$

Matrix equation for the Electric Field Samples

From (2.5), (2.7), and (2.8), the update equation for the E_x^n samples can then be written in matrix form as

$$\begin{aligned} \mathbf{\Lambda}'_{l,x} \mathbf{\Lambda}''_{S,x} \left(\frac{\mathbf{\Lambda}_{\varepsilon,x}}{\Delta t} + \frac{\mathbf{\Lambda}_{\sigma,x}}{2} \right) \mathbf{e}_x^{n+1} &= \mathbf{\Lambda}'_{l,x} \mathbf{\Lambda}''_{S,x} \left(\frac{\mathbf{\Lambda}_{\varepsilon,x}}{\Delta t} - \frac{\mathbf{\Lambda}_{\sigma,x}}{2} \right) \mathbf{e}_x^n \\ &+ \mathbf{\Lambda}'_{l,x} \left(\mathbf{C}_{yx}^T \mathbf{\Lambda}''_{l,y} \mathbf{h}_y^{n+\frac{1}{2}} + \mathbf{C}_{zx}^T \mathbf{\Lambda}''_{l,z} \mathbf{h}_z^{n+\frac{1}{2}} \right. \\ &+ \mathbf{L}_{xy,b} \mathbf{\Lambda}_{[\hat{x} \cdot \hat{n}_b \times \hat{y}]}'' \mathbf{\Lambda}''_{l,y,b} \mathbf{h}_{y,b}^{n+\frac{1}{2}} + \mathbf{L}_{xy,t} \mathbf{\Lambda}_{[\hat{x} \cdot \hat{n}_t \times \hat{y}]}'' \mathbf{\Lambda}''_{l,y,t} \mathbf{h}_{y,t}^{n+\frac{1}{2}} \\ &\left. + \mathbf{L}_{xz,s} \mathbf{\Lambda}_{[\hat{x} \cdot \hat{n}_s \times \hat{z}]}'' \mathbf{\Lambda}''_{l,z,s} \mathbf{h}_{z,s}^{n+\frac{1}{2}} + \mathbf{L}_{xz,n} \mathbf{\Lambda}_{[\hat{x} \cdot \hat{n}_n \times \hat{z}]}'' \mathbf{\Lambda}''_{l,z,n} \mathbf{h}_{z,n}^{n+\frac{1}{2}} \right). \end{aligned} \quad (2.19)$$

In (2.19), the diagonal matrix $\mathbf{\Lambda}''_{S,x}$ contains the areas of the internal secondary faces, which correspond to the conventional primary edges where E_x is sampled. These areas are equal to $\Delta y \Delta z$ for the internal primary edges (see (2.5) and Figure 2.2a), $\Delta y \Delta z / 2$ for the tangential primary edges on the faces of the region (see (2.7) and Figure 2.2b), and $\Delta y \Delta z / 4$ for the tangential primary edges on the edges of the region (see (2.8) and Figure 2.2c). Matrix coefficients $\mathbf{\Lambda}_{\varepsilon,x}$ and $\mathbf{\Lambda}_{\sigma,x}$ are diagonal matrices with the values of permittivity and conductivity, respectively, on the x -directed conventional primary edges where E_x is sampled. Vector $\mathbf{h}_{z,s}$ collects all z -directed hanging variables on the south boundary of the region, which appear in (2.7) and (2.8). Similarly, $\mathbf{h}_{y,b}$, $\mathbf{h}_{y,t}$, and $\mathbf{h}_{z,n}$ collect hanging variables associated with E_x samples on the bottom, top, and north faces, respectively. The coefficient matrices multiplying $\mathbf{h}_{z,s}$ are defined as follows and those corresponding to $\mathbf{h}_{y,b}$, $\mathbf{h}_{y,t}$, and $\mathbf{h}_{z,n}$ are defined analogously. Matrix $\mathbf{L}_{xz,s}$ consists of 0's and 1's that extract from the vector $\mathbf{h}_{z,s}$ the hanging variables corresponding to each tangential sample in $\mathbf{e}_x$. Matrix $\mathbf{\Lambda}_{[\hat{x} \cdot \hat{n}_s \times \hat{z}]}$ is the identity matrix, multiplied by $(\hat{x} \cdot \hat{n}_s \times \hat{z})$, where $\hat{n}_s$ is the outward normal unit vector to the south face of the boundary, namely $-\hat{y}$. Hence, $(\hat{x} \cdot \hat{n}_s \times \hat{z})$ equals to -1 . Matrix $\mathbf{\Lambda}''_{l,z,s}$ in (2.19) is a diagonal matrix containing the lengths of the tangential secondary edges where the hanging variables in $\mathbf{h}_{z,s}$ are sampled. For example, it contains Δz for the samples in $\mathbf{h}_{z,s}$ with indices $k = 2, 3, \dots, n_z$ (see (2.7) and Figure 2.2b) and $\Delta z / 2$ for the samples with indices $k = 1$ and $k = n_z + 1$ (see (2.8) and Figure 2.2c). The matrix equations for $\mathbf{e}_y$ and $\mathbf{e}_z$ are written similarly to (2.19).

We obtain the following update equation for the electric samples in the region by collecting (2.19) and similar equations for $\mathbf{e}_y$ and $\mathbf{e}_z$

$$\mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \left(\frac{\mathbf{\Lambda}_{\varepsilon}}{\Delta t} + \frac{\mathbf{\Lambda}_{\sigma}}{2} \right) \mathbf{e}^{n+1} = \mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \left(\frac{\mathbf{\Lambda}_{\varepsilon}}{\Delta t} - \frac{\mathbf{\Lambda}_{\sigma}}{2} \right) \mathbf{e}^n + \mathbf{\Lambda}'_l \mathbf{C}^T \mathbf{\Lambda}''_l \mathbf{h}^{n+\frac{1}{2}} + \mathbf{\Lambda}'_l \mathbf{L}_{\parallel} \mathbf{\Lambda}_{[\cdot \hat{n} \times]}'' \mathbf{\Lambda}''_{l,\parallel} \mathbf{h}_{\parallel}^{n+\frac{1}{2}}, \quad (2.20)$$

where

$$\mathbf{\Lambda}_S'' = \text{diag}(\mathbf{\Lambda}_{S,x}'', \mathbf{\Lambda}_{S,y}'', \mathbf{\Lambda}_{S,z}'') , \quad (2.21)$$

$$\mathbf{\Lambda}_\varepsilon = \text{diag}(\mathbf{\Lambda}_{\varepsilon,x}, \mathbf{\Lambda}_{\varepsilon,y}, \mathbf{\Lambda}_{\varepsilon,z}) , \quad (2.22)$$

$$\mathbf{\Lambda}_\sigma = \text{diag}(\mathbf{\Lambda}_{\sigma,x}, \mathbf{\Lambda}_{\sigma,y}, \mathbf{\Lambda}_{\sigma,z}) . \quad (2.23)$$

It should be noted that the transpose of $\mathbf{C}$ that appears in (2.12) serves as a discrete circulation/curl operator for $\mathbf{h}$ in (2.20). Similar symmetry properties of FIT and FDTD equations have been stated previously [127, 197]. Vector $\mathbf{h}_\parallel$ collects all of the hanging variables in the region⁵

$$\mathbf{h}_\parallel^{n+\frac{1}{2}} = \begin{bmatrix} \mathbf{h}_{\parallel,x}^{n+\frac{1}{2}} \\ \mathbf{h}_{\parallel,y}^{n+\frac{1}{2}} \\ \mathbf{h}_{\parallel,z}^{n+\frac{1}{2}} \end{bmatrix} , \quad (2.24)$$

where

$$\mathbf{h}_{\parallel,x}^{n+\frac{1}{2}} = \begin{bmatrix} \mathbf{h}_{x,s}^{n+\frac{1}{2}} \\ \mathbf{h}_{x,n}^{n+\frac{1}{2}} \\ \mathbf{h}_{x,b}^{n+\frac{1}{2}} \\ \mathbf{h}_{x,t}^{n+\frac{1}{2}} \end{bmatrix} , \quad \mathbf{h}_{\parallel,y}^{n+\frac{1}{2}} = \begin{bmatrix} \mathbf{h}_{y,w}^{n+\frac{1}{2}} \\ \mathbf{h}_{y,e}^{n+\frac{1}{2}} \\ \mathbf{h}_{y,b}^{n+\frac{1}{2}} \\ \mathbf{h}_{y,t}^{n+\frac{1}{2}} \end{bmatrix} , \quad \mathbf{h}_{\parallel,z}^{n+\frac{1}{2}} = \begin{bmatrix} \mathbf{h}_{z,w}^{n+\frac{1}{2}} \\ \mathbf{h}_{z,e}^{n+\frac{1}{2}} \\ \mathbf{h}_{z,s}^{n+\frac{1}{2}} \\ \mathbf{h}_{z,n}^{n+\frac{1}{2}} \end{bmatrix} . \quad (2.25)$$

Matrix $\mathbf{\Lambda}_{l,\parallel}''$ in (2.20) is given by

$$\mathbf{\Lambda}_{l,\parallel}'' = \text{diag} \left(\mathbf{\Lambda}_{l,\parallel,x}'', \mathbf{\Lambda}_{l,\parallel,y}'', \mathbf{\Lambda}_{l,\parallel,z}'' \right) , \quad (2.26)$$

where

$$\mathbf{\Lambda}_{l,\parallel,x}'' = \text{diag} \left(\mathbf{\Lambda}_{l,x,s}'', \mathbf{\Lambda}_{l,x,n}'', \mathbf{\Lambda}_{l,x,b}'', \mathbf{\Lambda}_{l,x,t}'' \right) , \quad (2.27)$$

$$\mathbf{\Lambda}_{l,\parallel,y}'' = \text{diag} \left(\mathbf{\Lambda}_{l,y,w}'', \mathbf{\Lambda}_{l,y,e}'', \mathbf{\Lambda}_{l,y,b}'', \mathbf{\Lambda}_{l,y,t}'' \right) , \quad (2.28)$$

$$\mathbf{\Lambda}_{l,\parallel,z}'' = \text{diag} \left(\mathbf{\Lambda}_{l,z,w}'', \mathbf{\Lambda}_{l,z,e}'', \mathbf{\Lambda}_{l,z,s}'', \mathbf{\Lambda}_{l,z,n}'' \right) . \quad (2.29)$$

Matrix $\mathbf{L}_\parallel$ has the following structure⁶:

$$\mathbf{L}_\parallel = \begin{bmatrix} \mathbf{0} & \mathbf{L}_{\parallel,xy} & \mathbf{L}_{\parallel,xz} \\ \mathbf{L}_{\parallel,yx} & \mathbf{0} & \mathbf{L}_{\parallel,yz} \\ \mathbf{L}_{\parallel,zx} & \mathbf{L}_{\parallel,zy} & \mathbf{0} \end{bmatrix} , \quad (2.30)$$

where

$$\mathbf{L}_{\parallel,xy} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{L}_{xy,b} & \mathbf{L}_{xy,t} \end{bmatrix} , \quad \mathbf{L}_{\parallel,xz} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{L}_{xz,s} & \mathbf{L}_{xz,n} \end{bmatrix} , \quad (2.31)$$

⁵The order of samples in $\mathbf{h}_\parallel$ is different from [193, 212]. The change was made for the sake of consistency with the other chapters. Ultimately, the order of samples should have no effect on the validity of the proposed theory and conclusions drawn from any of the numerical examples.

⁶In $\mathbf{L}_\parallel$, subscript $\parallel$ is used to denote the fact that the matrix left-multiplies a vector containing tangential hanging variables. In Chapter 3 an analogous matrix multiplying a vector of normal hanging variables will be defined.

$$\mathbf{L}_{\parallel, yx} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{L}_{yx,b} & \mathbf{L}_{yx,t} \end{bmatrix}, \quad \mathbf{L}_{\parallel, yz} = \begin{bmatrix} \mathbf{L}_{yz,w} & \mathbf{L}_{yz,e} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad (2.32)$$

$$\mathbf{L}_{\parallel, zx} = \begin{bmatrix} \mathbf{L}_{zx,s} & \mathbf{L}_{zx,n} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{L}_{\parallel, zy} = \begin{bmatrix} \mathbf{L}_{zy,w} & \mathbf{L}_{zy,e} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (2.33)$$

Matrix $\mathbf{\Lambda}_{[\cdot\hat{n}\times]}$ is given by

$$\mathbf{\Lambda}_{[\cdot\hat{n}\times]} = \text{diag} \left(\mathbf{\Lambda}_{[\cdot\hat{n}\times\hat{x}]}, \mathbf{\Lambda}_{[\cdot\hat{n}\times\hat{y}]}, \mathbf{\Lambda}_{[\cdot\hat{n}\times\hat{z}]} \right), \quad (2.34)$$

where

$$\mathbf{\Lambda}_{[\cdot\hat{n}\times\hat{x}]} = \text{diag} \left(\mathbf{\Lambda}_{[\hat{z}\cdot\hat{n}_s\times\hat{x}]}, \mathbf{\Lambda}_{[\hat{z}\cdot\hat{n}_n\times\hat{x}]}, \mathbf{\Lambda}_{[\hat{y}\cdot\hat{n}_b\times\hat{x}]}, \mathbf{\Lambda}_{[\hat{y}\cdot\hat{n}_t\times\hat{x}]} \right), \quad (2.35)$$

$$\mathbf{\Lambda}_{[\cdot\hat{n}\times\hat{y}]} = \text{diag} \left(\mathbf{\Lambda}_{[\hat{z}\cdot\hat{n}_w\times\hat{y}]}, \mathbf{\Lambda}_{[\hat{z}\cdot\hat{n}_e\times\hat{y}]}, \mathbf{\Lambda}_{[\hat{x}\cdot\hat{n}_b\times\hat{y}]}, \mathbf{\Lambda}_{[\hat{x}\cdot\hat{n}_t\times\hat{y}]} \right), \quad (2.36)$$

$$\mathbf{\Lambda}_{[\cdot\hat{n}\times\hat{z}]} = \text{diag} \left(\mathbf{\Lambda}_{[\hat{y}\cdot\hat{n}_w\times\hat{z}]}, \mathbf{\Lambda}_{[\hat{y}\cdot\hat{n}_e\times\hat{z}]}, \mathbf{\Lambda}_{[\hat{x}\cdot\hat{n}_s\times\hat{z}]}, \mathbf{\Lambda}_{[\hat{x}\cdot\hat{n}_n\times\hat{z}]} \right). \quad (2.37)$$

2.1.3 Dynamical System Formulation

Matrix equations (2.12) and (2.20) can now be written in the form of a dynamical system as follows

$$(\mathbf{R} + \mathbf{F})\mathbf{x}^{n+1} = (\mathbf{R} - \mathbf{F})\mathbf{x}^n + \mathbf{B}\mathbf{u}^{n+\frac{1}{2}}, \quad (2.38a)$$

$$\mathbf{y}^n = \mathbf{L}^T \mathbf{x}^n, \quad (2.38b)$$

where the state vector $\mathbf{x}^n$ contains the conventional FDTD samples in the region

$$\mathbf{x}^n = \begin{bmatrix} \mathbf{e}^n \\ \mathbf{h}^{n-\frac{1}{2}} \end{bmatrix}. \quad (2.39)$$

The hanging variables $\mathbf{h}_{\parallel}^{n+0.5}$ are not included in the state vector, and instead are regarded as the input of the FDTD region

$$\mathbf{u}^{n+\frac{1}{2}} = \mathbf{h}_{\parallel}^{n+\frac{1}{2}}. \quad (2.40)$$

The output vector $\mathbf{y}^n$ is given by

$$\mathbf{y}^n = \mathbf{e}_{\parallel}^n, \quad (2.41)$$

where $\mathbf{e}_{\parallel}$ contains the tangential boundary samples of electric field. This is ensured by choosing an appropriate expression for $\mathbf{L}$, given shortly. Matrices $\mathbf{R}$, $\mathbf{F}$, and $\mathbf{B}$ in (2.38a)–(2.38b) are obtained from (2.20), which forms the first set of equations, and (2.12), which forms the second set of equations,

$$\mathbf{R} = \begin{bmatrix} \mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \frac{\mathbf{\Lambda}_{\epsilon}}{\Delta t} & -\frac{1}{2} \mathbf{\Lambda}'_l \mathbf{C}^T \mathbf{\Lambda}''_l \\ -\frac{1}{2} \mathbf{\Lambda}''_l \mathbf{C} \mathbf{\Lambda}'_l & \mathbf{\Lambda}'_l \mathbf{\Lambda}'_S \frac{\mathbf{\Lambda}_{\mu}}{\Delta t} \end{bmatrix}, \quad (2.42)$$

$$\mathbf{F} = \begin{bmatrix} \mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \frac{\mathbf{\Lambda}_{\sigma}}{2} & -\frac{1}{2} \mathbf{\Lambda}'_l \mathbf{C}^T \mathbf{\Lambda}''_l \\ \frac{1}{2} \mathbf{\Lambda}''_l \mathbf{C} \mathbf{\Lambda}'_l & \mathbf{0} \end{bmatrix}, \quad (2.43)$$

$$\mathbf{B} = \begin{bmatrix} \mathbf{\Lambda}'_l \mathbf{L}_{\parallel} \mathbf{\Lambda}_{[\cdot\hat{n}\times]} \mathbf{\Lambda}''_{l,\parallel} \\ \mathbf{0} \end{bmatrix}. \quad (2.44)$$

Matrix $\mathbf{L}$ is defined as

$$\mathbf{L} = \begin{bmatrix} \mathbf{L}_{\parallel} \\ \mathbf{0} \end{bmatrix}. \quad (2.45)$$

In (2.38b), $\mathbf{L}^T$ selects the tangential electric field samples from the state vector $\mathbf{x}^n$. Each sample in $\mathbf{y}$ corresponds to a hanging variable in $\mathbf{u}$. For example, $E_x|_{i+0.5,1,k}$ sample in Figure 2.2b appears in $\mathbf{y} = \mathbf{e}_{\parallel}$ in the same position as the position in which the hanging variable $H_z|_{i+0.5,1,k}$ takes up in $\mathbf{u} = \mathbf{h}_{\parallel}$. Similarly, $E_x|_{i+0.5,1,n_z+1}$ sample in Figure 2.2c appears in two positions: those of $H_y|_{i+0.5,1,n_z+1}$ and $H_z|_{i+0.5,1,n_z+1}$ in $\mathbf{h}_{\parallel}$. The expressions of vectors and matrices provided in Appendix B ensure that this is the case. Remarkably, the dynamical system equations (2.38a)–(2.38b) have the same structure as in the 2-D case [182]. This fact allows us to generalize previous results valid in two dimensions to the most general 3-D case.

2.2 Dissipativity of a 3-D FDTD Region

In this section, we derive dissipativity conditions for the 3-D FDTD system (2.38a)–(2.38b) defined in Section 2.1 and explain their physical significance.

Definition 1. *According to the theory of dissipative systems [206], the discrete-time dynamical system (2.38a)–(2.38b) is dissipative with the supply rate $s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+0.5})$ if we can find a function $\mathcal{E}$ that satisfies the following properties:*

$$\mathcal{E}(\mathbf{x}^n) \geq 0, \quad \forall \mathbf{x}^n, \quad (2.46a)$$

$$\mathcal{E}(\mathbf{0}) = 0, \quad (2.46b)$$

$$\mathcal{E}(\mathbf{x}^{n+1}) - \mathcal{E}(\mathbf{x}^n) \leq s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+\frac{1}{2}}), \quad \forall \mathbf{u}^{n+\frac{1}{2}} \forall n. \quad (2.46c)$$

If (2.46c) is satisfied with equality, the system is said to be lossless or energy conserving.

The storage function $\mathcal{E}(\mathbf{x}^n)$ quantifies the energy stored in the FDTD region and the supply rate $s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+0.5})$ is the energy entering the region through its boundaries between time points n and $n+1$. The importance of having a lower bound on the energy (2.46a) for defining dissipative systems has been discussed in [205]. If (2.46c) holds without imposing (2.46a), the energy contained in the region is allowed to become infinitely negative. If that occurs, the region will supply an infinite amount of energy to the surrounding space, akin to a bottomless well. This would clearly indicate a violation in the principle of energy conservation.

We define the storage function as

$$\mathcal{E}(\mathbf{x}^n) = \frac{\Delta t}{2} (\mathbf{x}^n)^T \mathbf{R} \mathbf{x}^n. \quad (2.47)$$

An analogous definition for discrete energy was proposed in [197] for the finite integration technique. We propose to define the supply rate as

$$s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+\frac{1}{2}}) = \Delta t \frac{(\mathbf{y}^n + \mathbf{y}^{n+1})^T}{2} \mathbf{S} \mathbf{u}^{n+\frac{1}{2}}, \quad (2.48)$$

where

$$\mathbf{S} = \mathbf{\Lambda}'_{l,\parallel} \mathbf{\Lambda}_{[\cdot\hat{n}\times]} \mathbf{\Lambda}''_{l,\parallel}, \quad (2.49)$$

where $\mathbf{\Lambda}'_{l,\parallel}$ is given by

$$\mathbf{\Lambda}'_{l,\parallel} = \text{diag} \left(\mathbf{\Lambda}'_{l,z,s}, \mathbf{\Lambda}'_{l,z,n}, \mathbf{\Lambda}'_{l,y,b}, \mathbf{\Lambda}'_{l,y,t}, \mathbf{\Lambda}'_{l,z,w}, \mathbf{\Lambda}'_{l,z,e}, \mathbf{\Lambda}'_{l,x,b}, \mathbf{\Lambda}'_{l,x,t}, \right. \\ \left. \mathbf{\Lambda}'_{l,y,w}, \mathbf{\Lambda}'_{l,y,e}, \mathbf{\Lambda}'_{l,x,s}, \mathbf{\Lambda}'_{l,x,n} \right), \quad (2.50)$$

where, for example, $\mathbf{\Lambda}'_{l,x,s}$ contains on the diagonal the lengths of the conventional x -directed primary edges that are tangential to the south face of the boundary. Since uniform discretization is assumed, their value is simply Δx . The other eleven matrices in (2.50) are defined similarly. Expressions (2.47) and (2.48) generalize those from the 2-D case [182].

2.2.1 Physical Meaning of the Supply Rate and Storage Function

By substituting (2.39) and (2.42) into (2.47), we can write the storage function as

$$\mathcal{E}(\mathbf{x}^n) = \frac{1}{2}(\mathbf{e}^n)^T \mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \mathbf{\Lambda}_\varepsilon \mathbf{e}^n + \frac{\Delta t}{2} \left(\mathbf{h}^{n-\frac{1}{2}} \right)^T \left[\mathbf{\Lambda}''_l \mathbf{\Lambda}'_S \frac{\mathbf{\Lambda}_\mu}{\Delta t} \mathbf{h}^{n-\frac{1}{2}} - \mathbf{\Lambda}''_l \mathbf{C} \mathbf{\Lambda}'_l \mathbf{e}^n \right], \quad (2.51)$$

which, using (2.12) to simplify the term inside the square brackets, reduces to

$$\mathcal{E}(\mathbf{x}^n) = \frac{1}{2}(\mathbf{e}^n)^T \mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \mathbf{\Lambda}_\varepsilon \mathbf{e}^n + \frac{1}{2} \left(\mathbf{h}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}''_l \mathbf{\Lambda}'_S \mathbf{\Lambda}_\mu \mathbf{h}^{n+\frac{1}{2}}. \quad (2.52)$$

This expression for the stored energy has been proposed in [197] for FDTD/FIT. Equation (2.52) reveals the physical meaning of the storage function (2.47) as a discrete analogue of

$$\frac{1}{2} \iiint_V (\varepsilon_x E_x^2 + \varepsilon_y E_y^2 + \varepsilon_z E_z^2 + \mu_x H_x^2 + \mu_y H_y^2 + \mu_z H_z^2) dV, \quad (2.53)$$

which is the energy stored in electric and magnetic fields inside a given volume V [1].

To reveal the physical meaning of the supply rate, we can substitute (2.40), (2.41), and (2.49) into (2.48), obtaining

$$s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+\frac{1}{2}}) = \Delta t \frac{(\mathbf{e}^n_{\parallel} + \mathbf{e}^{n+1}_{\parallel})^T}{2} \mathbf{\Lambda}'_{l,\parallel} \mathbf{\Lambda}_{[\cdot\hat{n}\times]} \mathbf{\Lambda}''_{l,\parallel} \mathbf{h}^{n+\frac{1}{2}}. \quad (2.54)$$

The product $\mathbf{\Lambda}'_{l,\parallel} \mathbf{\Lambda}''_{l,\parallel}$ is a diagonal matrix containing the area of boundary faces associated with each hanging variable and the corresponding electric field sample. As a result, (2.54) is an approximation of the following integral

$$s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+\frac{1}{2}}) \approx \int_{n\Delta t}^{(n+1)\Delta t} \oint_{\partial V} \vec{E} \cdot \hat{n} \times \vec{H} dS dt = \int_{n\Delta t}^{(n+1)\Delta t} \oint_{\partial V} \vec{E} \times \vec{H} \cdot \hat{n}_{\text{inward}} dS dt, \quad (2.55)$$

where $\hat{n}_{\text{inward}} = -\hat{n}$ is the *inward* unit normal vector. The last expression in (2.55) is the flux of the Poynting vector [1] over the region's boundary, integrated from time point n to $n+1$. Therefore, supply rate (2.48) approximates the total energy that enters the region through its boundaries

between time n and time $n+1$.

2.2.2 Dissipativity Conditions for a 3-D FDTD Region

By substituting (2.47) and (2.48) into (2.46a)–(2.46c), we arrive at the following theorem, which gives simple conditions for the dissipativity of (2.38a)–(2.38b), which involve only the matrices in (2.38a)–(2.38b) and hence can be verified and enforced a priori.

Theorem 1. *A discrete-time dynamical system in the form (2.38a)–(2.38b) is dissipative with $\mathcal{E}(\mathbf{x}^n)$ in (2.47) as the storage function and $s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+0.5})$ in (2.48) as the supply rate if*

$$\mathbf{R} = \mathbf{R}^T \succ 0, \quad (2.56a)$$

$$\mathbf{F} + \mathbf{F}^T \succeq 0, \quad (2.56b)$$

$$\mathbf{L}\mathbf{S} = \mathbf{B}, \quad (2.56c)$$

where “ $\succ 0$ ” and “ $\succeq 0$ ” denote positive definite and positive semidefinite matrices, respectively. If, in addition to (2.56a) and (2.56c), condition (2.56b) is satisfied as

$$\mathbf{F} + \mathbf{F}^T = \mathbf{0}, \quad (2.57)$$

then the system is lossless.

Proof. We have proven the theorem in [182]. Here, the proof is reproduced with minor modifications. To prove the implication, assume that the dynamical system satisfies (2.56a)–(2.56c). Then, from (2.56a), condition (2.46a) is satisfied for any value of the state vector. Condition (2.46b) always holds due to the form of storage function (2.47).

Next, we expand the left hand side of (2.46c) using (2.47)

$$\mathcal{E}(\mathbf{x}^{n+1}) - \mathcal{E}(\mathbf{x}^n) = \frac{\Delta t}{2} (\mathbf{x}^{n+1})^T \mathbf{R} \mathbf{x}^{n+1} - \frac{\Delta t}{2} (\mathbf{x}^n)^T \mathbf{R} \mathbf{x}^n. \quad (2.58)$$

Using the symmetry of matrix $\mathbf{R}$,

$$\mathcal{E}(\mathbf{x}^{n+1}) - \mathcal{E}(\mathbf{x}^n) = \frac{\Delta t}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n)^T \mathbf{R} (\mathbf{x}^{n+1} - \mathbf{x}^n). \quad (2.59)$$

Rearranging (2.38a) as

$$\mathbf{R}(\mathbf{x}^{n+1} + \mathbf{x}^n) = -\mathbf{F}(\mathbf{x}^{n+1} + \mathbf{x}^n) + \mathbf{B}\mathbf{u}^{n+\frac{1}{2}} \quad (2.60)$$

and substituting the result into (2.59),

$$\mathcal{E}(\mathbf{x}^{n+1}) - \mathcal{E}(\mathbf{x}^n) = -\frac{\Delta t}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n)^T \mathbf{F}(\mathbf{x}^{n+1} + \mathbf{x}^n) + \frac{\Delta t}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n)^T \mathbf{B}\mathbf{u}^{n+\frac{1}{2}}. \quad (2.61)$$

Rewriting the first term in (2.59) using the fact that a scalar quantity remains unchanged by the

transpose operation, and rewriting the second term using (2.56c) and (2.38b),

$$\begin{aligned}\mathcal{E}(\mathbf{x}^{n+1}) - \mathcal{E}(\mathbf{x}^n) &= -\frac{\Delta t}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n)^T \frac{\mathbf{F} + \mathbf{F}^T}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n) + \frac{\Delta t}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n)^T \mathbf{L} \mathbf{S} \mathbf{u}^{n+\frac{1}{2}} \\ &= -\frac{\Delta t}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n)^T \frac{\mathbf{F} + \mathbf{F}^T}{2} (\mathbf{x}^{n+1} + \mathbf{x}^n) + \frac{\Delta t}{2} (\mathbf{y}^{n+1} + \mathbf{y}^n)^T \mathbf{S} \mathbf{u}^{n+\frac{1}{2}}.\end{aligned}\quad (2.62)$$

Moreover, from the definition of the supply rate (2.48),

$$\mathcal{E}(\mathbf{x}^{n+1}) - \mathcal{E}(\mathbf{x}^n) = -\Delta t \left(\frac{\mathbf{x}^{n+1} + \mathbf{x}^n}{2} \right)^T (\mathbf{F} + \mathbf{F}^T) \frac{\mathbf{x}^{n+1} + \mathbf{x}^n}{2} + s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+\frac{1}{2}}). \quad (2.63)$$

From (2.56b), the first term on the right hand side of (2.63) is non-positive, which implies (2.46c). Moreover, if (2.57) holds, then (2.46c) is satisfied with equality, making the system lossless by definition. $\square$

The following two theorems provide an insight into the physical meaning of (2.56a)–(2.56c) and (2.57).

Theorem 2. *Consider a region in Figures 2.2 and 2.3, described by the 3D FDTD equations (2.38a)–(2.38b).*

1. *Condition (2.56a) holds if and only if the time step is taken below the generalized CFL limit*

$$\Delta t < \Delta t_{CFL,gen} = \frac{2}{\sqrt{\rho(\mathbf{\Sigma}^T \mathbf{\Sigma})}}, \quad (2.64)$$

where⁷

$$\mathbf{\Sigma} = (\mathbf{\Lambda}_l')^{\frac{1}{2}} \mathbf{\Lambda}_\varepsilon^{-\frac{1}{2}} (\mathbf{\Lambda}_S'')^{-\frac{1}{2}} \mathbf{C}^T (\mathbf{\Lambda}_l'')^{\frac{1}{2}} (\mathbf{\Lambda}_\mu)^{-\frac{1}{2}} (\mathbf{\Lambda}_S')^{-\frac{1}{2}}. \quad (2.65)$$

2. *Condition (2.56b) holds if and only if the conductivity values on the conventional primary edges are non-negative. Condition (2.57) holds if and only if the conductivity values are zero.*
3. *Condition (2.56c) is true by construction.*

Proof. The first statement of the theorem can be shown by applying the Schur complement [213] to transform (2.56a) into

$$\begin{cases} \mathbf{\Lambda}_l' \mathbf{\Lambda}_S'' \frac{\mathbf{\Lambda}_\varepsilon}{\Delta t} \succ 0, \\ \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \frac{\mathbf{\Lambda}_\mu}{\Delta t} - \frac{\Delta t}{4} \mathbf{\Lambda}_l'' \mathbf{C} \mathbf{\Lambda}_l' (\mathbf{\Lambda}_\varepsilon)^{-1} (\mathbf{\Lambda}_S'')^{-1} \mathbf{C}^T \mathbf{\Lambda}_l'' \succ 0. \end{cases} \quad (2.66)$$

The first matrix inequality in (2.66) is always true because lengths, areas, and permittivity values in the region are positive. Applying a congruence with matrix $2(\Delta t \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \mathbf{\Lambda}_\mu)^{-\frac{1}{2}}$ to the left hand side of the second inequality in (2.66), we obtain a condition equivalent to (2.66) [96],

$$\frac{4}{\Delta t^2} \mathbf{I} - \mathbf{\Sigma}^T \mathbf{\Sigma} \succ 0, \quad (2.67)$$

⁷Notation $\rho(\cdot)$ indicates the spectral radius of a matrix, $(\cdot)^{\frac{1}{2}}$ and $(\cdot)^{-\frac{1}{2}}$ denote, respectively, the principal square root and the inverse of the principal square root [96].

which holds if and only if (2.64) is true.

To prove the second statement of the theorem, we expand $\mathbf{F} + \mathbf{F}^T$ as

$$\mathbf{F} + \mathbf{F}^T = \begin{bmatrix} \Lambda'_l \Lambda''_S \Lambda_\sigma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (2.68)$$

This matrix is positive semidefinite, as per condition (2.56b), if and only if the value of conductivity on each of the conventional primary edges is non-negative. The matrix is zero, as per condition (2.57), if and only if all of these conductivity values are zero. Matrix $\mathbf{F} + \mathbf{F}^T$ can thus be associated with energy loss via conductivity. In particular, rewriting (2.63) using (2.68),

$$\mathcal{E}(\mathbf{x}^{n+1}) - \mathcal{E}(\mathbf{x}^n) = -\Delta t \left(\frac{\mathbf{e}^{n+1} + \mathbf{e}^n}{2} \right)^T \Lambda'_l \Lambda''_S \Lambda_\sigma \frac{\mathbf{e}^{n+1} + \mathbf{e}^n}{2} + s(\mathbf{y}^n, \mathbf{y}^{n+1}, \mathbf{u}^{n+\frac{1}{2}}), \quad (2.69)$$

reveals that the first term in (2.63) is an approximation of the integral over the region of $-\sigma|\vec{E}|^2$, which quantifies Ohmic losses [1]. The role of conductivity as an energy loss mechanism in FDTD has also been pointed out in [197].

To show the validity of the third statement, we expand $\mathbf{L}\mathbf{S}$ as

$$\mathbf{L}\mathbf{S} = \begin{bmatrix} \mathbf{L}_\parallel \Lambda'_{l,\parallel} \Lambda_{[\hat{n} \times]}'' \Lambda''_{l,\parallel} \\ \mathbf{0} \end{bmatrix}. \quad (2.70)$$

Right-multiplication by $\Lambda'_{l,\parallel}$ has the same effect on $\mathbf{L}_\parallel$ as left-multiplication by Λ'_l . Hence,

$$\mathbf{L}_\parallel \Lambda'_{l,\parallel} = \Lambda'_l \mathbf{L}_\parallel, \quad (2.71)$$

and as a result $\mathbf{L}\mathbf{S} = \mathbf{B}$, which is the condition (2.56c). $\square$

Theorem 3.

$$\Delta t_{CFL} \leq \Delta t_{CFL,gen} \quad (2.72)$$

where $\Delta t_{CFL,gen}$ is the generalized CFL limit in (2.64) and Δt_{CFL} is the CFL limit (1.36), with $\varepsilon_{\min}$ being the smallest primary cell permittivity in the region and $\mu_{\min}$ the smallest primary cell permeability.

Proof. The proof is provided in Appendix D.1. The strategy is similar to [197], where an analogous result was shown for an isolated FIT/FDTD region. $\square$

When the CFL limit is violated, the energy stored in some cells is no longer bounded below by zero, which can make those cells capable of supplying unlimited amount of energy to the rest of the system, potentially causing instability. Conversely, under the CFL limit the FDTD discretization of the region respects the principle of energy dissipation (or energy conservation for the case of a lossless medium). This conclusion is consistent with the results presented by Edelvik, Schuhmann, and Weiland [197]. However, Theorems 1–3 generalize the theory to the case of a region that is not isolated, but is able to exchange energy with the surrounding space through the boundary. In contrast to the theory in [197], the results presented here allow characterizing the exchange of energy between different regions coupled together. As discussed in the next section, this analysis provides

a way to create new schemes by coupling the FDTD region to other models in an energy conserving fashion.

2.3 Systematic Method for Stability Enforcement

The developments in the previous sections provide a powerful approach to construct both simple and advanced FDTD schemes with guaranteed stability. The method is general since it is applicable to complicated setups involving multiple subsystems, such as FDTD grids with different resolution, multiple boundary conditions, circuit models, or reduced-order models. The approach involves the following steps:

1. Time-domain models are written as discrete-time dynamical systems with inputs and outputs representing quantities on their boundaries. For the case of FDTD models, the inputs are the hanging magnetic field variables introduced on the secondary edges tangential to the boundary and the outputs are the tangential electric field samples.
2. The models are coupled via their inputs and outputs. For example, in the case of the subgridding scheme derived in the next section, the equations between two FDTD region models are coupled by relating the tangential magnetic field samples (inputs) and the tangential electric fields samples (outputs) between the two grids of different resolution.
3. Stability is enforced by ensuring that all models in the system and equations coupling those models respect the principle of energy conservation or dissipation. For dynamical systems of the form (2.38a)–(2.38b) this can be done using (2.56a)–(2.56c).

It should be noted that, although we use a matrix formulation to investigate the dissipativity of FDTD equations, the final scheme can be implemented with scalar FDTD equations for optimal efficiency.

The proposed approach significantly simplifies the creation of new FDTD schemes with guaranteed stability, since dissipativity conditions can be imposed on individual models. A remarkable feature of the proposed approach is its modularity. Given a set of models and coupling equations that have been proven to conserve or dissipate energy, any of their combination is guaranteed to be stable. This feature is hoped to accelerate scientific research in the FDTD area, since it allows researchers to create new stable schemes without having to redo stability analysis for every change. In this thesis, a stable subgridding algorithm is derived based on the proposed theory. Moreover, in [212], a method was proposed for incorporating embedded reduced models of locally refined regions, stabilized beyond the CFL limit using the conditions in Theorem 1. Other examples of potential developments based on the proposed ideas include:

- stable algorithms for the transient analysis of coupled field-circuit problems, since the proposed dissipation theory extends to the FDTD method the dissipative modeling concepts developed earlier for lumped and distributed circuits [204, 214];
- stable schemes that hybridize FDTD with the method of moments or the finite element method in the time domain, since the proposed dissipation criterion can serve as a method-agnostic principle to achieve overall stability when coupling different methods;

- commercial FDTD solvers that can integrate user-provided models. Since FDTD stability can be easily compromised, a rigorous and modular stability theory is essential to make this feature viable. The proposed theory gives clear stability conditions that user-provided models must abide in order to avoid destabilizing the whole solver. Moreover, conditions (2.56a)-(2.56c) can be tested and eventually be enforced numerically in an automated fashion prior to the FDTD simulation.

2.4 Application to Stable 3-D FDTD Subgridding

As a demonstration of the stability framework in Section 2.3, we derive a stable subgridding algorithm for 3-D FDTD. In the proposed algorithm, a coarse grid is refined in selected regions to better resolve fine geometrical details. We denote the refinement factors as r_x , r_y , and r_z in the x , y , and z directions, respectively. The refinement factors are odd integers greater than one.

For simplicity, the proposed method is presented by considering the scenario in Figure 2.4, where the interface between the coarse and fine grids is planar. The case where the subgridding region is embedded inside of a coarse grid is discussed in Appendix C.

In order to develop a stable subgridding scheme with the method in Section 2.3, we view a subgridding algorithm as being composed of the following components:

- the coarse grid,
- the fine grid,
- the interpolation rule at the interface between the two grids, coupling the two FDTD models,
- the boundary conditions on the exterior faces of the two grids.

2.4.1 Interpolation Conditions

The interpolation rule between the coarse and fine grid fields is the core component of every subgridding algorithm. In the scenario of Figure 2.4, the interpolation rule has to properly relate the fields tangential to the north boundary of the coarse grid to the fields tangential to the south boundary of the fine grid. The interpolation rule is discussed in detail for the $E_{\parallel,z}$ - $H_{\parallel,x}$ pairs. The derivations for the $E_{\parallel,x}$ - $H_{\parallel,z}$ pairs can be done analogously.

As shown on the left side of Figure 2.4, we consider the portion of the interface surrounding one coarse electric field sample (denoted as E_z) and the corresponding hanging variable (denoted as $H_{\parallel,x}$). The fine samples that fall in the same region in the case of $r_x = r_z = 3$ are shown on the bottom right of Figure 2.4, where the circumflex notation ($\hat{\cdot}$) is used to indicate quantities pertaining to the fine grid. With that, the interpolation rule between the coarse and fine tangential variables on the fine and coarse grids is defined as follows. The transverse averages of the fine $\hat{E}_z$ and $\hat{H}_{\parallel,x}$ samples are all set to be equal to the coarse samples E_z and $H_{\parallel,x}$, respectively

$$E_z^n = \frac{1}{r_x} \sum_{m=1,2,3} \hat{E}_{zm}^n = \frac{1}{r_x} \sum_{m=4,5,6} \hat{E}_{zm}^n = \frac{1}{r_x} \sum_{m=7,8,9} \hat{E}_{zm}^n, \quad \forall n, \quad (2.73)$$

$$H_{\parallel,x}^{n+\frac{1}{2}} = \frac{1}{r_z} \sum_{m=1,4,7} \hat{H}_{\parallel,xm}^{n+\frac{1}{2}} = \frac{1}{r_z} \sum_{m=2,5,8} \hat{H}_{\parallel,xm}^{n+\frac{1}{2}} = \frac{1}{r_z} \sum_{m=3,6,9} \hat{H}_{\parallel,xm}^{n+\frac{1}{2}}, \quad \forall n, \quad (2.74)$$

where “transverse” for E_z refers to the x direction and “transverse” for $H_x^\parallel$ refers to the z direction. The fine electric field samples are equal to each other longitudinally (i.e. in the z direction)

$$\hat{E}_{z1}^n = \hat{E}_{z4}^n = \hat{E}_{z7}^n, \quad \forall n, \quad (2.75a)$$

$$\hat{E}_{z2}^n = \hat{E}_{z5}^n = \hat{E}_{z8}^n, \quad \forall n, \quad (2.75b)$$

$$\hat{E}_{z3}^n = \hat{E}_{z6}^n = \hat{E}_{z9}^n, \quad \forall n. \quad (2.75c)$$

The interpolation conditions (2.73), (2.74), and (2.75a)–(2.75c) can be equivalently written using the matrix notation as

$$E_z^n = \frac{\mathbf{T}_{r_x}^T}{r_x} \bar{\mathbf{e}}_z^n, \quad (2.76a)$$

$$\mathbf{T}_{r_x} H_{\parallel,x}^{n+\frac{1}{2}} = \left(\frac{\mathbf{T}_{r_z}^T}{r_z} \otimes \mathbf{I}_{r_x} \right) \hat{\mathbf{h}}_{\parallel,x}^{n+\frac{1}{2}}, \quad (2.76b)$$

$$\hat{\mathbf{e}}_z^n = (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) \bar{\mathbf{e}}_z^n, \quad (2.76c)$$

where $\hat{\mathbf{e}}_z$ and $\hat{\mathbf{h}}_{\parallel,x}$ denotes vectors containing the fine samples, ordered as shown in the figure (from 1 to 9 for the $r_x = r_z = 3$ example). Symbol $\otimes$ denotes the Kronecker product [96] and $\mathbf{T}_m$ is an $m \times 1$ matrix of ones. Vector $\bar{\mathbf{e}}_z$ was introduced to represent the unique samples of electric field on the fine side of the interface. The interpolation conditions (2.76a)–(2.76c) correspond to $1 + r_x + r_x r_z$ equations. This matches the number of additional variables introduced on the boundary, namely the total number of elements in $H_{\parallel,x}$, $\bar{\mathbf{e}}_z$, and $\hat{\mathbf{h}}_{\parallel,x}$. This is an important property that allows deriving unique update equations that enforce the interpolation conditions. The interpolation rule in two dimensions [182] drew inspiration from the reciprocity principle [179]. Conditions (2.73), (2.74), and (2.75a)–(2.75c) extend the conditions proposed in [182] to three dimensions.

2.4.2 Stability Proof

The stability of the proposed subgridding algorithm can be proven with the dissipation theory in Section 2.3. As was discussed at the beginning of Section 2.4, the proposed scheme is comprised of several components, which must all be dissipative in order to ensure stability. As proven in Section 2.2, the coarse and fine grids are dissipative under their respective CFL limits. For simplicity, we assume that a PEC boundary condition is imposed on the external faces of the boundaries of the two grids. In that scenario, there is no energy that can enter or exit the two grids through those surfaces, making that boundary condition a lossless component of the system. Other boundary conditions could be similarly analyzed using the proposed theory and their choice would have no implications on how the stable interpolation rule should be defined.

What is left to investigate is whether the interpolation rule is dissipative or not. We show that the interpolation rule in Section 2.4.1 is in fact lossless for any time step, by proving that the supply rate (2.54) into the fine grid and the supply rate into the coarse grid add up to zero. The supply rate of the interpolation rule is the summation of terms, each one associated to one hanging variable on its boundaries. Hence, the contributions of the x -directed hanging variables $H_{\parallel,x}$ and $\hat{\mathbf{h}}_{\parallel,x}$ in

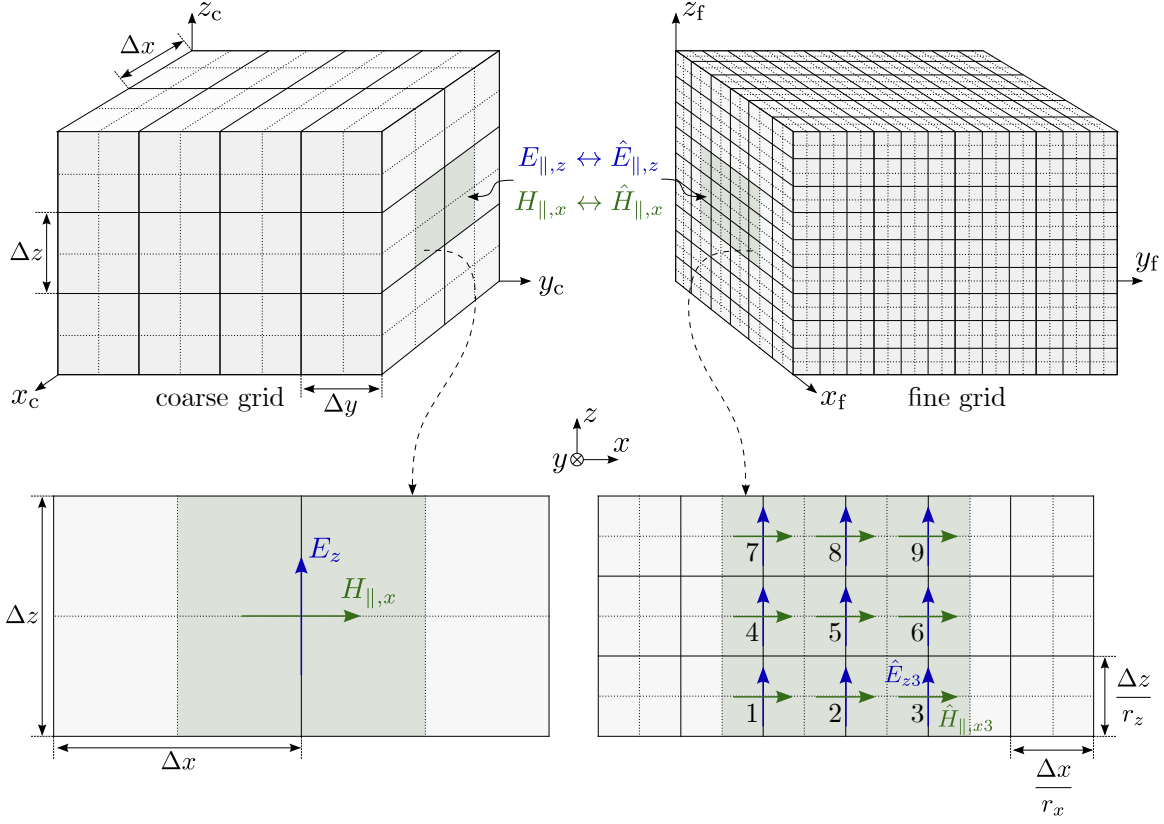


Figure 2.4: Coarse and fine sides of the subgridding interface in Section 2.4.

Figure 2.4 to the supply rate of the coarse and fine grid reads

$$\delta s^{n+\frac{1}{2}} = \Delta t \Delta x \Delta z \frac{E_z^n + E_z^{n+1}}{2} (\hat{z} \cdot \hat{n}_n \times \hat{x}) H_{\parallel,x}^{n+\frac{1}{2}}, \quad (2.77)$$

and

$$\delta \hat{s}^{n+\frac{1}{2}} = \Delta t \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \frac{(\hat{\mathbf{e}}_z^n + \hat{\mathbf{e}}_z^{n+1})^T}{2} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \hat{\mathbf{h}}_{\parallel,x}^{n+\frac{1}{2}}, \quad (2.78)$$

respectively. The circumflexes in $(\hat{z} \cdot \hat{n}_n \times \hat{x})$ and $(\hat{z} \cdot \hat{n}_s \times \hat{x})$ denote the unit vectors as per definitions in Section 2.1.2 and are not to be confused with the notation used for the fine grid quantities. Substituting (2.76c) into (2.78),

$$\delta \hat{s}^{n+\frac{1}{2}} = \Delta t \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \frac{(\bar{\mathbf{e}}_z^n + \bar{\mathbf{e}}_z^{n+1})^T}{2} (\hat{z} \cdot \hat{n}_s \times \hat{x}) (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \hat{\mathbf{h}}_{\parallel,x}^{n+\frac{1}{2}}. \quad (2.79)$$

Using (2.76b),

$$\delta \hat{s}^{n+\frac{1}{2}} = \Delta t \frac{\Delta x}{r_x} \Delta z \frac{(\bar{\mathbf{e}}_z^n + \bar{\mathbf{e}}_z^{n+1})^T}{2} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \mathbf{T}_{r_x} H_{\parallel,x}^{n+\frac{1}{2}}. \quad (2.80)$$

Lastly, from (2.76a),

$$\delta \hat{s}^{n+\frac{1}{2}} = \Delta t \Delta x \Delta z \frac{(E_z^n + E_z^{n+1})^T}{2} (\hat{z} \cdot \hat{n}_s \times \hat{x}) H_{\parallel,x}^{n+\frac{1}{2}}. \quad (2.81)$$

Comparing (2.81) to (2.77) and recalling that $(\hat{z} \cdot \hat{n}_s \times \hat{x}) + (\hat{z} \cdot \hat{n}_n \times \hat{x}) = 0$,

$$\delta \hat{s}^{n+\frac{1}{2}} + \delta s^{n+\frac{1}{2}} = 0. \quad (2.82)$$

Hence, the energy that leaves the coarse grid due to the samples E_z , $H_{\parallel,x}$ in Figure 2.4 during one time step is equal to the energy that enters the fine grid during that time step. Iterating the argument for all other coarse hanging variables and the corresponding groups of $r_x r_z$ hanging variables on the fine side, we conclude that the interpolation rule is lossless (i.e. conserves energy) and hence preserves the dissipativity of the subgridding scheme. Since the coarse and fine grids are dissipative under their own CFL limits, the entire scheme is stable under the CFL limit of the fine grid, which is the most restrictive CFL limit.

2.4.3 Update Equations at the Interface

In this section we discuss how the proposed subgridding algorithm can be implemented. For updating all field samples strictly inside the two grids, one can use conventional FDTD equations. The update equation for the field samples on the refinement interface is instead derived using the special equations on the boundary introduced in Section 2.1.1 and the interpolation rules (2.76a)–(2.76c).

As discussed in Section 2.1.1, the electric field sample E_z^n in Figure 2.4 on the north face of the coarse grid boundary satisfies

$$\begin{aligned} \Delta z \Delta x \frac{\Delta y}{2} \left(\frac{\varepsilon_z}{\Delta t} + \frac{\sigma_z}{2} \right) E_z^{n+1} &= \Delta z \Delta x \frac{\Delta y}{2} \left(\frac{\varepsilon_z}{\Delta t} - \frac{\sigma_z}{2} \right) E_z^n \\ &\quad + \Delta z \mathbf{C}^T \mathbf{\Lambda}_l'' \mathbf{h}^{n+\frac{1}{2}} + \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \Delta x H_{\parallel,x}^{n+\frac{1}{2}}, \end{aligned} \quad (2.83)$$

where, letting the indices of E_z be $(i, n_y + 1, k + 0.5)$,

$$\varepsilon_z = \varepsilon_z|_{i, n_y+1, k+\frac{1}{2}}, \quad (2.84)$$

$$\sigma_z = \sigma_z|_{i, n_y+1, k+\frac{1}{2}}, \quad (2.85)$$

$$\mathbf{h}^{n+\frac{1}{2}} = \begin{bmatrix} H_x|_{i, n_y+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} & H_y|_{i-\frac{1}{2}, n_y+1, k+\frac{1}{2}}^{n+\frac{1}{2}} & H_y|_{i+\frac{1}{2}, n_y+1, k+\frac{1}{2}}^{n+\frac{1}{2}} \end{bmatrix}^T, \quad (2.86)$$

$$\mathbf{\Lambda}_l'' = \text{diag} \left(\Delta x, \frac{\Delta y}{2}, \frac{\Delta y}{2} \right), \quad (2.87)$$

$$\mathbf{C} = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}^T. \quad (2.88)$$

Vector $\mathbf{h}$ and matrices $\mathbf{\Lambda}_l''$ and $\mathbf{C}$ are not to be confused with the corresponding matrices for the entire region. The former are sub-matrices of the full-region matrices. Similarly to (2.83), we can write the update equations for the nine fine $\hat{E}_z$ samples in Figure 2.4 as

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\frac{\hat{\mathbf{\Lambda}}_{\varepsilon,z}}{\Delta t} + \frac{\hat{\mathbf{\Lambda}}_{\sigma,z}}{2} \right) \hat{\mathbf{e}}_z^{n+1} &= \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\frac{\hat{\mathbf{\Lambda}}_{\varepsilon,z}}{\Delta t} - \frac{\hat{\mathbf{\Lambda}}_{\sigma,z}}{2} \right) \hat{\mathbf{e}}_z^n \\ &\quad + \frac{\Delta z}{r_z} \hat{\mathbf{C}}^T \hat{\mathbf{\Lambda}}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} + \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{h}}_{\parallel,x}^{n+\frac{1}{2}}. \end{aligned} \quad (2.89)$$

Next, we manipulate (2.83) and (2.89) using (2.76a)–(2.76c) to eliminate the hanging variables.

Multiplying (2.83) by $\mathbf{T}_{r_x}/r_x$ and using (2.76a), we have

$$\begin{aligned} \Delta z \frac{\Delta x}{r_x} \frac{\Delta y}{2} \frac{1}{r_x} \mathbf{T}_{r_x} \left(\frac{\varepsilon_z}{\Delta t} + \frac{\sigma_z}{2} \right) \mathbf{T}_{r_x}^T \bar{\mathbf{e}}_z^{n+1} &= \Delta z \frac{\Delta x}{r_x} \frac{\Delta y}{2} \frac{1}{r_x} \mathbf{T}_{r_x} \left(\frac{\varepsilon_x}{\Delta t} - \frac{\sigma_z}{2} \right) \mathbf{T}_{r_x}^T \bar{\mathbf{e}}_z^n \\ &+ \frac{\mathbf{T}_{r_x}}{r_x} \Delta z \mathbf{C}^T \mathbf{\Lambda}_l' \mathbf{h}^{n+\frac{1}{2}} + \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{r_x} \mathbf{T}_{r_x} H_{\parallel,x}^{n+\frac{1}{2}}. \end{aligned} \quad (2.90)$$

Substituting (2.76c) into (2.89) and multiplying the result by $(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x})$,

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \left(\frac{\hat{\Lambda}_{\varepsilon,z}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z}}{2} \right) (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) \bar{\mathbf{e}}_z^{n+1} &= \\ \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \left(\frac{\hat{\Lambda}_{\varepsilon,z}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z}}{2} \right) (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) \bar{\mathbf{e}}_z^n &+ \\ + (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \frac{\Delta z}{r_z} \hat{\mathbf{C}}^T \hat{\mathbf{\Lambda}}_l' \hat{\mathbf{h}}^{n+\frac{1}{2}} + \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \hat{\mathbf{h}}_{\parallel,x}^{n+\frac{1}{2}}. \end{aligned} \quad (2.91)$$

Adding (2.90) and (2.91) and cancelling the hanging variables using (2.76b) and the fact that $(\hat{z} \cdot \hat{n}_n \times \hat{x}) = -(\hat{z} \cdot \hat{n}_s \times \hat{x})$,

$$\begin{aligned} \Delta z \frac{\Delta x}{r_x} \left(\frac{\Delta y}{2} \frac{1}{r_x} \mathbf{T}_{r_x} \left(\frac{\varepsilon_z}{\Delta t} + \frac{\sigma_z}{2} \right) \mathbf{T}_{r_x}^T + \frac{\Delta y}{2r_y} \frac{1}{r_z} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \left(\frac{\hat{\Lambda}_{\varepsilon,z}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z}}{2} \right) (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) \right) \bar{\mathbf{e}}_z^{n+1} &= \\ \Delta z \frac{\Delta x}{r_x} \left(\frac{\Delta y}{2} \frac{1}{r_x} \mathbf{T}_{r_x} \left(\frac{\varepsilon_x}{\Delta t} - \frac{\sigma_z}{2} \right) \mathbf{T}_{r_x}^T + \frac{\Delta y}{2r_y} \frac{1}{r_z} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \left(\frac{\hat{\Lambda}_{\varepsilon,z}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z}}{2} \right) (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) \right) \bar{\mathbf{e}}_z^n &+ \\ + \frac{\mathbf{T}_{r_x}}{r_x} \Delta z \mathbf{C}^T \mathbf{\Lambda}_l' \mathbf{h}^{n+\frac{1}{2}} + (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \frac{\Delta z}{r_z} \hat{\mathbf{C}}^T \hat{\mathbf{\Lambda}}_l' \hat{\mathbf{h}}^{n+\frac{1}{2}}. \end{aligned} \quad (2.92)$$

Equation (2.92) relates the electric field vector $\bar{\mathbf{e}}_z^n$ (and therefore $\hat{\mathbf{e}}_z^n$) to the surrounding magnetic fields in both the coarse and the fine grids. We can see that the last step leading to (2.92) eliminates the hanging variables on the boundaries of the two grids, which serve as a temporary means to connect the two grids. Equation (2.92) is a recursive relation for the electric field samples at the interface, that can be used to compute $\bar{\mathbf{e}}_z^{n+1}$ knowing the electric and magnetic samples at the previous time points. Although (2.92) is not fully explicit in terms of $\bar{\mathbf{e}}_z^{n+1}$, the matrix left-multiplying this unknown vector is very small, having size $r_x \times r_x$. Moreover, it is a constant. Therefore, this matrix has to be inverted only once, leading to an explicit update equation for $\bar{\mathbf{e}}_z^{n+1}$, which will be used for all time steps.

Overall, the proposed subgridding scheme consists of the following steps.

1. Starting from E^n and $H^{n+0.5}$, we update all electric fields strictly inside fine and coarse regions using standard FDTD equations such as (2.5) (or equivalently (1.35)).
2. We update the electric fields on the interface between the two grids using (2.92) and a similar equation for $\bar{\mathbf{e}}_x^n$ samples. Once (2.92) is solved, the fine electric samples $\hat{E}_{z1}^{n+1}, \dots, \hat{E}_{z9}^{n+1}$ can be found using (2.76c) and the coarse electric field sample can be updated using (2.76a).
3. All conventional magnetic field samples are updated using the standard FDTD equations, such as (2.9) and (2.10) (equivalently, (1.33b)), to obtain the values at $n + 1.5$.

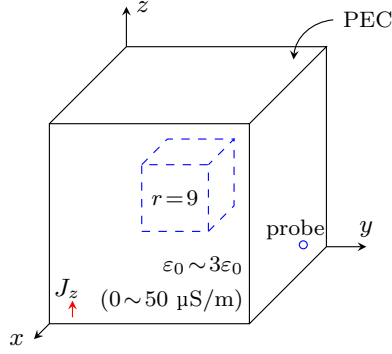


Figure 2.5: Setup of the stability verification test in Section 2.5.1.

It can be shown⁸ that the resulting update equations enforce the interpolation conditions (2.76a)–(2.76c).

The inclusion of material properties throughout the derivations rigorously guarantees stability in the case when arbitrary permittivity and conductivity values are assigned to cells on each side of the subgridding interface, including the case where material boundaries traverse the interface. This is in contrast, for example, to [156], where stability was only verified numerically.

2.5 Numerical Examples

The subgridding algorithm from Section 2.4 was implemented in MATLAB [215] in order to test the validity of the proposed theory for FDTD stability and to assess the accuracy of the proposed subgridding method. Time-consuming portions of the code, such as the update equations for the fields inside each grid, were written using vectorized operations.

2.5.1 Stability Verification

We verify the stability of the proposed algorithm by simulating the subgridding scenario shown in Figure 2.5 for a million time steps. The setup consists of a 12 cm × 12 cm × 12 cm cavity with PEC walls. The cavity is discretized with a uniform coarse mesh with $\Delta x = \Delta y = \Delta z = 1$ cm. A 4 cm × 4 cm × 4 cm subgridding region with refinement ratio $r = 9$ in all directions is placed at the center of the cavity. In order to test the correctness of stability analysis in the case when material properties vary, we randomly assign permittivity values to cells in both grids between the free space value ε_0 and $3\varepsilon_0$ using the `rand()` function in MATLAB. In another simulation, we test if losses are properly handled by introducing, in addition to the varying permittivity, random conductivity with the values distributed between zero and 5×10^{-5} S/m. We excite the cavity using a Gaussian pulse with half-width at half-maximum bandwidth of 3.53 GHz. Time step was set to 2.118 ps, which corresponds to 99% of the fine grid's CFL limit (1.36) with μ_0 and ε_0 as $\mu_{\min}$ and $\varepsilon_{\min}$, respectively.

From the results, presented in Figure 2.6, it can be seen that no signs of instability occur during the million time steps of the simulation, even if nonuniform materials traverse the coarse-fine

⁸Equations (2.83), (2.89), and (2.76a)–(2.76c) can be put into a linear system that has a unique solution. The aforementioned operations in deriving (2.92) can be viewed as a set of invertible row operations on the linear system. Hence, (2.92) yields the unique solution to the linear system, enforcing the interpolation conditions.

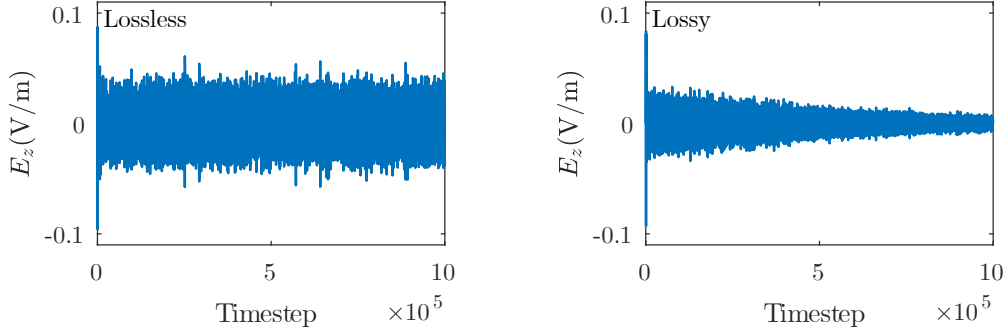


Figure 2.6: The z component of electric field at the probe in the stability verification test of Section 2.5.1.

interface. In addition to $r = 9$, we have performed the same test for a grid refinement value of 3, 5, and 7, obtaining stable simulations in all cases. This result validates the correctness of the stability analysis framework presented in this chapter. It should be noted that the lossless case puts the proposed developments to stringent test, since the simulated structure has no losses that could mask even a very small amount of energy gain resulting from a subgridding algorithm which is not perfectly dissipative. The results in the left panel of Figure 2.6 show that, in addition to not introducing any artificial gain, the proposed algorithm does not introduce artificial losses either, and faithfully reproduces the lossless nature of the actual structure.

2.5.2 Material Traverse

In this simulation, we test whether the proposed subgridding method produces consistent results when the subgridding interface traverses material boundaries, which is known to be a challenging scenario for subgridding schemes [216]. We use the setup shown in Figure 2.7, where the subgridding region is placed in three different ways: in Case 1 and Case 3, a $10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm}$ block of material traverses different sides of the interface. In the reference Case 2, the block is fully enclosed by the subgridding boundary. We perform the test for a lossless dielectric with $\epsilon_r = 3$, for a lossy dielectric with $\epsilon_r = 3$, $\sigma = 0.05 \text{ S/m}$, and for copper. The domain is terminated with an absorbing layer⁹. The coarse grid was set to $\Delta x = \Delta y = \Delta z = 1 \text{ cm}$ and the subgridding region was refined by a factor of $r = 3$ in each direction. A Gaussian source with half-width at half-maximum bandwidth of 1.02 GHz was used. The iteration time step was 99% of the CFL limit (1.36) dictated by the fine grid.

Figure 2.8 shows the electric field recorded at the probe in the three cases. The two cases with material traverse are in very good agreement with the reference simulation, showing that the proposed algorithm does not lose accuracy when different materials traverse the subgridding boundary. The test is successful for all three types of material.

⁹The implementation for the absorbing layer entailed selecting nonzero magnetic and electric conductivity values along the edges tangential to the boundary. Those values obeyed the relation $\sigma/\epsilon_0 = \sigma^*/\mu_0$ [133], where σ^* is the magnetic conductivity. This simplified absorbing layer was selected instead of the state of the art PML method [35] due to stability considerations. It should also be noted that, although the absorbing layer has not been analyzed with the dissipation theory, the stability of the proposed subgridding scheme is not contingent on what boundary condition is used. Moreover, the artificial losses introduced by this absorbing layer are unlikely to cause spurious production of energy.

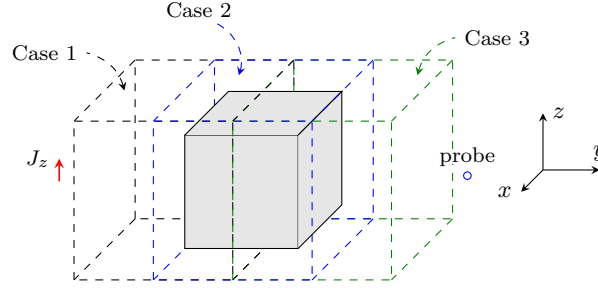
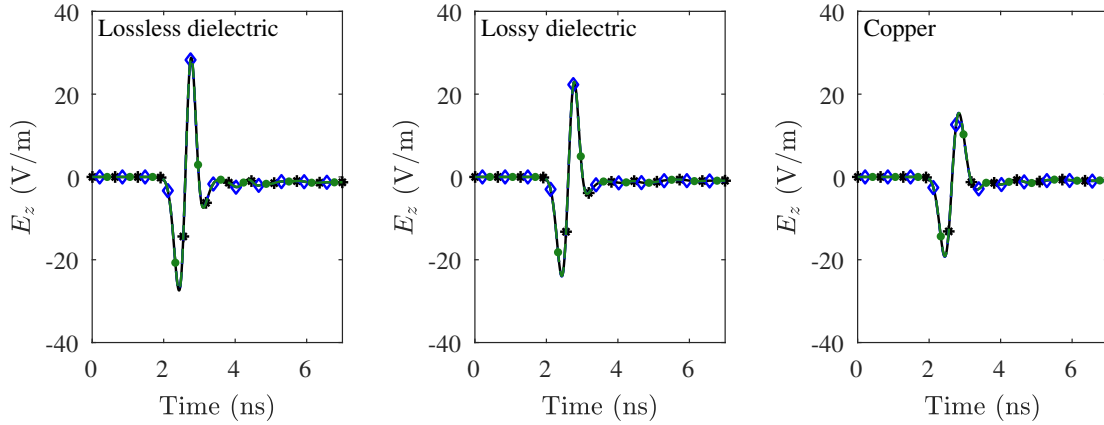


Figure 2.7: Setup of the material traverse test of Section 2.5.2.

Figure 2.8: The z component of electric field at the probe in material traverse test of Section 2.5.2 for Case 1 ($-*$), Case 2 ($-\diamond-$), and Case 3 ($- \bullet -$).

It should be noted that in [212], where the proposed subgridding was used to develop a reduced modeling scheme, fluctuations of the electromagnetic field were observed near the subgridding interface in Case 1. Similar fluctuations can be observed in this test for the subgridding region without the model order reduction. Those fluctuations manifest with and without the block of material traversing the interface. Typically, scenarios where current sources are placed close to the subgridding interface are more challenging. The fields that are affected by this phenomenon tend to be confined to a small region around the subgridding interface. Nevertheless, there are some scenarios, such as the one in Section 2.5.3, where this effect can negatively impact the results.

2.5.3 Waveguide Reflection

The setup for this example was taken from [22, 154] and is aimed at evaluating spurious reflections caused by the subgridding interface when a mode propagates in a rectangular waveguide. In particular, the waveguide has dimensions $20 \text{ mm} \times 7 \text{ mm}$ and is excited with a TE_{10} mode. The spatial resolution was set to $\Delta x = \Delta y = 1 \text{ mm}$ and $\Delta z = 2 \text{ mm}$. A subgridding region with dimensions $16 \text{ mm} \times 3 \text{ mm} \times 40 \text{ mm}$ and grid ratio $r = 3$ is placed in the center of the waveguide. The reflections were computed from 120 ns of the waveforms by subtracting the uniform coarse grid fields from the subgridding result¹⁰.

The resulting reflections are shown in Figure 2.9. The figure indicates presence of resonances,

¹⁰Okoniewski, Anderson, and Stuchly [217] describe a technique for computing the reflections in a single simulation.

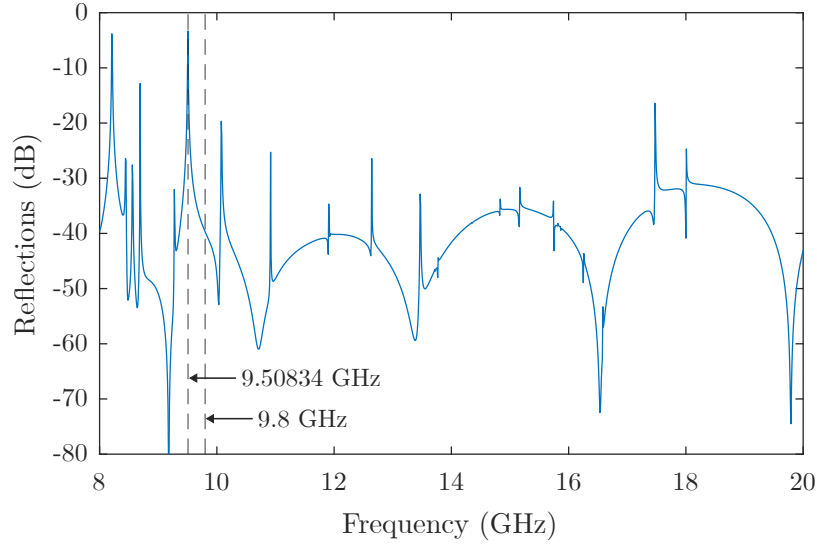


Figure 2.9: Relative reflected power due to the presence of subgridding region in the test of Section 2.5.3. The dashed vertical lines mark the frequencies used to produce the field plots in Figure 2.10.

which result in strong reflections at lower frequencies (8-11 GHz), with some resonance peaks also noticeable at higher frequencies. Figure 2.10 shows the fields when the grid is excited using a sinusoidal wave at 9.50834 GHz (the resonance marked in Figure 2.9) and a frequency that is away from that resonance, namely 9.8 GHz. The large fields are visible on the subgridding interface at 9.50834 GHz, indicating that the subgridding is capable of storing a significant amount of energy at that frequency.

The aforementioned resonances are not present in the results reported in [154] and [217]. Thus, this test suggests that there is a limitation in the accuracy of the proposed subgridding scheme. The advantage of the scheme is that the stable timestep is known analytically. In contrast, the stability analysis in [154] could only establish the existence of a stable time step, while its value had to be determined via simulation¹¹. The scheme in [217] is not provably stable.

2.5.4 Shielding Effectiveness

In this test, the subgridding algorithm in Section 2.4 is applied to investigate shielding effectiveness of the Mini-ITX computer enclosure in [177], shown in Figure 2.11. The shielding effectiveness at a given location for electric and magnetic field, respectively, is given by [218]

$$SE_E = -20 \log_{10} \left| \frac{E}{E_{\text{inc}}} \right|, \quad (2.93)$$

$$SE_H = -20 \log_{10} \left| \frac{H}{H_{\text{inc}}} \right|, \quad (2.94)$$

where E and H denote the amplitude of the electric and magnetic field and E_{inc} and H_{inc} are the corresponding amplitudes without the enclosure. In this test, the shielding effectiveness (2.93) at the

¹¹It was found to be the CFL limit of the fine grid [154].

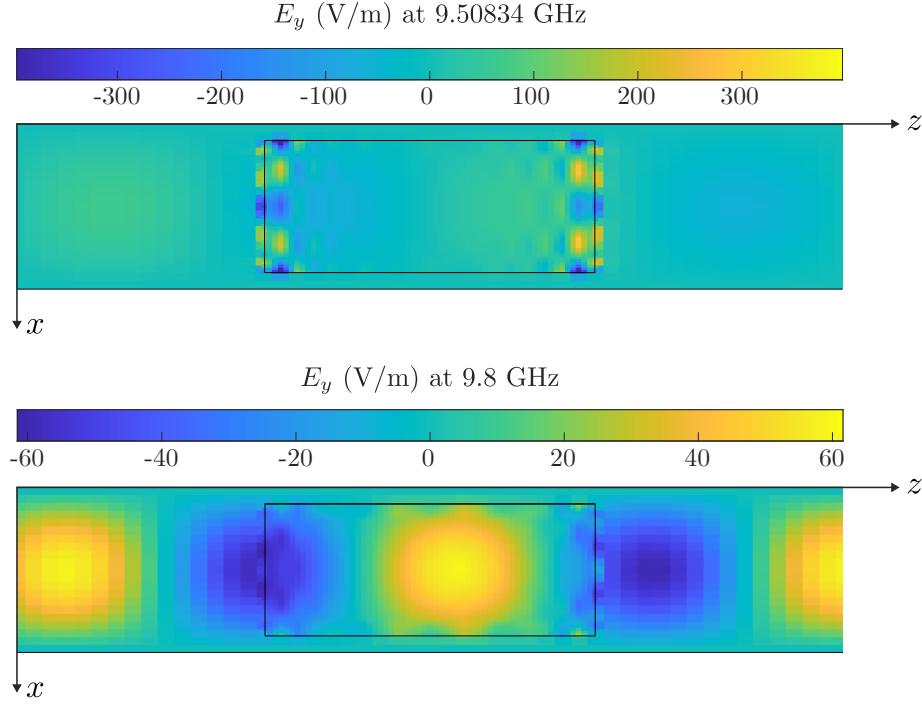


Figure 2.10: The y -component of electric field after 10 ns in the test of Section 2.5.3 when sinusoidal excitation is used at 9.50834 GHz (top panel) and 9.8 GHz (bottom panel).

center of the enclosure was estimated using the z component of electric field at (116, 108, 34) mm, relative to the inner corner of the enclosure. Likewise, SE_H in (2.94) was found from the y component of the magnetic field at (114, 108, 34) mm. The thickness of the metal was 0.8 mm. The main volume of the enclosure was discretized with a 4 mm grid step. A subgridding region of size 44 mm $\times$ 200 mm $\times$ 64 mm with a refinement ratio of $r_x = r_y = r_z = 5$ was used to resolve the structure of the 2.4 mm $\times$ 4 mm slots and the fields near them. As a reference, we run a simpler method for locally refining the grid, namely a non-uniform grid discussed in Section 1.5.1, such that a 20 mm $\times$ 184 mm $\times$ 60 mm area around the apertures is refined by varying the spatial step size. The walls of the enclosure have conductivity of 35 MS/m. As in [177], artificial losses were introduced to the system by setting conductivity inside the enclosure to 0.5 mS/m in order to shorten the time needed for the fields inside the enclosure to decay. The structure was simulated for 300 ns. The time step was chosen as 1.53 ps, which corresponds to 99% of the CFL limit (1.36) for a grid of resolution $\Delta = 0.8$ mm.

The simulated shielding effectiveness as a function of frequency is shown in Figure 2.12 for the nonuniform grid refinement and for the proposed subgridding. Very good correspondence between the two methods is observed for both SE_E and SE_H , with some small deviations at low frequency. At those frequencies, a very small amount of energy penetrates into the enclosure, aggravating the relative effect of spurious fields due to the subgridding interface. The speedup of the MATLAB implementation of both methods compared to a conventional uniform-grid FDTD is shown in Table 2.1. Subgridding achieved a speedup of 22.9 $\times$, while the nonuniform grid was only able to provide a speed up of 11.8 $\times$ due to the larger number of unknowns.

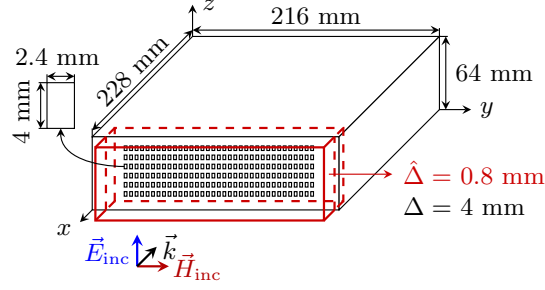


Figure 2.11: Simulation setup in the test of Section 2.5.4 (similar to [177]). Interior dimensions of the case are shown.

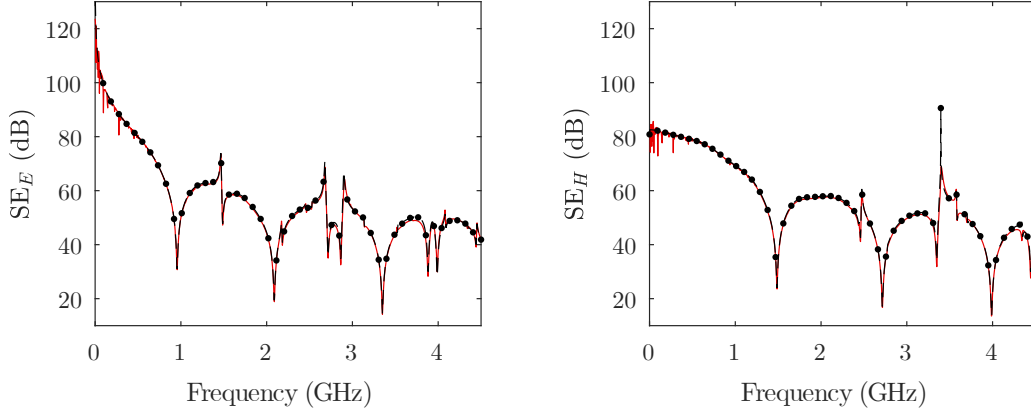


Figure 2.12: Simulated shielding effectiveness vs. frequency for nonuniform grid ($-\bullet-$) and proposed subgridding ($---$).

Table 2.1: Computational cost for different discretization in the shielding effectiveness example of Section 2.5.4.

Method	Uniform fine	Non-uniform	Proposed
Number of primary cells	64,879,056	4,773,384	1,906,960
Simulation time ^a (hours)	755.7	64.2	32.9
Speedup	Reference	11.8×	22.9×

^a Estimated from 40 time steps.

2.5.5 Meta-Screen

In this example, the proposed subgridding algorithm is used to study the three-slot meta-screen shown in Figure 2.13. The setup for this simulation is similar to [219], except for differences in discretization and screen material. In particular, we set the plate conductivity to 1.3×10^6 S/m and thickness of the plate to 6 mils (0.1524 mm), based on the value in [220]. The slot in the center is 13.2 mm by 1.2 mm and the two satellite slots on the sides are 17 mm by 0.6 mm, placed at a center-to-center distance of 3 mm away from the central slot. These dimensions were designed to achieve sub-wavelength focusing at 10 GHz [220]. A line of probes is placed 4.572 mm away from the meta-screen, which corresponds to approximately 15% of the wavelength at 10 GHz – distance at which measurements were performed in [220].

As in [219], we terminate the region before the meta-screen with PEC walls on the east and

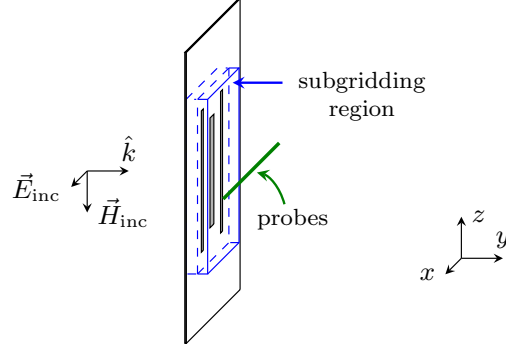


Figure 2.13: Setup of the meta-screen example of Section 2.5.5.

the west sides, and with perfect magnetic conductor walls on the top and the bottom sides. A five-cell absorbing layer is added on all sides in order to mimic an unbounded domain. The resulting simulation region dimensions were 25.2 mm $\times$ 19.2024 mm $\times$ 43.4 mm, including the absorbing layer.

The satellite slots have widths of only 2% of the wavelength, which makes the problem intrinsically multi-scale. Moreover, [219] reports resonance effects that make it difficult to resolve the structure near the design frequency. We choose this example to investigate the performance of the proposed subgridding scheme.

We set the coarse grid to $\Delta x = \Delta z = 0.7$ mm, and $\Delta y = 0.1524$ mm. In order to properly resolve the meta-screen structure and the fields, we refine a 9.8 mm $\times$ 2.5908 mm $\times$ 21 mm area around the slots with $r_x = r_z = 7$ and $r_y = 3$, using the proposed subgridding method. As a reference, we perform a conventional FDTD simulation with a nonuniform grid, where a region of dimensions 9.8 mm $\times$ 1.0668 mm $\times$ 21 mm is refined by $r_x = r_z = 7$ and $r_y = 3$ by means of varying Δx , Δy , and Δz . We also simulate the case where the entire structure is discretized with the coarse mesh.

A modulated Gaussian waveform with 10 GHz central frequency and 8.24 GHz half-width at half-maximum bandwidth was used to excite the structure with a sheet of x -directed uniform current density on the south side. The proposed method and FDTD with nonuniform discretization were run at $\Delta t = 0.1362$ ps, whereas the coarse grid simulation had the iteration time step of 0.4810 ps. These time steps correspond to 99% of the CFL limit of each mesh. The waveform was recorded for 5.5 ns, as in [219].

Figure 2.14 shows the amplitude of the x component of the electric field at 10 GHz, which was obtained with a Fourier transform of the time-domain waveforms. The results from subgridding are in very good agreement with the reference simulation. In contrast, the coarse resolution was insufficient to perform an accurate simulation of the meta-screen. This is not surprising, since any mesh with Δx larger than 0.3 mm or Δz larger than 0.1 mm cannot correctly resolve the meta-screen structure, let alone the fields around the slots.

Simulation times, as well as the number of primary FDTD cells are shown in Table 2.5.5. Subgridding almost doubles the speedup compared to the nonuniform grid refinement, which is already much faster than the simulation with uniform grid. This is a result of the lower number of unknowns associated with subgridding. The guarantee of stability of the proposed method allows for the reduction of simulation time without sacrificing reliability. Moreover, the ability of the method to handle material traverse allows placing the subgridding region across the meta-screen.

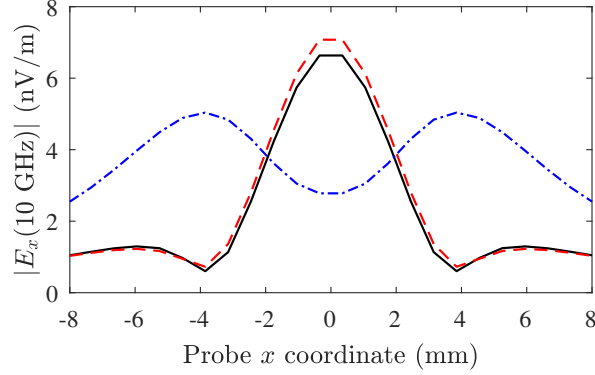


Figure 2.14: Magnitude of x -directed electric field along the line of probes in the focusing meta-screen example of Section 2.5.5 for uniform coarse grid (—), nonuniform grid (—), and the proposed subgridding method (—).

Table 2.2: Computational cost for different discretization in the meta-screen example of Section 2.5.5.

Method	Number of primary cells	Simulation time	Speedup
Coarse	281,232	0.18 hours	524.8
Fine	41,341,104	95.81 hours ^a	reference
Nonuniform	4,065,600	11.46 hours	8.4
Proposed	1,323,672	5.80 hours	16.5

^a Estimated runtime from 30 time steps.

2.6 Conclusion

In this chapter, the concepts of dissipative systems from control theory [205, 206] were extended to FDTD for the Maxwell equations [33, 35] in order to develop a modular and physically intuitive framework for creating new stable schemes. A key step in applying these concepts were the conditions on the energy conservation and dissipation formulated for an open region, which is in contrast to earlier works in FDTD and FIT [197, 202] that only studied these properties for an isolated region. The stability framework developed based on these theoretical results was applied to derive a provably stable subgridding scheme, which supports any odd grid refinement ratio and has material traverse capabilities. Moreover, the proposed scheme respects by construction the principle of energy dissipation or, for the case when there are no ohmic losses in the region, the principle of energy conservation.

Despite the aforementioned advantages of the proposed subgridding scheme, the results in Section 2.5.3 indicate that there is room to improve its accuracy. Nevertheless, this work is a step forward in developing a subgridding scheme that is accurate, energy conserving, and provably stable under a limit on the time step that is known a priori. If a more accurate relation can be found instead of the interpolation conditions in Section 2.4.1, only the stability proof in Section 2.4.2 would need to be revised, without any modification to the energy conservation analysis of the grids and without the need to determine the new stability limit on the time step. In addition to subgridding, the applicability of the proposed framework for constructing stable FDTD schemes has been demonstrated by developing a method for embedding reduced order models in 3D FDTD and extending

the stability limit of the resulting systems [212].

Energy Conservation Theory for FDTD with Potentials and Subgridding

Motivated by the possible applications in simulating quantum models [38, 48, 145], this chapter extends the proposed dissipation theory to FDTD for the Maxwell equations written in terms of potentials. As a starting point, we take the recently proposed FDTD-P scheme in [38] for inhomogeneous lossless media. Similarly to Chapter 2, we write the governing equations for a region in the form of discrete-time dynamical systems and study their energy conservation properties for the case of an open region, which can exchange energy through the boundary. Based on these results, we develop a provably stable and energy conserving subgridding algorithm for FDTD-P using an analogous approach to the previous chapter. To the best of our knowledge, subgridding schemes for potentials-based FDTD formulations have not been proposed before. In fact, no potentials-based FDTD method is yet comparable in its level of development to the traditional FDTD. The difficulty of ensuring stability is one of the major roadblocks in the advancement of these methods, which this chapter aims to address. The chapter is an expanded version of [221].

3.1 The FDTD-P Scheme [38]

The scheme in [38] discretizes the Maxwell equations in heterogeneous media written in terms of potentials. These equations can be derived following a similar procedure as the one described in Section 1.2 [41]. In particular, substituting (1.8) and (1.9) into (1.1a)–(1.1d), one obtains

$$\nabla \times (\mu^{-1} \nabla \times \vec{A}) = -\varepsilon \frac{\partial}{\partial t} \nabla \phi - \varepsilon \frac{\partial^2 \vec{A}}{\partial t^2} + \vec{J}_s, \quad (3.1)$$

$$-\nabla \cdot (\varepsilon \nabla \phi) - \frac{\partial}{\partial t} \nabla \cdot (\varepsilon \vec{A}) = \rho_v, \quad (3.2)$$

where $\vec{J}$ in (1.1b) is assumed to only have a contribution from the impressed current density¹ $\vec{J}_s$. The Lorenz gauge generalized for inhomogeneous media is given by [41, 42, 147]

$$\nabla \cdot (\varepsilon \vec{A}) + \chi \frac{\partial \phi}{\partial t} = 0, \quad (3.3)$$

where $\chi = \mu \varepsilon^2$. Using (3.3) to simplify (3.1)–(3.2) and rearranging, the wave equations governing the scalar and vector potentials become

$$\chi \frac{\partial^2 \phi}{\partial t^2} = \nabla \cdot \varepsilon \nabla \phi + \rho_v, \quad (3.4)$$

$$\varepsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\nabla \times (\mu^{-1} \nabla \times \vec{A}) + \varepsilon \nabla (\chi^{-1} \nabla \cdot (\varepsilon \vec{A})) + \vec{J}_s. \quad (3.5)$$

We consider a region divided into $n_x \times n_y \times n_z$ primary cells with dimensions $\Delta x \times \Delta y \times \Delta z$, shown in Figure 3.1. In [38], the discretization of (3.4) and (3.5) over this region is performed using second order centered finite differences. The scalar potential in (3.4) is sampled at the nodes of the primary cells (*primary nodes*) and integer time instances. With reference to Figure 3.1, the discretization of (3.4) reads

$$\begin{aligned} \chi|_{i,j,k} \frac{\phi|_{i,j,k}^{n+1} - 2\phi|_{i,j,k}^n + \phi|_{i,j,k}^{n-1}}{(\Delta t)^2} &= \frac{\varepsilon_x|_{i+\frac{1}{2},j,k} \left(\phi|_{i+1,j,k}^n - \phi|_{i,j,k}^n \right) - \varepsilon_x|_{i-\frac{1}{2},j,k} \left(\phi|_{i,j,k}^n - \phi|_{i-1,j,k}^n \right)}{(\Delta x)^2} \\ &+ \frac{\varepsilon_y|_{i,j+\frac{1}{2},k} \left(\phi|_{i,j+1,k}^n - \phi|_{i,j,k}^n \right) - \varepsilon_y|_{i,j-\frac{1}{2},k} \left(\phi|_{i,j,k}^n - \phi|_{i,j-1,k}^n \right)}{(\Delta y)^2} \\ &+ \frac{\varepsilon_z|_{i,j,k+\frac{1}{2}} \left(\phi|_{i,j,k+1}^n - \phi|_{i,j,k}^n \right) - \varepsilon_z|_{i,j,k-\frac{1}{2}} \left(\phi|_{i,j,k}^n - \phi|_{i,j,k-1}^n \right)}{(\Delta z)^2} + \rho_v|_{i,j,k}^n. \end{aligned} \quad (3.6)$$

Using (3.6), $\phi|^{n+1}$ can be computed from the values of ϕ at the previous time steps n and $n-1$. Equation (3.5) can be discretized similarly to (3.4) [38]. As shown in Figure 3.1, samples of the vector potential are defined at the primary edges at time instances shifted from integer by $\Delta t/2$. The resulting update equation for the x component of the vector potential is given by

$$\begin{aligned} \varepsilon_z|_{i+\frac{1}{2},j,k} \frac{A_x|_{i+\frac{1}{2},j,k}^{n+\frac{3}{2}} - 2A_x|_{i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + A_x|_{i+\frac{1}{2},j,k}^{n-\frac{1}{2}}}{(\Delta t)^2} &= \\ &- \left(\frac{\mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k} B_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - \mu_z^{-1}|_{i+\frac{1}{2},j-\frac{1}{2},k} B_z|_{i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} \right. \\ &- \left. \frac{\mu_y^{-1}|_{i+\frac{1}{2},j,k+\frac{1}{2}} B_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \mu_y^{-1}|_{i+\frac{1}{2},j,k-\frac{1}{2}} B_y|_{i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right) \\ &- \varepsilon_x|_{i+\frac{1}{2},j,k} \frac{\kappa|_{i+1,j,k}^{n+\frac{1}{2}} - \kappa|_{i,j,k}^{n+\frac{1}{2}}}{\Delta x} + J_{s,x}|_{i+\frac{1}{2},j,k}^{n+\frac{1}{2}}, \end{aligned} \quad (3.7)$$

where B_x , B_y , and B_z represent components of the curl of the vector potential, discretized using

¹A discussion of nonzero conductivity in potentials-based FDTD can be found in [146] for the case of homogeneous media.

central differences. For example,

$$B_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = \frac{A_y|_{i+1,j+\frac{1}{2},k}^{n+\frac{1}{2}} - A_y|_{i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x} - \frac{A_x|_{i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - A_x|_{i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y}. \quad (3.8)$$

Notation B_x , B_y , and B_z is chosen in order to recognize that the curl of the vector potential equals to the magnetic flux density. Quantity κ in (3.7) is given by

$$\begin{aligned} \kappa|_{i,j,k}^{n+\frac{1}{2}} = & -\frac{1}{\chi|_{i,j,k}} \left(\frac{\varepsilon_x|_{i+\frac{1}{2},j,k} A_x|_{i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \varepsilon_x|_{i-\frac{1}{2},j,k} A_x|_{i-\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta x} \right. \\ & + \frac{\varepsilon_y|_{i,j+\frac{1}{2},k} A_y|_{i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - \varepsilon_y|_{i,j-\frac{1}{2},k} A_y|_{i,j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} \\ & \left. + \frac{\varepsilon_z|_{i,j,k+\frac{1}{2}} A_z|_{i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \varepsilon_z|_{i,j,k-\frac{1}{2}} A_z|_{i,j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right), \quad (3.9) \end{aligned}$$

which is a discretization of $\kappa = -\chi^{-1} \nabla \cdot (\varepsilon \vec{A})$. In the continuous domain, κ equals to the time derivative of ϕ , as dictated by (3.3). Lastly, the charge density ρ_v in (3.6) is related to the current density $J_{s,x}$ in (3.7), as well as $J_{s,y}$ and $J_{s,z}$, through the discretized form of continuity equation (1.19) [38].

3.2 Dynamical System Formulation

In this section, a region that is discretized with FDTD-P equations [38] is written as a discrete-time state-space model similar to the model used in Chapter 2. To be more precise, two models are formulated: one describing the temporal evolution of the scalar potential and the other corresponding to the vector potential.

3.2.1 Continuous State Equations for the Scalar and Vector Potentials

PDEs (3.4) and (3.5) are second order in time. Since the formulation in Chapter 2 involves only first-order time derivatives, we make a change of variables to rewrite (3.4) and (3.5) in that form. For the case of (3.4), we introduce the following variables: $[\nabla\phi]$ for the gradient of the scalar potential and $[\partial_t\phi]$ for the temporal derivative of the scalar potential. The square brackets are used to indicate that we take the values of $\nabla\phi$ and $\partial_t\phi$ as unknowns. With that, (3.4) can be written in terms of $[\partial_t\phi]$ and $[\nabla\phi]$ as

$$\chi \frac{\partial}{\partial t} [\partial_t\phi] = \nabla \cdot \varepsilon [\nabla\phi], \quad (3.10a)$$

$$\varepsilon \frac{\partial}{\partial t} [\nabla\phi] = \varepsilon \nabla [\partial_t\phi], \quad (3.10b)$$

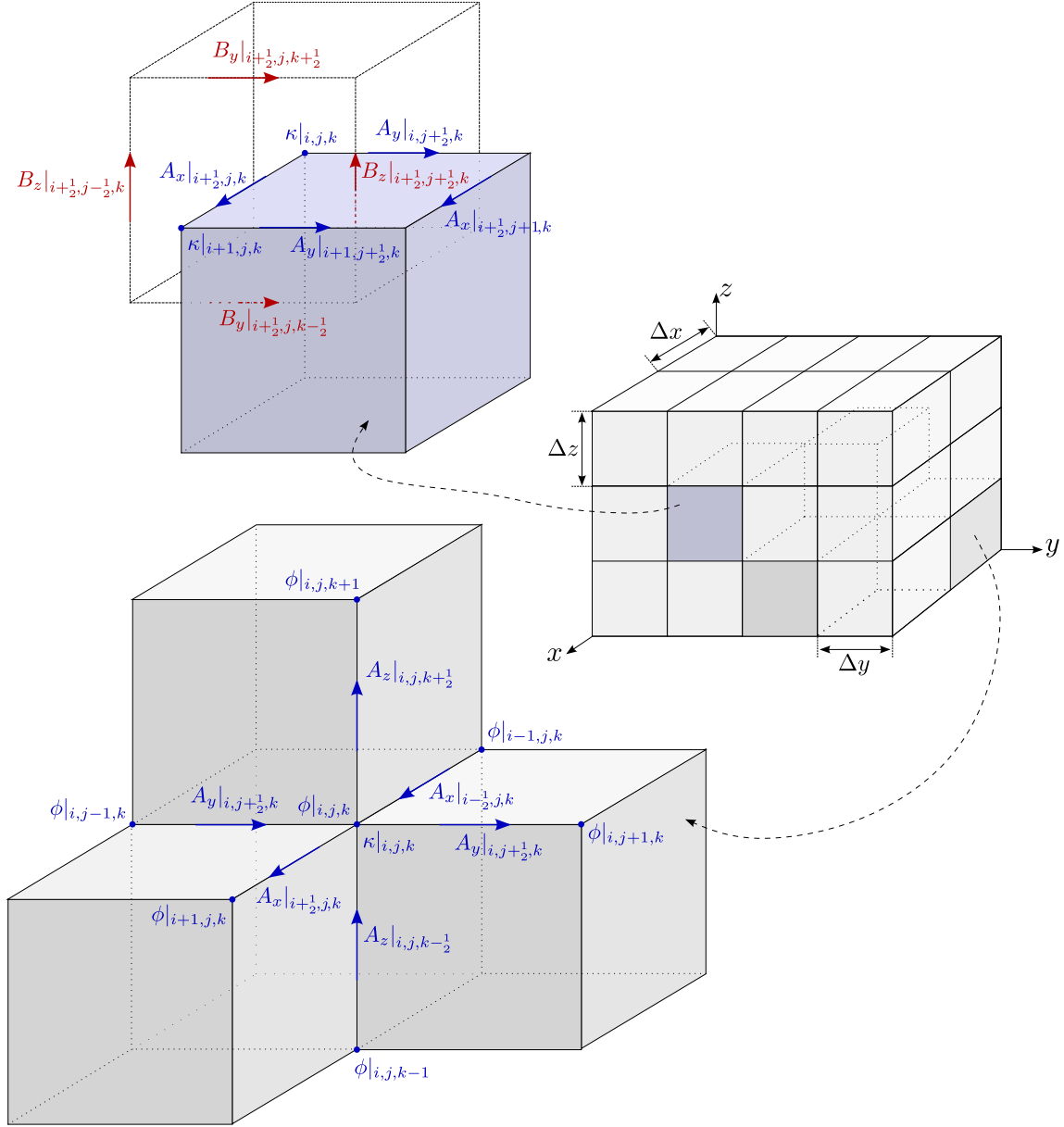


Figure 3.1: Illustration of the spatial discretization in FDTD-P [38], showing the samples involved in (3.6)–(3.9).

where it is assumed that no impressed current density exists, resulting in $\rho_v = 0$. Similarly, (3.5) can be written as

$$\mu^{-1} \frac{\partial \vec{B}}{\partial t} = \mu^{-1} \nabla \times [\partial_t \vec{A}], \quad (3.11a)$$

$$\varepsilon \frac{\partial [\partial_t \vec{A}]}{\partial t} = -\nabla \times \mu^{-1} \vec{B} - \varepsilon \nabla \kappa, \quad (3.11b)$$

$$\chi \frac{\partial \kappa}{\partial t} = -\nabla \cdot \varepsilon [\partial_t \vec{A}]. \quad (3.11c)$$

3.2.2 Discretized State Equations for the Scalar and Vector Potentials

In this section we define the discretized equations corresponding to (3.10a)–(3.10b) and (3.11a)–(3.11c). For nodes and edges inside the region, the resulting equations are equivalent to the scheme in [38], which was outlined in Section 3.1. However, on the boundary, we introduce special equations with hanging variables in a way similar to what was done in FDTD for the Maxwell equations written using fields.

Equations for the Scalar Potential

The variable $[\partial_t \phi]$ is sampled at the primary nodes and time instances $t = (n - 0.5)\Delta t$. The samples of $[\nabla \phi]$ that eventually become part of the state vector are defined at the conventional primary edges at the integer time instances $t = n\Delta t$. We refer to those samples as the conventional samples of $[\nabla \phi]$, since they can be computed from the scalar potential using the finite difference approximation, as was done in [38].

With these definitions, consider first the discretization of (3.10a). For the primary nodes strictly inside the region, such as the one shown in Figure 3.2a, the discretized equation is given by

$$\begin{aligned} \Delta x \Delta y \Delta z \frac{\chi|_{i,j,k}}{\Delta t} [\partial_t \phi]_{i,j,k}^{n+\frac{1}{2}} &= \Delta x \Delta y \Delta z \frac{\chi|_{i,j,k}}{\Delta t} [\partial_t \phi]_{i,j,k}^{n-\frac{1}{2}} \\ &\quad - \Delta y \Delta z \varepsilon_x|_{i-\frac{1}{2},j,k} [\partial_x \phi]_{i-\frac{1}{2},j,k}^n + \Delta y \Delta z \varepsilon_x|_{i+\frac{1}{2},j,k} [\partial_x \phi]_{i+\frac{1}{2},j,k}^n \\ &\quad - \Delta x \Delta z \varepsilon_y|_{i,j-\frac{1}{2},k} [\partial_y \phi]_{i,j-\frac{1}{2},k}^n + \Delta x \Delta z \varepsilon_y|_{i,j+\frac{1}{2},k} [\partial_y \phi]_{i,j+\frac{1}{2},k}^n \\ &\quad - \Delta x \Delta y \varepsilon_z|_{i,j,k-\frac{1}{2}} [\partial_z \phi]_{i,j,k-\frac{1}{2}}^n + \Delta x \Delta y \varepsilon_z|_{i,j,k+\frac{1}{2}} [\partial_z \phi]_{i,j,k+\frac{1}{2}}^n, \end{aligned} \quad (3.12)$$

where samples such as $[\partial_x \phi]_{i-0.5,j,k}^n$ are the conventional samples of the components of the gradient. The permittivity samples along the conventional primary edges, such as $\varepsilon_x|_{i+0.5,j,k}$ in (3.12), are obtained from their primary cell values in the way described in Section 2.1. The value of the parameter $\chi|_{i,j,k}$ is obtained by averaging the values in the eight² surrounding primary cells, which have the node (i, j, k) at one of their corners. As will become clear shortly, the values of the samples of the gradient of the scalar potential are equal to the finite difference approximations computed from the neighboring samples of ϕ . With that, (3.12) is equivalent to (3.6). The only differences are in the use of the first-order temporal finite difference and the multiplication factor $\Delta x \Delta y \Delta z$. The latter is the volume of the secondary cell containing the node (i, j, k) . We can interpret the right hand side of (3.12) as the discrete flux of $\varepsilon \nabla \phi$ through the faces of that secondary cell.

²Likewise, for the nodes strictly on one face of the boundary, the average involves four primary cells, and so on.

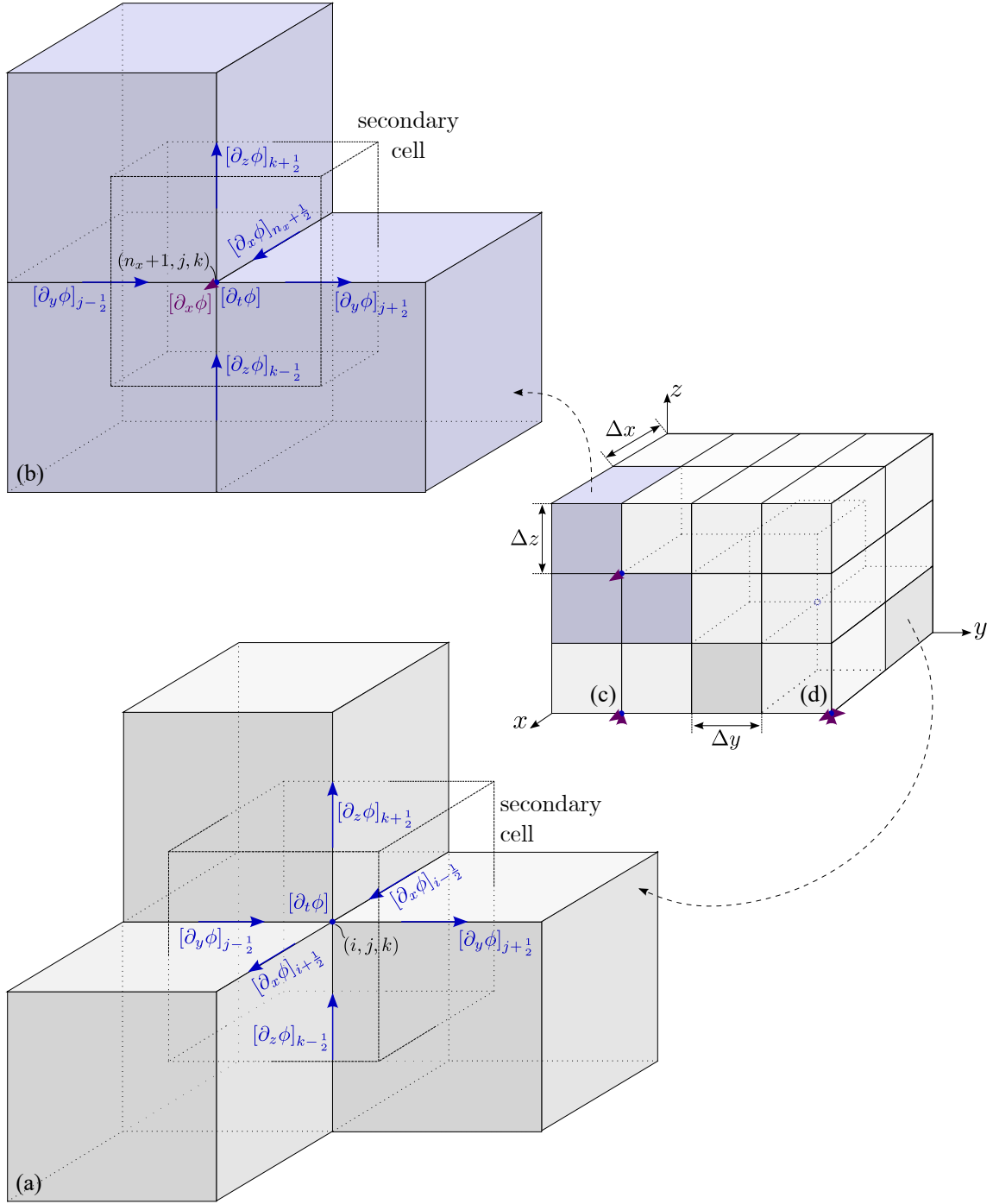


Figure 3.2: Samples involved in discretization of (3.10a). The hanging variables are indicated with purple. (a) illustrates equation (3.12). (b) illustrates (3.13). (c) labels the $[\partial_t \phi]$ sample and the hanging variables in (3.15), and (d) labels the corresponding quantities in (3.16). Repeating subscripts (i, j, k) and $(n_x + 1, j, k)$ are omitted.

For the nodes on the boundary, (3.12) would involve samples of $[\nabla\phi]$ outside the boundary. To avoid that, we write the modified equations by introducing hanging variables that represent the normal component of the gradient of the scalar potential on the boundary. For example, the equation for the $[\partial_t\phi]$ sample at a node on the east face of the region boundary shown in Figure 3.2b is given by

$$\begin{aligned} \frac{\Delta x}{2} \Delta y \Delta z \frac{\chi|_{n_x+1,j,k}}{\Delta t} [\partial_t\phi]_{n_x+1,j,k}^{n+\frac{1}{2}} &= \frac{\Delta x}{2} \Delta y \Delta z \frac{\chi|_{n_x+1,j,k}}{\Delta t} [\partial_t\phi]_{n_x+1,j,k}^{n-\frac{1}{2}} \\ &\quad - \Delta y \Delta z \varepsilon_x|_{n_x+\frac{1}{2},j,k} [\partial_x\phi]_{n_x+\frac{1}{2},j,k}^n + \Delta y \Delta z \varepsilon_x|_{n_x+1,j,k} [\partial_x\phi]_{n_x+1,j,k}^n \\ &\quad - \frac{\Delta x}{2} \Delta z \varepsilon_y|_{n_x+1,j-\frac{1}{2},k} [\partial_y\phi]_{n_x+1,j-\frac{1}{2},k}^n + \frac{\Delta x}{2} \Delta z \varepsilon_y|_{n_x+1,j+\frac{1}{2},k} [\partial_y\phi]_{n_x+1,j+\frac{1}{2},k}^n \\ &\quad - \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,k-\frac{1}{2}} [\partial_z\phi]_{n_x+1,j,k-\frac{1}{2}}^n + \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,k+\frac{1}{2}} [\partial_z\phi]_{n_x+1,j,k+\frac{1}{2}}^n, \end{aligned} \quad (3.13)$$

where the corresponding secondary cell has dimensions $\Delta x/2 \times \Delta y \times \Delta z$. Sample $[\partial_x\phi]_{n_x+1,j,k}$ is the hanging variable that is located at the additional primary edge placed on the boundary of the region. This type of edges will be referred to as the *normal primary edges*. The normal primary edge permittivity samples are averaged similarly to the conventional primary edges. For example, the normal sample in (3.13) is given by

$$\varepsilon_x|_{n_x+1,j,k} = \frac{\varepsilon|_{n_x+\frac{1}{2},j-\frac{1}{2},k-\frac{1}{2}} + \varepsilon|_{n_x+\frac{1}{2},j+\frac{1}{2},k-\frac{1}{2}} + \varepsilon|_{n_x+\frac{1}{2},j-\frac{1}{2},k+\frac{1}{2}} + \varepsilon|_{n_x+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}}}{4}. \quad (3.14)$$

The two types of primary edges (conventional and normal), as well as all other edges, faces, and cells associated with the primary and secondary grids are summarized in Appendix A.

Similarly to (3.13), for the primary nodes on the edges of the region, the update equation is written using two hanging variables. For example, for the bottom-east sample labeled as (c) in Figure 3.2, the equation reads

$$\begin{aligned} \frac{\Delta x}{2} \Delta y \frac{\Delta z}{2} \frac{\chi|_{n_x+1,j,1}}{\Delta t} [\partial_t\phi]_{n_x+1,j,1}^{n+\frac{1}{2}} &= \frac{\Delta x}{2} \Delta y \frac{\Delta z}{2} \frac{\chi|_{n_x+1,j,1}}{\Delta t} [\partial_t\phi]_{n_x+1,j,1}^{n-\frac{1}{2}} \\ &\quad - \Delta y \frac{\Delta z}{2} \varepsilon_x|_{n_x+\frac{1}{2},j,1} [\partial_x\phi]_{n_x+\frac{1}{2},j,1}^n + \Delta y \frac{\Delta z}{2} \varepsilon_x|_{n_x+1,j,1} [\partial_x\phi]_{n_x+1,j,1}^n \\ &\quad - \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1,j-\frac{1}{2},1} [\partial_y\phi]_{n_x+1,j-\frac{1}{2},1}^n + \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1,j+\frac{1}{2},1} [\partial_y\phi]_{n_x+1,j+\frac{1}{2},1}^n \\ &\quad - \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,1} [\partial_z\phi]_{n_x+1,j,1}^n + \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,\frac{3}{2}} [\partial_z\phi]_{n_x+1,j,\frac{3}{2}}^n. \end{aligned} \quad (3.15)$$

For the nodes on the corners of the region, the equation is written for the secondary cells of dimensions $\Delta x/2 \times \Delta y/2 \times \Delta z/2$ and each involves three hanging variables. Taking as an example

the bottom-north-east sample labeled as (d) in Figure 3.2,

$$\begin{aligned}
\frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\chi|_{n_x+1, n_y+1, 1}}{\Delta t} [\partial_t \phi]_{n_x+1, n_y+1, 1}^{n+\frac{1}{2}} &= \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\chi|_{n_x+1, n_y+1, 1}}{\Delta t} [\partial_t \phi]_{n_x+1, n_y+1, 1}^{n-\frac{1}{2}} \\
&- \frac{\Delta y}{2} \frac{\Delta z}{2} \varepsilon_x|_{n_x+\frac{1}{2}, n_y+1, 1} [\partial_x \phi]_{n_x+\frac{1}{2}, n_y+1, 1}^n + \frac{\Delta y}{2} \frac{\Delta z}{2} \varepsilon_x|_{n_x+1, n_y+1, 1} [\partial_x \phi]_{n_x+1, n_y+1, 1}^n \\
&- \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1, n_y+\frac{1}{2}, 1} [\partial_y \phi]_{n_x+1, n_y+\frac{1}{2}, 1}^n + \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1, n_y+1, 1} [\partial_y \phi]_{n_x+1, n_y+1, 1}^n \\
&- \frac{\Delta x}{2} \frac{\Delta y}{2} \varepsilon_z|_{n_x+1, n_y+1, 1} [\partial_z \phi]_{n_x+1, n_y+1, 1}^n + \frac{\Delta x}{2} \frac{\Delta y}{2} \varepsilon_z|_{n_x+1, n_y+1, \frac{3}{2}} [\partial_z \phi]_{n_x+1, n_y+1, \frac{3}{2}}^n. \quad (3.16)
\end{aligned}$$

Next, consider the discretization of (3.10b) at each of the conventional primary edges. For the x -directed edges strictly inside the region, including the normal edges that touch the boundary, we write the following equation

$$\begin{aligned}
\Delta x \Delta y \Delta z \frac{\varepsilon_x|_{i+\frac{1}{2}, j, k}}{\Delta t} [\partial_x \phi]_{i+\frac{1}{2}, j, k}^{n+1} &= \Delta x \Delta y \Delta z \frac{\varepsilon_x|_{i+\frac{1}{2}, j, k}}{\Delta t} [\partial_x \phi]_{i+\frac{1}{2}, j, k}^n \\
&+ \Delta y \Delta z \varepsilon_x|_{i+\frac{1}{2}, j, k} \left(-[\partial_t \phi]_{i, j, k}^{n+\frac{1}{2}} + [\partial_t \phi]_{i+1, j, k}^{n+\frac{1}{2}} \right), \quad (3.17)
\end{aligned}$$

where Δx is the length of the primary edge and $\Delta y \Delta z$ is the area of the secondary face through which the edge passes. The equation is obtained from

$$\frac{[\partial_x \phi]_{i+\frac{1}{2}, j, k}^{n+1} - [\partial_x \phi]_{i+\frac{1}{2}, j, k}^n}{\Delta t} = \frac{[\partial_t \phi]_{i+1, j, k}^{n+\frac{1}{2}} - [\partial_t \phi]_{i, j, k}^{n+\frac{1}{2}}}{\Delta x} \quad (3.18)$$

by multiplying both sides by the permittivity, the area of the corresponding primary edge, and the area of the corresponding secondary face. These modifications will allow writing the matrix equations in the form that facilitates the analysis of the energy conservation.

For the primary edges tangential to the boundary, a similar equation is written to (3.17), using the corresponding secondary face area. For example, for the edges tangential to the south face of the boundary,

$$\begin{aligned}
\Delta x \frac{\Delta y}{2} \Delta z \frac{\varepsilon_x|_{i+\frac{1}{2}, 1, k}}{\Delta t} [\partial_x \phi]_{i+\frac{1}{2}, 1, k}^{n+1} &= \Delta x \frac{\Delta y}{2} \Delta z \frac{\varepsilon_x|_{i+\frac{1}{2}, 1, k}}{\Delta t} [\partial_x \phi]_{i+\frac{1}{2}, 1, k}^n \\
&+ \frac{\Delta y}{2} \Delta z \varepsilon_x|_{i+\frac{1}{2}, 1, k} \left(-[\partial_t \phi]_{i, 1, k}^{n+\frac{1}{2}} + [\partial_t \phi]_{i+1, 1, k}^{n+\frac{1}{2}} \right). \quad (3.19)
\end{aligned}$$

Likewise, for the edges shared between the bottom and south faces of the boundary, the equation reads

$$\begin{aligned}
\Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_x|_{i+\frac{1}{2}, 1, 1}}{\Delta t} [\partial_x \phi]_{i+\frac{1}{2}, 1, 1}^{n+1} &= \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_x|_{i+\frac{1}{2}, 1, 1}}{\Delta t} [\partial_x \phi]_{i+\frac{1}{2}, 1, 1}^n \\
&+ \frac{\Delta y}{2} \frac{\Delta z}{2} \varepsilon_x|_{i+\frac{1}{2}, 1, 1} \left(-[\partial_t \phi]_{i, 1, 1}^{n+\frac{1}{2}} + [\partial_t \phi]_{i+1, 1, 1}^{n+\frac{1}{2}} \right). \quad (3.20)
\end{aligned}$$

Equations for the Vector Potential

Next, we consider the discretization of (3.11a)–(3.11c) for the vector potential. The conventional samples of magnetic flux density $\vec{B}$, i.e. those samples that will be part of the state vector, are

defined at the conventional secondary edges at the time instances $t = (n - 0.5)\Delta t$, similarly to the samples of the magnetic field intensity $\vec{H}$ in Chapter 2. The conventional samples of time-derivative of the vector potential $[\partial_t \vec{A}]$ are defined on the conventional primary edges at the integer time steps $t = n\Delta t$. Samples of κ are defined on the primary nodes at $t = (n - 0.5)\Delta t$.

The dynamical system equations corresponding to (3.11a) are written in a way analogous to those described in Section 2.1.1 for updating the magnetic field intensity $\vec{H}$. This can be done because the samples of $\vec{B}$ and $[\partial_t \vec{A}]$ are placed at the same locations as $\vec{H}$ and $\vec{E}$, respectively. For example, the equation for B_z samples with k indices ranging from 2 to n_z is written similarly to (2.9) as

$$\begin{aligned} \Delta z \Delta x \Delta y \frac{\mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta t} B_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} &= \Delta z \Delta x \Delta y \frac{\mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta t} B_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n-\frac{1}{2}} \\ &+ \Delta z \mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k} \left(\Delta x [\partial_t A_x]_{i+\frac{1}{2},j,k}^n - \Delta x [\partial_t A_x]_{i+\frac{1}{2},j+1,k}^n \right. \\ &\quad \left. - \Delta y [\partial_t A_y]_{i,j+\frac{1}{2},k}^n + \Delta y [\partial_t A_y]_{i+1,j+\frac{1}{2},k}^n \right), \end{aligned} \quad (3.21)$$

where μ^{-1} at the conventional secondary edges is defined the same way as in Section 2.1. For the samples on the top face of the boundary, the equation is analogous to (2.10)

$$\begin{aligned} \frac{\Delta z}{2} \Delta x \Delta y \frac{\mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},n_z+1}}{\Delta t} B_z|_{i+\frac{1}{2},j+\frac{1}{2},n_z+1}^{n+\frac{1}{2}} &= \frac{\Delta z}{2} \Delta x \Delta y \frac{\mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},n_z+1}}{\Delta t} B_z|_{i+\frac{1}{2},j+\frac{1}{2},n_z+1}^{n-\frac{1}{2}} \\ &+ \frac{\Delta z}{2} \mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},n_z+1} \left(\Delta x [\partial_t A_x]_{i+\frac{1}{2},j,n_z+1}^n - \Delta x [\partial_t A_x]_{i+\frac{1}{2},j+1,n_z+1}^n \right. \\ &\quad \left. - \Delta y [\partial_t A_y]_{i,j+\frac{1}{2},n_z+1}^n + \Delta y [\partial_t A_y]_{i+1,j+\frac{1}{2},n_z+1}^n \right), \end{aligned} \quad (3.22)$$

and similarly for the B_z samples on the bottom face of the boundary.

The discretized form of equation (3.11b) is defined analogously to (2.5), (2.7), and (2.8) for $\vec{E}$ in Section 2.1.1, except for the additional term in (3.11b) involving $\nabla \kappa$. In particular, for the $[\partial_t \vec{A}]$ samples at the internal conventional primary edges, such as the x -directed sample in Figure 3.3a, the equation reads

$$\begin{aligned} \Delta x \Delta y \Delta z \frac{\varepsilon_z|_{i+\frac{1}{2},j,k}}{\Delta t} [\partial_t A_x]_{i+\frac{1}{2},j,k}^{n+1} &= \Delta x \Delta y \Delta z \frac{\varepsilon_z|_{i+\frac{1}{2},j,k}}{\Delta t} [\partial_t A_x]_{i+\frac{1}{2},j,k}^n \\ &- \Delta x \left(\Delta y \mu_y^{-1}|_{i+\frac{1}{2},j,k-\frac{1}{2}} B_y|_{i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} - \Delta y \mu_y^{-1}|_{i+\frac{1}{2},j,k+\frac{1}{2}} B_y|_{i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right. \\ &\quad \left. - \Delta z \mu_z^{-1}|_{i+\frac{1}{2},j-\frac{1}{2},k} B_z|_{i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} + \Delta z \mu_z^{-1}|_{i+\frac{1}{2},j+\frac{1}{2},k} B_z|_{i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &\quad - \Delta y \Delta z \varepsilon_x|_{i+\frac{1}{2},j,k} \left(-\kappa|_{i,j,k}^{n+\frac{1}{2}} + \kappa|_{i+1,j,k}^{n+\frac{1}{2}} \right). \end{aligned} \quad (3.23)$$

This equation is equivalent to (3.7) for A_x samples at edges connecting internal nodes. Those are the samples for which $\vec{B}$ and κ samples involved in the update equations can be computed from (3.8) and (3.9), respectively. For the A_x samples on the south boundary, such as the one in Figure 3.3b,

we write

$$\begin{aligned}
\Delta x \frac{\Delta y}{2} \Delta z \frac{\varepsilon_z|_{i+\frac{1}{2},1,k}}{\Delta t} [\partial_t A_x]_{i+\frac{1}{2},1,k}^{n+1} &= \Delta x \frac{\Delta y}{2} \Delta z \frac{\varepsilon_z|_{i+\frac{1}{2},1,k}}{\Delta t} [\partial_t A_x]_{i+\frac{1}{2},1,k}^n \\
&\quad - \Delta x \left(\frac{\Delta y}{2} \mu_y^{-1}|_{i+\frac{1}{2},1,k-\frac{1}{2}} B_y|_{i+\frac{1}{2},1,k-\frac{1}{2}}^{n+\frac{1}{2}} - \frac{\Delta y}{2} \mu_y^{-1}|_{i+\frac{1}{2},1,k+\frac{1}{2}} B_y|_{i+\frac{1}{2},1,k+\frac{1}{2}}^{n+\frac{1}{2}} \right. \\
&\quad \left. - \Delta z \mu_z^{-1}|_{i+\frac{1}{2},1,k} B_z|_{i+\frac{1}{2},1,k}^{n+\frac{1}{2}} + \Delta z \mu_z^{-1}|_{i+\frac{1}{2},\frac{3}{2},k} B_z|_{i+\frac{1}{2},\frac{3}{2},k}^{n+\frac{1}{2}} \right) \\
&\quad - \frac{\Delta y}{2} \Delta z \varepsilon_z|_{i+\frac{1}{2},1,k} \left(-\kappa|_{i,1,k}^{n+\frac{1}{2}} + \kappa|_{i+1,1,k}^{n+\frac{1}{2}} \right), \quad (3.24)
\end{aligned}$$

where $B_z|_{i+0.5,1,k}$ is a hanging variable placed on the tangential secondary edge similarly to $H_z|_{i+0.5,1,k}$ in (2.7), in order to avoid using samples of the magnetic flux density beyond the boundary of the region and to facilitate coupling to other models. The reluctivity on the tangential secondary edge is averaged analogously to the permeability on the conventional secondary edges. Specifically,

$$\mu_z^{-1}|_{i+\frac{1}{2},1,k} = \frac{\mu_z^{-1}|_{i+\frac{1}{2},\frac{3}{2},k-\frac{1}{2}} + \mu_z^{-1}|_{i+\frac{1}{2},\frac{3}{2},k+\frac{1}{2}}}{2}. \quad (3.25)$$

For the $[\partial_t A_x]$ samples on the top-south edge shown in Figure 3.3c, the equation is given by

$$\begin{aligned}
\Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_z|_{i+\frac{1}{2},1,n_z+1}}{\Delta t} [\partial_t A_x]_{i+\frac{1}{2},1,n_z+1}^{n+1} &= \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_z|_{i+\frac{1}{2},1,n_z+1}}{\Delta t} [\partial_t A_x]_{i+\frac{1}{2},1,n_z+1}^n \\
&\quad - \Delta x \left(\frac{\Delta y}{2} \mu_y^{-1}|_{i+\frac{1}{2},1,n_z+\frac{1}{2}} B_y|_{i+\frac{1}{2},1,n_z+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{\Delta y}{2} \mu_y^{-1}|_{i+\frac{1}{2},1,n_z+1} B_y|_{i+\frac{1}{2},1,n_z+1}^{n+\frac{1}{2}} \right. \\
&\quad \left. - \frac{\Delta z}{2} \mu_z^{-1}|_{i+\frac{1}{2},1,n_z+1} B_z|_{i+\frac{1}{2},1,n_z+1}^{n+\frac{1}{2}} + \frac{\Delta z}{2} \mu_z^{-1}|_{i+\frac{1}{2},\frac{3}{2},n_z+1} B_z|_{i+\frac{1}{2},\frac{3}{2},n_z+1}^{n+\frac{1}{2}} \right) \\
&\quad - \frac{\Delta y}{2} \frac{\Delta z}{2} \varepsilon_z|_{i+\frac{1}{2},1,n_z+1} \left(-\kappa|_{i,1,n_z+1}^{n+\frac{1}{2}} + \kappa|_{i+1,1,n_z+1}^{n+\frac{1}{2}} \right). \quad (3.26)
\end{aligned}$$

Lastly, (3.11c) is discretized similarly to (3.10a). For the samples of κ on internal nodes, such as the node (i, j, k) in Figure 3.4a, we write an equation analogous to (3.12)

$$\begin{aligned}
\Delta x \Delta y \Delta z \frac{\chi|_{i,j,k}}{\Delta t} \kappa|_{i,j,k}^{n+\frac{1}{2}} &= \Delta x \Delta y \Delta z \frac{\chi|_{i,j,k}}{\Delta t} \kappa|_{i,j,k}^{n-\frac{1}{2}} \\
&\quad - \left(-\Delta y \Delta z \varepsilon_x|_{i-\frac{1}{2},j,k} [\partial_t A_x]_{i-\frac{1}{2},j,k}^n + \Delta y \Delta z \varepsilon_x|_{i+\frac{1}{2},j,k} [\partial_t A_x]_{i+\frac{1}{2},j,k}^n \right. \\
&\quad - \Delta x \Delta z \varepsilon_y|_{i,j-\frac{1}{2},k} [\partial_t A_y]_{i,j-\frac{1}{2},k}^n + \Delta x \Delta z \varepsilon_y|_{i,j+\frac{1}{2},k} [\partial_t A_y]_{i,j+\frac{1}{2},k}^n \\
&\quad \left. - \Delta x \Delta y \varepsilon_z|_{i,j,k-\frac{1}{2}} [\partial_t A_z]_{i,j,k-\frac{1}{2}}^n + \Delta x \Delta y \varepsilon_z|_{i,j,k+\frac{1}{2}} [\partial_t A_z]_{i,j,k+\frac{1}{2}}^n \right). \quad (3.27)
\end{aligned}$$

For samples on the east face of the boundary, such as the sample in Figure 3.4b, the equation is

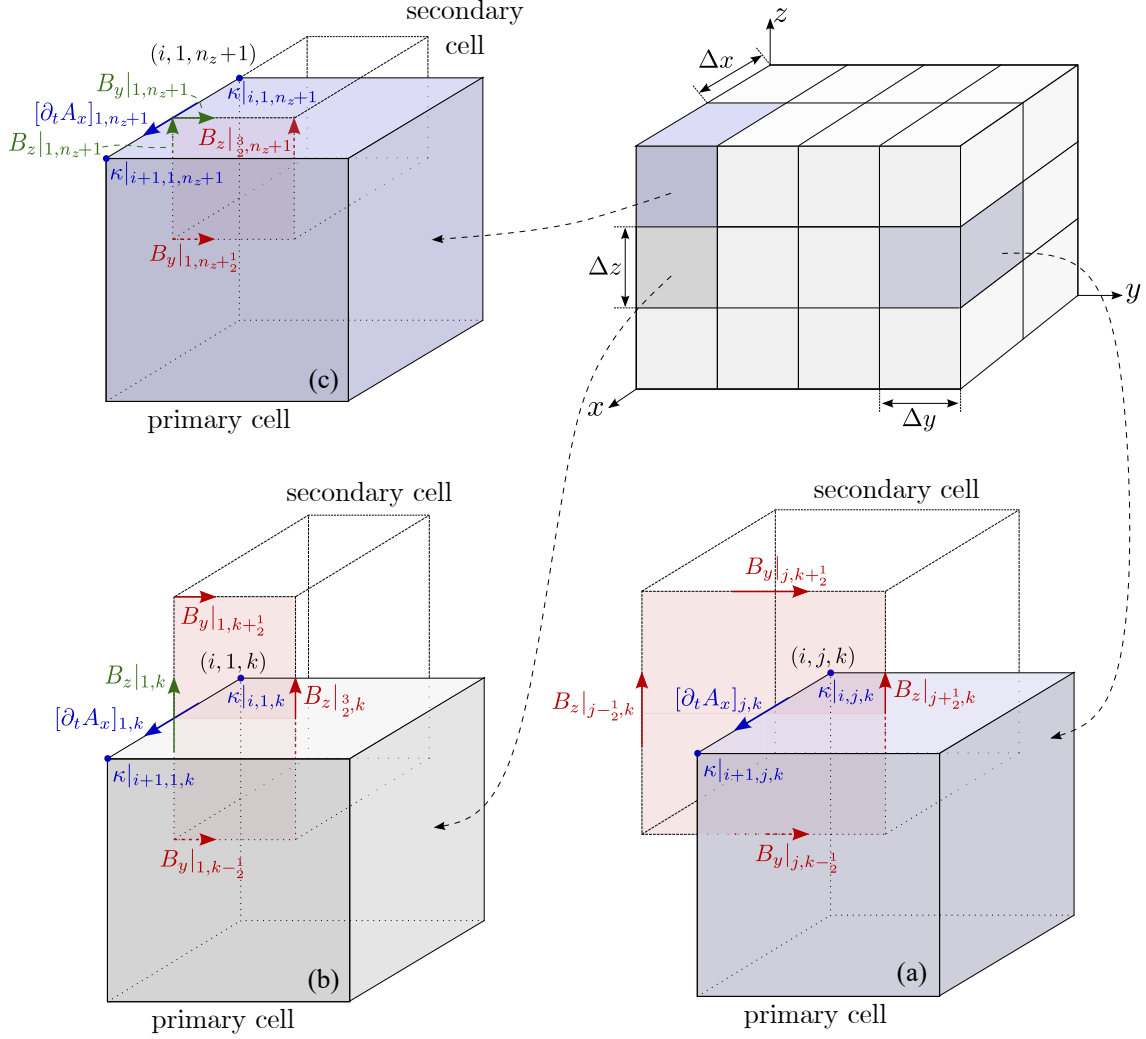


Figure 3.3: Samples involved in (a) equation (3.23), (b) equation (3.24), and (c) equation (3.26). The common subscript $i + 0.5$ is omitted. Green is used to indicate the tangential magnetic hanging variable.

analogous to (3.13), namely

$$\begin{aligned}
 \frac{\Delta x}{2} \Delta y \Delta z \frac{\chi|_{n_x+1,j,k}}{\Delta t} \kappa|_{n_x+1,j,k}^{n+\frac{1}{2}} &= \frac{\Delta x}{2} \Delta y \Delta z \frac{\chi|_{n_x+1,j,k}}{\Delta t} \kappa|_{n_x+1,j,k}^{n-\frac{1}{2}} \\
 &\quad - \left(-\Delta y \Delta z \varepsilon_x|_{n_x+\frac{1}{2},j,k} [\partial_t A_x]_{n_x+\frac{1}{2},j,k}^n + \Delta y \Delta z \varepsilon_x|_{n_x+1,j,k} [\partial_t A_x]_{n_x+1,j,k}^n \right. \\
 &\quad - \frac{\Delta x}{2} \Delta z \varepsilon_y|_{n_x+1,j-\frac{1}{2},k} [\partial_t A_y]_{n_x+1,j-\frac{1}{2},k}^n + \frac{\Delta x}{2} \Delta z \varepsilon_y|_{n_x+1,j+\frac{1}{2},k} [\partial_t A_y]_{n_x+1,j+\frac{1}{2},k}^n \\
 &\quad \left. - \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,k-\frac{1}{2}} [\partial_t A_z]_{n_x+1,j,k-\frac{1}{2}}^n + \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,k+\frac{1}{2}} [\partial_t A_z]_{n_x+1,j,k+\frac{1}{2}}^n \right), \quad (3.28)
 \end{aligned}$$

where $[\partial_t A_x]_{n_x+1,j,k}$ is a hanging variable placed at the corresponding normal primary edge. For samples on the bottom-east edge and the bottom-north-east corner of the region boundary, the

equations read, respectively,

$$\begin{aligned} \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\chi|_{n_x+1,j,1}}{\Delta t} \kappa|_{n_x+1,j,1}^{n+\frac{1}{2}} &= \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\chi|_{n_x+1,j,1}}{\Delta t} \kappa|_{n_x+1,j,1}^{n-\frac{1}{2}} \\ &\quad - \left(-\Delta y \frac{\Delta z}{2} \varepsilon_x|_{n_x+\frac{1}{2},j,1} [\partial_t A_x]_{n_x+\frac{1}{2},j,1}^n + \Delta y \frac{\Delta z}{2} \varepsilon_x|_{n_x+1,j,1} [\partial_t A_x]_{n_x+1,j,1}^n \right. \\ &\quad - \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1,j-\frac{1}{2},1} [\partial_t A_y]_{n_x+1,j-\frac{1}{2},1}^n + \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1,j+\frac{1}{2},1} [\partial_t A_y]_{n_x+1,j+\frac{1}{2},1}^n \\ &\quad \left. - \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,1} [\partial_t A_z]_{n_x+1,j,1}^n + \frac{\Delta x}{2} \Delta y \varepsilon_z|_{n_x+1,j,\frac{3}{2}} [\partial_t A_z]_{n_x+1,j,\frac{3}{2}}^n \right), \quad (3.29) \end{aligned}$$

$$\begin{aligned} \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\chi|_{n_x+1,n_y+1,1}}{\Delta t} \kappa|_{n_x+1,n_y+1,1}^{n+\frac{1}{2}} &= \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\chi|_{n_x+1,n_y+1,1}}{\Delta t} \kappa|_{n_x+1,n_y+1,1}^{n-\frac{1}{2}} \\ &\quad - \left(-\frac{\Delta y}{2} \frac{\Delta z}{2} \varepsilon_x|_{n_x+\frac{1}{2},n_y+1,1} [\partial_t A_x]_{n_x+\frac{1}{2},n_y+1,1}^n + \frac{\Delta y}{2} \frac{\Delta z}{2} \varepsilon_x|_{n_x+1,n_y+1,1} [\partial_t A_x]_{n_x+1,n_y+1,1}^n \right. \\ &\quad - \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1,n_y+\frac{1}{2},1} [\partial_t A_y]_{n_x+1,n_y+\frac{1}{2},1}^n + \frac{\Delta x}{2} \frac{\Delta z}{2} \varepsilon_y|_{n_x+1,n_y+1,1} [\partial_t A_y]_{n_x+1,n_y+1,1}^n \\ &\quad \left. - \frac{\Delta x}{2} \frac{\Delta y}{2} \varepsilon_z|_{n_x+1,n_y+1,1} [\partial_t A_z]_{n_x+1,n_y+1,1}^n + \frac{\Delta x}{2} \frac{\Delta y}{2} \varepsilon_z|_{n_x+1,n_y+1,\frac{3}{2}} [\partial_t A_z]_{n_x+1,n_y+1,\frac{3}{2}}^n \right). \quad (3.30) \end{aligned}$$

3.2.3 Dynamical System Equations in Compact Matrix Form

Next, we write the equations in Section 3.2.2 in a compact matrix form and formulate those equations as dynamical systems by identifying the state vectors, the inputs, and the outputs. These dynamical systems will allow extending the results from [205, 206] and Chapter 2 to FDTD with potentials.

Dynamical System for the Scalar Potential

Equations such as (3.12), (3.13), (3.15), (3.16), (3.17), (3.19), and (3.20) can be written in a compact matrix form as

$$\mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \frac{\mathbf{\Lambda}_\varepsilon}{\Delta t} [\nabla \Phi]^{n+1} = \mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \frac{\mathbf{\Lambda}_\varepsilon}{\Delta t} [\nabla \Phi]^n - \mathbf{\Lambda}''_S \mathbf{\Lambda}_\varepsilon \mathbf{D}^T [\partial_t \Phi]^{n+\frac{1}{2}}, \quad (3.31a)$$

$$\mathbf{\Lambda}''_V \frac{\mathbf{\Lambda}_\chi}{\Delta t} [\partial_t \Phi]^{n+\frac{1}{2}} = \mathbf{\Lambda}''_V \frac{\mathbf{\Lambda}_\chi}{\Delta t} [\partial_t \Phi]^{n-\frac{1}{2}} + \mathbf{D} \mathbf{\Lambda}''_S \mathbf{\Lambda}_\varepsilon [\nabla \Phi]^n + \mathbf{L}_\perp \mathbf{\Lambda}_{[\hat{n}]} \mathbf{\Lambda}''_{S,\dagger} \mathbf{\Lambda}_{\varepsilon,\perp} [\nabla \Phi]^n, \quad (3.31b)$$

where $[\nabla \Phi]$ and $[\nabla \Phi]_\perp$ are vectors collecting the samples of the gradient of the scalar potential along the conventional primary edges and normal primary edges, respectively. Vector $[\partial_t \Phi]$ collects the $[\partial_t \phi]$ samples at the primary nodes. Diagonal matrix $\mathbf{\Lambda}''_V$ contains the volumes of secondary cells associated with the samples in $[\partial_t \Phi]$. For example, it contains $\Delta x \Delta y \Delta z$ in positions corresponding to the internal nodes, $\Delta x \Delta y \Delta z / 2$ in positions corresponding to the nodes on a single face of the boundary, and so on. Matrix $\mathbf{\Lambda}_\chi$ contains the values of χ on the primary nodes. Matrix $\mathbf{D}$ contains 1's, -1's, and 0's needed for the computation of discrete divergence or flux. Its rows correspond to the primary nodes (or, equivalently, secondary cells) and its columns correspond to the conventional primary edges (equivalently, internal secondary faces). The negative transpose of this matrix, $-\mathbf{D}^T$, corresponds to the discrete gradient of quantities sampled at the primary nodes, such as $[\partial_t \phi]$. Matrix $\mathbf{L}_\perp$ contains 0's and 1's, meant to extract from $[\nabla \Phi]_\perp$ the hanging variables collocated with the boundary samples in $[\partial_t \Phi]$. Matrix $\mathbf{\Lambda}_{[\hat{n}]}$ contains 1 or -1 on the diagonal, depending on the sign of

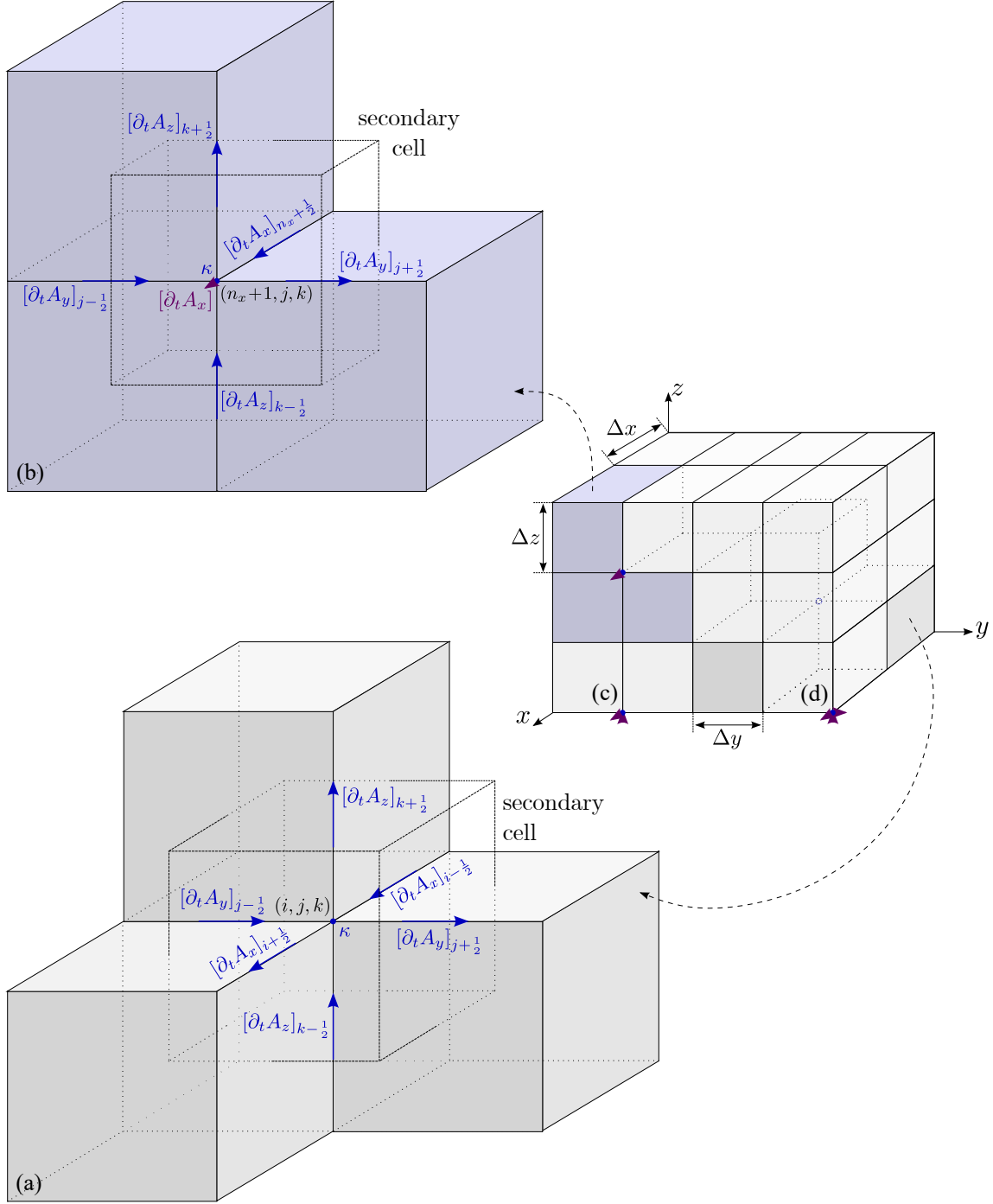


Figure 3.4: Samples involved in discretization of (3.11c). The hanging variables are shown in purple. (a) illustrates equation (3.27). (b) illustrates (3.28). (c) labels the $[\partial_t \phi]$ sample and the hanging variables in (3.29), and (d) labels the corresponding quantities in (3.30). The common subscripts (i, j, k) and $(n_x + 1, j, k)$ are omitted.

the dot product between the *outward* normal vector $\hat{n}$ to the boundary and the Cartesian unit vector associated with the additional normal primary edge where the hanging variable is located. Matrix $\Lambda''_{S,\dagger}$ contains the areas of boundary secondary faces and $\Lambda_{\varepsilon,\perp}$ contains the permittivity values on the edges associated with the hanging variables in $[\nabla\Phi]_{\perp}$. The expressions for the matrices in (3.31a)–(3.31b) are given in Appendix B.

The vectors $[\nabla\Phi]^n$ and $[\partial_t\Phi]^{n-0.5}$ are viewed as the state variables of the dynamical system and $[\nabla\Phi]_{\perp}^{n+0.5}$ is viewed as the input of the system. The output of the system is a vector containing the samples of $[\partial_t\phi]^{n-0.5}$ located on the boundary. Collecting (3.31a)–(3.31b) into a single matrix equation, we obtain

$$(\mathbf{R}_{\phi} + \mathbf{F}_{\phi})\mathbf{x}_{\phi}^{n+\frac{1}{2}} = (\mathbf{R}_{\phi} - \mathbf{F}_{\phi})\mathbf{x}_{\phi}^{n-\frac{1}{2}} + \mathbf{B}_{\phi}\mathbf{u}_{\phi}^n, \quad (3.32a)$$

and adding the output equation,

$$\mathbf{y}_{\phi}^{n-\frac{1}{2}} = \mathbf{L}_{\phi}^T \mathbf{x}_{\phi}^{n-\frac{1}{2}}. \quad (3.32b)$$

The state, input, and output vectors in (3.32a)–(3.32b) are given by

$$\mathbf{x}_{\phi}^{n-\frac{1}{2}} = \begin{bmatrix} [\nabla\Phi]^n \\ [\partial_t\Phi]^{n-\frac{1}{2}} \end{bmatrix}, \quad \mathbf{u}_{\phi}^n = [\nabla\Phi]_{\perp}^n, \quad \mathbf{y}_{\phi}^{n-\frac{1}{2}} = [\partial_t\Phi]_{\dagger}^{n-\frac{1}{2}}, \quad (3.33)$$

where $[\partial_t\Phi]_{\dagger}$ is a vector defined as

$$[\partial_t\Phi]_{\dagger}^{n-\frac{1}{2}} = \mathbf{L}_{\perp}^T [\partial_t\Phi]^{n-\frac{1}{2}}, \quad (3.34)$$

which collects the $[\partial_t\phi]$ samples located on the boundary, with each element corresponding to an element in $[\nabla\Phi]_{\perp}$. Matrices $\mathbf{R}_{\phi}$, $\mathbf{F}_{\phi}$, $\mathbf{B}_{\phi}$, and $\mathbf{L}_{\phi}$ in (3.32a)–(3.32b) are given by

$$\mathbf{R}_{\phi} = \begin{bmatrix} \Lambda'_l \Lambda''_S \frac{\Lambda_{\varepsilon}}{\Delta t} & \frac{1}{2} \Lambda_{\varepsilon} \Lambda''_S \mathbf{D}^T \\ \frac{1}{2} \mathbf{D} \Lambda''_S \Lambda_{\varepsilon} & \Lambda''_V \frac{\Lambda_{\chi}}{\Delta t} \end{bmatrix}, \quad (3.35)$$

$$\mathbf{F}_{\phi} = \begin{bmatrix} \mathbf{0} & \frac{1}{2} \Lambda_{\varepsilon} \Lambda''_S \mathbf{D}^T \\ -\frac{1}{2} \mathbf{D} \Lambda''_S \Lambda_{\varepsilon} & \mathbf{0} \end{bmatrix}, \quad (3.36)$$

$$\mathbf{B}_{\phi} = \begin{bmatrix} \mathbf{0} \\ \mathbf{L}_{\perp} \mathbf{S}_{\perp} \end{bmatrix}, \quad (3.37)$$

$$\mathbf{L}_{\phi} = \begin{bmatrix} \mathbf{0} \\ \mathbf{L}_{\perp} \end{bmatrix}, \quad (3.38)$$

where $\mathbf{S}_{\perp}$ is given by

$$\mathbf{S}_{\perp} = \Lambda_{[\hat{n}\cdot]} \Lambda''_{S,\dagger} \Lambda_{\varepsilon,\perp}. \quad (3.39)$$

It should be noted that by writing the equations for the scalar potential in the form (3.32a)–(3.32b) we are able to apply the approach in Section 2.2 to study the energy conservation properties of this system.

Dynamical System for the Vector Potential

In an analogous way, we write the following compact matrix form for the discretized equations for vector potential, such as (3.21)–(3.24), (3.26)–(3.30):

$$\Lambda_l'' \Lambda_S' \Lambda_{1/\mu} \mathbf{b}^{n+\frac{1}{2}} = \Lambda_l'' \Lambda_S' \Lambda_{1/\mu} \mathbf{b}^{n-\frac{1}{2}} + \Lambda_l'' \Lambda_{1/\mu} \mathbf{C} \Lambda_l' [\partial_t \mathbf{a}]^n, \quad (3.40a)$$

$$\begin{aligned} \Lambda_l' \Lambda_S'' \frac{\Lambda_\varepsilon}{\Delta t} [\partial_t \mathbf{a}]^{n+1} &= \Lambda_l' \Lambda_S'' \frac{\Lambda_\varepsilon}{\Delta t} [\partial_t \mathbf{a}]^n \\ &\quad - \Lambda_l' \mathbf{C}^T \Lambda_l'' \Lambda_{1/\mu} \mathbf{b}^{n+\frac{1}{2}} + \Lambda_S'' \Lambda_\varepsilon \mathbf{D}^T \boldsymbol{\kappa}^{n+\frac{1}{2}} - \Lambda_l' \mathbf{L}_\parallel \Lambda_{[\hat{n} \times]} \Lambda_{l,\parallel}'' \Lambda_{1/\mu,\parallel} \mathbf{b}_\parallel^{n+\frac{1}{2}}, \end{aligned} \quad (3.40b)$$

$$\Lambda_V'' \frac{\Lambda_\chi}{\Delta t} \boldsymbol{\kappa}^{n+\frac{1}{2}} = \Lambda_V'' \frac{\Lambda_\chi}{\Delta t} \boldsymbol{\kappa}^{n-\frac{1}{2}} - \mathbf{D} \Lambda_S'' \Lambda_\varepsilon [\partial_t \mathbf{a}]^n - \mathbf{L}_\perp \Lambda_{[\hat{n} \cdot]} \Lambda_{S,\dagger}'' \Lambda_{\varepsilon,\perp} [\partial_t \mathbf{a}]_\perp^n. \quad (3.40c)$$

The state vector of the dynamical system is formed by $\mathbf{b}^{n-0.5}$, $[\partial_t \mathbf{a}]^n$, and $\boldsymbol{\kappa}^{n-0.5}$. The input vector consists of $\mathbf{b}_\parallel^{n+0.5}$ and $[\partial_t \mathbf{a}]_\perp^n$. The output consists of the vectors $[\partial_t \mathbf{a}]_\parallel^n$ and $\boldsymbol{\kappa}_\dagger^{n-0.5}$ defined as

$$[\partial_t \mathbf{a}]_\parallel^n = \mathbf{L}_\parallel^T [\partial_t \mathbf{a}]^n, \quad (3.41)$$

$$\boldsymbol{\kappa}_\dagger^{n-\frac{1}{2}} = \mathbf{L}_\perp^T \boldsymbol{\kappa}^{n-\frac{1}{2}}. \quad (3.42)$$

From (3.40a)–(3.40c) and (3.41)–(3.42), we obtain the equations for the discrete-time dynamical system corresponding to the vector potential

$$(\mathbf{R}_A + \mathbf{F}_A) \mathbf{x}_A^{n+\frac{1}{2}} = (\mathbf{R}_A - \mathbf{F}_A) \mathbf{x}_A^{n-\frac{1}{2}} + \mathbf{B}_A \mathbf{u}_A^n, \quad (3.43a)$$

$$\mathbf{y}_A^{n-\frac{1}{2}} = \mathbf{L}_A^T \mathbf{x}_A^{n-\frac{1}{2}}, \quad (3.43b)$$

where

$$\mathbf{x}_A^{n-\frac{1}{2}} = \begin{bmatrix} \mathbf{b}^{n-\frac{1}{2}} \\ [\partial_t \mathbf{a}]^n \\ \boldsymbol{\kappa}^{n-\frac{1}{2}} \end{bmatrix}, \quad \mathbf{u}_A^n = \begin{bmatrix} \mathbf{b}_\parallel^{n+\frac{1}{2}} \\ [\partial_t \mathbf{a}]_\perp^n \end{bmatrix}, \quad \mathbf{y}_A^{n-\frac{1}{2}} = \begin{bmatrix} [\partial_t \mathbf{a}]_\parallel^n \\ \boldsymbol{\kappa}_\dagger^{n-\frac{1}{2}} \end{bmatrix}, \quad (3.44)$$

and

$$\mathbf{R}_A = \begin{bmatrix} \Lambda_l'' \Lambda_S' \frac{\Lambda_{1/\mu}}{\Delta t} & \frac{1}{2} \Lambda_{1/\mu} \Lambda_l'' \mathbf{C} \Lambda_l' & \mathbf{0} \\ \frac{1}{2} \Lambda_l' \mathbf{C}^T \Lambda_l'' \Lambda_{1/\mu} & \Lambda_l' \Lambda_S'' \frac{\Lambda_\varepsilon}{\Delta t} & -\frac{1}{2} \Lambda_\varepsilon \Lambda_S'' \mathbf{D}^T \\ \mathbf{0} & -\frac{1}{2} \mathbf{D} \Lambda_S'' \Lambda_\varepsilon & \Lambda_V'' \frac{\Lambda_\chi}{\Delta t} \end{bmatrix}, \quad (3.45)$$

$$\mathbf{F}_A = \begin{bmatrix} \mathbf{0} & -\frac{1}{2} \Lambda_{1/\mu} \Lambda_l'' \mathbf{C} \Lambda_l' & \mathbf{0} \\ \frac{1}{2} \Lambda_l' \mathbf{C}^T \Lambda_l'' \Lambda_{1/\mu} & \mathbf{0} & -\frac{1}{2} \Lambda_\varepsilon \Lambda_S'' \mathbf{D}^T \\ \mathbf{0} & \frac{1}{2} \mathbf{D} \Lambda_S'' \Lambda_\varepsilon & \mathbf{0} \end{bmatrix}, \quad (3.46)$$

$$\mathbf{B}_A = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ -\mathbf{L}_\parallel \mathbf{S}_\parallel & \mathbf{0} \\ \mathbf{0} & -\mathbf{L}_\perp \mathbf{S}_\perp \end{bmatrix}, \quad (3.47)$$

$$\mathbf{L}_A = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{L}_{\parallel} & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_{\perp} \end{bmatrix}. \quad (3.48)$$

Matrix $\mathbf{S}_{\parallel}$ in (3.47) is given by

$$\mathbf{S}_{\parallel} = \mathbf{\Lambda}'_{l,\parallel} \mathbf{\Lambda}_{[\cdot \hat{n} \times]} \mathbf{\Lambda}''_{l,\parallel} \mathbf{\Lambda}_{1/\mu,\parallel}. \quad (3.49)$$

We used equality (2.71) to write $\mathbf{B}_A$ in the form (3.47).

3.3 Energy Conservation Analysis

This section investigates the energy conservation properties of the dynamical systems (3.32a)–(3.32b) and (3.43a)–(3.43b). Specifically, it provides expressions for the contribution of the scalar and vector potentials to the energy stored in the region. With those definitions, a priori energy conservation conditions are derived. Analogously to Chapter 2, we find that the CFL stability limit (1.36) can be interpreted as the conservation of energy condition for the potentials-based FDTD formulation.

3.3.1 Definitions of the Storage Functions and Supply Rates

We choose expressions analogous to those in Chapter 2 for the storage function and supply rate of each of the two dynamical systems (3.32a)–(3.32b) and (3.43a)–(3.43b), namely

$$\mathcal{E}_{\phi}(\mathbf{x}_{\phi}^{n-\frac{1}{2}}) = \frac{\Delta t}{2} \left(\mathbf{x}_{\phi}^{n-\frac{1}{2}} \right)^T \mathbf{R}_{\phi} \mathbf{x}_{\phi}^{n-\frac{1}{2}} \quad (3.50)$$

and

$$\mathcal{E}_A(\mathbf{x}_A^{n-\frac{1}{2}}) = \frac{\Delta t}{2} \left(\mathbf{x}_A^{n-\frac{1}{2}} \right)^T \mathbf{R}_A \mathbf{x}_A^{n-\frac{1}{2}} \quad (3.51)$$

for the storage functions and

$$s_{\phi}(\mathbf{y}_{\phi}^{n-\frac{1}{2}}, \mathbf{y}_{\phi}^{n+\frac{1}{2}}, \mathbf{u}_{\phi}^n) = \Delta t \frac{\left(\mathbf{y}_{\phi}^{n-\frac{1}{2}} + \mathbf{y}_{\phi}^{n+\frac{1}{2}} \right)^T}{2} \mathbf{S}_{\phi} \mathbf{u}_{\phi}^n, \quad (3.52)$$

$$s_A(\mathbf{y}_A^{n-\frac{1}{2}}, \mathbf{y}_A^{n+\frac{1}{2}}, \mathbf{u}_A^n) = \Delta t \frac{\left(\mathbf{y}_A^{n-\frac{1}{2}} + \mathbf{y}_A^{n+\frac{1}{2}} \right)^T}{2} \mathbf{S}_A \mathbf{u}_A^n \quad (3.53)$$

for the supply rates, where $\mathbf{S}_{\phi}$ and $\mathbf{S}_A$ are given by

$$\mathbf{S}_{\phi} = \mathbf{S}_{\perp}, \quad (3.54)$$

$$\mathbf{S}_A = \text{diag}(-\mathbf{S}_{\parallel}, -\mathbf{S}_{\perp}). \quad (3.55)$$

3.3.2 Physical Meaning of the Storage Functions and Supply Rates

This section elaborates on the physical meaning of the expressions for the storage functions (3.50)–(3.51) and the supply rates (3.52)–(3.53). In particular, it discusses how these quantities relate to the continuous expressions for the stored energy and supplied power, written in terms of the scalar and vector potentials.

Continuous Expressions for Energy and Power

For a region V governed by (3.10a)–(3.10b) and (3.11a)–(3.11c), stored energy is given by [70, 222]

$$\mathcal{E} = \mathcal{E}_A - \mathcal{E}_\phi, \quad (3.56)$$

with

$$\mathcal{E}_\phi(t) = \iiint_V w_\phi dV, \quad \mathcal{E}_A(t) = \iiint_V w_A dV, \quad (3.57)$$

where the energy density contribution w_ϕ , associated with the scalar potential, is given by [70, 222]

$$w_\phi = \frac{1}{2}\varepsilon[\nabla\phi] \cdot [\nabla\phi] + \frac{1}{2}\chi[\partial_t\phi]^2, \quad (3.58)$$

and the contribution of the vector potential w_A is given by [70]

$$w_A = \frac{1}{2}\mu^{-1}\vec{B} \cdot \vec{B} + \frac{1}{2}\varepsilon[\partial_t\vec{A}] \cdot [\partial_t\vec{A}] + \frac{1}{2}\chi\kappa^2. \quad (3.59)$$

The power that enters the region through the boundary is given by [222]

$$s(t) = s_A(t) - s_\phi(t), \quad (3.60)$$

with

$$s_\phi(t) = -\oint_{\partial V} \vec{S}_\phi \cdot \hat{n} dS, \quad s_A(t) = -\oint_{\partial V} \vec{S}_A \cdot \hat{n} dS, \quad (3.61)$$

where $\hat{n}$ is the outward normal vector to the boundary and ∂V is the boundary of the region. Vector $\vec{S}_\phi$ is the contribution of the scalar potential to the power flux density and is given by [222]

$$\vec{S}_\phi = -\varepsilon[\partial_t\phi][\nabla\phi]. \quad (3.62)$$

The contribution of the vector potential $\vec{S}_A$ can be defined as

$$\vec{S}_A = \mu^{-1}\vec{B} \times [\partial_t\vec{A}] + \varepsilon\kappa[\partial_t\vec{A}]. \quad (3.63)$$

The power associated with the vector potential $s_A(t)$ can also be broken down into the following contributions, which we will refer to in the subsequent discussion

$$s_A(t) = s_{A,B}(t) + s_{A,\kappa}(t), \quad (3.64)$$

where

$$s_{A,B}(t) = -\oint_{\partial V} \vec{S}_{A,B} \cdot \hat{n} dS, \quad s_{A,\kappa}(t) = -\oint_{\partial V} \vec{S}_{A,\kappa} \cdot \hat{n} dS, \quad (3.65)$$

and

$$\vec{S}_{A,B} = \mu^{-1}\vec{B} \times [\partial_t\vec{A}], \quad \vec{S}_{A,\kappa} = \varepsilon\kappa[\partial_t\vec{A}]. \quad (3.66)$$

We are not aware of any works where (3.63) appears as the power flux density associated with the vector potential. A different expression was proposed in [222] for the case of homogeneous media. The validity of (3.63) was confirmed by energy conservation arguments. In particular, one

can show [222, 223] that the total energy density $w_A - w_\phi$ and the power flux density $\vec{S}_A - \vec{S}_\phi$ obey

$$\frac{\partial}{\partial t}(w_A - w_\phi) = -\nabla \cdot (S_A - S_\phi), \quad (3.67)$$

which is a necessary property dictated by the principle of energy conservation in the continuous domain³. To see the significance of this property, one can write (3.67) in the integral form, obtaining

$$\frac{d\mathcal{E}}{dt} = s(t), \quad (3.68)$$

which states that the rate of change of energy stored in the region equals to the power entering the region through the boundary.

Relation of the Storage Functions and Supply Rates to the Continuous Expressions

Next, we relate the proposed expressions for the storage functions and the supply rates to (3.57) and (3.61). Consider first the storage function expressions (3.50) and (3.51). Expanding (3.50) with (3.35) and the definition of $\mathbf{x}_\phi^{n-0.5}$ in (3.33), we have

$$\begin{aligned} \mathcal{E}_\phi(\mathbf{x}_\phi^{n-\frac{1}{2}}) &= \frac{\Delta t}{2} \left(([\nabla \Phi]^n)^T \Lambda'_l \Lambda''_S \frac{\Lambda_\varepsilon}{\Delta t} [\nabla \Phi]^n \right. \\ &\quad \left. + ([\nabla \Phi]^n)^T \Lambda_\varepsilon \Lambda''_S \mathbf{D}^T [\partial_t \Phi]^{n-\frac{1}{2}} + \left([\partial_t \Phi]^{n-\frac{1}{2}} \right)^T \Lambda''_V \frac{\Lambda_\chi}{\Delta t} [\partial_t \Phi]^{n-\frac{1}{2}} \right). \end{aligned} \quad (3.69)$$

Using (3.31a) and simplifying, one obtains

$$\mathcal{E}_\phi(\mathbf{x}_\phi^{n-\frac{1}{2}}) = \frac{1}{2} ([\nabla \Phi]^n)^T \Lambda'_l \Lambda''_S \Lambda_\varepsilon [\nabla \Phi]^{n-1} + \frac{1}{2} \left([\partial_t \Phi]^{n-\frac{1}{2}} \right)^T \Lambda''_V \Lambda_\chi [\partial_t \Phi]^{n-\frac{1}{2}}. \quad (3.70)$$

The latter can be interpreted as an approximation of $\mathcal{E}_\phi(t)$ in (3.57) at time instance $t = (n-0.5)\Delta t$.

Expanding (3.51) in a similar way,

$$\begin{aligned} \mathcal{E}_A(\mathbf{x}_A^{n-\frac{1}{2}}) &= \frac{\Delta t}{2} \left(\left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \Lambda''_l \Lambda'_S \frac{\Lambda_{1/\mu}}{\Delta t} \mathbf{b}^{n-\frac{1}{2}} + ([\partial_t \mathbf{a}]^n)^T \Lambda'_l \Lambda''_S \frac{\Lambda_\varepsilon}{\Delta t} [\partial_t \mathbf{a}]^n \right. \\ &\quad \left. + \left(\boldsymbol{\kappa}^{n-\frac{1}{2}} \right)^T \Lambda''_V \frac{\Lambda_\chi}{\Delta t} \boldsymbol{\kappa}^{n-\frac{1}{2}} + ([\partial_t \mathbf{a}]^n)^T \Lambda'_l \mathbf{C}^T \Lambda''_l \Lambda_{1/\mu} \mathbf{b}^{n-\frac{1}{2}} - ([\partial_t \mathbf{a}]^n)^T \Lambda_\varepsilon \Lambda''_S \mathbf{D}^T \boldsymbol{\kappa}^{n-\frac{1}{2}} \right). \end{aligned} \quad (3.71)$$

Using (3.40b) and simplifying,

$$\begin{aligned} \mathcal{E}_A(\mathbf{x}_A^{n-\frac{1}{2}}) &= \frac{1}{2} \left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \Lambda''_l \Lambda'_S \Lambda_{1/\mu} \mathbf{b}^{n-\frac{1}{2}} + \frac{1}{2} ([\partial_t \mathbf{a}]^n)^T \Lambda'_l \Lambda''_S \Lambda_\varepsilon [\partial_t \mathbf{a}]^{n-1} \\ &\quad + \frac{1}{2} \left(\boldsymbol{\kappa}^{n-\frac{1}{2}} \right)^T \Lambda''_V \Lambda_\chi \boldsymbol{\kappa}^{n-\frac{1}{2}} - \frac{\Delta t}{2} ([\partial_t \mathbf{a}]^n)^T \Lambda'_l \mathbf{L}_\parallel \Lambda_{[\cdot \hat{n} \times]} \Lambda''_{l,\parallel} \Lambda_{1/\mu}^\parallel \mathbf{b}_\parallel^{n-\frac{1}{2}}. \end{aligned} \quad (3.72)$$

³It should be noted that in [224] arguments are provided in favor of defining the energy density and the power flux density as $w_A + w_\phi$ and $\vec{S}_A + \vec{S}_\phi$, respectively. This change would not affect the validity of (3.67). However, the integration of $w_A + w_\phi$ over the entire space does not produce the same total energy as (2.53), which is one of the properties of $w_A - w_\phi$ [70].

As the time step approaches zero, one can neglect the last term in (3.72),

$$\mathcal{E}_A \left(\mathbf{x}_A^{n-\frac{1}{2}} \right) \approx \frac{1}{2} \left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \mathbf{\Lambda}_{1/\mu} \mathbf{b}^{n-\frac{1}{2}} + \frac{1}{2} ([\partial_t \mathbf{a}]^n)^T \mathbf{\Lambda}_l' \mathbf{\Lambda}_S'' \mathbf{\Lambda}_\varepsilon [\partial_t \mathbf{a}]^{n-1} + \frac{1}{2} \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_V'' \mathbf{\Lambda}_\chi \mathbf{\kappa}^{n-\frac{1}{2}}, \quad (3.73)$$

revealing that (3.51) is an approximation of $\mathcal{E}_A(t)$ in (3.57).

Expression (3.73) can be viewed as an approximation of $\mathcal{E}_A(t)$ in at time $t = (n - 0.5)\Delta t$, since the first and last terms involve samples at $n - 0.5$ and the second term involves samples at n and $n - 1$. One can also relate (3.51) to an analogous approximation at time $t = n\Delta t$ by writing the storage function as

$$\begin{aligned} \mathcal{E}_A \left(\mathbf{x}_A^{n-\frac{1}{2}} \right) &= \frac{\Delta t}{2} \left(\left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \frac{\mathbf{\Lambda}_{1/\mu}}{\Delta t} \mathbf{b}^{n-\frac{1}{2}} + ([\partial_t \mathbf{a}]^n)^T \mathbf{\Lambda}_l' \mathbf{\Lambda}_S'' \frac{\mathbf{\Lambda}_\varepsilon}{\Delta t} [\partial_t \mathbf{a}]^n \right. \\ &\quad \left. + \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_V'' \frac{\mathbf{\Lambda}_\chi}{\Delta t} \mathbf{\kappa}^{n-\frac{1}{2}} + \left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_{1/\mu} \mathbf{\Lambda}_l'' \mathbf{C} \mathbf{\Lambda}_l' [\partial_t \mathbf{a}]^n - \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{D} \mathbf{\Lambda}_S'' \mathbf{\Lambda}_\varepsilon [\partial_t \mathbf{a}]^n \right) \end{aligned} \quad (3.74)$$

and using (3.40a) and (3.40c) to obtain

$$\begin{aligned} \mathcal{E}_A \left(\mathbf{x}_A^{n-\frac{1}{2}} \right) &= \frac{\Delta t}{2} \left(\left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \frac{\mathbf{\Lambda}_{1/\mu}}{\Delta t} \mathbf{b}^{n+\frac{1}{2}} + ([\partial_t \mathbf{a}]^n)^T \mathbf{\Lambda}_l' \mathbf{\Lambda}_S'' \frac{\mathbf{\Lambda}_\varepsilon}{\Delta t} [\partial_t \mathbf{a}]^n \right. \\ &\quad \left. + \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_V'' \frac{\mathbf{\Lambda}_\chi}{\Delta t} \mathbf{\kappa}^{n-\frac{1}{2}} - \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{D} \mathbf{\Lambda}_S'' \mathbf{\Lambda}_\varepsilon [\partial_t \mathbf{a}]^n \right) \\ &= \frac{\Delta t}{2} \left(\left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \frac{\mathbf{\Lambda}_{1/\mu}}{\Delta t} \mathbf{b}^{n+\frac{1}{2}} + ([\partial_t \mathbf{a}]^n)^T \mathbf{\Lambda}_l' \mathbf{\Lambda}_S'' \frac{\mathbf{\Lambda}_\varepsilon}{\Delta t} [\partial_t \mathbf{a}]^n \right. \\ &\quad \left. + \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_V'' \frac{\mathbf{\Lambda}_\chi}{\Delta t} \mathbf{\kappa}^{n+\frac{1}{2}} + \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{L}_\perp \mathbf{\Lambda}_{[\hat{n} \cdot]} \mathbf{\Lambda}_{S,\dagger}'' \mathbf{\Lambda}_{\varepsilon,\perp} [\partial_t \mathbf{a}]_\perp^n \right), \end{aligned} \quad (3.75)$$

which simplifies to

$$\begin{aligned} \mathcal{E}_A \left(\mathbf{x}_A^{n-\frac{1}{2}} \right) &= \frac{1}{2} \left(\mathbf{b}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_l'' \mathbf{\Lambda}_S' \mathbf{\Lambda}_{1/\mu} \mathbf{b}^{n+\frac{1}{2}} + \frac{1}{2} ([\partial_t \mathbf{a}]^n)^T \mathbf{\Lambda}_l' \mathbf{\Lambda}_S'' \mathbf{\Lambda}_\varepsilon [\partial_t \mathbf{a}]^n \\ &\quad + \frac{1}{2} \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{\Lambda}_V'' \mathbf{\Lambda}_\chi \mathbf{\kappa}^{n+\frac{1}{2}} + \frac{\Delta t}{2} \left(\mathbf{\kappa}^{n-\frac{1}{2}} \right)^T \mathbf{L}_\perp \mathbf{\Lambda}_{[\hat{n} \cdot]} \mathbf{\Lambda}_{S,\dagger}'' \mathbf{\Lambda}_{\varepsilon,\perp} [\partial_t \mathbf{a}]_\perp^n. \end{aligned} \quad (3.76)$$

Neglecting the last term, (3.76) can be seen as an approximation of $\mathcal{E}_A(t)$ at $t = n\Delta t$.

Next, consider the supply rate expressions (3.52) and (3.53). Rewriting (3.52) using (3.33) and (3.54),

$$s_\phi \left(\mathbf{y}_\phi^{n-\frac{1}{2}}, \mathbf{y}_\phi^{n+\frac{1}{2}}, \mathbf{u}_\phi^n \right) = \Delta t \frac{\left([\partial_t \Phi]_\dagger^{n-\frac{1}{2}} + [\partial_t \Phi]_\dagger^{n+\frac{1}{2}} \right)^T}{2} \mathbf{S}_\perp [\nabla \Phi]_\perp^n, \quad (3.77)$$

where $\mathbf{S}_\perp$, given by (3.39), contains the products of the areas of the boundary secondary faces, the permittivity values, and the dot products between $\hat{n}$ and the unit vectors associated with each hanging variable $[\nabla \Phi]_\perp$. Hence, (3.52) is a discrete counterpart of the energy entering the region during one time step due to the scalar potential,

$$s_\phi \left(\mathbf{y}_\phi^{n-\frac{1}{2}}, \mathbf{y}_\phi^{n+\frac{1}{2}}, \mathbf{u}_\phi^n \right) \approx \int_{(n-\frac{1}{2})\Delta t}^{(n+\frac{1}{2})\Delta t} s_\phi(t) dt, \quad (3.78)$$

where $s_\phi(t)$ is the continuous expression in (3.61). It should be pointed out that (3.78) contributes to the overall supply rate with a negative sign, as per (3.60).

Similarly, (3.53) can be expanded as

$$s_A \left(\mathbf{y}_A^{n-\frac{1}{2}}, \mathbf{y}_A^{n+\frac{1}{2}}, \mathbf{u}_A^n \right) = -\Delta t \frac{\left([\partial_t \mathbf{a}]_\parallel^n + [\partial_t \mathbf{a}]_\parallel^{n+1} \right)^T}{2} \mathbf{S}_\parallel \mathbf{b}_\parallel^{n+\frac{1}{2}} - \Delta t \frac{\left(\kappa_\dagger^{n-\frac{1}{2}} + \kappa_\dagger^{n+\frac{1}{2}} \right)^T}{2} \mathbf{S}_\perp [\partial_t \mathbf{a}]_\perp^n. \quad (3.79)$$

Recalling the expressions of $\mathbf{S}_\perp$ in (3.39) and $\mathbf{S}_\parallel$ in (3.49), we can interpret (3.53) as an approximation of the following expression:

$$\begin{aligned} s_A \left(\mathbf{y}_A^{n-\frac{1}{2}}, \mathbf{y}_A^{n+\frac{1}{2}}, \mathbf{u}_A^n \right) &\approx - \int_{n\Delta t}^{(n+1)\Delta t} \oint_{\partial V} \mu^{-1} [\partial_t \vec{A}] \cdot \hat{n} \times \vec{B} \, dS \, dt - \int_{(n-\frac{1}{2})\Delta t}^{(n+\frac{1}{2})\Delta t} \oint_{\partial V} \varepsilon \kappa [\partial_t \vec{A}] \cdot \hat{n} \, dS \, dt \\ &= - \int_{n\Delta t}^{(n+1)\Delta t} \oint_{\partial V} \mu^{-1} \vec{B} \times [\partial_t \vec{A}] \cdot \hat{n} \, dS \, dt - \int_{(n-\frac{1}{2})\Delta t}^{(n+\frac{1}{2})\Delta t} \oint_{\partial V} \varepsilon \kappa [\partial_t \vec{A}] \cdot \hat{n} \, dS \, dt. \end{aligned} \quad (3.80)$$

The two last terms can be related to $s_{A,B}(t)$ and $s_{A,\kappa}(t)$ in (3.65) as

$$s_A \left(\mathbf{y}_A^{n-\frac{1}{2}}, \mathbf{y}_A^{n+\frac{1}{2}}, \mathbf{u}_A^n \right) \approx - \int_{n\Delta t}^{(n+1)\Delta t} s_{A,B}(t) \, dt - \int_{(n-\frac{1}{2})\Delta t}^{(n+\frac{1}{2})\Delta t} s_{A,\kappa}(t) \, dt, \quad (3.81)$$

which is the power entering the region during one time step due to the vector potential.

3.3.3 Energy Conservation Conditions

The region described by the continuous equations (3.10a)–(3.10b) and (3.11a)–(3.11c) exhibits no energy losses, since zero conductivity has been assumed in deriving those equations. The following theorems provide conditions for which this property is preserved when (3.10a)–(3.10b) and (3.11a)–(3.11c) are discretized using FDTD-P.

Theorem 4. *Consider a region in Figure 3.2, described by the discrete-time dynamical system equations (3.32a)–(3.32b). The system is energy conserving if the time step is below the generalized CFL limit*

$$\Delta t < \Delta t_{CFL,gen} = \frac{2}{\sqrt{\rho(\Sigma_\phi^T \Sigma_\phi)}}, \quad (3.82)$$

where

$$\Sigma_\phi = (\Lambda_S'')^{\frac{1}{2}} \Lambda_\varepsilon^{\frac{1}{2}} (\Lambda_l')^{-\frac{1}{2}} \mathbf{D}^T \Lambda_\chi^{-\frac{1}{2}} (\Lambda_V'')^{-\frac{1}{2}}. \quad (3.83)$$

Proof. The dynamical system (3.32a)–(3.32b), the storage function (3.50), and the supply rate (3.52) have the same form as (2.38a)–(2.38b), (2.47), and (2.48), respectively. Hence, Theorem 1 applies. This means that the system (3.32a)–(3.32b) is energy conserving if the following conditions hold

$$\mathbf{R}_\phi = \mathbf{R}_\phi^T \succ 0, \quad (3.84)$$

$$\mathbf{F}_\phi + \mathbf{F}_\phi^T = 0, \quad (3.85)$$

$$\mathbf{L}_\phi \mathbf{S}_\phi = \mathbf{B}_\phi. \quad (3.86)$$

From (3.36), (3.85) is true by construction. Likewise, from (3.38), (3.54), and (3.37), condition

(3.86) always holds. Condition (3.84) can be written using (3.35) and the Schur complement [213] as

$$\begin{cases} \Lambda'_l \Lambda''_S \frac{\Lambda_\varepsilon}{\Delta t} \succ 0, \\ \Lambda''_V \frac{\Lambda_\chi}{\Delta t} - \frac{\Delta t}{4} \mathbf{D} \Lambda''_S \Lambda_\varepsilon (\Lambda'_l)^{-1} \mathbf{D}^T \succ 0. \end{cases} \quad (3.87)$$

The first condition in (3.87) is satisfied as long as the permittivity values are positive at each of the conventional primary edges. The second condition can be rewritten by performing a congruence transformation with an invertible matrix $2(\Delta t \Lambda_\chi \Lambda''_V)^{-1/2}$ on the left and right hand sides of the inequality [96]

$$\frac{4}{(\Delta t)^2} \mathbf{I} - \Sigma_\phi^T \Sigma_\phi \succ 0. \quad (3.88)$$

Condition (3.88) holds if and only if (3.82) holds. $\square$

Theorem 5. *Consider a region in Figures 3.3 and 3.4, described by the discrete-time dynamical system equations (3.43a)–(3.43b). The system is energy conserving if the time step is taken below the following generalized CFL limit*

$$\Delta t < \Delta t_{CFL,gen} = \frac{2}{\sqrt{\rho(\Sigma_A^T \Sigma_A)}}, \quad (3.89)$$

where

$$\Sigma_A = \begin{bmatrix} (\Lambda'_S)^{-\frac{1}{2}} \Lambda_{1/\mu}^{\frac{1}{2}} (\Lambda'_l)^{\frac{1}{2}} \mathbf{C} (\Lambda'_l)^{\frac{1}{2}} (\Lambda''_S)^{-\frac{1}{2}} \Lambda_\varepsilon^{-\frac{1}{2}} \\ (\Lambda''_V)^{-\frac{1}{2}} \Lambda_\chi^{-\frac{1}{2}} \mathbf{D} (\Lambda''_S)^{\frac{1}{2}} \Lambda_\varepsilon^{\frac{1}{2}} (\Lambda'_l)^{-\frac{1}{2}} \end{bmatrix}. \quad (3.90)$$

Proof. From Theorem 1, the system (3.43a)–(3.43b) is energy conserving if

$$\mathbf{R}_A = \mathbf{R}_A^T \succ 0, \quad (3.91)$$

$$\mathbf{F}_A + \mathbf{F}_A^T = 0, \quad (3.92)$$

$$\mathbf{L}_A \mathbf{S}_A = \mathbf{B}_A. \quad (3.93)$$

Condition (3.92) for matrix $\mathbf{F}_A$ in (3.46) is always true. The same is the case for (3.93) for $\mathbf{L}_A$, $\mathbf{S}_A$, and $\mathbf{B}_A$ in (3.48), (3.55), and (3.47). Condition (3.91) for $\mathbf{R}_A$ in (3.45) can be written using the Schur complement [213] as

$$\begin{cases} \Lambda'_l \Lambda''_S \frac{\Lambda_{1/\mu}}{\Delta t} \succ 0, \\ \left[\begin{array}{cc} \Lambda'_l \Lambda''_S \frac{\Lambda_\varepsilon}{\Delta t} - \frac{\Delta t}{4} \Lambda'_l \mathbf{C}^T (\Lambda'_S)^{-1} \Lambda_{1/\mu} \Lambda''_l \mathbf{C} \Lambda'_l & -\frac{1}{2} \Lambda_\varepsilon \Lambda''_S \mathbf{D}^T \\ -\frac{1}{2} \mathbf{D} \Lambda''_S \Lambda_\varepsilon & \Lambda''_V \frac{\Lambda_\chi}{\Delta t} \end{array} \right] \succ 0. \end{cases} \quad (3.94)$$

The first inequality in (3.94) is satisfied if reluctivity is positive at all conventional secondary edges. Using the Schur complement again, the second inequality in (3.94) becomes

$$\begin{cases} \Lambda''_V \frac{\Lambda_\chi}{\Delta t} \succ 0, \\ \Lambda'_l \Lambda''_S \frac{\Lambda_\varepsilon}{\Delta t} - \frac{\Delta t}{4} \Lambda'_l \mathbf{C}^T (\Lambda'_S)^{-1} \Lambda_{1/\mu} \Lambda''_l \mathbf{C} \Lambda'_l - \frac{\Delta t}{4} \Lambda_\varepsilon \Lambda''_S \mathbf{D}^T (\Lambda''_V)^{-1} \Lambda_\chi^{-1} \mathbf{D} \Lambda''_S \Lambda_\varepsilon \succ 0. \end{cases} \quad (3.95)$$

The first condition in (3.95) is true, as long as χ samples are positive at all of the primary nodes, which is true if the permittivity and permeability values are positive. Applying a congruence with $2(\Delta t \mathbf{\Lambda}'_l \mathbf{\Lambda}''_S \mathbf{\Lambda}_\varepsilon)^{-1/2}$ to the second condition, we obtain

$$\begin{aligned} & \frac{4}{(\Delta t)^2} \mathbf{I} - \mathbf{\Lambda}_\varepsilon^{-\frac{1}{2}} (\mathbf{\Lambda}''_S)^{-\frac{1}{2}} (\mathbf{\Lambda}'_l)^{\frac{1}{2}} \mathbf{C}^T (\mathbf{\Lambda}'_S)^{-1} \mathbf{\Lambda}_{1/\mu} \mathbf{\Lambda}''_l \mathbf{C} (\mathbf{\Lambda}'_l)^{\frac{1}{2}} (\mathbf{\Lambda}''_S)^{-\frac{1}{2}} \mathbf{\Lambda}_\varepsilon^{-\frac{1}{2}} \\ & - (\mathbf{\Lambda}'_l)^{-\frac{1}{2}} \mathbf{\Lambda}_\varepsilon^{\frac{1}{2}} (\mathbf{\Lambda}''_S)^{\frac{1}{2}} \mathbf{D}^T (\mathbf{\Lambda}''_V)^{-1} \mathbf{\Lambda}_\chi^{-1} \mathbf{D} (\mathbf{\Lambda}'_S)^{\frac{1}{2}} \mathbf{\Lambda}_\varepsilon^{\frac{1}{2}} (\mathbf{\Lambda}'_l)^{-\frac{1}{2}} \succ 0, \end{aligned} \quad (3.96)$$

which can be written as

$$\frac{4}{(\Delta t)^2} \mathbf{I} - \mathbf{\Sigma}_A^T \mathbf{\Sigma}_A \succ 0. \quad (3.97)$$

Condition (3.97) holds if and only if (3.89) holds. $\square$

The generalized CFL limits in (3.82) and (3.89) are tight, in a sense that choosing a time step beyond those limits results in violation of energy conservation. In particular, if (3.82) is exceeded, matrix $\mathbf{R}_\phi$ has at least one negative eigenvalue. As a result, the storage function (3.50) becomes negative for some state vectors $\mathbf{x}_\phi^n$, violating (2.46a) in the definition of energy conserving systems. Likewise, exceeding (3.89) results in violation of (2.46a) by the storage function (3.51) corresponding to the vector potential. The following theorem provides a sufficient condition on the time step that can be used in practice to ensure energy conservation of (3.32a)–(3.32b) and (3.43a)–(3.43b).

Theorem 6. *Let Δt_{CFL} be the CFL limit (1.36), with $\varepsilon_{\min}$ being the smallest primary cell permittivity in the region and $\mu_{\min}$ the smallest primary cell permeability. Then*

1. *For the case of the dynamical system (3.32a)–(3.32b) governing the scalar potential,*

$$\Delta t_{CFL} \leq \Delta t_{CFL,gen}, \quad (3.98)$$

where $\Delta t_{CFL,gen}$ is the generalized CFL limit in (3.82).

2. *For the case of the dynamical system (3.43a)–(3.43b) governing the vector potential,*

$$\Delta t_{CFL} \leq \Delta t_{CFL,gen}, \quad (3.99)$$

where $\Delta t_{CFL,gen}$ is the generalized CFL limit in (3.89).

Proof. See Appendix D.2. $\square$

Similarly to the conventional fields-based FDTD, the CFL limit (1.36) has been understood as a stability condition for different FDTD schemes involving potentials [38, 145, 147], with a proof provided for the scheme in [145]. Theorems 4–6 reveal the role of the CFL condition in ensuring the energy conservation of FDTD-P. To the best of our knowledge, this role of the CFL condition has not been recognized for the potentials-based FDTD. Similarly to Chapter 2, the proposed theory of energy conservation for an open region will form a basis for a modular stability framework, which can be used for creating new FDTD schemes based on potentials. In Section 3.4, we apply this approach to develop a stable subgridding algorithm for FDTD-P.

3.3.4 Numerical Investigation of Energy Conservation

Prior to demonstrating the application of the theory proposed in Section 3.3, we verify its correctness numerically for an open region discretized with FDTD-P. The region under consideration has a side length $a = 10$ cm and contains vacuum. We consider the following particular solution of the wave equations (3.4) and (3.5), obtained using separation of variables [1, 183]:

$$\phi = C_\phi \sin(k_x x) \sin(k_y y) \sin(k_z z) \cos\left(\omega t + \frac{\pi}{3}\right), \quad (3.100)$$

$$\vec{A} = C_A \cos(k_x x) \sin(k_y y) \sin(k_z z) \sin\left(\omega t + \frac{\pi}{3}\right) \hat{x}, \quad (3.101)$$

where

$$k_x = \frac{3\pi}{a}, \quad k_y = \frac{\pi}{a}, \quad k_z = \frac{\pi}{a}, \quad (3.102)$$

$$\omega = \sqrt{\frac{\varepsilon_0}{\chi_0}(k_x^2 + k_y^2 + k_z^2)}, \quad (3.103)$$

where $C_A = 10^{-9}$ Tm, $C_\phi = -C_A \varepsilon_0 k_x / (\chi_0 \omega)$, ε_0 and χ_0 are values of free-space ε and χ . From (3.57), (3.100), and (3.101), one can find the energy stored in the region for the scalar and vector potential

$$\mathcal{E}_\phi(t) = \frac{1}{16} \chi_0 C_\phi^2 \omega^2 a^3, \quad \mathcal{E}_A(t) = \frac{1}{16} \varepsilon_0 C_A^2 \omega^2 a^3. \quad (3.104)$$

In order to analyze this scenario with FDTD-P, the region was divided into $n_x = 90$ and $n_y = n_z = 30$ cells. The potentials were updated based on the equations⁴ in Section 3.2. The initial conditions on the variables were set in accordance with (3.100)–(3.101)⁵. The hanging variables on the boundary ($[\nabla\phi]_\perp$, $\vec{B}_\parallel$, and $\vec{A}_\perp$) were likewise set by solving for their values from (3.100)–(3.101). For instance, the $[\partial_x \phi]^n$ on the west boundary was found by differentiating (3.100) with respect to x and evaluating the result at $x = 0$ and $t = n\Delta t$.

The top row of Figure 3.5 shows the evolution of the potentials over time for time steps 0.1% below and 0.1% above the CFL limit (1.36). The simulation run below the CFL limit shows no sign of instability. The simulation performed above the CFL limit is unstable, as a result of the algorithm violating the principle of energy conservation. The two plots in the middle row of Figure 3.5 confirm that the storage functions for scalar and vector potentials are approximately equal to their continuous counterparts (3.104).

From (2.46c) for the lossless dynamical systems corresponding to the scalar and vector potentials, the change in the stored energy at every time step equals to the energy supplied through the boundary via (3.52) and (3.53). In other words, the storage function should be equal to the energy initially present in the region at the beginning of the simulation plus the total energy that has entered the region through the boundaries. This is consistent with the observations for the test performed under the CFL limit in the second row of Figure 3.5, where the two curves match with a maximum error comparable to machine precision: 1.8×10^{-15} pJ and 2.1×10^{-15} pJ for the scalar and vector potentials, respectively. Interestingly, when the CFL limit is violated (the last row of Figure 3.5), the two curves deviate substantially. In that scenario, the violation of energy conservation causes

⁴To be precise, the implementation is based on the equivalent formulation involving second order finite differences in time, similar to (3.6)–(3.9). The same is the case for the implementation of the proposed subgridding method.

⁵The analytic solution was sampled to obtain the initial conditions on $[\partial_t \phi]^{-0.5}$, $\phi|^{-0.5}$, $[\partial_t \vec{A}]^{-0.5}$, $[\partial_t \vec{A}]^0$. All other variables were initialized based on those values. For example, $\phi|^{-1}$, was computed from $\phi|^{-0.5}$ and $[\partial_t \phi]^{-0.5}$ as $\phi|^{-1} = \phi|^{-0.5} - \Delta t [\partial_t \phi]^{-0.5}$, initializing the update procedure for $\phi|^{-n}$.

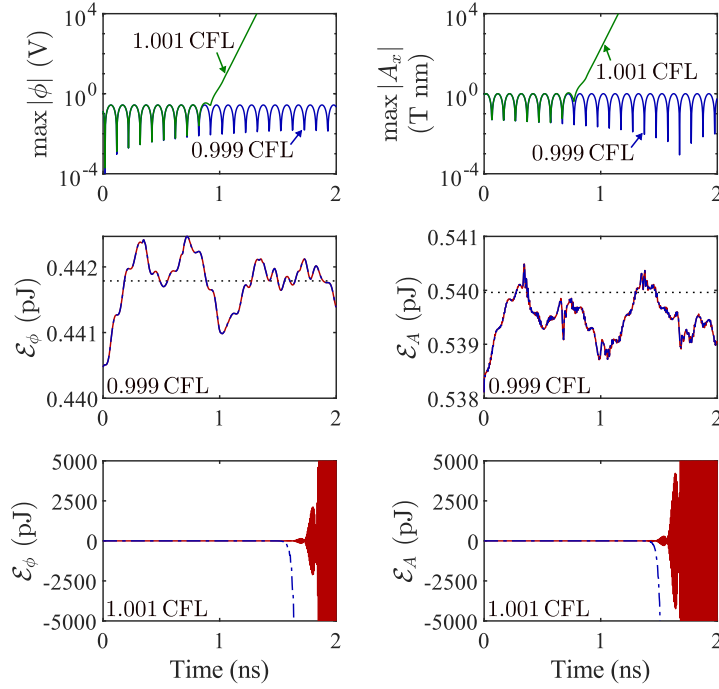


Figure 3.5: Results of the tests in Section 3.3.4. Top row: maximum absolute values of potentials over the region. Middle row: storage functions (3.50) and (3.51) (—) and the sum of initial and supplied energy (—) for $\Delta t = 0.999\Delta t_{\text{CFL}}$. The dotted black lines show the exact values of energy stored in the scalar and vector potential, evaluated from (3.104). Last row: storage functions (3.50) and (3.51) (—) and the sum of initial and supplied energy (—) for $\Delta t = 1.001\Delta t_{\text{CFL}}$.

the values of the unknowns to diverge as the iterations progress. When this occurs, the values represented by the two curves in the last row of Figure 3.5 can no longer be computed accurately due to the finite machine precision, which explains the large derivation.

3.4 Subgridding Scheme

In order to construct a stable subgridding scheme, we follow the same approach as in Chapter 2. Here, the planar interface in Figures 3.6, 3.7, and 3.8 is considered, where the north face of the coarse grid is connected to the south face of the fine grid. Treatment of the scenario where the fine grid is embedded inside of the coarse grid is similar to that described in Appendix C for the fields-based FDTD.

3.4.1 Interpolation Conditions

The interpolation conditions are imposed on the inputs and outputs of the dynamical systems (3.32a)–(3.32b) and (3.43a)–(3.43b), namely:

- the normal component of the gradient of the scalar potential $[\nabla\phi]_{\perp}$ and the time derivative of the scalar potential on the boundary $[\partial_t\phi]_{\dagger}$,

- the tangential component of the magnetic flux density $\vec{B}_\parallel$ and the tangential component of the time derivative of the vector potential $[\partial_t \vec{A}]_\parallel$,
- the normal component of the time derivative of the vector potential $[\partial_t \vec{A}]_\perp$ and the quantity $\kappa_\dagger$.

With reference to Figure 3.6, the following conditions are imposed between $[\nabla\phi]_\perp$ and $[\partial_t\phi]_\dagger$ on the coarse and fine sides of the interface

$$\hat{\Lambda}_{\varepsilon,\perp,y}[\widehat{\partial_y\Phi}]_\perp^n = \mathbf{T}_{r_x r_z} \varepsilon_{\perp,y} [\partial_y \phi]_\perp^n, \quad (3.105a)$$

$$[\partial_t \phi]^{n-\frac{1}{2}} = \frac{\mathbf{T}_{r_x r_z}^T}{r_x r_z} [\widehat{\partial_t \Phi}]^{n-\frac{1}{2}}, \quad (3.105b)$$

where $[\widehat{\partial_y\Phi}]_\perp$ is a vector collecting the $r_x r_z$ hanging variables in the shaded area of the fine grid in Figure 3.6, with the samples ordered as indicated in the figure. Vector $[\widehat{\partial_t\Phi}]$ collects the fine samples of the time derivative of the scalar potential shown in Figure 3.6. Scalars $[\partial_y\phi]_\perp$ and $[\partial_t\phi]$ represent the corresponding coarse samples. Matrix $\hat{\Lambda}_{\varepsilon,\perp,y}$ contains permittivity samples on the $r_x r_z$ normal primary edges of the fine grid and $\varepsilon_{\perp,y}$ is the corresponding sample of the coarse grid. Hence, condition (3.105a) states that, for the values of the normal component of $\varepsilon \nabla \phi$, the fine samples are all equated to the coarse sample. Condition (3.105b) states that the coarse $[\partial_t\phi]$ sample is equal to the average of the corresponding fine samples.

The interpolation conditions for the variables $[\partial_t \vec{A}]_\perp$ and κ are similar. With reference to Figure 3.7, the interpolation conditions are given by

$$\hat{\Lambda}_{\varepsilon,\perp,y}[\widehat{\partial_t \mathbf{a}_y}]_\perp^n = \mathbf{T}_{r_x r_z} \varepsilon_{\perp,y} [\partial_t A_y]_\perp^n, \quad (3.106a)$$

$$\kappa^{n-\frac{1}{2}} = \frac{\mathbf{T}_{r_x r_z}^T}{r_x r_z} \hat{\mathbf{k}}^{n-\frac{1}{2}}, \quad (3.106b)$$

where $\hat{\Lambda}_{\varepsilon,\perp,y}$ and $\varepsilon_{\perp,y}$ are the same quantities as those appearing in (3.105a)–(3.105b) and $[\widehat{\partial_t \mathbf{a}_y}]_\perp$, $\hat{\mathbf{k}}$, $[\partial_t A_y]_\perp$, and κ are defined analogously to $[\widehat{\partial_y\Phi}]_\perp$, $[\widehat{\partial_t\Phi}]$, $[\partial_y\phi]_\perp$, and $[\partial_t\phi]$, respectively. Conditions (3.105a)–(3.105b) and (3.106a)–(3.106b) together impose that the flux of the vector $\vec{D}$ through the shaded area on the boundary of the coarse grid in Figures 3.6 and 3.7 is the same as that through the shaded area of the fine grid boundary. This is consistent with the fact that the physical system being modeled has no free surface charges at the interface between the two regions.

The interpolation conditions between $\vec{B}_\parallel$ and $[\partial_t \vec{A}]_\parallel$ are imposed over patches such as the one shown in Figure 3.8 for the case of $B_{\parallel,x}$ and $[\partial_t A_z]_\parallel$ components. Those patches are the same as the ones discussed in Section 2.4 and the interpolation conditions are analogous to those imposed between $\vec{H}_{\parallel,x}$ and $\vec{E}_{\parallel,z}$ in Chapter 2, namely

$$[\partial_t A_z]^n = \frac{\mathbf{T}_{r_x}^T}{r_x} [\widehat{\partial_t \mathbf{a}_z}]^n, \quad (3.107a)$$

$$\mathbf{T}_{r_x} \mu_{\parallel,x}^{-1} B_{\parallel,x}^{n+\frac{1}{2}} = \left(\frac{\mathbf{T}_{r_z}^T}{r_z} \otimes \mathbf{I}_{r_x} \right) \hat{\Lambda}_{1/\mu,\parallel,x} \hat{\mathbf{b}}_{\parallel,x}^{n+\frac{1}{2}}, \quad (3.107b)$$

$$[\widehat{\partial_t \mathbf{a}_z}]^n = (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) [\widehat{\partial_t \mathbf{a}_z}]^n, \quad (3.107c)$$

where $\hat{\Lambda}_{1/\mu,\parallel,x}$ contains the reluctivity samples on the tangential secondary edges in the shaded area

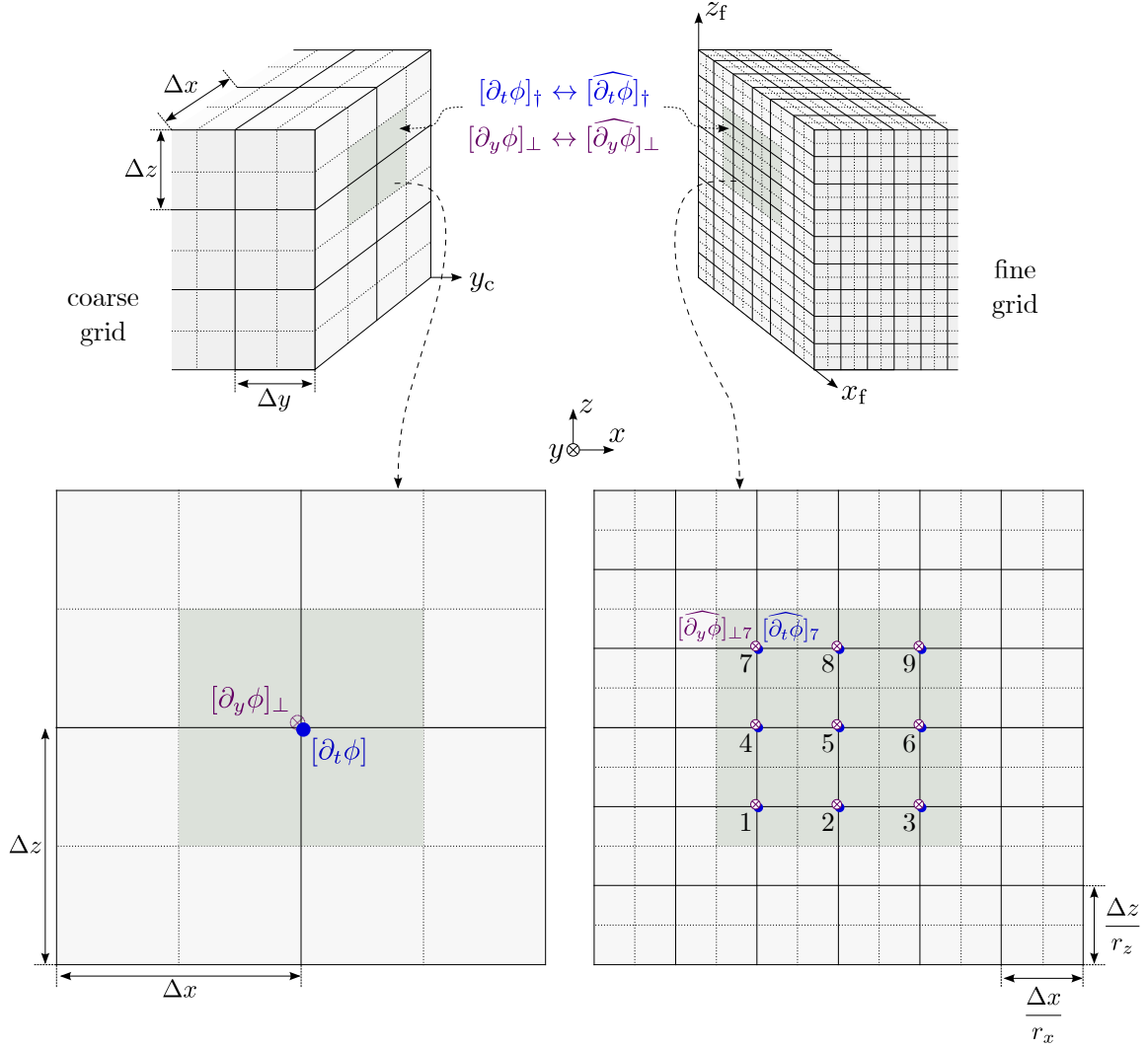


Figure 3.6: Coarse and fine sides of the subgridding interface in Section 3.4 and the samples involved in (3.105a)–(3.105b).

of the fine grid boundary in Figure 3.8 and $\mu_{\parallel,x}^{-1}$ is the corresponding coarse sample. Vectors $[\partial_t A_z]$, $[\widehat{\partial_t a_z}]$, $[\widehat{\partial_t a_z}]$, $B_{\parallel,x}$, $\hat{\mathbf{b}}_{\parallel,x}$ are defined analogously to the corresponding quantities in (2.76a)–(2.76c).

3.4.2 Stability Proof

This section provides the proof of stability of the interpolation rules proposed in Section 3.4.1 for a time step below the most restrictive value of the generalized CFL limits (3.82) and (3.89) of the two grids.

Stability of the Updates of the Scalar Potential

Assuming the aforementioned time step, by Theorem 4, the dynamical system (3.32a)–(3.32b) corresponding to each of the two grids is energy conserving. Thus, in order to ensure stability of the updates of the scalar potential, it is sufficient to show that interpolation rule (3.105a)–(3.105b),

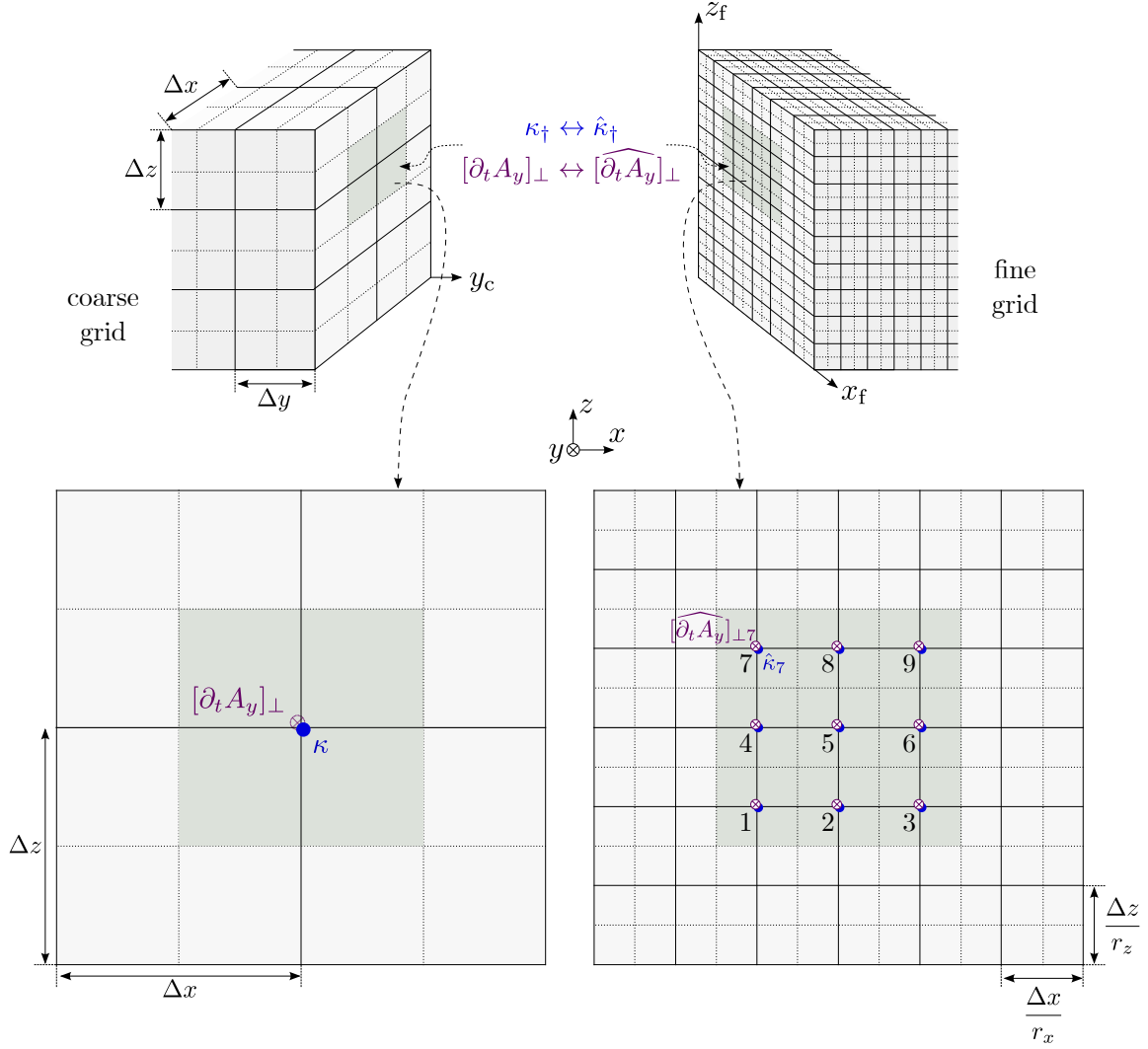


Figure 3.7: Coarse and fine sides of the subgridding interface in Section 3.4 and the samples involved in (3.106a)–(3.106b).

which couples the dynamical systems for each of the grids, is unable to generate energy on its own. This is done by analyzing the contributions of the shaded areas in Figure 3.6 to the supply rate (3.52) for each of the two grids.

The contribution of the coarse samples on the interface in Figure 3.6 to the supply rate of the coarse grid (3.52), or equivalently (3.77), is

$$\delta s \left([\partial_t \phi]^{n-\frac{1}{2}}, [\partial_t \phi]^{n+\frac{1}{2}}, [\partial_y \phi]_{\perp}^n \right) = \Delta t \frac{[\partial_t \phi]^{n-\frac{1}{2}} + [\partial_t \phi]^{n+\frac{1}{2}}}{2} (\hat{n}_n \cdot \hat{y}) \Delta x \Delta z \varepsilon_{\perp, y} [\partial_y \phi]_{\perp}^n, \quad (3.108)$$

where the circumflexes in $(\hat{n}_n \cdot \hat{y})$ denote the unit vectors. The corresponding contribution of the

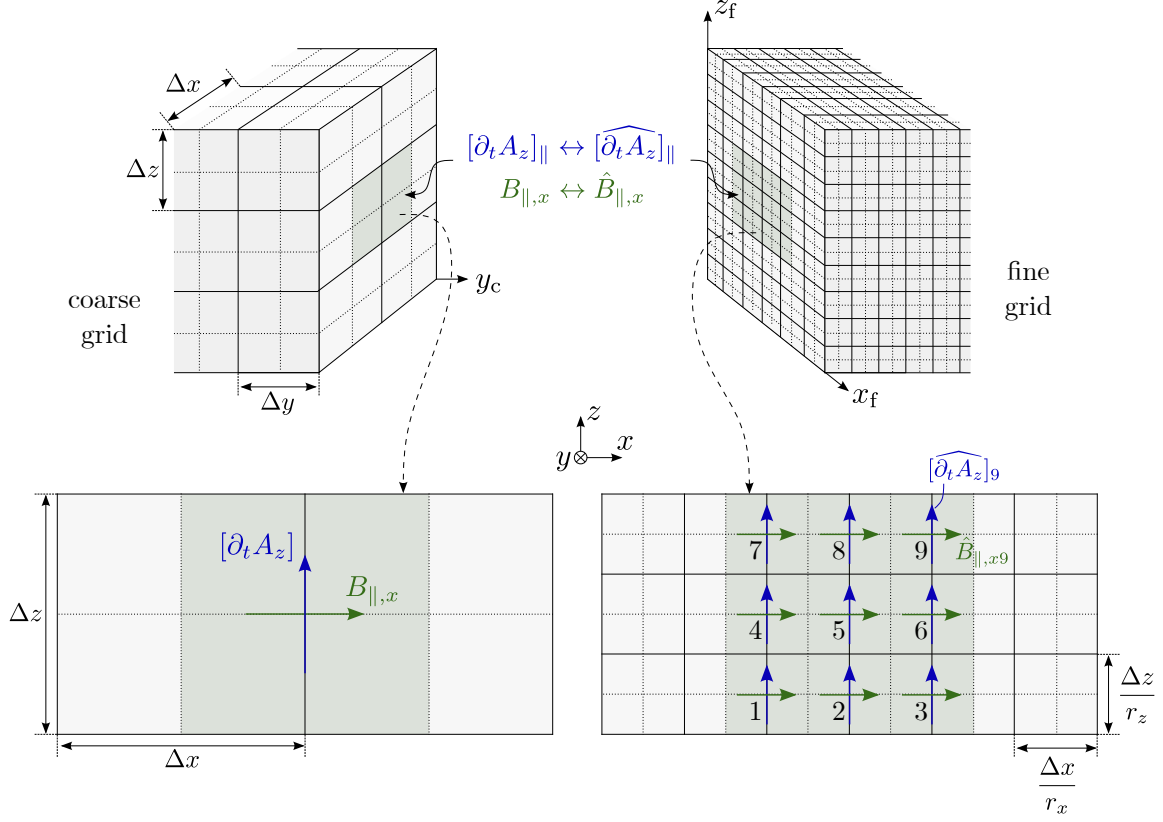


Figure 3.8: Coarse and fine sides of the subgridding interface in Section 3.4 and the samples involved in (3.107a)–(3.107c).

fine samples to the supply rate into the fine grid is

$$\delta \hat{s} \left([\widehat{\partial_t \Phi}]^{n-\frac{1}{2}}, [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}}, [\widehat{\partial_y \Phi}]_{\perp}^n \right) = \Delta t \frac{\left([\widehat{\partial_t \Phi}]^{n-\frac{1}{2}} + [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}} \right)^T}{2} (\hat{n}_s \cdot \hat{y}) \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \hat{\mathbf{A}}_{\epsilon, \perp, y} [\widehat{\partial_y \Phi}]_{\perp}^n. \quad (3.109)$$

From (3.105a),

$$\delta \hat{s} \left([\widehat{\partial_t \Phi}]^{n-\frac{1}{2}}, [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}}, [\widehat{\partial_y \Phi}]_{\perp}^n \right) = \Delta t \frac{\left([\widehat{\partial_t \Phi}]^{n-\frac{1}{2}} + [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}} \right)^T}{2} (\hat{n}_s \cdot \hat{y}) \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \mathbf{T}_{r_x r_z \epsilon_{\perp, y}} [\partial_y \phi]_{\perp}^n, \quad (3.110)$$

which, using (3.105b), becomes

$$\delta \hat{s} \left([\widehat{\partial_t \Phi}]^{n-\frac{1}{2}}, [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}}, [\widehat{\partial_y \Phi}]_{\perp}^n \right) = \Delta t \frac{[\partial_t \phi]^{n-\frac{1}{2}} + [\partial_t \phi]^{n+\frac{1}{2}}}{2} (\hat{n}_s \cdot \hat{y}) \Delta x \Delta z \epsilon_{\perp, y} [\partial_y \phi]_{\perp}^n. \quad (3.111)$$

Since $(\hat{n}_s \cdot \hat{y}) + (\hat{n}_n \cdot \hat{y}) = 0$, from (3.108) and (3.111),

$$\delta \hat{s} \left([\widehat{\partial_t \Phi}]^{n-\frac{1}{2}}, [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}}, [\widehat{\partial_y \Phi}]_{\perp}^n \right) + \delta s \left([\partial_t \phi]^{n-\frac{1}{2}}, [\partial_t \phi]^{n+\frac{1}{2}}, [\partial_y \phi]_{\perp}^n \right) = 0. \quad (3.112)$$

Thus, the interpolation rule contributes no net energy to the dynamical systems (3.32a)–(3.32b) corresponding to the fine and coarse grids. Any energy that is entering either of the two grids is a result of the same amount of energy leaving the other grid. This means that the interpolation rule respects the principle of energy conservation. Hence, the proposed subgridding scheme ensures stable updates of the scalar potential under the most restrictive generalized CFL condition (3.82). From Theorem 6, a sufficient rule in practice is to select the time step below the smallest CFL limit (1.36) of the two grids.

Stability of the Updates of the Vector Potential

In a similar way, we show that the interpolation rules (3.106a)–(3.106b) and (3.107a)–(3.107c) conserve energy. From Theorem 5, the dynamical system (3.43a)–(3.43b) for each of the grids is energy conserving under the aforementioned time step. Hence, in order to ensure that the updates of the vector potential are stable with the proposed interpolation rules, all that remains to be shown is that those interpolation rules also respect the principle of energy conservation. This is done by analyzing the contribution of the shaded areas in Figures 3.7–3.8 to the supply rate (3.53), or equivalently (3.79), of the two grids.

For the samples in Figure 3.7, their contributions to the supply rates (3.79) of the coarse and fine grids, respectively, are given by

$$\delta s^n \left(\kappa^{n-\frac{1}{2}}, \kappa^{n+\frac{1}{2}}, [\partial_t A_y]_{\perp}^n \right) = -\Delta t \frac{\kappa^{n-\frac{1}{2}} + \kappa^{n+\frac{1}{2}}}{2} (\hat{n}_n \cdot \hat{y}) \Delta x \Delta z \varepsilon_{\perp, y} [\partial_t A_y]_{\perp}^n, \quad (3.113)$$

$$\delta \hat{s} \left(\hat{\kappa}^{n-\frac{1}{2}}, \hat{\kappa}^{n+\frac{1}{2}}, [\widehat{\partial_t \mathbf{a}_y}]_{\perp}^n \right) = -\Delta t \frac{\left(\hat{\kappa}^{n-\frac{1}{2}} + \hat{\kappa}^{n+\frac{1}{2}} \right)^T}{2} (\hat{n}_s \cdot \hat{y}) \frac{\Delta y}{r_y} \frac{\Delta z}{r_z} \hat{\mathbf{A}}_{\varepsilon, \perp, y} [\widehat{\partial_t \mathbf{a}_y}]_{\perp}^n. \quad (3.114)$$

Using (3.106a) and (3.106b) we can show that

$$\delta s^n \left(\hat{\kappa}^{n-\frac{1}{2}}, \hat{\kappa}^{n+\frac{1}{2}}, [\widehat{\partial_t \mathbf{a}_y}]_{\perp}^n \right) + \delta s^n \left(\kappa^{n-\frac{1}{2}}, \kappa^{n+\frac{1}{2}}, [\partial_t A_y]_{\perp}^n \right) = 0. \quad (3.115)$$

Continuing this reasoning for the shaded areas in Figure 3.8, the contributions of the samples on the coarse and fine grids to their respective supply rates (3.79) are given by

$$\delta s \left([\partial_t A_z]^n, [\partial_t A_z]^{n+1}, B_{\parallel, x}^{n+\frac{1}{2}} \right) = -\Delta t \frac{[\partial_t A_z]^n + [\partial_t A_z]^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \Delta x \mu_{\parallel, x}^{-1} B_{\parallel, x}^{n+\frac{1}{2}}, \quad (3.116)$$

$$\begin{aligned} \delta \hat{s} \left([\widehat{\partial_t \mathbf{a}_z}]^n, [\widehat{\partial_t \mathbf{a}_z}]^{n+1}, \hat{\mathbf{b}}_{\parallel, x}^{n+\frac{1}{2}} \right) = \\ -\Delta t \frac{\left([\widehat{\partial_t \mathbf{a}_z}]^n + [\widehat{\partial_t \mathbf{a}_z}]^{n+1} \right)^T}{2} \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{A}}_{1/\mu, \parallel, x} \hat{\mathbf{b}}_{\parallel, x}^{n+\frac{1}{2}}. \end{aligned} \quad (3.117)$$

From (3.107c),

$$\begin{aligned} \delta \hat{s} \left([\widehat{\partial_t \mathbf{a}_z}]^n, [\widehat{\partial_t \mathbf{a}_z}]^{n+1}, \hat{\mathbf{b}}_{\parallel, x}^{n+\frac{1}{2}} \right) = \\ - \Delta t \frac{([\overline{\partial_t \mathbf{a}_z}]^n + [\overline{\partial_t \mathbf{a}_z}]^{n+1})^T}{2} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{A}}_{1/\mu, \parallel, x} \hat{\mathbf{b}}_{\parallel, x}^{n+\frac{1}{2}}. \end{aligned} \quad (3.118)$$

Using (3.107b),

$$\begin{aligned} \delta \hat{s} \left([\widehat{\partial_t \mathbf{a}_z}]^n, [\widehat{\partial_t \mathbf{a}_z}]^{n+1}, \hat{\mathbf{b}}_{\parallel, x}^{n+\frac{1}{2}} \right) = \\ - \Delta t \frac{([\overline{\partial_t \mathbf{a}_z}]^n + [\overline{\partial_t \mathbf{a}_z}]^{n+1})^T}{2} \mathbf{T}_{r_x} \Delta z (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \mu_{\parallel, x}^{-1} B_{\parallel, x}^{n+\frac{1}{2}}. \end{aligned} \quad (3.119)$$

Finally, using (3.107a), we obtain

$$\delta \hat{s} \left([\widehat{\partial_t \mathbf{a}_z}]^n, [\widehat{\partial_t \mathbf{a}_z}]^{n+1}, \hat{\mathbf{b}}_{\parallel, x}^{n+\frac{1}{2}} \right) = - \Delta t \frac{[\partial_t A_z]^n + [\partial_t A_z]^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_s \times \hat{x}) \Delta x \mu_{\parallel, x}^{-1} B_{\parallel, x}^{n+\frac{1}{2}}, \quad (3.120)$$

from which we similarly conclude that

$$\delta \hat{s} \left([\widehat{\partial_t \mathbf{a}_z}]^n, [\widehat{\partial_t \mathbf{a}_z}]^{n+1}, \hat{\mathbf{b}}_{\parallel, x}^{n+\frac{1}{2}} \right) + \delta s \left([\partial_t A_z]^n, [\partial_t A_z]^{n+1}, B_{\parallel, x}^{n+\frac{1}{2}} \right) = 0. \quad (3.121)$$

Due to the relations such as (3.115) and (3.121), the interpolation rule ensures that there is no net energy supplied to the two grids. Therefore, the updates of the vector potential are stable when the time step respects the generalized CFL limit (3.89) in each of the two grids. From Theorem 6, the most restrictive value of the conventional CFL limit (1.36) can be used in practice.

3.4.3 Update Equations

This section outlines the update equations for the quantities on the subgridding interface that implicitly enforce the interpolation conditions in Section 3.4.1.

Update Equations for $[\partial_t \phi]$

The update equations for the time derivative of the scalar potential are governed by the interpolation conditions (3.105a)–(3.105b) and dynamical system equations of Section 3.2 at the primary nodes on the boundaries of each of the grids. For the coarse sample shown in Figure 3.6, the dynamical system equation can be written as

$$\begin{aligned} \Delta x \frac{\Delta y}{2} \Delta z \frac{\chi}{\Delta t} [\partial_t \phi]^{n+\frac{1}{2}} = \Delta x \frac{\Delta y}{2} \Delta z \frac{\chi}{\Delta t} [\partial_t \phi]^{n-\frac{1}{2}} \\ + \mathbf{D} \mathbf{\Lambda}_S'' \mathbf{\Lambda}_\varepsilon [\nabla \boldsymbol{\phi}]^n + (\hat{n}_n \cdot \hat{y}) \Delta x \Delta z \varepsilon_{\perp, y} [\partial_y \phi]_{\perp}^n, \end{aligned} \quad (3.122)$$

where, letting (i, j, k) be the indices of the node where $[\partial_t \phi]$ is sampled,

$$\chi = \chi|_{i,j,k}, \quad (3.123)$$

$$\mathbf{D} = \begin{bmatrix} -1 & 1 & -1 & -1 & 1 \end{bmatrix}, \quad (3.124)$$

$$\mathbf{\Lambda}_S'' = \text{diag} \left(\frac{\Delta y}{2} \Delta z, \frac{\Delta y}{2} \Delta z, \Delta x \Delta z, \Delta x \frac{\Delta y}{2}, \Delta x \frac{\Delta y}{2} \right), \quad (3.125)$$

$$\mathbf{\Lambda}_\varepsilon = \text{diag} \left(\varepsilon_x|_{i-\frac{1}{2},j,k}, \varepsilon_x|_{i+\frac{1}{2},j,k}, \varepsilon_y|_{i,j-\frac{1}{2},k}, \varepsilon_y|_{i,j,k-\frac{1}{2}}, \varepsilon_z|_{i,j,k+\frac{1}{2}} \right), \quad (3.126)$$

$$[\nabla \Phi]^n = \begin{bmatrix} [\partial_x \phi]_{i-\frac{1}{2},j,k}^n & [\partial_x \phi]_{i+\frac{1}{2},j,k}^n & [\partial_y \phi]_{i,j-\frac{1}{2},k}^n & [\partial_y \phi]_{i,j,k-\frac{1}{2}}^n & [\partial_z \phi]_{i,j,k+\frac{1}{2}}^n \end{bmatrix}^T. \quad (3.127)$$

The corresponding equation for the samples of the fine grid is given by

$$\begin{aligned} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \frac{\Delta z}{r_z} \frac{\hat{\Lambda}_\chi}{\Delta t} [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}} &= \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \frac{\Delta z}{r_z} \frac{\hat{\Lambda}_\chi}{\Delta t} [\partial_t \Phi]^{n-\frac{1}{2}} \\ &\quad + \hat{\mathbf{D}} \hat{\Lambda}_S'' \hat{\Lambda}_\varepsilon [\widehat{\nabla \Phi}]^n + (\hat{n}_s \cdot \hat{y}) \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \hat{\Lambda}_{\varepsilon,\perp,y} [\partial_y \Phi]_{\perp}^n. \end{aligned} \quad (3.128)$$

From (3.122), (3.128), and (3.105a)–(3.105b), we obtain the following update equation for the samples on the fine grid boundary in Figure 3.6:

$$\begin{aligned} \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \left(\frac{\Delta y}{2} \frac{1}{r_x r_z} \mathbf{T}_{r_x r_z} \frac{\chi}{\Delta t} \mathbf{T}_{r_x r_z}^T + \frac{\Delta y}{2r_y} \frac{\hat{\Lambda}_\chi}{\Delta t} \right) [\widehat{\partial_t \Phi}]^{n+\frac{1}{2}} \\ = \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \left(\frac{\Delta y}{2} \frac{1}{r_x r_z} \mathbf{T}_{r_x r_z} \frac{\chi}{\Delta t} \mathbf{T}_{r_x r_z}^T + \frac{\Delta y}{2r_y} \frac{\hat{\Lambda}_\chi}{\Delta t} \right) [\partial_t \Phi]^{n-\frac{1}{2}} \\ + \frac{\mathbf{T}_{r_x r_z}}{r_x r_z} \mathbf{D} \mathbf{\Lambda}_S'' \mathbf{\Lambda}_\varepsilon [\nabla \Phi]^n + \hat{\mathbf{D}} \hat{\Lambda}_S'' \hat{\Lambda}_\varepsilon [\widehat{\nabla \Phi}]^n. \end{aligned} \quad (3.129)$$

The coarse grid sample $[\partial_t \phi]$ can be updated using (3.105b) once $[\widehat{\partial_t \Phi}]^{n+0.5}$ is computed.

Update Equations for κ

In a similar manner, one can obtain the update equation for the fine samples $\hat{\kappa}$ in Figure 3.7

$$\begin{aligned} \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \left(\frac{\Delta y}{2} \frac{1}{r_x r_z} \mathbf{T}_{r_x r_z} \frac{\chi}{\Delta t} \mathbf{T}_{r_x r_z}^T + \frac{\Delta y}{2r_y} \frac{\hat{\Lambda}_\chi}{\Delta t} \right) \hat{\kappa}^{n+\frac{1}{2}} = \\ \frac{\Delta x}{r_x} \frac{\Delta z}{r_z} \left(\frac{\Delta y}{2} \frac{1}{r_x r_z} \mathbf{T}_{r_x r_z} \frac{\chi}{\Delta t} \mathbf{T}_{r_x r_z}^T + \frac{\Delta y}{2r_y} \frac{\hat{\Lambda}_\chi}{\Delta t} \right) \hat{\kappa}^{n-\frac{1}{2}} \\ - \frac{\mathbf{T}_{r_x r_z}}{r_x r_z} \mathbf{D} \mathbf{\Lambda}_S'' \mathbf{\Lambda}_\varepsilon [\partial_t \mathbf{a}]^n - \hat{\mathbf{D}} \hat{\Lambda}_S'' \hat{\Lambda}_\varepsilon [\widehat{\partial_t \mathbf{a}_z}]^n, \end{aligned} \quad (3.130)$$

where χ , $\mathbf{D}$, $\mathbf{\Lambda}_S''$, and $\mathbf{\Lambda}_\varepsilon$ have been defined in (3.123), (3.124), (3.125), and (3.126), respectively, and the rest of the quantities are defined analogously as those in (3.129). The coarse sample can be updated using (3.106b).

Update Equations for $[\partial_t \vec{A}]$

The procedure for deriving the update equation for the $[\partial_t \vec{A}]$ samples in Figure 3.8 is similar to the one used to derive (2.92). The resulting equation for the $[\partial_t \vec{a}_z]$ samples is given by

$$\begin{aligned} \Delta z \frac{\Delta x}{r_x} \left(\frac{\Delta y}{2} \frac{1}{r_x} \mathbf{T}_{r_x} \frac{\varepsilon_z}{\Delta t} \mathbf{T}_{r_x}^T + \frac{\Delta y}{2r_y} \frac{1}{r_z} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \frac{\hat{\Lambda}_{\varepsilon,z}}{\Delta t} (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) \right) [\partial_t \vec{a}_z]^{n+1} = \\ \Delta z \frac{\Delta x}{r_x} \left(\frac{\Delta y}{2} \frac{1}{r_x} \mathbf{T}_{r_x} \frac{\varepsilon_z}{\Delta t} \mathbf{T}_{r_x}^T + \frac{\Delta y}{2r_y} \frac{1}{r_z} (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \frac{\hat{\Lambda}_{\varepsilon,z}}{\Delta t} (\mathbf{T}_{r_z} \otimes \mathbf{I}_{r_x}) \right) [\partial_t \vec{a}_z]^n \\ - \frac{\mathbf{T}_{r_x}}{r_x} \Delta z \mathbf{C}^T \hat{\Lambda}_l'' \hat{\Lambda}_{1/\mu} \mathbf{b}^{n+\frac{1}{2}} - (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \frac{\Delta z}{r_z} \hat{\mathbf{C}}^T \hat{\Lambda}_l'' \hat{\Lambda}_{1/\mu} \mathbf{b}^{n+\frac{1}{2}} \\ + \frac{\mathbf{T}_{r_x}}{r_x} \Delta x \frac{\Delta y}{2} \varepsilon_z \mathbf{D}^T \mathbf{k}^{n+\frac{1}{2}} + (\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{r_x}) \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \hat{\Lambda}_{\varepsilon,z} \hat{\mathbf{D}}^T \mathbf{k}^{n+\frac{1}{2}}, \quad (3.131) \end{aligned}$$

where the matrices and vectors are defined analogously to those in (2.92). The coarse and fine samples can be obtained from (3.107a) and (3.107c), respectively.

3.5 Numerical Examples

The subgridding algorithm proposed in Section 3.4 was implemented in MATLAB [215]. Several tests were performed in order to verify the stability of the scheme and assess its accuracy. The simulated structures were excited via the impressed electric current density $\vec{J}_s$ using the procedure in [38]. The charge density samples, such as $\rho_v|_{i,j,k}^n$ in (3.6), were updated from the values of the current density using the discretized continuity equation, as discussed in [38].

3.5.1 Stability Verification

In this example, we consider the scenario in Figure 3.9, similar to the one in Section 2.5.1. The simulated structure is a cube with a side length of 24 cm. On the boundary, the scalar potential and the tangential component of the vector potential are set to zero, which enforces a PEC boundary condition [41]. The main grid has a resolution of $\Delta x = \Delta y = \Delta z = 2$ cm. An 8 cm $\times$ 8 cm $\times$ 8 cm subgridding region with the grid ratio of $r = 5$ in each direction is placed in the center of the structure. The relative permeability values in each primary cell of the main and subgridding regions were randomly set between ε_0 and $2\varepsilon_0$ using the `rand` function in MATLAB. Three current sources were placed at the primary edges close to the bottom-south-west corner of the region, with

$$J_x|_{\frac{3}{2},2,2}(t) = \frac{I_0}{\Delta y \Delta z} g(t), \quad (3.132)$$

$$J_y|_{2,\frac{3}{2},2}(t) = \frac{I_0}{\Delta y \Delta z} g(t), \quad (3.133)$$

$$J_z|_{2,2,\frac{3}{2}}(t) = \frac{I_0}{\Delta y \Delta z} g(t), \quad (3.134)$$

Table 3.1: The values of the CFL limit (1.36) for the coarse and fine grids in the stability verification example of Section 3.5.1.

Grid	Smallest primary cell permittivity $\varepsilon_{\min}$	CFL limit
Coarse	$1.00052 \varepsilon_0$	38.5267 ps
Fine	$1.00023 \varepsilon_0$	7.7042 ps

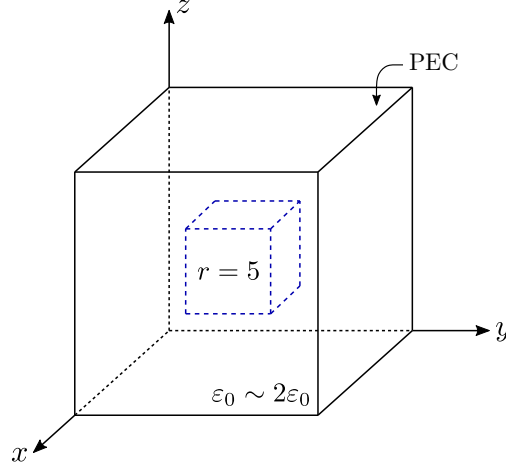


Figure 3.9: Setup of the stability verification test in Section 3.5.1. Sources and probes are placed near the bottom-south-west corner of the region, as described in the text.

where the subscripts denote the spatial indices of the primary edges where the sources were placed, I_0 was equal to 1 A. Waveform $g(t)$ is given by

$$g(t) = u_s(t - t_1) - 2u_s(t - t_2) + u_s(t - t_3), \quad (3.135)$$

where t_1 , t_2 , and t_3 are 0.677 ns, 1.344 ns, 2.011 ns and $u_s(\cdot)$ a unit step with the edges smoothed out, as per recommendation in [35]. The initial conditions on the charge density were set to zero. The values of the CFL limit (1.36) associated with each grid are shown in Table 3.1. The time step was selected as 7.7035 ps, which was the most restrictive of the two CFL limits, scaled by 0.9999. The structure was updated for one million time steps. Figure 3.10 shows the recorded values of the scalar potential at the location $(\Delta x, \Delta y, \Delta z)$ and the x component of the vector potential at $(1.5\Delta x, \Delta y, \Delta z)$. As evident from the figure, no instability occurs after a million time steps, confirming that the algorithm is stable, even when the material properties have an arbitrary spatial variation.

3.5.2 Resolving Spherical Objects

In this test, we apply the proposed subgridding scheme to compute the potentials at the probe in the setup in Figure 3.11. In this scenario, the wave from a short line source impinges on an object composed of eight spheres of radius 1 cm. The relative permittivity of the bottom four spheres and the top four spheres is 4 and 8, respectively. The probes for measuring ϕ , $\vec{A}$, $\vec{E}$, and $\vec{H}$ are placed at the center of the object. The distances between different objects in the setup are indicated in

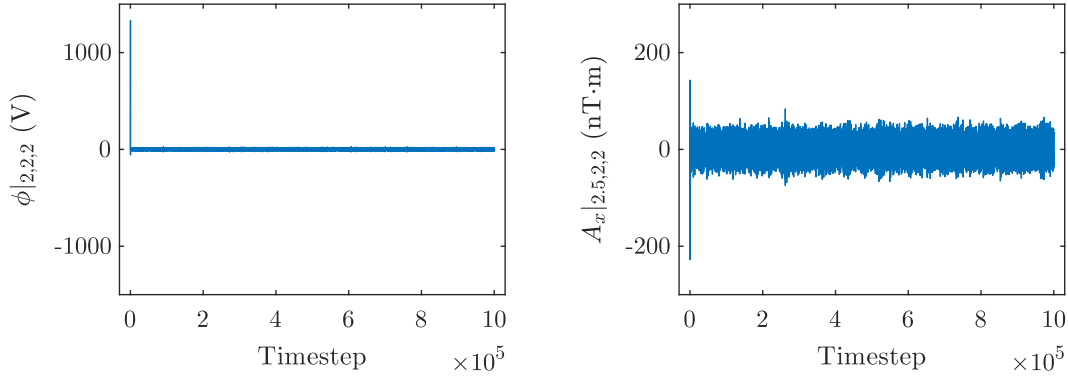


Figure 3.10: Values recorded at the probes in the stability verification test in Section 3.5.1.

Figure 3.13. The excitation is generated by a 5 cm-long line current course, carrying a current

$$I(t) = I_0 g(t), \quad (3.136)$$

where I_0 is 1 A and $g(t)$ is a Gaussian with a time-domain maximum of one and a half-width at half-maximum in the frequency domain of 1.77 GHz.

The resolution of the main grid in this test was $\Delta x = \Delta y = \Delta z = 1$ cm. A subgridding region with grid ratio of 3 was defined around the object, as shown in Figure 3.11. As a reference, we also performed two simulations with a uniform fine grid and a uniform coarse grid, with cell dimensions of $(1/3)$ cm $\times$ $(1/3)$ cm $\times$ $(1/3)$ cm and 1 cm $\times$ 1 cm $\times$ 1 cm, respectively. In each simulation, the time step was taken as 0.9999 of the most restrictive CFL limit value (1.36) in the simulation domain. The region was terminated using an 8-cell PML described in [38]. The simulations were carried with a termination time of 30 ns. The electric field was computed from the surrounding samples of the vector and scalar potentials by means of a finite difference approximation of (1.8). The magnetic field was obtained from the magnetic flux density, which was one of the quantities computed during the FDTD-P update procedure. In each of the three simulations (subgridding, uniform coarse, and uniform fine) the probe was at the center of a z -directed primary edge, where only some of the quantities of interest were available, namely E_z and A_z . For the remaining quantities, averaging was performed using the values recorded at the nearest available sampling locations. The resulting values were converted into the frequency domain via Fourier transform and normalized by the Fourier transform of the source waveform $g(t)$.

The magnitude and phase of the normalized frequency domain results are shown in Figures 3.12, 3.13, 3.14, and 3.15, where $\mathcal{F}$ denotes the Fourier transform. The x components of $\vec{A}$ and $\vec{E}$ and the y and z components of $\vec{H}$ were omitted, as their values were negligible due to the symmetry of the problem. The vertical line in the plots indicates the frequency at which the free space wavelength is equal to 20 coarse spatial steps, below which the main grid should have sufficient resolution to avoid the effects of dispersion [35]. From Figure 3.12, showing the results of the scalar potential calculation, the subgridding algorithm had excellent agreement with the uniform fine grid reference. The results of the vector potential in Figure 3.13 somewhat deviate from the reference simulation, with the deviations becoming more pronounced at the frequencies beyond $\lambda = 20\Delta x$. Nevertheless, the algorithm achieves substantial improvement in accuracy with respect to the coarse grid simulation.

Table 3.2: Computational cost for different discretization in the example of Section 3.5.2.

Method	Number of primary cells	Simulation time	Speedup
Coarse	217,464	1.3 minutes	46.6
Fine	3,210,856	58.5 minutes	reference
Proposed	258,648	9.8 minutes	6.0

As can be seen from Figure 3.14, the computation of the y component of electric field suffered from a noticeable error between approximately 0.9 and 1.9 GHz. Upon further investigation, it was found that in that range of frequencies, the phase difference between the time derivative of A_y and the y component of the gradient of ϕ , which are the two components comprising E_y , were close to 180° ⁶. It should be noted that $\partial\vec{A}/\partial t$ and $\nabla\phi$ are computed using two separate update procedures. Hence, the addition of these two quantities can incur a significant relative error when they are close to being 180° out of phase, even if the relative error due to each update procedure is small. Therefore, caution must be taken when applying the proposed algorithm for finding the electric field. When considering the phase of E_y in Figure 3.14 (bottom-left panel), it should be noted that, while there was a significant discrepancy around 1.07 GHz between the subgridding and the fine grid results, the difference in the phase at other frequencies was comparable to a multiple of 360° . For the case of the magnetic field in Figure 3.15, excellent correspondence with the reference simulation was observed.

Figure 3.16 shows relative errors compared to the fine grid simulation, which were computed as

$$\frac{|\mathcal{F}\{\phi\} - \mathcal{F}\{\phi_{\text{fine}}\}|}{|\mathcal{F}\{\phi_{\text{fine}}\}|} \quad (3.137)$$

for the scalar potential, as⁷

$$\frac{\|\mathcal{F}\{\vec{A}\} - \mathcal{F}\{\vec{A}_{\text{fine}}\}\|_2}{\|\mathcal{F}\{\vec{A}_{\text{fine}}\}\|_2} \quad (3.138)$$

for the vector potential, and similarly for the electric and magnetic field. The results for both the coarse and subgridding discretization exhibited increase of the relative error in the scalar potential and electric field at lower frequencies. This can be explained by the decrease in magnitude of these quantities as the frequency decreases, making it challenging for both discretization methods to capture their values accurately. For the case of ϕ , $\vec{A}$, and $\vec{H}$, the proposed subgridding achieved a lower relative error compared to the coarse grid simulation for most frequencies. For $\vec{A}$ and $\vec{H}$ at lower frequencies, the error in the proposed method stayed roughly constant as frequency decreased, whereas the relative error of the coarse grid decayed, allowing the coarse discretization to outperform the proposed method at those frequencies. For the case of electric field, we observe a degraded accuracy of the proposed subgridding around 1.07 GHz, which can be attributed to the deviations in the amplitude and phase of the y component in Figure 3.14. Nevertheless, the proposed method improved accuracy in the electric field at other frequencies. The computational cost comparison between the uniform coarse, uniform fine, and subgridding discretization is shown in Table 3.2.

⁶For the case of the subgridding simulation, the phase difference was between 160° and 210° , crossing 180° at three different frequencies in that range.

⁷The 2-norm of a complex vector quantity $\vec{M}$ is computed as $\sqrt{\vec{M}^* \cdot \vec{M}}$ [96].

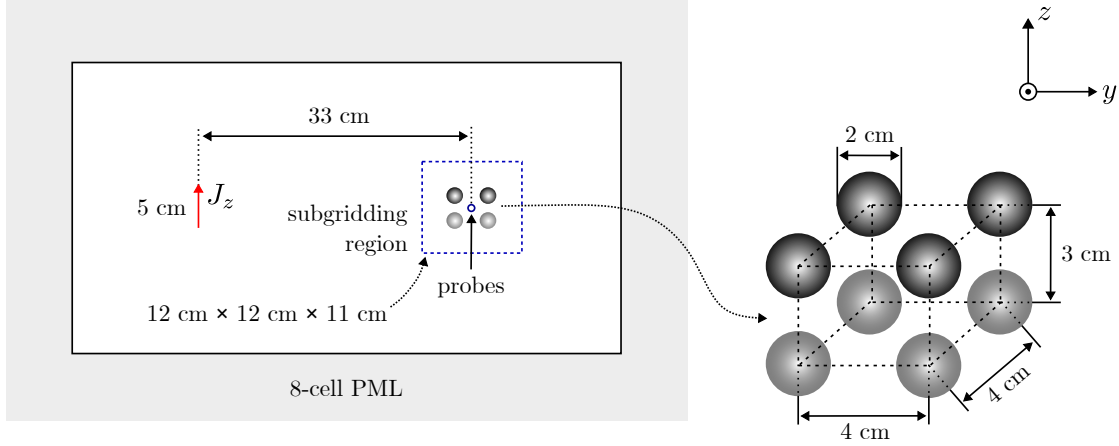


Figure 3.11: Setup of the test in Section 3.5.2.

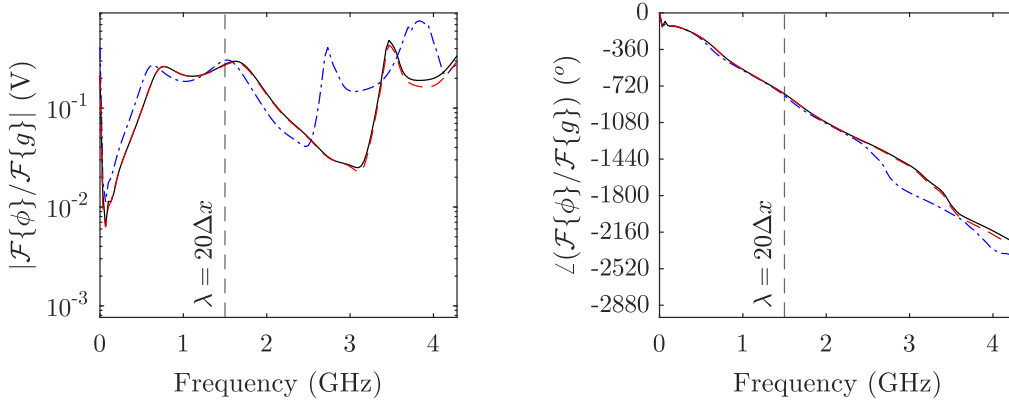


Figure 3.12: Normalized Fourier transform of the scalar potential recorded at the probe in the test in Section 3.5.2 for the coarse grid (—), the fine grid (—), and the proposed subgridding method (—).

3.5.3 Material Traverse

In this example we consider the same setup as the one described in Section 2.5.2, where the subgridding region is placed in three different ways shown in Figure 2.7. In this section, the material considered is a lossless dielectric with permittivity $3\epsilon_0$. Some changes were made to the setup of Section 2.5.2 in order to facilitate observation of the behavior of the scalar potential. In particular, the source used to excite the simulation was elongated to 4 cm. In addition, the probes were shifted up by 2 cm with respect to the mid-plane of the object and placed at the nearest available sampling locations.

The recorded values at the probes are shown in Figures 3.17, 3.18, 3.19, and 3.20. The results agree well between the three cases, indicating that the proposed subgridding algorithm supports material traverse.

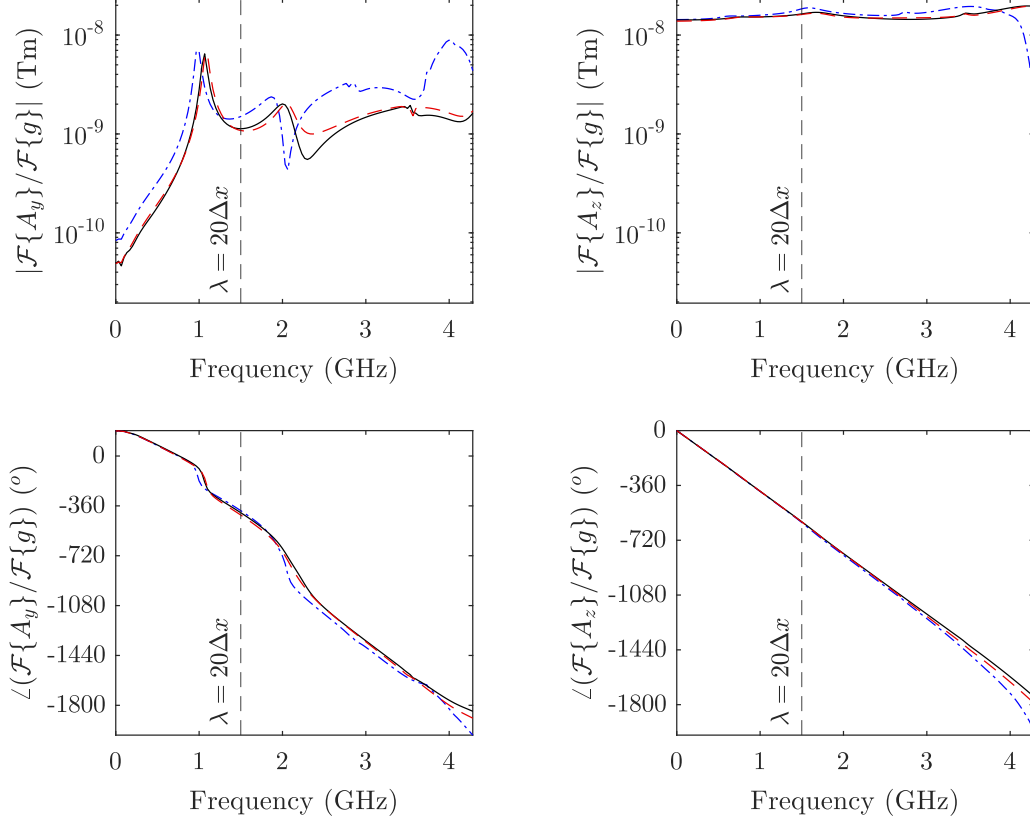


Figure 3.13: Normalized Fourier transform of the y and z components of the vector potential recorded at the probe in the test in Section 3.5.2 for the coarse grid (—), the fine grid (—), and the proposed subgridding method (—).

3.5.4 Waveguide Reflection

The setup of this example is the same as the one described in Section 2.5.3, originally taken from [22, 154]. The incident and reflected power was calculated from potentials based on (3.60). The resulting reflections are shown in Figure 3.21. The reflections are small and do not exhibit resonant behavior appearing in the test in Section 2.5.3. The reflections are higher than in some of the schemes reported in the literature for the fields-based FDTD [22, 154]. However, to the best of our knowledge, potentials-based subgridding schemes are not currently available. Moreover, due to the use of the proposed stability analysis framework, this subgridding scheme is energy conserving and stable under a CFL limit that is known analytically.

3.6 Conclusion

In this chapter, the dissipation theory in Chapter 2 was extended to FDTD with potentials. In particular, it was shown that the update equations of the FDTD-P method [38] can be cast in the same discrete-time dynamical system form as the fields-based FDTD equations. Using a similar approach to Chapter 2, we proposed suitable discrete expressions to compute the energy stored in the scalar and vector potential present in a region, as well as the power that enters the region through its boundary. Through these equations, we identified the generalized CFL limit under which the

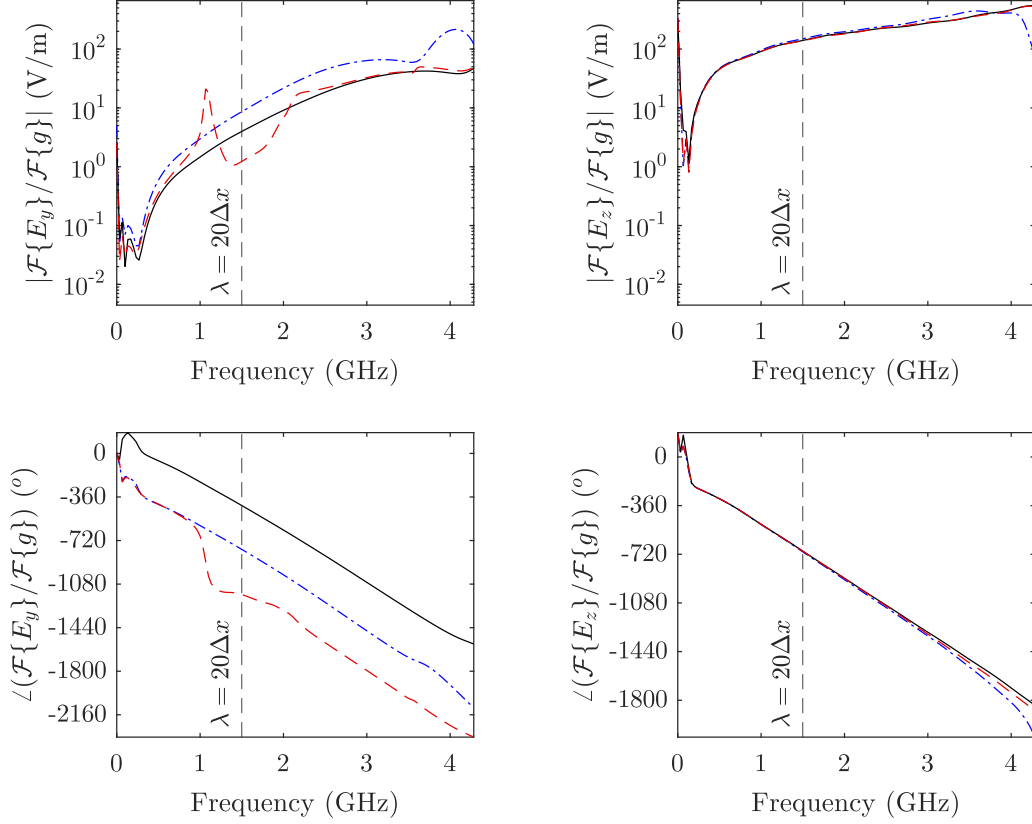


Figure 3.14: Normalized Fourier transform of the y and z components of electric field recorded at the probe in the test in Section 3.5.2 for the coarse grid (—), the fine grid (—), and the proposed subgridding method (—).

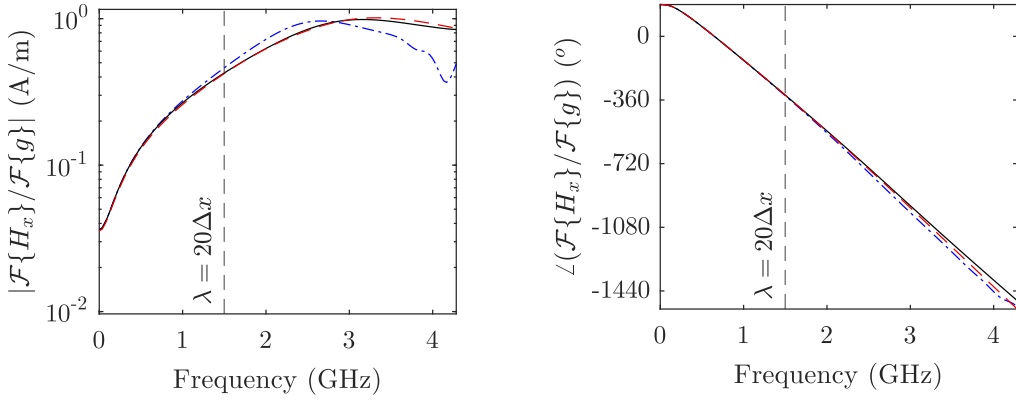


Figure 3.15: Normalized Fourier transform of the x component of the magnetic field recorded at the probe in the test in Section 3.5.2 for the coarse grid (—), the fine grid (—), and the proposed subgridding method (—).

update equations of this method conserve energy in the discrete time domain. Similarly to the fields-based FDTD, the conventional CFL limit (1.36) was rigorously shown to be a sufficient condition, which can be used in practice. Moreover, based on the proposed theory, we derive a provably stable subgridding algorithm with potentials, which respects the principle of energy conservation

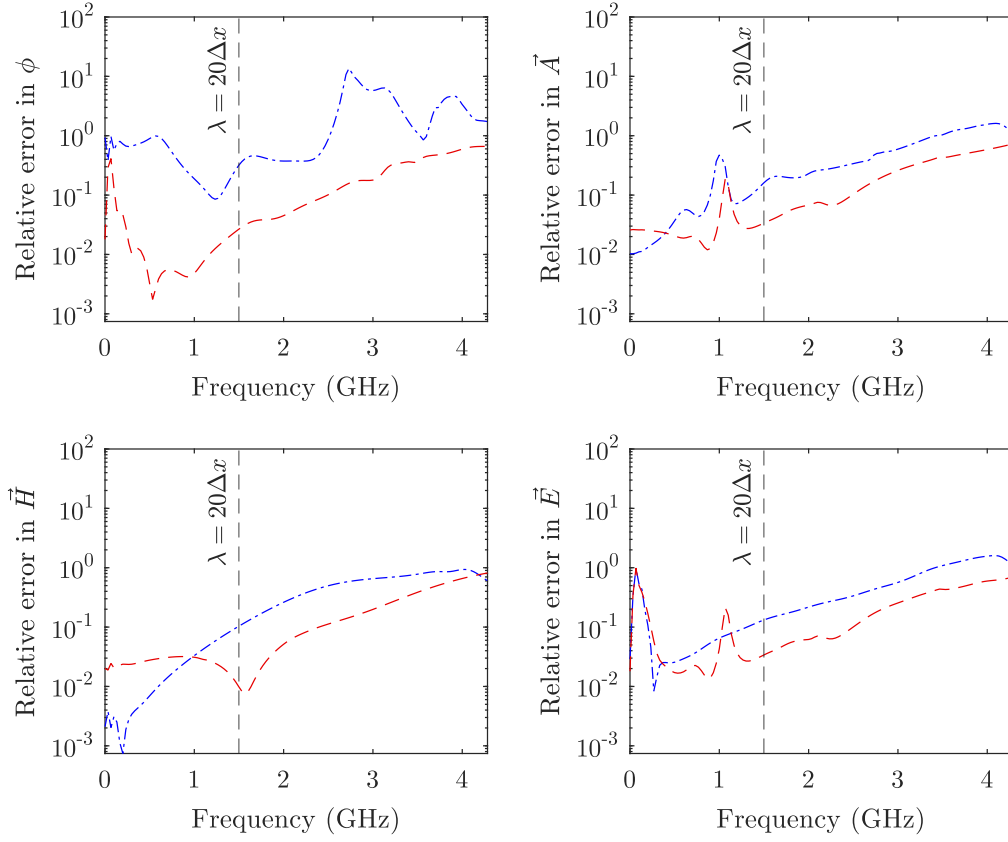


Figure 3.16: Relative errors in the test in Section 3.5.2 for the coarse grid (—) and the proposed subgridding method (---) with respect to the fine grid result. The errors are computed using (3.137) and (3.138) for the scalar and vector potentials and similarly for the electric and magnetic fields.

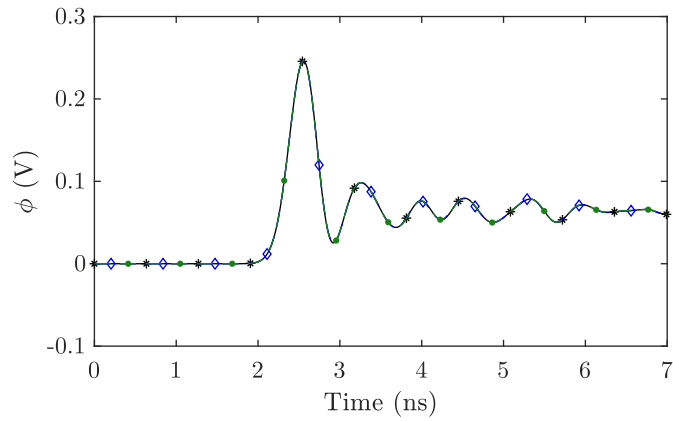


Figure 3.17: The scalar potential at the probe in material traverse test of Section 3.5.3 for Case 1 (—*), Case 2 (—◇—), and Case 3 (—●—).

by construction. The numerical examples confirm the accuracy of the algorithm when computing potentials or magnetic field. It was found that the computation of electric field can be inaccurate

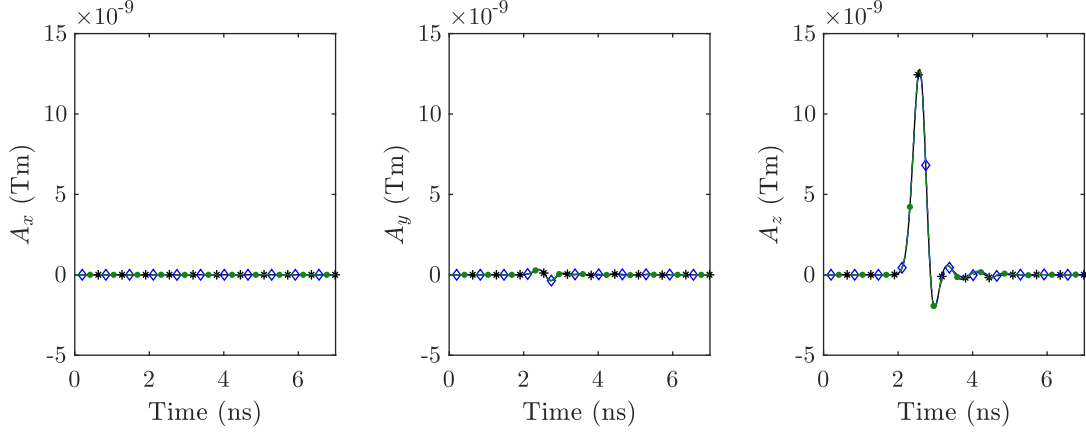


Figure 3.18: The vector potential at the probe in material traverse test of Section 3.5.3 for Case 1 ($\text{---}\star\text{---}$), Case 2 ($\text{---}\diamond\text{---}$), and Case 3 ($\text{---}\bullet\text{---}$).

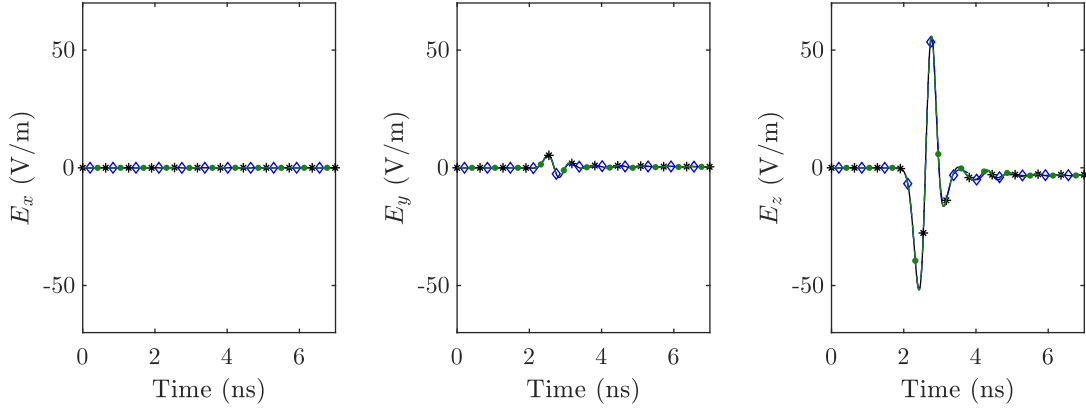


Figure 3.19: The electric field at the probe in material traverse test of Section 3.5.3 for Case 1 ($\text{---}\star\text{---}$), Case 2 ($\text{---}\diamond\text{---}$), and Case 3 ($\text{---}\bullet\text{---}$).

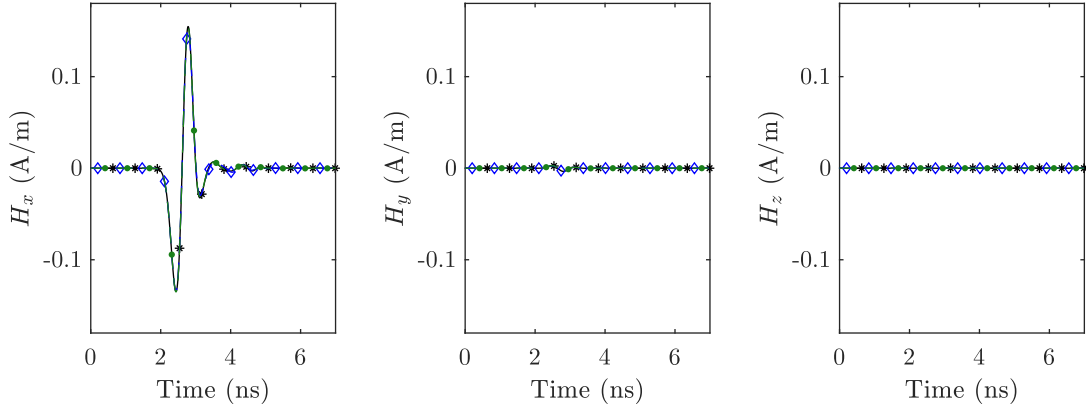


Figure 3.20: The magnetic field at the probe in material traverse test of Section 3.5.3 for Case 1 ($\text{---}\star\text{---}$), Case 2 ($\text{---}\diamond\text{---}$), and Case 3 ($\text{---}\bullet\text{---}$).

when the scalar and vector potential contributions are close to cancelling out.

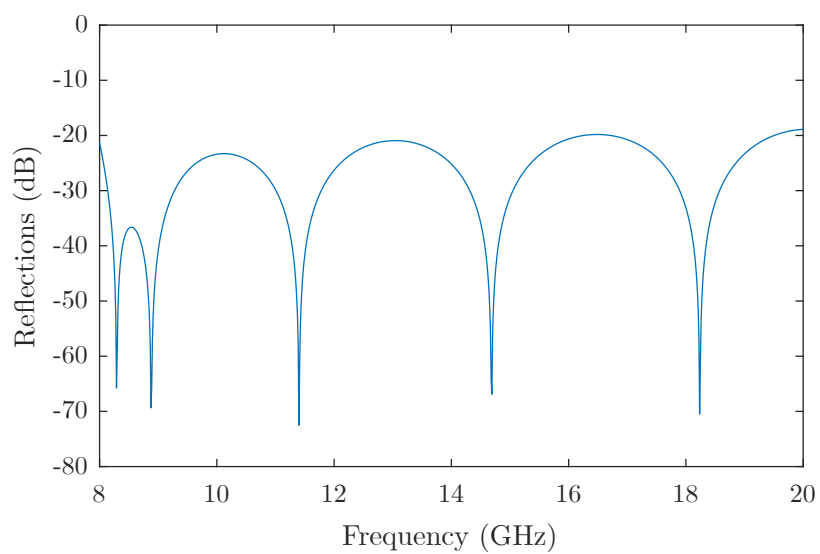


Figure 3.21: Relative reflected power due to the presence of subgridding region in the test of Section 3.5.4.

Probability and Energy Conservation Theory for the FDTD Scheme for Solving the Schrödinger Equation

This chapter proposes a theory of probability and energy conservation for the FDTD-Q scheme [36, 135], outlined in Section 1.5, for solving the Schrödinger equation. In the continuous domain, solutions of the Schrödinger equation respect the principle of conservation for both probability and energy. When one considers the entire space, the total probability of finding a particle must be constant and, with proper normalization, equal to one [2]. Similarly, the amount of energy (or more precisely the expectation value of energy) associated with the particle in a time-invariant potential must stay constant [2, 225, 226]. The probability and energy would also stay constant when one considers a region that is isolated from the surrounding space, for example an infinite potential well [2, 225]. In general, when the Schrödinger equation is discretized, the conservation properties are not guaranteed to be preserved, much like in the case of the Maxwell equations and other PDEs. This is problematic for two reasons. Firstly, if the total probability of finding the particle deviates from one over the course of the simulation, the computation of any observable can become skewed as the wavefunction loses the proper normalization. Secondly, generation of spurious probability or energy can cause instability of the algorithm.

In contrast to FDTD for the Maxwell equations, the connection between the conservation properties and stability has not been recognized for the case of the leap-frog FDTD-Q scheme for the Schrödinger equation considered in this chapter¹. Nevertheless, some relevant results can be found in [36] regarding the probability conservation properties of the scheme in one dimension. In particular, two approximations have been proposed for the probability density, which were argued, though without detailed proof, to conserve the total probability [36]. Regarding energy conservation, we are not aware of any work that studies this property specifically in the context of leap-frog FDTD-Q. However, investigations on the conservation of energy have been carried out for other numerical methods for the linear time-dependent Schrödinger equation [85, 136, 226, 229]. Approaches based on sym-

¹It should be noted that the conservation argument for stability has appeared in the literature on other time-domain numerical methods for the linear and nonlinear Schrödinger equations [226–228].

plectic integrators [230, 231] have been proposed for solving the Schrödinger equation [137, 165, 229, 232, 233]. Symplectic algorithms can be constructed to conserve energy [229, 230] by preserving the symplectic structure of the continuous equations. Symplectic integrators give rise to a wide class of methods with different temporal discretization, including the leap-frog approach [137, 230, 232]. However, to the best of our knowledge, they have not been applied to analyze the conservation properties of the FDTD-Q scheme considered in this chapter.

The works in the literature on conservation properties of numerical methods for the Schrödinger equation typically assume either zero [36] or periodic boundary conditions [36, 136, 229]. Their results cannot describe a scenario where multiple regions are coupled together and allowed to exchange energy, which is an essential part of being able to apply the probability and energy conservation principle to stability analysis of complex schemes. In this chapter, we derive the conditions under which a region discretized via FDTD-Q conserves probability and energy. Based on these results, we propose a method for creating schemes that conserve probability and energy by construction, which ensures stability and consistency with the fundamental physical principles. This chapter is a modified version of [234, 235].

4.1 Equations for the Region

We consider a region with $n_x \times n_y \times n_z$ primary cells shown in Figure 4.1, discretized using the FDTD-Q equations (1.38a)–(1.38b) repeated below [108, 135]

$$\begin{aligned} \hbar \frac{\psi_R|_{i,j,k}^{n+1} - \psi_R|_{i,j,k}^n}{\Delta t} = & -\frac{\hbar^2}{2m} \left(\frac{\psi_I|_{i+1,j,k}^{n+\frac{1}{2}} - 2\psi_I|_{i,j,k}^{n+\frac{1}{2}} + \psi_I|_{i-1,j,k}^{n+\frac{1}{2}}}{(\Delta x)^2} + \frac{\psi_I|_{i,j+1,k}^{n+\frac{1}{2}} - 2\psi_I|_{i,j,k}^{n+\frac{1}{2}} + \psi_I|_{i,j-1,k}^{n+\frac{1}{2}}}{(\Delta y)^2} \right. \\ & \left. + \frac{\psi_I|_{i,j,k+1}^{n+\frac{1}{2}} - 2\psi_I|_{i,j,k}^{n+\frac{1}{2}} + \psi_I|_{i,j,k-1}^{n+\frac{1}{2}}}{(\Delta z)^2} \right) + U|_{i,j,k} \psi_I|_{i,j,k}^{n+\frac{1}{2}}, \quad (4.1a) \end{aligned}$$

$$\begin{aligned} \hbar \frac{\psi_I|_{i,j,k}^{n+\frac{1}{2}} - \psi_I|_{i,j,k}^{n-\frac{1}{2}}}{\Delta t} = & \frac{\hbar^2}{2m} \left(\frac{\psi_R|_{i+1,j,k}^n - 2\psi_R|_{i,j,k}^n + \psi_R|_{i-1,j,k}^n}{(\Delta x)^2} + \frac{\psi_R|_{i,j+1,k}^n - 2\psi_R|_{i,j,k}^n + \psi_R|_{i,j-1,k}^n}{(\Delta y)^2} \right. \\ & \left. + \frac{\psi_R|_{i,j,k+1}^n - 2\psi_R|_{i,j,k}^n + \psi_R|_{i,j,k-1}^n}{(\Delta z)^2} \right) - U|_{i,j,k} \psi_R|_{i,j,k}^n. \quad (4.1b) \end{aligned}$$

Similarly to the earlier chapters, we begin by writing a self-contained model for the region with special update equations on the boundary involving hanging variables [155]. As will become clear in the subsequent discussion, the hanging variables will help quantify the rate at which the region exchanges the probability and energy with the surrounding space [193].

4.1.1 Equations at Each Node

With reference to Figure 4.1a, the equation for the real part of the wavefunction at the primary nodes strictly inside the region is given by

$$\begin{aligned} \hbar \Delta x \Delta y \Delta z \frac{\psi_R|_{i,j,k}^{n+1} - \psi_R|_{i,j,k}^n}{\Delta t} = & -\frac{\hbar^2}{2m} \left(-\Delta y \Delta z \frac{\psi_I|_{i,j,k}^{n+\frac{1}{2}} - \psi_I|_{i-1,j,k}^{n+\frac{1}{2}}}{\Delta x} + \Delta y \Delta z \frac{\psi_I|_{i+1,j,k}^{n+\frac{1}{2}} - \psi_I|_{i,j,k}^{n+\frac{1}{2}}}{\Delta x} \right. \\ & - \Delta x \Delta z \frac{\psi_I|_{i,j,k}^{n+\frac{1}{2}} - \psi_I|_{i,j-1,k}^{n+\frac{1}{2}}}{\Delta y} + \Delta x \Delta z \frac{\psi_I|_{i,j+1,k}^{n+\frac{1}{2}} - \psi_I|_{i,j,k}^{n+\frac{1}{2}}}{\Delta y} \\ & \left. - \Delta x \Delta y \frac{\psi_I|_{i,j,k}^{n+\frac{1}{2}} - \psi_I|_{i,j,k-1}^{n+\frac{1}{2}}}{\Delta z} + \Delta x \Delta y \frac{\psi_I|_{i,j,k+1}^{n+\frac{1}{2}} - \psi_I|_{i,j,k}^{n+\frac{1}{2}}}{\Delta z} \right) + \Delta x \Delta y \Delta z U|_{i,j,k} \psi_I|_{i,j,k}^{n+\frac{1}{2}}, \quad (4.2) \end{aligned}$$

which is obtained by multiplying (4.1a) on both sides by the volume of the secondary cell containing the primary node (i, j, k) , namely $\Delta x \Delta y \Delta z$. The terms in the brackets on the right hand side of (4.2) correspond to the discrete flux of $\nabla \psi_I$ through the surface of the secondary cell.

For the ψ_R samples on a single face of the boundary, the equation is written in a similar fashion as (3.13) with the use of hanging variables, which are introduced in order to avoid using samples beyond the boundary of the region. For instance, for the sample on the east face shown in Figure 4.1b, the equation reads

$$\begin{aligned} \hbar \frac{\Delta x}{2} \Delta y \Delta z \frac{\psi_R|_{n_x+1,j,k}^{n+1} - \psi_R|_{n_x+1,j,k}^n}{\Delta t} = & -\frac{\hbar^2}{2m} \left(-\Delta y \Delta z \frac{\psi_I|_{n_x+1,j,k}^{n+\frac{1}{2}} - \psi_I|_{n_x,j,k}^{n+\frac{1}{2}}}{\Delta x} + \Delta y \Delta z [\partial_x \psi_I]_{n_x+1,j,k}^{n+\frac{1}{2}} \right. \\ & - \frac{\Delta x}{2} \Delta z \frac{\psi_I|_{n_x+1,j,k}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,j-1,k}^{n+\frac{1}{2}}}{\Delta y} + \frac{\Delta x}{2} \Delta z \frac{\psi_I|_{n_x+1,j+1,k}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,j,k}^{n+\frac{1}{2}}}{\Delta y} \\ & - \frac{\Delta x}{2} \Delta y \frac{\psi_I|_{n_x+1,j,k}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,j,k-1}^{n+\frac{1}{2}}}{\Delta z} + \frac{\Delta x}{2} \Delta y \frac{\psi_I|_{n_x+1,j,k+1}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,j,k}^{n+\frac{1}{2}}}{\Delta z} \left. \right) \\ & + \frac{\Delta x}{2} \Delta y \Delta z U|_{n_x+1,j,k} \psi_I|_{n_x+1,j,k}^{n+\frac{1}{2}}, \quad (4.3) \end{aligned}$$

where $[\partial_x \psi_I]_{n_x+1,j,k}$ is the hanging variable placed on the boundary, representing the x component of the gradient of ψ_I on the normal primary edge shown in purple in Figure 4.1b. Similarly to (3.13), (4.3) is written over the secondary cell with dimensions $\Delta x/2 \times \Delta y \times \Delta z$.

For the samples on the bottom-east edge of the boundary, the equation is written similarly to

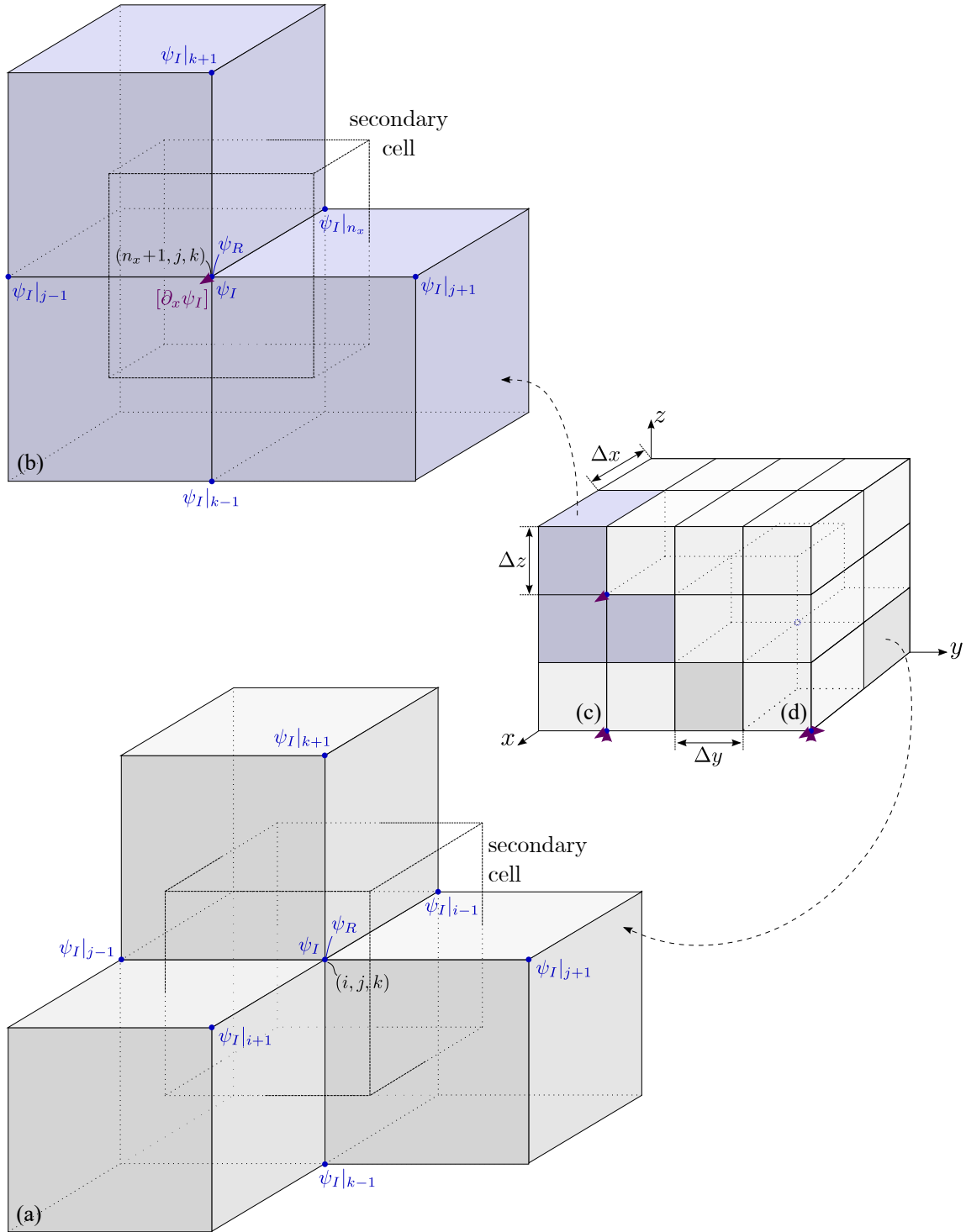


Figure 4.1: Samples involved in discretization of the Schrödinger equation for updating the real part of the wavefunction. The hanging variables are shown in purple. (a) illustrates equation (4.2). (b) illustrates (4.3). (c) labels the ψ_R sample and the hanging variables in (4.4), and (d) labels the corresponding quantities in (4.5). The common subscripts (i, j, k) and $(n_x + 1, j, k)$ are omitted.

(3.15), yielding

$$\begin{aligned}
\hbar \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\psi_R|_{n_x+1,j,1}^{n+1} - \psi_R|_{n_x+1,j,1}^n}{\Delta t} = \\
- \frac{\hbar^2}{2m} \left(- \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\psi_I|_{n_x+1,j,1}^{n+\frac{1}{2}} - \psi_I|_{n_x,j,1}^{n+\frac{1}{2}}}{\Delta x} + \frac{\Delta y}{2} \frac{\Delta z}{2} [\partial_x \psi_I]_{n_x+1,j,1}^{n+\frac{1}{2}} \right. \\
- \frac{\Delta x}{2} \frac{\Delta z}{2} \frac{\psi_I|_{n_x+1,j,1}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,j-1,1}^{n+\frac{1}{2}}}{\Delta y} + \frac{\Delta x}{2} \frac{\Delta z}{2} \frac{\psi_I|_{n_x+1,j+1,1}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,j,1}^{n+\frac{1}{2}}}{\Delta y} \\
\left. - \frac{\Delta x}{2} \frac{\Delta y}{2} [\partial_z \psi_I]_{n_x+1,j,1}^{n+\frac{1}{2}} + \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\psi_I|_{n_x+1,j,2}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,j,1}^{n+\frac{1}{2}}}{\Delta z} \right) \\
+ \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} U|_{n_x+1,j,1} \psi_I|_{n_x+1,j,1}^{n+\frac{1}{2}}. \quad (4.4)
\end{aligned}$$

Likewise, the equation for the bottom-north-east corner of the region is written similarly to (3.16)

$$\begin{aligned}
\hbar \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\psi_R|_{n_x+1,n_y+1,1}^{n+1} - \psi_R|_{n_x+1,n_y+1,1}^n}{\Delta t} = \\
- \frac{\hbar^2}{2m} \left(- \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\psi_I|_{n_x+1,n_y+1,1}^{n+\frac{1}{2}} - \psi_I|_{n_x,n_y+1,1}^{n+\frac{1}{2}}}{\Delta x} + \frac{\Delta y}{2} \frac{\Delta z}{2} [\partial_x \psi_I]_{n_x+1,n_y+1,1}^{n+\frac{1}{2}} \right. \\
- \frac{\Delta x}{2} \frac{\Delta z}{2} \frac{\psi_I|_{n_x+1,n_y+1,1}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,n_y,1}^{n+\frac{1}{2}}}{\Delta y} + \frac{\Delta x}{2} \frac{\Delta z}{2} [\partial_y \psi_I]_{n_x+1,n_y+1,1}^{n+\frac{1}{2}} \\
\left. - \frac{\Delta x}{2} \frac{\Delta y}{2} [\partial_z \psi_I]_{n_x+1,n_y+1,1}^{n+\frac{1}{2}} + \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\psi_I|_{n_x+1,n_y+1,2}^{n+\frac{1}{2}} - \psi_I|_{n_x+1,n_y+1,1}^{n+\frac{1}{2}}}{\Delta z} \right) \\
+ \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} U|_{n_x+1,n_y+1,1} \psi_I|_{n_x+1,n_y+1,1}^{n+\frac{1}{2}}. \quad (4.5)
\end{aligned}$$

Equations for the imaginary part of the wavefunction are written analogously to (4.2)–(4.5), using (4.1b) as a starting point.

4.1.2 Compact Matrix Form

The equations in Section 4.1.1, such as (4.2)–(4.5), are written in a compact matrix form as

$$\hbar \mathbf{\Lambda}_V'' \frac{\boldsymbol{\Psi}_R^{n+1} - \boldsymbol{\Psi}_R^n}{\Delta t} = - \frac{\hbar^2}{2m} \left(- \mathbf{D} \mathbf{\Lambda}_S'' (\mathbf{\Lambda}_l')^{-1} \mathbf{D}^T \boldsymbol{\Psi}_I^{n+\frac{1}{2}} + \mathbf{L}_\perp \mathbf{\Lambda}_{[\hat{n} \cdot]} \mathbf{\Lambda}_{S,\dagger}'' [\nabla \boldsymbol{\Psi}_I]_\perp^{n+\frac{1}{2}} \right) + \mathbf{\Lambda}_V'' \mathbf{\Lambda}_U \boldsymbol{\Psi}_I^{n+\frac{1}{2}} \quad (4.6)$$

and an analogous matrix form can be written for the imaginary part of the wavefunction. In (4.6), vectors $\boldsymbol{\Psi}_R$ and $\boldsymbol{\Psi}_I$ contain samples of ψ_R and ψ_I at the primary nodes and vector $[\nabla \boldsymbol{\Psi}_I]_\perp$ collects the hanging variables at the normal primary edges. Diagonal matrix $\mathbf{\Lambda}_U$ contains the values of the potential U at the primary nodes. The two terms in the brackets on the right hand side of (4.6) correspond to the discrete outward flux of $\nabla \psi_I$ through the boundary of each of the secondary cells. To see why this is the case, we can single out from the first term in the brackets the finite-difference approximation of $\nabla \psi_I$ at the conventional primary edges

$$[\nabla \boldsymbol{\Psi}_I]^{n+\frac{1}{2}} = -(\mathbf{\Lambda}_l')^{-1} \mathbf{D}^T \boldsymbol{\Psi}_I^{n+\frac{1}{2}}. \quad (4.7)$$

Likewise, the second term in the brackets involves the hanging variables, which are the samples of $\nabla\psi_I$ at the normal primary edges. Similarly to $[\nabla\psi_I]$ in (4.7), we also define a compact notation for the vector containing the finite difference approximation of $\nabla\psi_R$ at the conventional primary edges as

$$[\nabla\psi_R]^n = -(\Lambda'_l)^{-1}\mathbf{D}^T\psi_R^n, \quad (4.8)$$

which will be useful in the subsequent discussion.

Equation (4.6) and its counterpart for the imaginary part of the wavefunction can be written more compactly as

$$\hbar\Lambda_V''\frac{\psi_R^{n+1}-\psi_R^n}{\Delta t} = \mathbf{H}\psi_I^{n+\frac{1}{2}} - \mathbf{H}_\perp[\nabla\psi_I]_\perp^{n+\frac{1}{2}}, \quad \forall n = 0, 1, \dots, n_t - 1, \quad (4.9a)$$

$$\hbar\Lambda_V''\frac{\psi_I^{n+\frac{1}{2}}-\psi_I^{n-\frac{1}{2}}}{\Delta t} = -\mathbf{H}\psi_R^n + \mathbf{H}_\perp[\nabla\psi_R]_\perp^n, \quad \forall n = 0, 1, \dots, n_t - 1, \quad (4.9b)$$

where n_t is the number of time steps and $\mathbf{H}$ and $\mathbf{H}_\perp$ are given by

$$\mathbf{H} = \frac{\hbar^2}{2m}\mathbf{D}\Lambda_S''(\Lambda'_l)^{-1}\mathbf{D}^T + \Lambda_V''\Lambda_U, \quad (4.10)$$

$$\mathbf{H}_\perp = \frac{\hbar^2}{2m}\mathbf{L}_\perp\Lambda_{[\hat{n}]}''\Lambda_{S,\dagger}'' . \quad (4.11)$$

Equations (4.9a)–(4.9b) can also be written as a single matrix equation,

$$\hbar\mathcal{P}\frac{\psi^{n+1}-\psi^n}{\Delta t} = (\mathbf{J}_1 \otimes \mathbf{H})\frac{\psi^{n+1}+\psi^n}{2} - (\mathbf{J}_1 \otimes \mathbf{H}_\perp)[\nabla\psi]_\perp^{n+\frac{1}{2}}, \quad n = 0, 1, \dots, n_t - 1, \quad (4.12)$$

where

$$\psi^n = \begin{bmatrix} \psi_R^n \\ \psi_I^{n-\frac{1}{2}} \end{bmatrix}, \quad [\nabla\psi]_\perp^{n+\frac{1}{2}} = \begin{bmatrix} [\nabla\psi_R]_\perp^n \\ [\nabla\psi_I]_\perp^{n+\frac{1}{2}} \end{bmatrix}, \quad (4.13)$$

$$\mathcal{P} = \mathbf{I}_2 \otimes \Lambda_V'' - \frac{\Delta t}{2\hbar}|\mathbf{J}_1| \otimes \mathbf{H} = \begin{bmatrix} \Lambda_V'' & -\frac{\Delta t}{2\hbar}\mathbf{H} \\ -\frac{\Delta t}{2\hbar}\mathbf{H} & \Lambda_V'' \end{bmatrix}, \quad (4.14)$$

where $\mathbf{J}_m$ is a $2m \times 2m$ matrix of the form

$$\mathbf{J}_m = \begin{bmatrix} \mathbf{0} & \mathbf{I}_m \\ -\mathbf{I}_m & \mathbf{0} \end{bmatrix} \quad (4.15)$$

and brackets “ $|\cdot|$ ” in (4.14) denote an element-wise absolute value. Matrix $\mathcal{P}$ in (4.14) will lead to an expression for the probability of finding the particle in the region. Equation (4.12) can be seen as a discrete-time dynamical system that approximates the solution of the Schrödinger equation describing the evolution in time of the real and imaginary parts of the wavefunction.

4.2 Probability Conservation

In this section we propose expressions for the total probability in the region and for the probability current leaving the region through the boundary. We show that these expressions satisfy probability

conservation under a condition on Δt , which is recognized to be a generalized CFL limit. Furthermore, we prove that the conventional CFL limit (1.39) is a sufficient condition for the probability conservation.

4.2.1 Total Probability and Probability Current

In the continuous domain, the probability of finding a particle in a volume V is given by [2]

$$\mathcal{P}(t) = \iiint_V |\psi|^2 dV = \iiint_V (\psi_R^2 + \psi_I^2) dV. \quad (4.16)$$

The probability can leave the region through the boundary ∂V at the rate dictated by the outward probability current

$$\mathcal{I}_P(t) \approx \oint_{\partial V} \vec{J}_P(x, y, z, t) \cdot \hat{n} dS, \quad (4.17)$$

where $\hat{n}$ is the outward normal vector and $\vec{J}_P(x, y, z, t)$ is the probability current density, also known in the literature as the particle current density [2]

$$\vec{J}_P = \frac{i\hbar}{2m}(\psi \nabla \psi^* - \psi^* \nabla \psi) = \frac{\hbar}{m}(\psi_R \nabla \psi_I - \psi_I \nabla \psi_R). \quad (4.18)$$

In order to analyze the probability conservation properties of FDTD-Q, we define discrete expressions that approximate (4.16) and (4.17).

The most obvious approach to defining the probability associated with ψ^n in (4.13) would be to discretize the integral in (4.16) with the use of ψ_R^n and $\psi_I^{n-0.5}$ as

$$\begin{aligned} \mathcal{P}_{\text{simple}}^n = & \sum_{i=1}^{n_x+1} \sum_{j=1}^{n_y+1} \sum_{k=1}^{n_z+1} \Delta V''|_{i,j,k} \left((\psi_R^n|_{i,j,k})^2 + (\psi_I^{n-\frac{1}{2}}|_{i,j,k})^2 \right) = (\psi_R^n)^T \mathbf{\Lambda}_V'' \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{\Lambda}_V'' \psi_I^{n-\frac{1}{2}}, \end{aligned} \quad (4.19)$$

where $\Delta V''|_{i,j,k}$ is the volume of the secondary cell associated with the node (i, j, k) . However, due to the leap-frog nature of FDTD-Q updates, this simple expression does not respect the principle of probability conservation even for an isolated region, as will become evident from Section 4.5. Instead, we propose an expression for the total probability that involves the matrix $\mathcal{P}$, which appears in equation (4.12) describing the time evolution of the wavefunction.

Definition 2 (Total Probability). *The total probability of finding the particle in a region described by FDTD-Q equations (4.9a)–(4.9b) is given by*

$$\mathcal{P}^n = (\psi^n)^T \mathcal{P} \psi^n, \quad n = 0, 1, \dots, n_t. \quad (4.20)$$

This definition has a similar form to the expression of the storage function (2.47) in Chapter 2 and the discrete energy in the finite integration technique for the Maxwell equations [197].

To see the connection between (4.20) and (4.16), we expand (4.20) using the definitions of ψ^n

and $\mathcal{P}$ in (4.13) and (4.14) as

$$\begin{aligned} \mathcal{P}^n = (\psi_R^n)^T \Lambda_V'' \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \Lambda_V'' \psi_I^{n-\frac{1}{2}} - \frac{\Delta t}{\hbar} (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_R^n = \\ (\psi_R^n)^T \Lambda_V'' \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \left(\Lambda_V'' \psi_I^{n-\frac{1}{2}} - \frac{\Delta t}{\hbar} \mathbf{H} \psi_R^n \right), \end{aligned} \quad (4.21)$$

and with the use of (4.9b) obtain

$$\mathcal{P}^n = (\psi_R^n)^T \Lambda_V'' \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \Lambda_V'' \psi_I^{n+\frac{1}{2}} - \frac{\Delta t}{\hbar} (\psi_I^{n-\frac{1}{2}})^T \mathbf{H}_\perp [\nabla \psi_R]_\perp^n. \quad (4.22)$$

When Δt is small, the last term in (4.22) can be neglected and $\mathcal{P}^n$ can be approximated as

$$\mathcal{P}^n \approx (\psi_R^n)^T \Lambda_V'' \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \Lambda_V'' \psi_I^{n+\frac{1}{2}}, \quad (4.23)$$

which involves the samples of the real part of the wavefunction at time $t = n\Delta t$ and the product of the samples of the imaginary part at $(n - 0.5)\Delta t$ and $(n + 0.5)\Delta t$. Hence, $\mathcal{P}^n$ can be seen as an approximation of $\mathcal{P}(t)$ in (4.16) at $t = n\Delta t$. Alternatively, $\mathcal{P}^n$ can be interpreted as an approximation of $\mathcal{P}(t)$ performed at $t = (n - 0.5)\Delta t$. To see this, we can use (4.9a) instead of (4.9b) to write $\mathcal{P}^n$ as

$$\mathcal{P}^n = (\psi_R^n)^T \Lambda_V'' \psi_R^{n-1} + (\psi_I^{n-\frac{1}{2}})^T \Lambda_V'' \psi_I^{n-\frac{1}{2}} - \frac{\Delta t}{\hbar} (\psi_R^n)^T \mathbf{H}_\perp [\nabla \psi_I]_\perp^{n-\frac{1}{2}}, \quad (4.24)$$

to obtain

$$\mathcal{P}^n \approx (\psi_R^n)^T \Lambda_V'' \psi_R^{n-1} + (\psi_I^{n-\frac{1}{2}})^T \Lambda_V'' \psi_I^{n-\frac{1}{2}}. \quad (4.25)$$

The combination of staggered samples in (4.23) and (4.25) has been used by Visscher [36] to propose expressions for the probability density in 1D leap-frog FDTD-Q, where periodic or zero Dirichlet boundary conditions were assumed. Visscher argued that those expressions ensure that the total probability stays constant with time. For the case of zero Dirichlet or zero Neumann boundary conditions, the last term in (4.22) and the last term in (4.24) become zero and $\mathcal{P}^n$ reduces to (4.23) and (4.25), respectively, becoming analogous to either of the two expressions proposed in [36]. Hence, Definition 2 can be thought of as a generalization of (4.23) and (4.25) for a 3D region that could be either isolated or open to the flow of probability through the boundary. The proposed Definition 2 has an advantage of being a quadratic form, which will be important for the analysis of the conservation properties [193, 197].

Next, we propose an expression approximating (4.17) in FDTD-Q. This expression quantifies the probability current through the boundary of the region and, as will be shown in Section 4.2.2, respects the principle of probability conservation.

Definition 3 (Probability Current). *The probability current flowing out of a region described by FDTD-Q equations (4.9a)–(4.9b) is given by*

$$\mathcal{I}_P^{n+\frac{1}{2}} = \frac{2}{\hbar} \left(\frac{\psi^{n+1} + \psi^n}{2} \right)^T (\mathbf{J}_1 \otimes \mathbf{H}_\perp) [\nabla \psi]_\perp^{n+\frac{1}{2}}, \quad \forall n = 0, 1, \dots, n_t - 1. \quad (4.26)$$

In order to recognize that (4.26) approximates (4.17), we expand (4.26) using (4.11), (4.13), and

(4.15) to obtain

$$\mathcal{I}_P^{n+\frac{1}{2}} = \frac{\hbar}{m} \left(\frac{\psi_{R,\dagger}^{n+1} + \psi_{R,\dagger}^n}{2} \right)^T \mathbf{\Lambda}_{[\hat{n}]} \mathbf{\Lambda}_{S,\dagger}'' [\nabla \psi_I]_{\perp}^{n+\frac{1}{2}} - \frac{\hbar}{m} \left(\frac{\psi_{I,\dagger}^{n+\frac{1}{2}} + \psi_{I,\dagger}^{n-\frac{1}{2}}}{2} \right)^T \mathbf{\Lambda}_{[\hat{n}]} \mathbf{\Lambda}_{S,\dagger}'' [\nabla \psi_R]_{\perp}^n, \quad (4.27)$$

where

$$\psi_{R,\dagger}^n = \mathbf{L}_{\perp}^T \psi_R^n, \quad \psi_{I,\dagger}^{n+\frac{1}{2}} = \mathbf{L}_{\perp}^T \psi_I^{n+\frac{1}{2}} \quad (4.28)$$

are vectors collecting the samples of the real and imaginary parts of the wavefunction at the primary nodes on the boundary. This way, expressions of the form $(\psi_{R,\dagger})^T \mathbf{\Lambda}_{[\hat{n}]} \mathbf{\Lambda}_{S,\dagger}'' [\nabla \psi_I]_{\perp}$ or $(\psi_{I,\dagger})^T \mathbf{\Lambda}_{[\hat{n}]} \mathbf{\Lambda}_{S,\dagger}'' [\nabla \psi_R]_{\perp}$ are summations of terms representing the contribution of each boundary secondary face to the outward flux of $\psi_R \nabla \psi_I$ or $\psi_I \nabla \psi_R$, respectively, which appear in (4.18). Hence, the first term in (4.27) approximates the flux of $(\hbar/m) \psi_R \nabla \psi_I$ at $t = (n + 0.5)\Delta t$ and the second term approximates the flux of $-(\hbar/m) \psi_I \nabla \psi_R$ at $t = n\Delta t$, which are the fluxes involved in (4.17).

4.2.2 Probability Conservation

The following definition provides the conditions that constitute the principle of probability conservation for the FDTD-Q model of the region in Figure 4.1.

Definition 4. Consider the region in Figure 4.1 governed by the FDTD-Q equations (4.9a)–(4.9b). Let $\mathcal{P}^n$ and $\mathcal{I}^{n+0.5}$ be, respectively, the total probability and the probability current flowing out of the region. The model is said to conserve probability if $\mathcal{P}^n$ is bounded below by zero

$$\mathcal{P}^n \geq 0, \quad \forall n = 0, \dots, n_t \quad (4.29)$$

for any initial condition ψ^0 and any set of values of the hanging variables $[\nabla \psi]_{\perp}^{n+0.5}$, and if the rate of change of the total probability is equal to the rate at which the probability is entering the region through the boundary via the probability current $\mathcal{I}_P^{n+0.5}$

$$\frac{\mathcal{P}^{n+1} - \mathcal{P}^n}{\Delta t} = -\mathcal{I}_P^{n+\frac{1}{2}}, \quad \forall n = 0, 1, \dots, n_t - 1. \quad (4.30)$$

These conditions are analogous to the conditions for a lossless system in the context of dissipative systems theory in [205, 206] and in the context of FDTD for the Maxwell equations in Chapter 2. The following theorem provides a restriction on the time step in FDTD-Q which ensures that the proposed expressions for the total probability (4.20) and probability current (4.26) satisfy Definition 4.

Theorem 7. The region described by FDTD-Q equations (4.9a)–(4.9b), with the total probability (4.20) and the probability current (4.26) conserves probability if the time step is taken below the following generalized CFL limit

$$\Delta t < \Delta t_{CFL,gen} = \frac{2}{\rho(\mathbf{\Sigma})}, \quad (4.31)$$

where

$$\mathbf{\Sigma} = \frac{1}{\hbar} (\mathbf{\Lambda}_V'')^{-\frac{1}{2}} \mathbf{H} (\mathbf{\Lambda}_V'')^{-\frac{1}{2}}. \quad (4.32)$$

It should be noted that exceeding the generalized CFL limit (4.31) results in violation of probability conservation condition (4.29) for some initial conditions. The proof of the theorem is provided shortly, after elaborating on its meaning. When the generalized CFL condition (4.31) holds, the largest amount of probability that the region can supply to the surrounding space via $\mathcal{I}_P^{n+0.5}$ over the course of the simulation is equal to the probability stored in the region at the beginning of the simulation [205]. In contrast, when the time step exceeds the generalized CFL limit, there is no bound on how much probability can leave the region. Hence, violation of the generalized CFL condition allows the region to provide an infinite amount of spurious probability to the surrounding space. This behavior is unphysical and distorts the calculation of any observable. The matrices involved in the expression for the generalized CFL limit in (4.31) depend only on the cell dimensions and on the potential profile, similarly to the conventional CFL limit (1.39). Moreover, the following relation holds between the conventional and generalized CFL limits (1.39) and (4.31). The proof, in part, draws inspiration from the approach that was used in [197] in the context of electromagnetic energy.

Theorem 8. *Consider a region described by (4.9a)–(4.9b). Let Δt_{CFL} be the CFL limit in (1.39) with $|U|_{\max}$ being the largest absolute value of potential across all primary nodes in the region. Let $\Delta t_{CFL,gen}$ be the generalized CFL limit in (4.31). Then,*

$$\Delta t_{CFL} \leq \Delta t_{CFL,gen}. \quad (4.33)$$

Proof. See Appendix D.3. □

Hence, the CFL limit (1.39) can be used in place of (4.31) as a sufficient condition to ensure probability conservation. Theorems 7 and 8 give a new meaning to the CFL condition (1.39) in FDTD-Q, which so far has been only understood as a condition for stability of the scheme. Specifically, we recognize that the same condition ensures the conservation of probability in a general open region. Next, the proof of Theorem 7 is provided, starting with the following lemma.

Lemma 1. *Matrix $\mathcal{P}$ in (4.14) has the following property:*

$$\mathcal{P} \succ 0 \quad \Longleftrightarrow \quad \Delta t < \Delta t_{CFL,gen}. \quad (4.34)$$

Proof. Consider matrix $\mathcal{P}$ defined by (4.14). Using the properties of the Schur complement [213], the condition $\mathcal{P} \succ 0$ holds if and only if

$$\begin{cases} \Lambda_V'' \succ 0, \\ \Lambda_V'' - \left(\frac{\Delta t}{2\hbar}\right)^2 \mathbf{H}(\Lambda_V'')^{-1} \mathbf{H} \succ 0. \end{cases} \quad (4.35)$$

The first inequality in (4.35) holds for any time step. The second inequality can be simplified by writing the equivalent condition [96]

$$(\Lambda_V'')^{-\frac{1}{2}} \left(\Lambda_V'' - \left(\frac{\Delta t}{2\hbar}\right)^2 \mathbf{H}(\Lambda_V'')^{-1} \mathbf{H} \right) (\Lambda_V'')^{-\frac{1}{2}} \succ 0, \quad (4.36)$$

which reduces to

$$\mathbf{I} - \left(\frac{\Delta t}{2}\right)^2 \boldsymbol{\Sigma}^2 \succ 0, \quad (4.37)$$

where $\boldsymbol{\Sigma}$ is defined in (4.32). The matrix inequality (4.37) holds if and only if the time step satisfies

$$\Delta t < \frac{2}{\rho(\boldsymbol{\Sigma})}, \quad (4.38)$$

proving (4.34). $\square$

Proof of Theorem 7. From Definition 2 and Lemma 1, condition (4.29) holds if the time step is taken below the generalized CFL limit (4.31). Condition (4.30) is true by construction, which can be shown by expanding the left hand side using Definition 2

$$\frac{\mathcal{P}^{n+1} - \mathcal{P}^n}{\Delta t} = \frac{(\boldsymbol{\psi}^{n+1})^T \mathcal{P} \boldsymbol{\psi}^n - (\boldsymbol{\psi}^n)^T \mathcal{P} \boldsymbol{\psi}^n}{\Delta t} = 2 \frac{(\boldsymbol{\psi}^{n+1} + \boldsymbol{\psi}^n)^T}{2} \mathcal{P} \frac{\boldsymbol{\psi}^{n+1} - \boldsymbol{\psi}^n}{\Delta t} \quad (4.39)$$

and using (4.12) to obtain

$$\frac{\mathcal{P}^{n+1} - \mathcal{P}^n}{\Delta t} = \frac{2}{\hbar} \frac{(\boldsymbol{\psi}^{n+1} + \boldsymbol{\psi}^n)^T}{2} (\mathbf{J}_1 \otimes \mathbf{H}) \frac{\boldsymbol{\psi}^{n+1} - \boldsymbol{\psi}^n}{2} - \frac{2}{\hbar} \frac{(\boldsymbol{\psi}^{n+1} + \boldsymbol{\psi}^n)^T}{2} (\mathbf{J}_1 \otimes \mathbf{H}_\perp) [\nabla \boldsymbol{\psi}]_\perp^{n+\frac{1}{2}}. \quad (4.40)$$

The first term on the right hand side of (4.40) is zero, since $\mathbf{J}_1 \otimes \mathbf{H}$ is a skew-symmetric matrix. The second term equals $-\mathcal{I}_P^{n+0.5}$ in (4.26), which proves (4.30). $\square$

4.3 Energy Conservation

In this section, expressions are proposed for the total energy in the region and the power supplied through its boundary. Conditions are provided under which these expressions satisfy the principle of energy conservation. We find that energy conservation can be demonstrated under the generalized CFL limit (4.31) if the total probability (4.20) is bounded from above. It is further argued that the existence of the upper bound on probability is guaranteed as long as the model of the space outside the region conserves probability as well.

4.3.1 Total Energy and Supplied Power

The following expression can be used to describe the energy associated with the region

$$\mathcal{H}(t) = \iiint_V W(x, y, z, t) dV, \quad (4.41)$$

where W is the energy density² [225]

$$W(x, y, z, t) = \frac{\hbar^2}{2m} \nabla \boldsymbol{\psi}^* \cdot \nabla \boldsymbol{\psi} + U \boldsymbol{\psi}^* \boldsymbol{\psi} = \frac{\hbar^2}{2m} \nabla \boldsymbol{\psi}_R \cdot \nabla \boldsymbol{\psi}_R + \frac{\hbar^2}{2m} \nabla \boldsymbol{\psi}_I \cdot \nabla \boldsymbol{\psi}_I + U \boldsymbol{\psi}_R^2 + U \boldsymbol{\psi}_I^2. \quad (4.42)$$

²Different expressions can be chosen to represent the kinetic energy contribution W_{kin} to the energy density (4.42). In this work we choose $W_{\text{kin}}^{(1)} = \frac{\hbar^2}{2m} \nabla \boldsymbol{\psi}^* \cdot \nabla \boldsymbol{\psi}$, which appears in [225, 226, 236]. Expression $W_{\text{kin}}^{(2)} = -\frac{\hbar^2}{2m} \boldsymbol{\psi}^* \nabla^2 \boldsymbol{\psi}$ from [2, 236] is one possible alternative. When considering the entire space in (4.41), the two expressions $W_{\text{kin}}^{(1)}$ and $W_{\text{kin}}^{(2)}$ can be shown to give the same value of $\mathcal{H}(t)$ [236]. However, this is not the case when a finite region V is considered. Various expressions for W_{kin} , including $W_{\text{kin}}^{(1)}$ and $W_{\text{kin}}^{(2)}$, have been studied in [236].

The corresponding power entering the region through the boundary is given by

$$s(t) = \oint_S \vec{S}(x, y, z, t) \cdot (-\hat{n}) dS, \quad (4.43)$$

where $\vec{S}$ is the energy flux density [225]

$$\begin{aligned} \vec{S}(x, y, z, t) = \frac{i\hbar}{2m} & \left(\left(-\frac{\hbar^2}{2m} \nabla^2 \psi + U\psi \right) \nabla \psi^* - \left(-\frac{\hbar^2}{2m} \nabla^2 \psi + U\psi \right)^* \nabla \psi \right) = \\ & -\frac{\hbar^2}{m} \frac{\partial \psi_R}{\partial t} \nabla \psi_R - \frac{\hbar^2}{m} \frac{\partial \psi_I}{\partial t} \nabla \psi_I. \end{aligned} \quad (4.44)$$

The proposed definitions of the total energy and energy flux in an FDTD-Q region serve as discrete counterparts of (4.41) and (4.43).

Similarly to the case of probability, one could define the total energy in FDTD-Q by directly discretizing the volume integration and the gradient of the wavefunction in (4.41), arriving at

$$\begin{aligned} \mathcal{H}_{\text{simple}}^n = \frac{\hbar^2}{2m} & ([\nabla \psi_R]^n)^T \mathbf{\Lambda}_S'' \mathbf{\Lambda}_l' [\nabla \psi_R]^n + \frac{\hbar^2}{2m} ([\nabla \psi_I]^{n-\frac{1}{2}})^T \mathbf{\Lambda}_S'' \mathbf{\Lambda}_l' [\nabla \psi_I]^{n-\frac{1}{2}} \\ & + (\psi_R^n)^T \mathbf{\Lambda}_V'' \mathbf{\Lambda}_U \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{\Lambda}_V'' \mathbf{\Lambda}_U \psi_I^{n-\frac{1}{2}}, \end{aligned} \quad (4.45)$$

where $[\nabla \psi_R]^n$ and $[\nabla \psi_I]^{n-0.5}$ are defined in (4.8) and (4.7), respectively. The treatment of temporally staggered samples in (4.45) in computing energy can be found, for example, in [140]. Using (4.10), (4.45) can be written more compactly as

$$\mathcal{H}_{\text{simple}}^n = (\psi_R^n)^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}}. \quad (4.46)$$

Similarly to $\mathcal{P}_{\text{simple}}^n$, $\mathcal{H}_{\text{simple}}^n$ does not respect the energy conservation principle, as will be demonstrated in Section 4.5. To address this issue, we propose expressions for the total energy and the corresponding supplied power for which the energy conservation can be rigorously proven.

Definition 5 (Total Energy). *The total energy stored in a region described by FDTD-Q equations (4.9a)–(4.9b) is given by*

$$\begin{aligned} \mathcal{H}^n = (\psi_R^n)^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}} & + \Delta t \frac{(\psi_R^n - \psi_R^{n-1})^T}{\Delta t} (\hbar \mathbf{\Lambda}_V'') \frac{\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}}}{\Delta t}, \\ n = 1, 2, \dots, n_t - 1. \end{aligned} \quad (4.47)$$

The expression for $\mathcal{H}^n$ consists of $\mathcal{H}_{\text{simple}}^n$ and a term proportional to Δt , which vanishes when the time step approaches zero. In that respect, the expression (4.47) somewhat resembles the expressions for energy in FDTD for the Maxwell equations in Chapters 2 and 3 and in [197]. The last term in (4.47) is needed to ensure that the principle of energy conservation is respected, as will be shown in the subsequent discussion.

Following similar steps to Section 4.2.1, one can use (4.9a) to rewrite $\mathcal{H}^n$ as

$$\mathcal{H}^n = (\psi_R^n)^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n+\frac{1}{2}} - \Delta t \left(\mathbf{H}_\perp [\nabla \psi_I]_\perp^{n-\frac{1}{2}} \right)^T \frac{\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}}}{\Delta t}. \quad (4.48)$$

Assuming Δt is small,

$$\mathcal{H}^n \approx (\psi_R^n)^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n+\frac{1}{2}}, \quad (4.49)$$

which can be seen as an approximation of (4.41) at $t = n\Delta t$. Expression for the energy $\mathcal{H}^n$ in (4.47) can also be interpreted as an approximation of (4.41) at $t = (n - 0.5)\Delta t$ by rewriting (4.47) using (4.9b) as

$$\mathcal{H}^n = (\psi_R^{n-1})^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}} + \Delta t \frac{(\psi_R^n - \psi_R^{n-1})^T}{\Delta t} \mathbf{H}_\perp [\nabla \psi_R]_\perp^n \quad (4.50)$$

and neglecting the last term to obtain

$$\mathcal{H}^n \approx (\psi_R^{n-1})^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}}. \quad (4.51)$$

Definition 6 (Supplied Power). *The power supplied through the boundary to a region described by (4.9a)–(4.9b) is given by*

$$s^{n+\frac{1}{2}} = 2 \frac{(\psi_R^{n+1} - \psi_R^n)^T}{\Delta t} \mathbf{H}_\perp \frac{[\nabla \psi_R]_\perp^{n+1} + [\nabla \psi_R]_\perp^n}{2} + 2 \frac{(\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}})^T}{\Delta t} \mathbf{H}_\perp \frac{[\nabla \psi_I]_\perp^{n+\frac{1}{2}} + [\nabla \psi_I]_\perp^{n-\frac{1}{2}}}{2}, \quad n = 1, 2, \dots, n_t - 2. \quad (4.52)$$

In order to reveal the similarity between (4.52) and (4.43), we expand (4.52) using (4.11) and (4.28) to obtain

$$s^{n+\frac{1}{2}} = \frac{\hbar^2}{m} \left(\frac{\psi_{R,\dagger}^{n+1} - \psi_{R,\dagger}^n}{\Delta t} \right)^T \mathbf{\Lambda}_{[\hat{n}\cdot]}'' \mathbf{\Lambda}_{S,\dagger}'' \frac{[\nabla \psi_R]_\perp^{n+1} + [\nabla \psi_R]_\perp^n}{2} + \frac{\hbar^2}{m} \left(\frac{\psi_{I,\dagger}^{n+\frac{1}{2}} - \psi_{I,\dagger}^{n-\frac{1}{2}}}{\Delta t} \right)^T \mathbf{\Lambda}_{[\hat{n}\cdot]}'' \mathbf{\Lambda}_{S,\dagger}'' \frac{[\nabla \psi_I]_\perp^{n+\frac{1}{2}} + [\nabla \psi_I]_\perp^{n-\frac{1}{2}}}{2}. \quad (4.53)$$

The first term in (4.53) is an approximation of the part of (4.43) associated with ψ_R at $t = (n + 0.5)\Delta t$. The second term is an approximation of the part of (4.43) associated with ψ_I at $t = n\Delta t$.

4.3.2 Energy conservation

The energy conservation properties of (4.47) and (4.52) are verified in a similar way to probability conservation. In particular, using concepts from dissipative systems theory [205, 206], we investigate the conditions for which the energy is bounded from below and show that the rate of change of energy equals the power supplied to the region through the boundary.

Theorem 9. *Consider a region described by FDTD-Q equations (4.9a)–(4.9b). Assume that $\Delta t < \Delta t_{CFL,gen}$ in (4.31) and that the total probability (4.20) is bounded from above by some finite value $\mathcal{P}_{\max}$*

$$\mathcal{P}^n \leq \mathcal{P}_{\max}, \quad \forall n = 0, 1, \dots, n_t. \quad (4.54)$$

For this region, total energy (4.47) is bounded from below as follows

$$\mathcal{H}^n \geq \Delta x \Delta y \Delta z \frac{\mathcal{P}_{\max}}{\lambda_{\min}(\mathcal{P})} \left(\min \left(\min_{i,j,k} U|_{i,j,k}, 0 \right) - \frac{4\hbar}{\Delta t} \right), \quad \forall n = 1, 2, \dots, n_t - 1, \quad (4.55)$$

where $\lambda_{\min}$ denotes the smallest eigenvalue of a symmetric real matrix. Moreover, the total energy (4.47) and the supplied power (4.52) satisfy the following relation:

$$\frac{\mathcal{H}^{n+1} - \mathcal{H}^n}{\Delta t} = s^{n+\frac{1}{2}}, \quad \forall n = 1, 2, \dots, n_t - 2. \quad (4.56)$$

The proof of Theorem 9 will be provided shortly. Prior to that, a lemma is given which shows that (4.54) can be assumed if the model of the space outside the region obeys the principle of probability conservation.

Lemma 2. Consider the region in Figure 4.1 described by (4.9a)–(4.9b). Let the time step be taken below the generalized CFL limit (4.31). Let $\tilde{\mathcal{P}}^n$ be the probability associated with the space outside the region and let $\tilde{\mathcal{I}}_P^{n+0.5}$ be the probability current leaving that space. Let $\tilde{\mathcal{P}}^n$ and $\tilde{\mathcal{I}}_P^{n+0.5}$ satisfy the principle of probability conservation in Definition 4. Furthermore, assume that the probability current leaving the region equals the current entering the space surrounding the region:

$$\mathcal{I}_P^{n+\frac{1}{2}} = -\tilde{\mathcal{I}}_P^{n+\frac{1}{2}}, \quad \forall n = 0, 1, \dots, n_t - 1. \quad (4.57)$$

Then the probability $\mathcal{P}^n$ associated with the region in Figure 4.1 has a finite a priori upper bound

$$\mathcal{P}^n \leq \mathcal{P}_{\max}, \quad \forall n = 0, 1, \dots, n_t, \quad (4.58)$$

given by $\mathcal{P}_{\max} = \mathcal{P}^0 + \tilde{\mathcal{P}}^0$.

Proof. From Theorem 7, $\mathcal{P}^n$ and $\mathcal{I}_P^{n+0.5}$ satisfy (4.30). Likewise, (4.30) holds for the space surrounding the region. Adding (4.30) for the region and for the surrounding space and using (4.57),

$$\frac{(\mathcal{P}^{n+1} + \tilde{\mathcal{P}}^{n+1}) - (\mathcal{P}^n + \tilde{\mathcal{P}}^n)}{\Delta t} = -\mathcal{I}_P^{n+\frac{1}{2}} - \tilde{\mathcal{I}}_P^{n+\frac{1}{2}} = 0. \quad (4.59)$$

From (4.59),

$$\mathcal{P}^n + \tilde{\mathcal{P}}^n = \mathcal{P}^0 + \tilde{\mathcal{P}}^0. \quad (4.60)$$

Subtracting $\tilde{\mathcal{P}}^n$ from both sides of (4.60),

$$\mathcal{P}^n = \mathcal{P}^0 + \tilde{\mathcal{P}}^0 - \tilde{\mathcal{P}}^n, \quad (4.61)$$

and using (4.29) for $\tilde{\mathcal{P}}^n$, we conclude

$$\mathcal{P}^n \leq \mathcal{P}^0 + \tilde{\mathcal{P}}^0, \quad (4.62)$$

proving (4.58). $\square$

Typically, the simulation setup would be such that $\mathcal{P}_{\max}$ in Lemma 2 equals to one. A region terminated in zero Dirichlet or zero Neumann boundary conditions would constitute a trivial case of the lemma, with $\tilde{\mathcal{P}}^n = 0$ and $\tilde{\mathcal{I}}_P^{n+0.5} = 0$. Section 4.4.2 considers in more detail an example of a

setup in Lemma 2 consisting of two connected regions, as well as the boundary conditions at their interface that ensure (4.57). Next, the proof of Theorem 9 is given.

Proof of Theorem 9. The first part of the proof is to derive the lower bound (4.55). Under the generalized CFL limit (4.31), all eigenvalues of $\mathcal{P}$ are strictly positive (Lemma 1). This allows us to derive an upper bound on $\|\psi^n\|_2$, which will then be used to derive (4.55):

$$\lambda_{\min}(\mathcal{P})\|\psi^n\|_2^2 \leq (\psi^n)^T \mathcal{P} \psi^n \leq \mathcal{P}_{\max}, \quad (4.63)$$

which yields

$$\|\psi^n\|_2 \leq \sqrt{\frac{\mathcal{P}_{\max}}{\lambda_{\min}(\mathcal{P})}}. \quad (4.64)$$

The first two terms in (4.47) can be written using $\mathbf{H}$ in (4.10) and ψ^n in (4.13) as

$$\begin{aligned} (\psi_R^n)^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}} = \\ (\psi^n)^T \left(\mathbf{I}_2 \otimes \frac{\hbar^2}{2m} \mathbf{D} \Lambda_S'' (\Lambda_I')^{-1} \mathbf{D}^T \right) \psi^n + (\psi^n)^T (\mathbf{I}_2 \otimes \Lambda_V'' \Lambda_U) \psi^n. \end{aligned} \quad (4.65)$$

The first term on the right hand side of (4.65) is non-negative. The second term is bounded from below by zero if $\min_{i,j,k} U|_{i,j,k} \geq 0$ and by $\Delta x \Delta y \Delta z \min_{i,j,k} U|_{i,j,k} \|\psi^n\|_2^2$ otherwise. Thus, with the use of (4.64), the first two terms in (4.47) are bounded from below as

$$(\psi_R^n)^T \mathbf{H} \psi_R^n + (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}} \geq \Delta x \Delta y \Delta z \min_{i,j,k} \left(\min U|_{i,j,k}, 0 \right) \frac{\mathcal{P}_{\max}}{\lambda_{\min}(\mathcal{P})}. \quad (4.66)$$

Using the properties of matrix 2-norms [96] and the bound (4.64) on the norm of ψ^n , the last term in (4.47) can be shown to have the following lower bound:

$$\begin{aligned} \Delta t \frac{(\psi_R^n - \psi_R^{n-1})^T}{\Delta t} (\hbar \Lambda_V'') \frac{\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}}}{\Delta t} \\ \geq -\frac{\hbar}{\Delta t} \Delta x \Delta y \Delta z (\|\psi_R^n\|_2 + \|\psi_R^{n-1}\|_2) \left(\|\psi_I^{n+\frac{1}{2}}\|_2 + \|\psi_I^{n-\frac{1}{2}}\|_2 \right) \\ \geq -\Delta x \Delta y \Delta z \frac{4\hbar}{\Delta t} \frac{\mathcal{P}_{\max}}{\lambda_{\min}(\mathcal{P})}. \end{aligned} \quad (4.67)$$

Inequality (4.55) then follows from (4.66) and (4.67).

The second part of the proof is to show (4.56). For that, we use (4.47) to expand the left hand side of (4.56) as

$$\begin{aligned} \frac{\mathcal{H}^{n+1} - \mathcal{H}^n}{\Delta t} = \frac{(\psi_R^{n+1})^T \mathbf{H} \psi_R^{n+1} - (\psi_R^n)^T \mathbf{H} \psi_R^n}{\Delta t} + \frac{(\psi_I^{n+\frac{1}{2}})^T \mathbf{H} \psi_I^{n+\frac{1}{2}} - (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}}}{\Delta t} \\ + \frac{(\psi_R^{n+1} - \psi_R^n)^T}{\Delta t} (\hbar \Lambda_V'') \frac{\psi_I^{n+\frac{3}{2}} - \psi_I^{n+\frac{1}{2}}}{\Delta t} - \frac{(\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}})^T}{\Delta t} (\hbar \Lambda_V'') \frac{\psi_R^n - \psi_R^{n-1}}{\Delta t}, \end{aligned} \quad (4.68)$$

where the last term, being a scalar, has been written as its own transpose. The first two terms on

the right hand side of (4.68) can be written as

$$\frac{(\psi_R^{n+1})^T \mathbf{H} \psi_R^{n+1} - (\psi_R^n)^T \mathbf{H} \psi_R^n}{\Delta t} = \frac{(\psi_R^{n+1} - \psi_R^n)^T}{\Delta t} (\mathbf{H} \psi_R^{n+1} + \mathbf{H} \psi_R^n), \quad (4.69)$$

$$\frac{(\psi_I^{n+\frac{1}{2}})^T \mathbf{H} \psi_I^{n+\frac{1}{2}} - (\psi_I^{n-\frac{1}{2}})^T \mathbf{H} \psi_I^{n-\frac{1}{2}}}{\Delta t} = \frac{(\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}})^T}{\Delta t} (\mathbf{H} \psi_I^{n+\frac{1}{2}} + \mathbf{H} \psi_I^{n-\frac{1}{2}}). \quad (4.70)$$

With that, we can write (4.68) as

$$\begin{aligned} \frac{\mathcal{H}^{n+1} - \mathcal{H}^n}{\Delta t} &= \frac{(\psi_R^{n+1} - \psi_R^n)^T}{\Delta t} \left(\mathbf{H} \psi_R^n + \mathbf{H} \psi_R^{n+1} + \hbar \Lambda_V'' \frac{\psi_I^{n+\frac{3}{2}} - \psi_I^{n+\frac{1}{2}}}{\Delta t} \right) \\ &\quad + \frac{(\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}})^T}{\Delta t} \left(\mathbf{H} \psi_I^{n+\frac{1}{2}} + \mathbf{H} \psi_I^{n-\frac{1}{2}} - \hbar \Lambda_V'' \frac{\psi_R^n - \psi_R^{n-1}}{\Delta t} \right). \end{aligned} \quad (4.71)$$

Using (4.9a) and (4.9b), the left hand side of (4.56) can thus be written as

$$\begin{aligned} \frac{\mathcal{H}^{n+1} - \mathcal{H}^n}{\Delta t} &= \frac{(\psi_R^{n+1} - \psi_R^n)^T}{\Delta t} \left(-\hbar \Lambda_V'' \frac{\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}}}{\Delta t} + \mathbf{H}_\perp [\nabla \psi_R]_\perp^n + \mathbf{H}_\perp [\nabla \psi_R]_\perp^{n+1} \right) \\ &\quad + \frac{(\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}})^T}{\Delta t} \left(\hbar \Lambda_V'' \frac{\psi_R^{n+1} - \psi_R^n}{\Delta t} + \mathbf{H}_\perp [\nabla \psi_I]_\perp^{n+\frac{1}{2}} + \mathbf{H}_\perp [\nabla \psi_I]_\perp^{n-\frac{1}{2}} \right) \\ &= \frac{(\psi_R^{n+1} - \psi_R^n)^T}{\Delta t} (\mathbf{H}_\perp [\nabla \psi_R]_\perp^n + \mathbf{H}_\perp [\nabla \psi_R]_\perp^{n+1}) \\ &\quad + \frac{(\psi_I^{n+\frac{1}{2}} - \psi_I^{n-\frac{1}{2}})^T}{\Delta t} (\mathbf{H}_\perp [\nabla \psi_I]_\perp^{n+\frac{1}{2}} + \mathbf{H}_\perp [\nabla \psi_I]_\perp^{n-\frac{1}{2}}), \end{aligned} \quad (4.72)$$

which is equal to the right hand side of (4.56). □

4.4 A New Framework to Create FDTD-Q Schemes with Guaranteed Stability

The theory proposed in this chapter provides a rigorous way to construct new stable FDTD schemes for the Schrödinger equation in a modular and constructive fashion. The approach is based on ensuring that the models and the coupling between them respect the principle of probability conservation. Here, as a proof of concept, the application of the conservation approach is demonstrated using two examples: (1) a region isolated to the flow of probability current and (2) two regions that are coupled via boundary conditions. Application to more advanced scenarios is left for the future work.

4.4.1 Region with Zero Probability Current Through the Boundary

Lemma 3. *Consider a region described by FDTD-Q equations (4.9a)–(4.9b). Assume that $\Delta t < \Delta t_{\text{CFL,gen}}$ in (4.31) and that*

$$\mathcal{I}_P^{n+\frac{1}{2}} = 0, \quad \forall n = 0, 2, \dots, n_t - 1. \quad (4.73)$$

Then

$$\|\psi^n\|_2 \leq \sqrt{\kappa(\mathcal{P})} \|\psi^0\|_2, \quad \forall n = 0, 1, \dots, n_t, \quad (4.74)$$

where $\kappa(\mathcal{P})$ is the condition number of $\mathcal{P}$.

Examples of boundary conditions where no net probability current exists include zero Dirichlet ($\psi = 0$) and zero Neumann ($[\nabla\psi]_\perp = 0$) boundary conditions.

Proof of Lemma 3. Assume $\Delta t < \Delta t_{\text{CFL,gen}}$. From Theorem 7 and from (4.73), $\mathcal{P}^n$ stays constant over the course of the simulation, giving

$$(\psi^n)^T \mathcal{P} \psi^n = (\psi^0)^T \mathcal{P} \psi^0, \quad \forall n = 0, 1, \dots, n_t. \quad (4.75)$$

This equality allows relating the norm of the vector ψ^n to its norm at $n = 0$. Specifically,

$$\lambda_{\min}(\mathcal{P}) \|\psi^n\|_2^2 \leq (\psi^n)^T \mathcal{P} \psi^n \quad (4.76)$$

and

$$(\psi^0)^T \mathcal{P} \psi^0 \leq \lambda_{\max}(\mathcal{P}) \|\psi^0\|_2^2. \quad (4.77)$$

Combining (4.75), (4.76), and (4.77),

$$\lambda_{\min}(\mathcal{P}) \|\psi^n\|_2^2 \leq \lambda_{\max}(\mathcal{P}) \|\psi^0\|_2^2. \quad (4.78)$$

Since, from Lemma 1, $\lambda_{\max}(\mathcal{P})$ is strictly positive,

$$\|\psi^n\|_2^2 \leq \frac{\lambda_{\max}(\mathcal{P})}{\lambda_{\min}(\mathcal{P})} \|\psi^0\|_2^2. \quad (4.79)$$

The ratio of eigenvalues for a symmetric positive definite matrix is the condition number [96], which proves the statement of the lemma. $\square$

Hence, the 2-norm of the vector ψ^n , which contains the samples of the real and imaginary parts of the wavefunction, has a bound that is known prior to running the simulation. The existence of such bound guarantees stability. From Theorem 8, the conventional CFL limit Δt_{CFL} in (1.39) can be used in place of $\Delta t_{\text{CFL,gen}}$ in Lemma 3. Hence, Lemma 3 in conjunction with Theorem 8 provides an alternative proof for the previously known result [108, 140] on FDTD-Q stability under the CFL limit (1.39). Generalized stability limits that could be relaxed to simpler conditions such as (1.39) have been derived in the past using the iteration matrix approach for a method similar to FDTD-Q [26] and for FDTD for the Maxwell equations [35, 189].

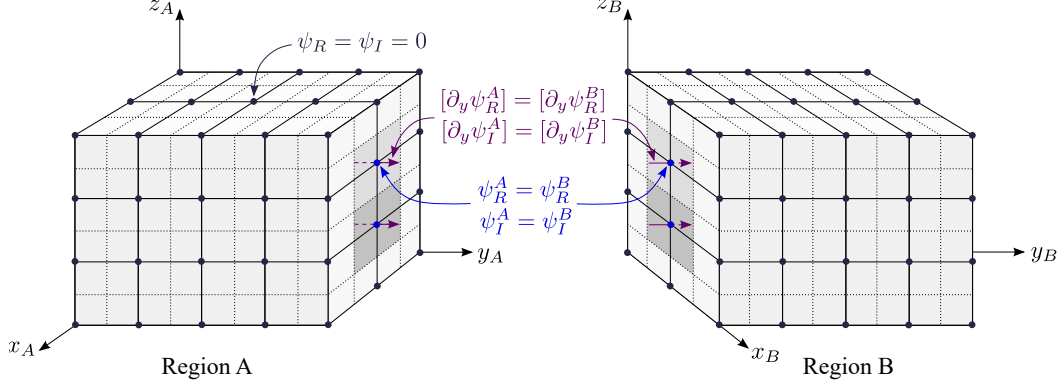


Figure 4.2: Two FDTD-Q regions joined together by equating the quantities on the common boundary as described in (4.80a)–(4.80d).

4.4.2 Connection of FDTD-Q Regions

This section discusses how the proposed theory can be used to couple FDTD-Q models in a stable manner. We consider a simple but representative scenario in Figure 4.2, involving two adjacent regions discretized using FDTD-Q with the same grid size and time step. The two regions need to be appropriately coupled at the interface to maintain probability conservation and hence stability of the scheme. The north face of the first region (Region A) is adjacent to the south face of the second region (Region B). To couple the two regions, we equate the samples of the wavefunction and the hanging variables at the interface between Region A and Region B

$$\psi_R^A|_{i,n_y^A+1,k}^n = \psi_R^B|_{i,1,k}^n, \quad 2 \leq i \leq n_x, \quad 2 \leq k \leq n_z, \quad (4.80a)$$

$$\psi_I^A|_{i,n_y^A+1,k}^{n-\frac{1}{2}} = \psi_I^B|_{i,1,k}^{n-\frac{1}{2}}, \quad 2 \leq i \leq n_x, \quad 2 \leq k \leq n_z, \quad (4.80b)$$

$$[\partial_y \psi_R^A]_{i,n_y^A+1,k}^n = [\partial_y \psi_R^B]_{i,1,k}^n, \quad 2 \leq i \leq n_x, \quad 2 \leq k \leq n_z, \quad (4.80c)$$

$$[\partial_y \psi_I^A]_{i,n_y^A+1,k}^{n+\frac{1}{2}} = [\partial_y \psi_I^B]_{i,1,k}^{n+\frac{1}{2}}, \quad 2 \leq i \leq n_x, \quad 2 \leq k \leq n_z, \quad (4.80d)$$

where “A” and “B” denote the region a quantity corresponds to. At all other nodes on the boundary of the two regions, zero Dirichlet boundary condition is imposed. Next, we derive the update equations resulting from (4.80a)–(4.80d) and show that the coupled scheme is stable if the generalized CFL limit (4.31) is satisfied in each of the regions.

Update Equations at the Interface

From Section 4.1, the equation for the real part of the wavefunction at the primary nodes on the north face of Region A is given by

$$\begin{aligned}
\hbar \Delta x \frac{\Delta y}{2} \Delta z \frac{\psi_R^A|_{i,n_y^A+1,k}^{n+1} - \psi_R^A|_{i,n_y^A+1,k}^n}{\Delta t} = \\
- \frac{\hbar^2}{2m} \left(- \frac{\Delta y}{2} \Delta z \frac{\psi_I^A|_{i,n_y^A+1,k}^{n+\frac{1}{2}} - \psi_I^A|_{i-1,n_y^A+1,k}^{n+\frac{1}{2}}}{\Delta x} + \frac{\Delta y}{2} \Delta z \frac{\psi_I^A|_{i+1,n_y^A+1,k}^{n+\frac{1}{2}} - \psi_I^A|_{i,n_y^A+1,k}^{n+\frac{1}{2}}}{\Delta x} \right. \\
- \Delta x \Delta z \frac{\psi_I^A|_{i,n_y^A+1,k}^{n+\frac{1}{2}} - \psi_I^A|_{i,n_y^A,k}^{n+\frac{1}{2}}}{\Delta y} + \Delta x \Delta z [\partial_y \psi_I^A]_{i,n_y^A+1,k}^{n+\frac{1}{2}} \\
\left. - \Delta x \frac{\Delta y}{2} \frac{\psi_I^A|_{i,n_y^A+1,k}^{n+\frac{1}{2}} - \psi_I^A|_{i,n_y^A+1,k-1}^{n+\frac{1}{2}}}{\Delta z} + \Delta x \frac{\Delta y}{2} \frac{\psi_I^A|_{i,n_y^A+1,k+1}^{n+\frac{1}{2}} - \psi_I^A|_{i,n_y^A+1,k}^{n+\frac{1}{2}}}{\Delta z} \right) \\
+ \Delta x \frac{\Delta y}{2} \Delta z U^A|_{i,n_y^A+1,k} \psi_I^A|_{i,n_y^A+1,k}^{n+\frac{1}{2}}. \quad (4.81)
\end{aligned}$$

The corresponding equation for the south face of Region B is

$$\begin{aligned}
\hbar \Delta x \frac{\Delta y}{2} \Delta z \frac{\psi_R^B|_{i,1,k}^{n+1} - \psi_R^B|_{i,1,k}^n}{\Delta t} = \\
- \frac{\hbar^2}{2m} \left(- \frac{\Delta y}{2} \Delta z \frac{\psi_I^B|_{i,1,k}^{n+\frac{1}{2}} - \psi_I^B|_{i-1,1,k}^{n+\frac{1}{2}}}{\Delta x} + \frac{\Delta y}{2} \Delta z \frac{\psi_I^B|_{i+1,1,k}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}}{\Delta x} \right. \\
- \Delta x \Delta z [\partial_y \psi_I^B]_{i,1,k}^{n+\frac{1}{2}} + \Delta x \Delta z \frac{\psi_I^B|_{i,2,k}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}}{\Delta y} \\
\left. - \Delta x \frac{\Delta y}{2} \frac{\psi_I^B|_{i,1,k}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k-1}^{n+\frac{1}{2}}}{\Delta z} + \Delta x \frac{\Delta y}{2} \frac{\psi_I^B|_{i,1,k+1}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}}{\Delta z} \right) \\
+ \Delta x \frac{\Delta y}{2} \Delta z U^B|_{i,1,k} \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}. \quad (4.82)
\end{aligned}$$

Adding (4.81) and (4.82) and using (4.80a), (4.80b), and (4.80d),

$$\begin{aligned}
\hbar \Delta x \Delta y \Delta z \frac{\psi_R^B|_{i,1,k}^{n+1} - \psi_R^B|_{i,1,k}^n}{\Delta t} = \\
- \frac{\hbar^2}{2m} \left(- \Delta y \Delta z \frac{\psi_I^B|_{i,1,k}^{n+\frac{1}{2}} - \psi_I^B|_{i-1,1,k}^{n+\frac{1}{2}}}{\Delta x} + \Delta y \Delta z \frac{\psi_I^B|_{i+1,1,k}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}}{\Delta x} \right. \\
- \Delta x \Delta z \frac{\psi_I^B|_{i,1,k}^{n+\frac{1}{2}} - \psi_I^A|_{i,n_y^A,k}^{n+\frac{1}{2}}}{\Delta y} + \Delta x \Delta z \frac{\psi_I^B|_{i,2,k}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}}{\Delta y} \\
\left. - \Delta x \Delta y \frac{\psi_I^B|_{i,1,k}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k-1}^{n+\frac{1}{2}}}{\Delta z} + \Delta x \Delta y \frac{\psi_I^B|_{i,1,k+1}^{n+\frac{1}{2}} - \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}}{\Delta z} \right) \\
+ \Delta x \Delta y \Delta z \frac{U^A|_{i,n_y^A+1,k} + U^B|_{i,1,k}}{2} \psi_I^B|_{i,1,k}^{n+\frac{1}{2}}, \quad (4.83)
\end{aligned}$$

which has the exact same form as the FDTD-Q equation for internal nodes (4.2), with the potential taken as the average of the two adjacent nodes³. The same can be shown for the equations corresponding to imaginary part of the wavefunction.

Stability Analysis

Assume that the time step satisfies the generalized CFL limit (4.31) for each region

$$\Delta t < \min(\Delta t_{\text{CFL, gen}}^A, \Delta t_{\text{CFL, gen}}^B). \quad (4.84)$$

From Theorem 7, the probability in each region evolves according to

$$\frac{\mathcal{P}_A^{n+1} - \mathcal{P}_A^n}{\Delta t} = -\mathcal{I}_{P,A}^{n+\frac{1}{2}}, \quad \frac{\mathcal{P}_B^{n+1} - \mathcal{P}_B^n}{\Delta t} = -\mathcal{I}_{P,B}^{n+\frac{1}{2}}. \quad (4.85)$$

Adding the two equations in (4.85),

$$\frac{(\mathcal{P}_A^{n+1} + \mathcal{P}_B^{n+1}) - (\mathcal{P}_A^n + \mathcal{P}_B^n)}{\Delta t} = -\mathcal{I}_{P,A}^{n+\frac{1}{2}} - \mathcal{I}_{P,B}^{n+\frac{1}{2}}. \quad (4.86)$$

As can be seen from (4.27), the conditions (4.80a)–(4.80d) equating the wavefunction and the hanging variables on the adjacent boundaries of the two regions ensure that the probability currents on the right hand side of (4.86) cancel out. Hence,

$$\mathcal{P}_A^n + \mathcal{P}_B^n = \mathcal{P}_A^0 + \mathcal{P}_B^0. \quad (4.87)$$

From Theorem 7, condition (4.84) implies that $\mathcal{P}_A^n$, $\mathcal{P}_B^n$ are both non-negative. Hence, from (4.87), $\mathcal{P}_A^n$, and $\mathcal{P}_B^n$ are each at most $\mathcal{P}_A^0 + \mathcal{P}_B^0$. Repeating the reasoning in the proof of Lemma 3,

$$\|\psi_A^n\|_2^2 \leq \frac{\mathcal{P}_A^0 + \mathcal{P}_B^0}{\lambda_{\min}(\mathcal{P})}, \quad \|\psi_B^n\|_2^2 \leq \frac{\mathcal{P}_A^0 + \mathcal{P}_B^0}{\lambda_{\min}(\mathcal{P})}. \quad (4.88)$$

This means that the system is stable, as the values of the wavefunction samples cannot grow without bound. Section 4.5.4 investigates the consequences of taking the time step beyond (4.84). In particular, we show that violating the generalized CFL limit in a region allows that region to provide infinite probability to the surrounding space and thus destabilize the simulation. Lastly, under condition (4.84), the coupled scheme can be shown to also conserve energy using similar arguments.

This example, although simple, shows how with the proposed theory one can create composite FDTD schemes obtained by coupling different models discretizing the Schrödinger equation. If each of the models satisfies the conservation of probability and the models are coupled in the probability-conserving manner, the resulting scheme will by construction satisfy the probability conservation. The resulting scheme will be stable under the most restrictive value of the generalized CFL limit (4.84), which is known a priori. The choice of coupling between the models, such as the boundary conditions (4.80a)–(4.80d), is essential for ensuring probability conservation and stability. In general,

³The derivation of (4.83) is analogous to the treatment of material interfaces in FDTD for electromagnetics [35]. It should also be noted that the similarity between (4.83) at the interface and (4.2) for internal nodes in FDTD-Q makes existing results on FDTD-Q stability [108, 140] applicable to this particular scenario and the specific boundary conditions (4.80a)–(4.80d) selected at the interface. However, as will be discussed next, the proposed approach could be applied to developing other schemes and this example serves as an illustration.

Table 4.1: Error of probability and energy expressions in the infinite well example in Section 4.5.1 ($a = 30$ nm, $\Delta t = 0.999\Delta t_{\text{CFL}}$, $n_t\Delta t = 28.76$ ps).

$n_x = n_y = n_z$	$\max_n \mathcal{P}^n - 1 $	$\max_n \mathcal{P}_{\text{simple}}^n - 1 $	$\min_n \mathcal{H}^n - E_1 /E_1$	$\max_n \mathcal{H}_{\text{simple}}^n - E_1 /E_1$
10	8.88×10^{-16}	2.51×10^{-2}	8.20×10^{-3}	3.19×10^{-2}
20	5.55×10^{-16}	6.19×10^{-3}	2.05×10^{-3}	8.15×10^{-3}
30	2.22×10^{-15}	2.74×10^{-3}	9.14×10^{-4}	3.64×10^{-3}
40	1.55×10^{-15}	1.54×10^{-3}	5.14×10^{-4}	2.05×10^{-3}
50	3.44×10^{-15}	9.87×10^{-4}	3.29×10^{-4}	1.31×10^{-3}

a different coupling scheme is not guaranteed to be probability-conserving.

The proposed approach allows for a systematic, modular, and constructive strategy to create stable composite FDTD schemes for the Schrödinger equation with a guarantee of probability and energy conservation. The method could be applied to developing more advanced schemes in the future. For example, subgridding scenarios [22, 194] could be analyzed as a connection of grids of different resolution, as was done in Chapters 2 and 3 for FDTD schemes for the Maxwell equations. The proposed conservation-based approach also provides an intuitive physical interpretation of the root cause of instability, attributing it to the generation of spurious energy or probability.

4.5 Numerical Examples

In order to investigate the validity of the results in Sections 4.2 and 4.3, the proposed method was implemented in MATLAB [215] and the time evolution of numerical probability and energy was investigated for different scenarios.

4.5.1 Infinite Well

In this test, the validity of the proposed theory is verified for an isolated region. In particular, we consider an electron trapped in a potential well with the potential U equal to zero inside the region and tending to infinity outside the region. The infinite potential well results in zero Dirichlet boundary conditions [2]. The region has a side length of $a = 30$ nm in each dimension.

The region was discretized into $n_x = n_y = n_z = 30$ primary cells. The CFL limit (1.39) and the generalized CFL limit (4.31) were both equal to 2.879 fs, with the relative difference (1.370×10^{-15}) comparable to machine precision. The time step Δt was taken as $0.999 \Delta t_{\text{CFL}}$. The initial conditions $\psi_R|_0$ and $\psi_I|^{-0.5}$ were set by sampling the following particular solution of the Schrödinger equation [2]

$$\psi(x, y, z, t) = A \sin(k_x x) \sin(k_y y) \sin(k_z z) \exp\left(-i\left(\frac{E_1}{\hbar}t + \frac{\pi}{3}\right)\right) \quad (4.89)$$

at the primary nodes. In (4.89),

$$k_x = k_y = k_z = \frac{\pi}{a}, \quad (4.90)$$

$$E_1 = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2), \quad (4.91)$$

and A is the real positive normalization constant ensuring that $\mathcal{P}^0$ in (4.20) is equal to 1. This choice of normalization will be discussed shortly.

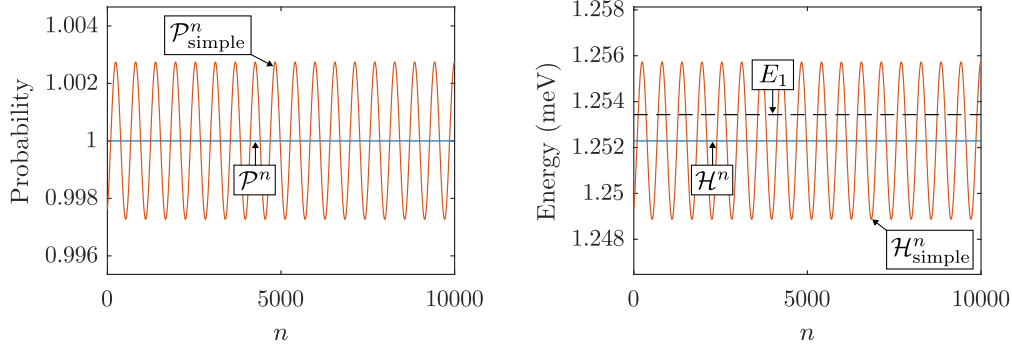


Figure 4.3: Infinite well example in Section 4.5.1 for $n_x = n_y = n_z = 30$: (a) the total probability $\mathcal{P}^n$ computed with the proposed formula (4.20) and $\mathcal{P}_{\text{simple}}^n$ computed with the simpler expression (4.19) (b) the total energy $\mathcal{H}^n$ in (4.47) and $\mathcal{H}_{\text{simple}}^n$ in (4.46). The dashed line shows the analytic value of energy (4.91).

Since the region is isolated from the flow of power and probability current, the energy and probability contained inside the region are constant in the continuous domain. From the blue curves in Figure 4.3, it is evident that the proposed expressions for probability and energy respect this property in the discrete domain, in accordance with Theorem 7 and Theorem 9. Indeed, the range of values of $\mathcal{P}^n$ and $\mathcal{H}^n$ was 3.997×10^{-15} and 3.228 aeV, which could be explained by finite machine precision. In contrast, the values of $\mathcal{P}_{\text{simple}}^n$ and $\mathcal{H}_{\text{simple}}^n$ shown in red in the figure exhibit some fluctuations. Moreover, the values of $\mathcal{P}_{\text{simple}}^n$ exceed 1 in some of the time intervals, which is inconsistent with physics.

The continuous expression for energy (4.41) can be shown to equal $E_1 = 1.2534$ meV, which is plotted in Figure 4.3b with the dashed line. The value of the proposed expression, $\mathcal{H}^n = 1.2523$ meV, is in good agreement with E_1 , corresponding to a relative error of 9.14×10^{-4} . Table 4.1 shows deviations of the values of different probability and energy expressions from the analytic solution. The values were obtained for different grid resolutions, while maintaining the infinite well geometry and keeping the ratio $\Delta t / \Delta t_{\text{CFL}}$ constant. As evident from the table, the discrepancy between $\mathcal{H}^n$ and E_1 can be reduced by refining the cell size and the corresponding time step. The error in $\mathcal{P}_{\text{simple}}^n$ and $\mathcal{H}_{\text{simple}}^n$ also reduces with improved spatial resolution. However, both $\mathcal{P}_{\text{simple}}^n$ and $\mathcal{H}_{\text{simple}}^n$ exhibit larger errors than the proposed expressions.

Hence the proposed expressions of probability and energy provide a more accurate approximation of the analytical quantities and respect the principle of probability and energy conservation, in contrast to $\mathcal{P}_{\text{simple}}^n$ and $\mathcal{H}_{\text{simple}}^n$. The proposed expression of the total probability is very convenient for computing the normalization constant A in (4.89), since the value of $\mathcal{P}^n$ stays constant in an isolated region.

4.5.2 Reflection from a Potential Barrier

In this test the scenario in Figure 4.4 is considered, where a Gaussian wavepacket impinges on a potential barrier. The expression for the incident pulse is [2]

$$\psi_{\text{inc}}(x, t) = \sum_p A_p e^{-i(\omega_p t - k_p(x - x_0))}, \quad (4.92)$$

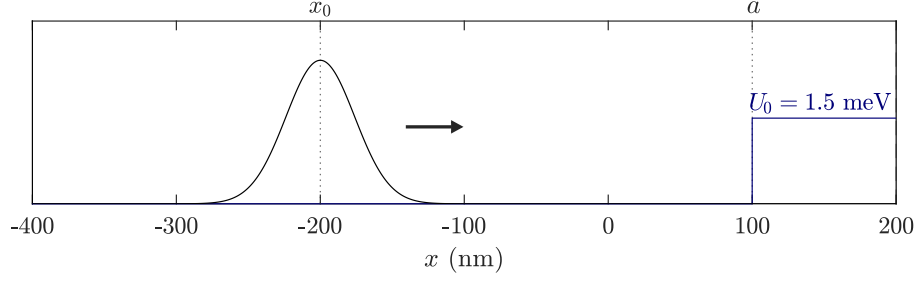


Figure 4.4: Scenario considered in Section 4.5.2, where a Gaussian wavepacket impinges on a potential barrier.

where p is an integer index, x_0 is the center of the wavepacket at $t = 0$, k_p are evenly spaced real scalars, ω_p are the corresponding angular frequencies given by $\hbar k_p^2/(2m)$, and the coefficients A_p are given by

$$A_p = \exp\left(-\frac{1}{4}\left(\frac{k_p - \bar{k}}{\sigma_k}\right)^2\right), \quad (4.93)$$

where $\bar{k}$ determines the center of the wavepacket in the k -space and σ_k determines its width. The wavepacket impinges on a barrier with potential given by

$$U(x) = \begin{cases} 0, & x < a, \\ \frac{U_0}{2}, & x = a, \\ U_0, & x > a. \end{cases} \quad (4.94)$$

The solution can be found in quantum mechanics textbooks [2] and is given by

$$\psi(x, t) = \begin{cases} \psi_{\text{inc}}(x, t) + \psi_{\text{ref}}(x, t), & x < a, \\ \psi_{\text{tran}}(x, t) & x > a, \end{cases} \quad (4.95)$$

where ψ_{ref} and ψ_{tran} are reflected and transmitted waves

$$\psi_{\text{ref}}(x, t) = \sum_p A_p R_p e^{-i(\omega_p t - k_p(2a - x_0 - x))}, \quad (4.96)$$

$$\psi_{\text{tran}}(x, t) = \sum_p A_p T_p e^{-i(\omega_p t - k_p(a - x_0) - K_p(x - a))}, \quad (4.97)$$

where $K_p = \sqrt{2m(\hbar\omega_p - U_0)}/\hbar$, which can be real or imaginary, depending on the sign of $\hbar\omega_p - U_0$. When K_p is real, the corresponding wave in ψ_{tran} propagates forward in the $x > a$ region. When K_p is imaginary, the wave decays with x and hence cannot propagate. The reflection and transmission coefficients are given by

$$R_p = \frac{k_p - K_p}{k_p + K_p}, \quad T_p = \frac{2k_p}{k_p + K_p}. \quad (4.98)$$

In this test, m is the mass of an electron, $x_0 = -200$ nm, $\bar{k} = 2\pi/\bar{\lambda}$, where $\bar{\lambda}$ is 30 nm, and $\sigma_k = \bar{k}/10$. The values of k_p range between $k_{\min} = \bar{k} - 10\sigma_k$ and $k_{\max} = \bar{k} + 10\sigma_k$, with a spacing of $\Delta k = \sigma_k/100$. The height of the potential barrier is $U_0 = 1.5$ meV and $a = 100$ nm.

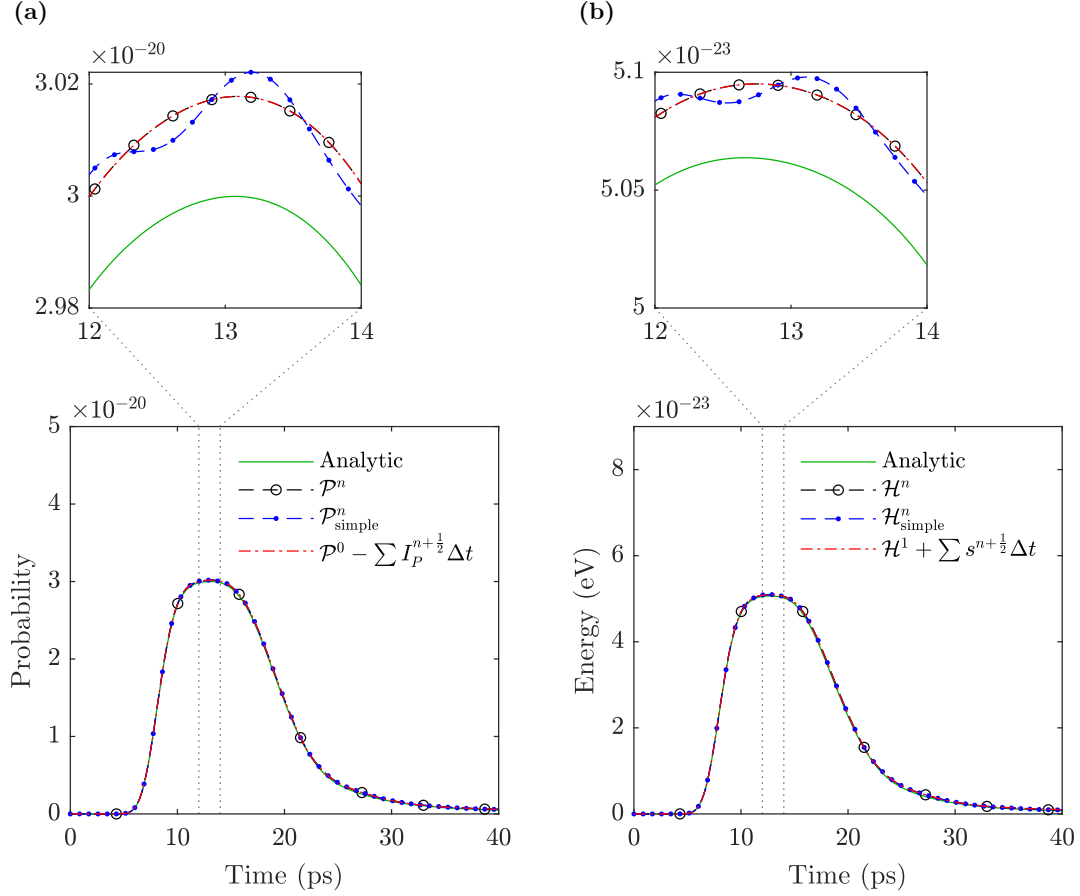


Figure 4.5: Results of the test in Section 4.5.2: (a) total probability in the region (b) total energy associated with the region.

We select the space from $x = 0$ to $x = 200$ nm to be modeled with the proposed method. The dimensions of the region were selected as 2 nm in the y and z dimensions. The region was discretized into cells of size $\Delta x = \Delta y = \Delta z = 1$ nm. The initial conditions in the region were set by sampling the analytical solution (4.95). The update equations on the boundary were the modified FDTD-Q equations described in Section 4.1.1, such as (4.3)–(4.5), which involved the hanging variables. In order to obtain the values of the hanging variables, expressions were found for the normal component of the gradient of the analytical solution (4.95). The expressions were evaluated at integer time points $n\Delta t$ for the real part and at $(n+0.5)\Delta t$ for the imaginary part. The values of the hanging variables were zero on all faces of the boundary except for the east and west, where their values were uniform in the y and z direction. This, in essence, rendered the problem one-dimensional, which was the reason why choosing the y and z dimensions to be small (2 nm) was possible. The generalized CFL limit in (4.31) was $\Delta t_{\text{CFL,gen}} = 2.869968$ fs. The CFL limit Δt_{CFL} in (1.39) was slightly lower (2.869915 fs), which is in agreement with the theory. The simulation time step was set to $0.999\Delta t_{\text{CFL}}$.

Figure 4.5a shows the probability of finding the particle in the region obtained via analytical computation, via the proposed expression $\mathcal{P}^n$ in (4.20), and the simple expression $\mathcal{P}_{\text{simple}}^n$ in (4.19). The analytic values were obtained from (4.16), applied to the analytical solution (4.95). The integration

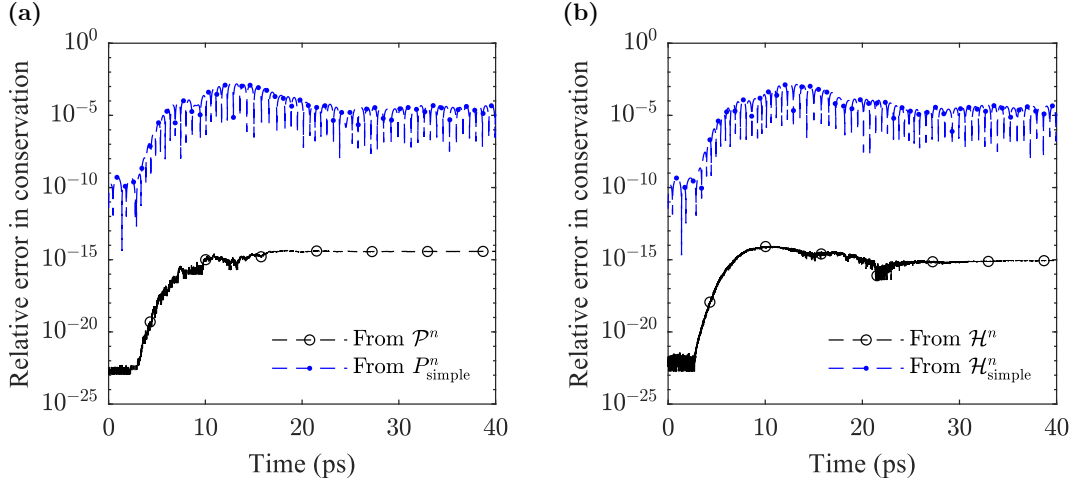


Figure 4.6: Results of the test in Section 4.5.2: (a) relative error in (4.99) (b) relative error in (4.100).

in (4.16) was approximated via a Riemann sum using the midpoint rule with 1000 intervals between $x = 0$ nm and $x = 200$ nm. When using the analytical solution as a reference, the accuracy of $\mathcal{P}^n$ and $\mathcal{P}_{\text{simple}}^n$ was comparable. The maximum relative errors were 8.827×10^{-3} for $\mathcal{P}^n$, 8.978×10^{-3} for $\mathcal{P}_{\text{simple}}^n$. Figure 4.5b shows the energy stored in the region computed using $\mathcal{H}^n$ and $\mathcal{H}_{\text{simple}}^n$, as well as the analytic values approximated via the Riemann sum. The accuracy of $\mathcal{H}^n$ and $\mathcal{H}_{\text{simple}}^n$ was also comparable, with a maximum relative error associated with $\mathcal{H}^n$ of 1.188×10^{-2} and the error associated with $\mathcal{H}_{\text{simple}}^n$ of 1.197×10^{-2} .

As predicted by Theorem 7, the total probability $\mathcal{P}^n$ was non-negative, with the smallest value of 3.184×10^{-27} . The smallest value of energy $\mathcal{H}^n$ was 4.812×10^{-30} eV, which was higher than the bound of -6.193×10^{-17} eV predicted by Theorem 9 when taking $\mathcal{P}_{\text{max}}$ in (4.55) to be the highest value of $\mathcal{P}^n$ over the course of the simulation.

Based on Theorem 7, the values of the proposed expression $\mathcal{P}^n$ must satisfy

$$\mathcal{P}^n = \mathcal{P}^0 - \sum_{n'=0}^{n-1} \mathcal{I}_P^{n'+\frac{1}{2}} \Delta t, \quad (4.99)$$

where the right hand side is the sum of the initial probability and the probability that has entered the region due to the flow of the probability current. The equality in (4.99) is evident from Figure 4.5a, showing both the left and right hand sides of (4.99). The relative error between the two sides of (4.99) is plotted in Figure 4.6a. The largest relative error was 4.514×10^{-15} , which is comparable to the machine precision and hence corroborates the prediction. The relative error in Figure 4.6a was normalized to the largest value of the theoretical probability over all time steps $\max \{\mathcal{P}(n\Delta t)\} = 3.000 \times 10^{-20}$. In contrast to the proposed expression for the total probability, when using $\mathcal{P}_{\text{simple}}^n$ in place of $\mathcal{P}^n$ in (4.99), the left and right hand sides of the relation are no longer equal, as can be seen from Figure 4.6a. The maximum value of the relative error was 1.519×10^{-3} , which is much larger than the error in (4.99) associated with $\mathcal{P}^n$. Hence, at least when $\mathcal{I}_P^{n+0.5}$ is used as the representation of the probability current, $\mathcal{P}_{\text{simple}}^n$ does not possess the conservation properties exhibited by $\mathcal{P}^n$. These errors in conservation when using the simpler expression $\mathcal{P}_{\text{simple}}^n$

also manifested themselves in the small fluctuations visible in the inset of Figure 4.5a.

Similarly, from Figure 4.5b and Figure 4.6b one can observe that the following relation is satisfied by the proposed expressions for the total energy and supplied power:

$$\mathcal{H}^n = \mathcal{H}^1 + \sum_{n'=1}^{n-1} s^{n'+\frac{1}{2}} \Delta t, \quad (4.100)$$

with the largest relative error of 9.111×10^{-15} , which is comparable to machine precision. In contrast, replacing $\mathcal{H}^n$ with $\mathcal{H}_{\text{simple}}^n$ results in the relative error of 1.428×10^{-3} between the left and right hand sides of (4.100). The normalization constant for the relative error in Figure 4.6b was $\max\{\mathcal{H}(n\Delta t)\} = 5.064 \times 10^{-23}$ eV. Small fluctuations that were exhibited by $\mathcal{P}_{\text{simple}}^n$ can also be observed for $\mathcal{H}_{\text{simple}}^n$ in the inset of Figure 4.5b. Hence, the results confirm that both $\mathcal{P}^n$ and $\mathcal{H}^n$ satisfy (4.99) and (4.100) and thus (4.30) and (4.56), respectively. This property makes the proposed expressions preferable to $\mathcal{P}_{\text{simple}}^n$ and $\mathcal{H}_{\text{simple}}^n$.

4.5.3 Harmonic Oscillator

In this test, we carry out similar analysis to Section 4.5.2 for a three-dimensional quantum harmonic oscillator scenario that was studied in [26, 38]. The scenario was originally chosen in [38] to resemble a quantum dot [237]. The potential profile for the harmonic oscillator is given by [26]

$$U(x, y, z) = \frac{1}{2} m_{\text{eff}} \omega_0^2 (x^2 + y^2 + z^2), \quad (4.101)$$

where ω_0 is the angular frequency and m_{eff} is the effective mass of the particle [31], which takes on the role of the mass m in the Schrödinger equation (1.20) when modeling behavior of particles in crystal lattices. With this potential profile, a superposition of solutions known as *coherent state* mimics the behaviour of a classical harmonic oscillator and is given by [2, 26]

$$\psi(x, y, z, t) = f(x, t) f(y, t) f(z, t), \quad (4.102)$$

where

$$f(v, t) = \sum_{p=0}^{p_{\text{max}}} C_{Np} \exp\left(-i\left(p + \frac{1}{2}\right)\omega_0 t\right) \psi_p\left(\sqrt{\frac{m_{\text{eff}}\omega_0}{\hbar}}v\right), \quad (4.103)$$

where p is the mode number corresponding to each direction, and p_{max} tends to infinity. The coefficients C_{Np} are given by [2]

$$C_{Np} = \sqrt{\frac{N^p \exp(-N)}{p!}}, \quad (4.104)$$

where N is a real scalar related to the energy of the oscillating particle as [2]

$$E_{\text{total}} = 3\left(N + \frac{1}{2}\right)\hbar\omega_0. \quad (4.105)$$

As in [26, 38], ω_0 is chosen as 1.984×10^{15} rad/s and m_{eff} as 0.023 of the mass of an electron. We select N as 2 and consider the modes up to $p_{\text{max}} = 16$.

Five different simulations were performed. The difference between the simulations was the region

Table 4.2: Normalized error with respect to the analytical solution for the test in Section 4.5.3.

L	Err. for $\mathcal{P}^n$	Err. for $\mathcal{P}_{\text{simple}}^n$	Norm. err. for $\mathcal{H}^n$	Norm. err. for $\mathcal{H}_{\text{simple}}^n$
2.253 nm	4.921×10^{-4}	4.911×10^{-4}	7.494×10^{-4}	7.518×10^{-4}
4.505 nm	9.101×10^{-4}	9.112×10^{-4}	1.527×10^{-3}	1.531×10^{-3}
6.758 nm	1.449×10^{-3}	1.446×10^{-3}	1.698×10^{-3}	1.701×10^{-3}
9.010 nm	5.656×10^{-3}	5.655×10^{-3}	6.547×10^{-3}	6.545×10^{-3}
11.263 nm	3.667×10^{-3}	3.665×10^{-3}	5.189×10^{-3}	5.187×10^{-3}

of space chosen to be modelled via FDTD-Q. In all simulations the region was a cube of side length L around the minimum point of the potential (4.101), where L was taken on the values of 2.253 nm, 4.505 nm, 6.758 nm, 9.010 nm, and 11.263 nm. The initial conditions and hanging variables were determined similarly to the way it was done in Section 4.5.2. The spatial discretization was chosen as $\Delta x = \Delta y = \Delta z = 0.5631 \text{ \AA}$.

Figures 4.7a and 4.7b show the probability and energy in the region for $L = 2.253 \text{ nm}$ and $L = 11.263 \text{ nm}$, computed using $\mathcal{P}^n$ in (4.20), $\mathcal{P}_{\text{simple}}^n$ in (4.19), the analytical expression for the probability in (4.16), $\mathcal{H}^n$ in (4.47), $\mathcal{H}_{\text{simple}}^n$ in (4.46), and the analytical expression for the energy in (4.41). Good agreement with the analytical solution is observed for both simple and proposed methods. For the case of $L = 11.263 \text{ nm}$, most of the probability density is located within the region as the particle oscillates. For the smaller region with $L = 2.253 \text{ nm}$, most of the probability associated with the particle is outside the region at the beginning of the simulation. The particle then moves into the region, although less than half of the total probability is encompassed by the region.

Table 4.2 shows the maximum relative error of $\mathcal{P}^n$, $\mathcal{P}_{\text{simple}}^n$, $\mathcal{H}^n$, and $\mathcal{H}_{\text{simple}}^n$ with respect to their continuous counterparts (4.16) and (4.41). The error in energy was normalized by E_{total} in (4.105), which was equal to 9.794 eV. The two methods (simple and proposed), had similar errors. In order to further compare the accuracy of these methods, we plot the normalized errors corresponding to Figures 4.7a and 4.7b over the course of the simulation. It should be noted that the dominant contribution to the errors varies with time. At the beginning of the simulation, the errors are primarily caused by the approximation of probability and energy, since the wavefunction has not deviated significantly from the analytical solution. In contrast, as the simulation progresses, the error in the wavefunction increases, contributing significantly to the errors in the computed probability and energy. The resulting normalized errors are shown in Figures 4.7c and 4.7d. Consistently with the results of Table 4.2, the errors were comparable for the two approaches of computing the probability and energy. However, near the beginning of the simulation with $L = 11.263 \text{ nm}$ in Figure 4.7d, the errors of the proposed expressions were significantly lower than those of the simple expressions. For the case of $L = 2.253 \text{ nm}$ in Figure 4.7c, the errors near the beginning of the simulation were very low for both methods, since most of the probability and energy associated with the region was close to zero at that time.

Similarly to Section 4.5.2, we compute the error associated with the probability and energy conservation relations (4.99) and (4.100), normalizing the error in energy by E_{total} . The results are summarized in Table 4.3. Consistently with the previous results, the proposed expressions obey the probability and energy conservation relations within the limits of the machine precision, which is not the case for the simple expressions.

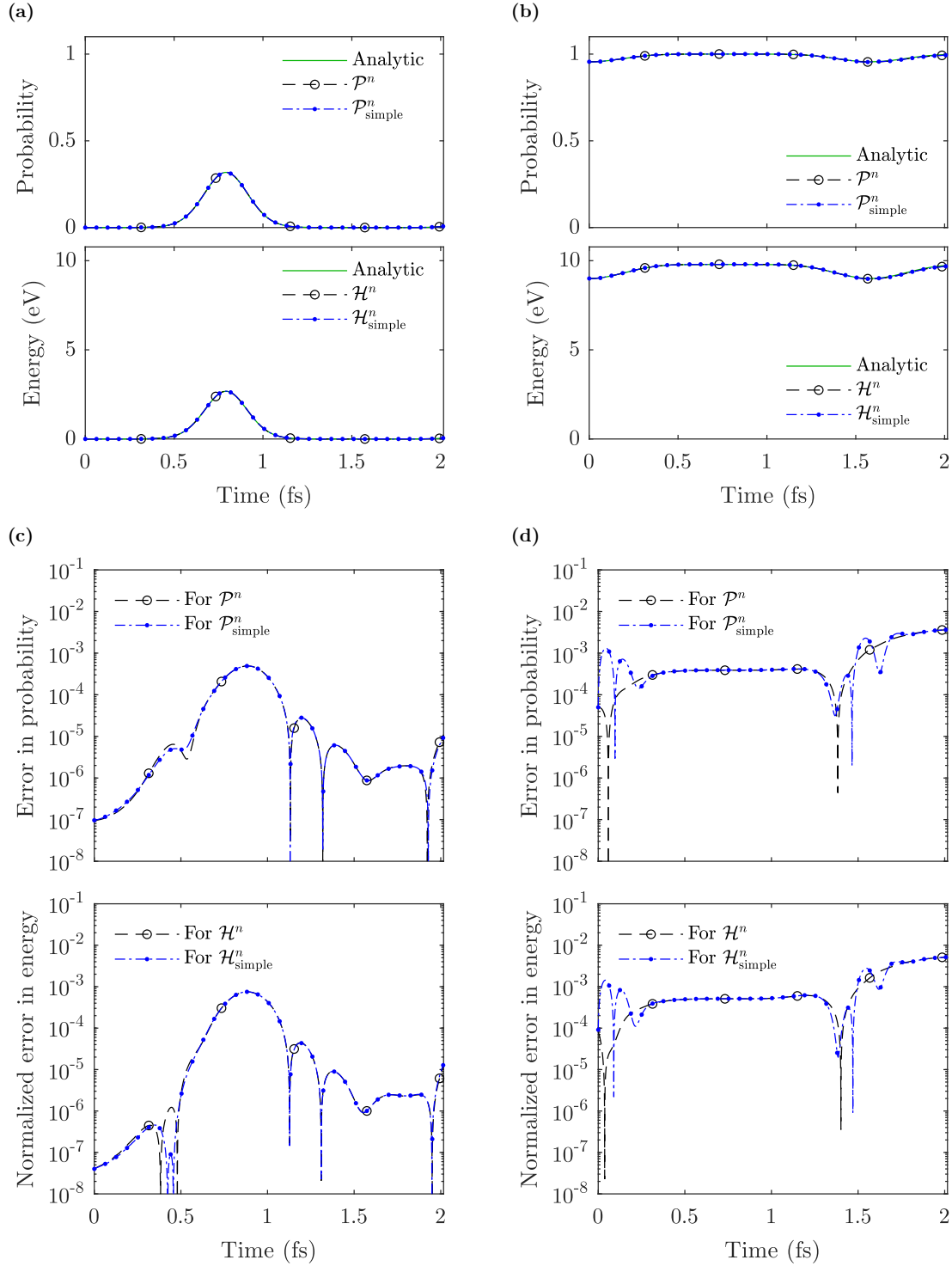


Figure 4.7: Results of the test in Section 4.5.3: (a) total probability and total energy in the region for $L = 2.253$ nm (b) the probability and energy for $L = 11.263$ nm (c) normalized error with respect to analytical solution for $L = 2.253$ nm (d) normalized error with respect to analytical solution for $L = 11.263$ nm.

Table 4.3: Normalized error in probability and energy conservation for the test in Section 4.5.3.

L	Err. for $\mathcal{P}^n$	Err. for $\mathcal{P}_{\text{simple}}^n$	Norm. err. for $\mathcal{H}^n$	Norm. err. for $\mathcal{H}_{\text{simple}}^n$
2.253 nm	1.619×10^{-15}	3.929×10^{-6}	6.045×10^{-15}	3.984×10^{-6}
4.505 nm	1.082×10^{-14}	8.199×10^{-6}	1.718×10^{-14}	1.316×10^{-5}
6.758 nm	7.216×10^{-15}	1.587×10^{-4}	1.399×10^{-14}	1.188×10^{-4}
9.010 nm	1.166×10^{-14}	7.969×10^{-4}	1.043×10^{-14}	8.286×10^{-4}
11.263 nm	6.317×10^{-14}	1.301×10^{-3}	2.381×10^{-14}	1.457×10^{-3}

4.5.4 Proton Tunneling

In this section we consider the scenario illustrated in Figure 4.8, which was taken from [61] and can be used as a simplified model for hydrogen transfer reactions. The problem consists of three regions: reactant (r), barrier (b), and product (p), and a proton that can transfer between the reactant and product regions via tunneling. The size of the reactant, barrier, and product regions is $l_x^R \times l_y \times l_z$, $l_x^B \times l_y \times l_z$, and $l_x^P \times l_y \times l_z$, respectively. The barrier region has a potential of U_0 and the other two regions have the potential equal to zero. Dirichlet zero boundary conditions are imposed on the external boundaries of the regions. On the interfaces, the continuity of the wavefunction and of the normal component of its gradient is imposed. Using separation of variables, the analytical solution in each region can be found as [61]

$$\psi^{\text{R,B,P}}(x, y, z, t) = \sum_{m=1}^N M_m f_m^{\text{R,B,P}}(x) g_m(y) h_m(z) \exp\left(-i \frac{E_m}{\hbar} t\right),$$

$$0 \leq x \leq l_x^{\text{R,B,P}}, \quad 0 \leq y \leq l_y, \quad 0 \leq z \leq l_z, \quad (4.106)$$

where N is the number of eigenfunctions considered. The expression for $f_m^{\text{R,B,P}}(x)$ in each region is

$$f_m^{\text{R}}(x) = A_m \sin(k_{xm} x), \quad (4.107a)$$

$$f_m^{\text{B}}(x) = B_m \cosh\left(K_{xm} \left(x - \frac{l_x^{\text{B}}}{2}\right)\right) + C_m \sinh\left(K_{xm} \left(x - \frac{l_x^{\text{B}}}{2}\right)\right), \quad (4.107b)$$

$$f_m^{\text{P}}(x) = D_m \sin(k_{xm} (x - l_x^{\text{P}})), \quad (4.107c)$$

where the choice between $B_m = 0$ and $C_m = 0$ determines the symmetry of the eigenfunction. The coefficients A_m , B_m , C_m , and D_m are chosen to ensure that the wavefunction is continuous across the region interfaces and that

$$\int_{\text{R}} |f_m^{\text{R}}(x)|^2 dx + \int_{\text{B}} |f_m^{\text{B}}(x)|^2 dx + \int_{\text{P}} |f_m^{\text{P}}(x)|^2 dx = 1. \quad (4.108)$$

The expressions for $g_m(y)$ and $h_m(z)$ are given by

$$g_m(y) = \sqrt{\frac{2}{l_y}} \sin(k_{ym} y), \quad h_m(z) = \sqrt{\frac{2}{l_z}} \sin(k_{zm} z). \quad (4.109)$$

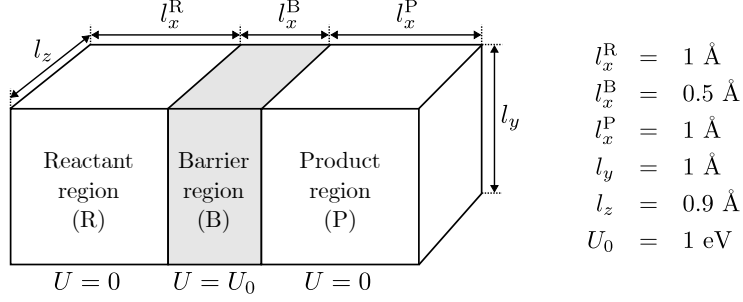


Figure 4.8: The scenario from [61] considered in Section 4.5.4, modeling proton tunneling through a potential barrier with $U = U_0$.

The values of k_{ym} and k_{zm} are such that the zero Dirichlet boundary conditions are satisfied for $y = l_y$ and $z = l_z$. The values of k_{xm} and K_{xm} are related through

$$E_{xm} = \frac{\hbar^2 k_{xm}^2}{2m_P} = U_0 - \frac{\hbar^2 K_{xm}^2}{2m_P}, \quad (4.110)$$

where $m_P = 1$ dalton. The value of E_{xm} can be obtained by imposing the continuity of the x derivative of the wavefunction across the region interfaces. The energy E_m in (4.106) corresponding to each eigenfunction is given by

$$E_m = E_{xm} + E_{ym} + E_{zm}, \quad (4.111)$$

where

$$E_{ym} = \frac{\hbar^2 k_{ym}^2}{2m_P}, \quad E_{zm} = \frac{\hbar^2 k_{zm}^2}{2m_P}. \quad (4.112)$$

Following [61], the coefficients M_m in (4.106) are assumed to have the following dependence on the temperature T :

$$M_m = \frac{M'_m}{\sqrt{\sum_{m=1}^N |M'_m|^2}}, \quad (4.113)$$

where

$$M'_m = \exp\left(-\frac{E_m}{2Tk_B} + i\delta_m\right), \quad (4.114)$$

k_B is the Boltzmann constant, and δ_m are phases that can be chosen arbitrarily.

The temperature and the number of eigenfunctions were taken from [61] as $T = 298$ K and $N = 8$, respectively. The dimensions of the regions, also from [61], are shown in Figure 4.8. Unlike [61], we only consider a particular solution with the phases δ_m in Table 4.4, as opposed to an ensemble of tunneling systems with many sets of randomized phases δ_m . In [61], the model was further extended to include the effect of thermal vibrations.

We use this scenario to examine the conservation of probability and energy when multiple FDTD-Q models are connected and to study the transfer of these quantities from one region's model to another. Three FDTD-Q models were defined: one discretizing the reactant region, another discretizing the barrier, and the third discretizing the product region. The theoretical treatment of this scenario is described in Section 4.4.2 for two regions and an extension to three regions is

Table 4.4: Description of the solution modes considered in the scenario in Section 4.5.4 (See [61] for Details).

m	δ_m	E_{xm} (meV)	Symmetry of $f_m^B(x)$	k_{ym}	k_{zm}
1	0.15π	18.858713602402805	$C_m = 0$	π/l_y	π/l_z
2	0.95π	18.858842481348997	$B_m = 0$	π/l_y	π/l_z
3	0.25π	75.369931622707597	$C_m = 0$	π/l_y	π/l_z
4	1.1π	75.370616971460279	$B_m = 0$	π/l_y	π/l_z
5	0	E_{x1}	$C_m = 0$	$2\pi/l_y$	π/l_z
6	1.3π	E_{x2}	$B_m = 0$	$2\pi/l_y$	π/l_z
7	0	E_{x1}	$C_m = 0$	π/l_y	$2\pi/l_z$
8	0.7π	E_{x2}	$B_m = 0$	π/l_y	$2\pi/l_z$

Table 4.5: CFL limit (1.39) and generalized CFL limit (4.31) in each region in the test of Section 4.5.4.

Region	Reactant	Barrier	Product
$\Delta t_{\text{CFL,gen}}$ (as)	58.318879645754564	55.844895610996367	58.318879645754564
Δt_{CFL} (as)	58.318879645754819	55.844895610996282	58.318879645754819

analogous. The cell dimensions in each region were $\Delta x = \Delta y = \Delta z = (1/30) \text{ \AA}$. The three FDTD-Q models were coupled by equating the wavefunction values and hanging variables on the interfaces, as described in Section 4.4.2. The CFL limits and the generalized CFL limits in each region are shown in Table 4.5. Their values were the same, apart from round-off error⁴. The simulation time step was taken as 0.999 of the CFL limit of the barrier region ($\Delta t \approx 55.79 \text{ as}$), which automatically satisfied the CFL limit condition in the reactant and product regions. This time step ensures stability, according to equation 26 in [140] or based on the discussion in Section 4.4.2. The initial conditions were set by sampling the analytical solution (4.106) and normalizing the result to achieve the total probability in the three regions $\mathcal{P}_R^0 + \mathcal{P}_B^0 + \mathcal{P}_P^0 = 1$.

One simulation run was performed to study the first 1 ps of the particle's evolution and a longer simulation was done to observe the behavior of the system over 35 ns. In the longer simulation, probability and energy were computed once in ten time steps ($10\Delta t = 0.5579 \text{ fs}$). This allowed reducing the memory usage while providing sufficient information, considering that the shortest period of a mode in (4.106) was 29.26 fs. The results are shown in Figure 4.9. During the first 1 ps, the simulated results match well with the analytical prediction. For the 35 ns simulation, the results exhibit some deviation from the analytical solution. However, this is expected due to dispersion errors caused by the finite discretization of the Schrödinger equation. It should be noted that the duration of the 35 ns simulation was several orders of magnitude longer than the period of each mode in (4.106), making it challenging to manage dispersion⁵. The vast disparity of the temporal scales present in the problem is evident from Figure 4.10, showing the energy in the reactant region with the time axis plotted on a logarithmic scale. The energy oscillates rapidly with a small amplitude around a value that gradually changes over a much larger period of time. As can be seen from Figure 4.10, the simulated values of energy faithfully reproduce the analytical solution for many oscillations. However, the dispersion errors eventually cause the wavefunction to deviate significantly

⁴This is consistent with the results in Table D.1 in Appendix D.3, where it was found that the conventional CFL limit of a region with constant non-negative potential equals to its generalized CFL limit.

⁵The periods of the modes in Table 4.4 ranged from 29.3 fs to 63.7 fs. Hence, the results in Figure 4.9 and Figure 4.10 for 35 ns represent hundreds of thousands of oscillations.

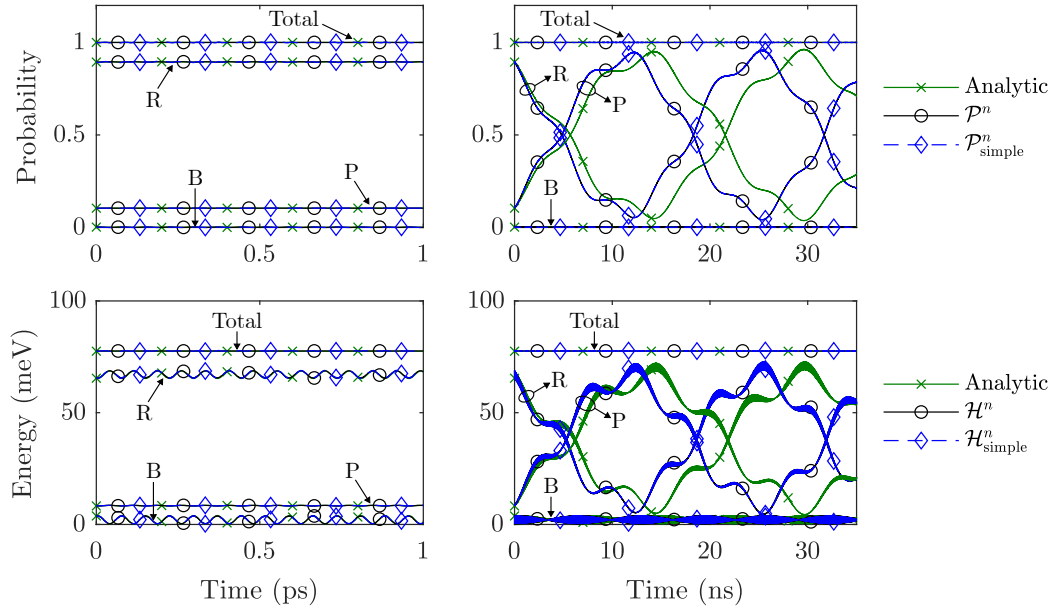


Figure 4.9: Energy and probability computed in the test of Section 4.5.4. Left panels: 1 ps simulation. Right panels: 35 ns simulation. The green lines show analytic values of probability (4.16) and energy (4.41) in each region (reactant, product, and barrier), as well as the total probability and energy in the three regions. The black lines show the values of the proposed expressions (4.20) and (4.47), namely $\mathcal{P}_R^n$, $\mathcal{P}_B^n$, $\mathcal{P}_P^n$, and $\mathcal{P}_R^n + \mathcal{P}_B^n + \mathcal{P}_P^n$ in the two top panels and $\mathcal{H}_R^n$, $\mathcal{H}_B^n$, $\mathcal{H}_P^n$, and $\mathcal{H}_R^n + \mathcal{H}_B^n + \mathcal{H}_P^n$ in the two bottom panels. The blue lines show the corresponding quantities evaluated using the simple expressions (4.19) and (4.46).

from the analytical value, which in turn causes the visible deviations in Figure 4.9 and Figure 4.10. Nevertheless, the overall trend of the solution is correctly reproduced by the simulation. Based on Figure 4.9 and Figure 4.10, the accuracy of $\mathcal{P}_{\text{simple}}^n$ and $\mathcal{H}_{\text{simple}}^n$ for each region was comparable to that of $\mathcal{P}^n$ and $\mathcal{H}^n$.

From Theorem 9, using 1 as $\mathcal{P}_{\text{max}}$, the energy $\mathcal{H}^n$ cannot take on values below -1.608 keV for the reactant or product regions and below -54.10 keV for the barrier region. The values of energy in each region were positive, which is in agreement with these lower bounds. Probability in each region was also positive, which is consistent with Theorem 7.

Based on the discussion in Section 4.4.2, the sum of the total probabilities over the three regions must be equal to its initial value. Similarly, one can show an analogous result for the sum of the total energies. In order to assess these predictions, we compute the relative deviation of the probability and energy from their initial values as

$$|(\mathcal{P}_R^n + \mathcal{P}_B^n + \mathcal{P}_P^n) - (\mathcal{P}_R^0 + \mathcal{P}_B^0 + \mathcal{P}_P^0)|, \quad n = 0, 10, 20, \dots, \quad (4.115)$$

$$\frac{1}{\mathcal{H}(n\Delta t)} |(\mathcal{H}_R^n + \mathcal{H}_B^n + \mathcal{H}_P^n) - (\mathcal{H}_R^1 + \mathcal{H}_B^1 + \mathcal{H}_P^1)|, \quad n = 1, 11, 21, \dots, \quad (4.116)$$

where $\mathcal{H}(t)$ is the energy associated with the analytical solution and is equal to 77.5 meV. The largest values of (4.115) and (4.116) were 4.13×10^{-14} and 4.16×10^{-14} , respectively. As expected,

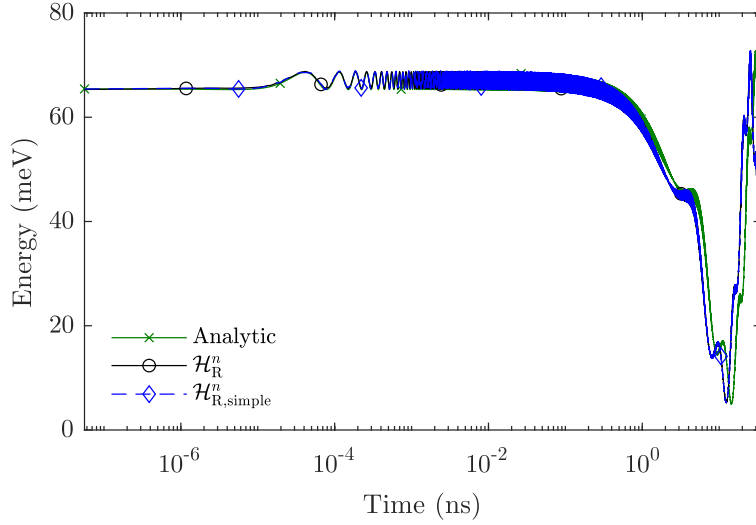


Figure 4.10: Energy (4.47) in the reactant region computed in the test of Section 4.5.4 for the 35 ns simulation.

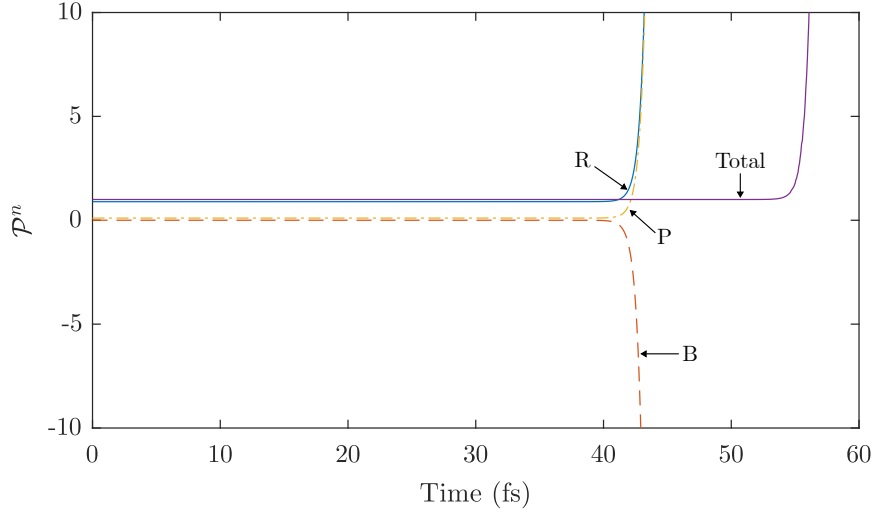


Figure 4.11: Results of the test in Section 4.5.4 for the time step exceeding the CFL limit $\Delta t_{\text{CFL}}^{\text{B}}$ and the generalized CFL limit $\Delta t_{\text{CFL, gen}}^{\text{B}}$ in the barrier region.

these fluctuations are extremely small and can be attributed to the finite machine precision. For comparison, we perform the same computation for $\mathcal{P}_{\text{simple}}^n$ and $\mathcal{H}_{\text{simple}}^n$. The largest values corresponding to (4.115) and (4.116) were, respectively, 3.92×10^{-3} and 4.21×10^{-3} , meaning that the simple expressions do not abide the conservation of probability and energy.

In order to illustrate a possible mechanism for the breakdown of conservation properties beyond the CFL limit, we also ran a simulation with a time step that violates the CFL limit condition in the barrier region ($\Delta t = 1.005 \Delta t_{\text{CFL}}^{\text{B}} = 0.9624 \Delta t_{\text{CFL}}^{\text{R}} = 0.9624 \Delta t_{\text{CFL}}^{\text{P}}$). The resulting probability is shown in Figure 4.11. Beyond the CFL limit, the inequality (4.29) can be violated in the barrier region and the probability associated with that region is permitted to indefinitely grow negative. This

allows the barrier region to provide an infinite amount of probability to the reactant and product regions, causing their probabilities to indefinitely grow positive. The total probability remains unchanged, until the exponential growth of the wavefunction values due to instability eventually causes the total probability to deviate from 1, as a result of the finite machine precision.

4.6 Conclusions

In this chapter, we analyze the probability and energy conservation of the FDTD-Q scheme for solving the Schrödinger equation. Existing works on the conservation properties of numerical schemes for the Schrödinger equation consider a region that is terminated with periodic or zero Dirichlet boundary conditions, which do not allow any net exchange of probability and energy with the surrounding space. In contrast, we considered a general scenario where a region can either be terminated using boundary conditions or form a portion of a larger domain, where probability and energy may enter or leave the region through the boundary. We proposed expressions for the total probability and energy contained in a region discretized using FDTD-Q, as well as expressions for the probability current and supplied power. Using these expressions, we showed that the FDTD-Q method conserves probability under the CFL limit that has been traditionally used for ensuring stability. Furthermore, we have shown that the CFL condition ensures that the energy is conserved, under an assumption that the region is coupled to other probability-conserving models.

The proposed expressions for computing probability and energy were compared to a more straightforward approach and several advantages were found. First, the proposed expressions respect the probability and energy conservation exactly. The exact conservation properties avoid spurious fluctuations exhibited by the values of the simpler expressions, which are especially evident when a system is isolated. Second, the proposed expressions for probability and energy tend to be more accurate than the simple expressions. Lastly, in isolated systems, the value of the proposed probability is convenient for normalizing initial values, since it is guaranteed to stay constant.

The proposed theory can serve as a basis for a stability analysis and enforcement framework in scenarios where the region is coupled to other models that can exchange energy and/or probability with the region. As a proof of concept, we considered the case of multiple regions with the same FDTD-Q discretization that are coupled to each other. We envision other possible applications, such as creation of stable subgridding schemes [22, 194] and stable incorporation of reduced order models [195], which have been demonstrated for the case of FDTD for the Maxwell equations [111, 193, 212]. Other possible applications include stability analysis of advanced boundary conditions [26, 135, 138, 139]. Moreover, multi-physics simulations [145, 171, 199, 238, 239] can also be considered as a type of scenario where an exchange occurs between the energy associated with the quantum particle and the energy stored in another form, such as the energy of electromagnetic field.

Conclusions

Conservation of energy is a fundamental physical principle respected by the Maxwell equations describing the electromagnetic field and by the Schrödinger equation governing the temporal evolution of the wavefunction associated with a quantum particle. The Schrödinger equation also respects the principle of probability conservation, which dictates that the probability of finding the particle in a certain region evolves according to the rate at which the probability current enters or leaves the region through the boundary. Improper discretization of partial differential equations can lead to the loss of these conservation properties. In this thesis, a theory is developed to allow construction of new numerical methods based on FDTD in a way that the principle of conservation is respected in the discrete domain. An important outcome of the theory is a modular stability analysis framework for creating provably stable schemes by coupling models that conserve or dissipate energy or, for the case of the Schrödinger equation, models that conserve probability.

This work takes inspiration from the theory of dissipative systems [205, 206]. The first step in our approach is to consider a region of space discretized using FDTD equations, which can exchange the energy or probability with the surrounding space via unknowns on the boundary. Using the conventional FDTD equations [33, 36, 38, 135] and modified FDTD equations introduced on the boundary, we define self-contained models for the region, whose probability and energy conservation and dissipation properties can be analyzed using the methods from dissipative systems theory [205, 206]. We provide expressions to quantify the energy contained in the region and the rate at which the energy enters or leaves the region through the boundary. For the case of the Schrödinger equation, we also provide expressions for the probability and its current through the boundary of the region. With these expressions, we derive conditions that can be used to ensure that the models for the region respect the conservation laws in the discrete domain. These conditions provide bounds on the time steps, which are generalizations of the well known CFL limits for stability [35, 140].

These results provide a way to create new schemes by coupling models that respect the conservation laws. As a demonstration, we create a subgridding scheme for the conventional fields-based FDTD by coupling two regions discretized with different resolution. A remarkable feature of the proposed approach is that the time step limits produced as a result of the conservation analysis of individual regions are by construction applicable to the coupled scheme. In [212], the theory was also applied to allow embedding reduced macromodels of refined regions in FDTD and extending their CFL limits to improve efficiency by avoiding excessive temporal refinement imposed by the

refined grids. In a similar way, we develop a subgridding scheme for the potentials-based FDTD. For the case of FDTD for the Schrödinger equation, we provide a proof-of-concept demonstration using regions of the same resolution and show how the probability conservation results can be applied to ensure stability of coupled schemes. The proposed approach also results in an alternative proof of stability for an isolated FDTD region under the conventional CFL limit [140]. We hope that the proposed results will facilitate creation of new numerical methods by the FDTD community.

The research in this thesis can be continued in several directions. Analogously to [111, 212], the model order reduction and CFL limit extension techniques can be developed for FDTD with potentials, using the subgridding scheme in Section 3.4 as a starting point. Another research direction is to generalize the dissipation theory in Chapters 2 and 3 to dispersive media [35, 240]. Moreover, in the energy conservation theory and stability analysis in Chapter 3 the medium was assumed to have zero conductivity. In principle, the assumption could be lifted, similarly to the dissipation theory in Chapter 2. Moreover, it is interesting to investigate subgridding schemes alternative to Section 2.4, with the goal of addressing the spurious resonances in Section 2.5.3, while retaining the stability, energy conservation, and material traverse properties of the scheme. Likewise, it is possible that the subgridding scheme in Section 3.4 could be modified to achieve lower reflections and higher accuracy when computing the electric field. Analogously to FDTD for Maxwell equations, the theory in Chapter 4 should allow creating stable schemes for FDTD-Q, including subgridding, model order reduction, and CFL limit extension. Lastly, we hope that the approach proposed in this work can be extended to FDTD schemes for simultaneous solution of the Maxwell and Schrödinger equations.

5.1 List of Publications

The following is a list of publications on the proposed research.

Journal Papers

- F. Bekmambetova, X. Zhang, and P. Triverio, “A dissipation theory for three-dimensional FDTD with application to stability analysis and subgridding,” *IEEE Trans. Antennas Propag.*, vol. 66, no. 12, pp. 7156–7170, 2018¹.
- F. Bekmambetova and P. Triverio, “A dissipation theory for potentials-based FDTD for lossless inhomogeneous media,” *IEEE Antennas Wirel. Propag. Lett.*, vol. 21, no. 3, pp. 486–490, 2022.
- F. Bekmambetova and P. Triverio, “Calculation and conservation of probability and energy in the numerical solution of the Schrödinger equation with the finite-difference time-domain method,” *IEEE Trans. Microw. Theory Tech.*, 2023 (in press).

Conference Papers and Abstracts

- F. Bekmambetova and P. Triverio, “A dissipation theory for 3-D FDTD with application to stable subgridding,” *IEEE AP-S Symp. Antennas Propag.*, San Diego, CA, July 9-14, 2017, pp. 2401–2402.

¹We have later published a correction [207].

- F. Bekmambetova, X. Zhang, and P. Triverio, “Acceleration of shielding effectiveness analysis using stable FDTD subgridding,” *26th IEEE Conf. Elect. Perform. Electron. Packag. Syst.*, San Jose, CA, Oct. 15-18, 2017, pp. 1–3.
- F. Bekmambetova and P. Triverio, “A dissipation theory for creating new stable FDTD algorithms with potentials”, USNC-URSI Nat. Radio Sci. Meet., Singapore (Online), Dec. 4-10, 2021.
- F. Bekmambetova, and P. Triverio, “A framework for creating stable FDTD schemes for the Schrödinger equation that conserve probability and energy,” USNC-URSI Radio Sci. Meet., Portland, OR, July 23–28, 2023 (accepted).

Publications on Research Prior to September 2016

Journal Papers

- F. Bekmambetova and X. Zhang and P. Triverio, “A dissipative systems theory for FDTD with application to stability analysis and subgridding,” *IEEE Trans. Antennas Propag.*, vol. 65, no. 2, pp. 751–762, 2017.
- X. Zhang and F. Bekmambetova and P. Triverio, “A Stable FDTD method with embedded reduced-order models,” *IEEE Trans. Antennas Propag.*, vol. 66, no. 2, pp. 827–837, 2018.

Conference Papers

- F. Bekmambetova, X. Zhang, and P. Triverio, “Accelerating Electromagnetic Simulations with Human Models through FDTD Subgridding and CFL Limit Extension,” USNC-URSI Nat. Radio Sci. Meet., Fajardo, Puerto Rico, June 26 - July 1, 2016.
- X. Zhang, F. Bekmambetova, and P. Triverio, “A Dissipative Control Approach to Ensure Stability in Advanced FDTD Schemes,” USNC-URSI Nat. Radio Sci. Meet., Fajardo, Puerto Rico, June 26 - July 1, 2016.
- F. Bekmambetova and X. Zhang and P. Triverio, “A Passivity Approach to FDTD Stability with Application to Interconnect Modeling,” 25th IEEE Conf. Elect. Perform. Electron. Packag. and Syst., San Diego, CA, Oct. 23-26, 2016, pp. 207-210 (Best Student Paper Award).
- X. Zhang and F. Bekmambetova and P. Triverio, “Reduced Order Modeling in FDTD with Provable Stability beyond the CFL Limit,” 25th IEEE Conf. Elect. Perform. of Electron. Packag. Syst., San Diego, CA, Oct. 23-26, 2016, pp. 99-102.

Appendix A

Primary and Secondary Grid Entities

For a cuboid region considered in this thesis, the entities associated with the primary and secondary grid [35] are defined as follows:

- *Primary cells* (Figure A.1): the cells of dimensions $\Delta x \times \Delta y \times \Delta z$ into which the region is divided. There are $n_x n_y n_z$ cells in the region. The primary cells are denoted using a prime (').
- *Primary faces* (Figure A.2): the faces of the primary cells. Their number of primary faces oriented in the $+x$, $+y$, and $+z$ directions, respectively, is $(n_x + 1)n_y n_z$, $n_x(n_y + 1)n_z$, and $n_x n_y(n_z + 1)$. The primary faces are denoted with a prime.
- *Primary edges* (Figure A.3):
 - *Conventional primary edges*: the edges of the primary cells. There are $n_x(n_y + 1)(n_z + 1)$, $(n_x + 1)n_y(n_z + 1)$, and $(n_x + 1)(n_y + 1)n_z$ edges oriented in the $+x$, $+y$, and $+z$ directions, respectively. These edges are denoted using a prime.
 - *Normal primary edges*: additional edges placed on the boundary of the region, oriented at 90° to the boundary. These edges can be thought of as having zero length. There are $(n_y + 1)(n_z + 1)$ normal primary edges on the west face of the boundary ($i = 1$) and $(n_y + 1)(n_z + 1)$ on the east face of the boundary ($i = n_x + 1$), all oriented in the $+x$ direction. The number of normal primary edges on the other faces of the boundary is analogous. These edges are denoted using a prime and an up tack symbol ($\perp$).
- *Primary nodes* (Figure A.4): nodes of the primary cells. The amount of the primary nodes in the region is $(n_x + 1)(n_y + 1)(n_z + 1)$. They are denoted using a prime.
- *Secondary cells* (Figure A.5): cells of the secondary grid. The internal secondary cells have dimensions $\Delta x \times \Delta y \times \Delta z$. The cells adjacent to the west and east faces of the boundary are halved in the x dimension, and similarly for the cells adjacent to the other faces of the boundary. The cells adjacent to two faces of the boundary are halved in two dimensions. Likewise, the cells adjacent to three faces of the boundary are halved in all three dimensions. There are $(n_x + 1)(n_y + 1)(n_z + 1)$ secondary cells in the region. They are denoted using a double prime ('').

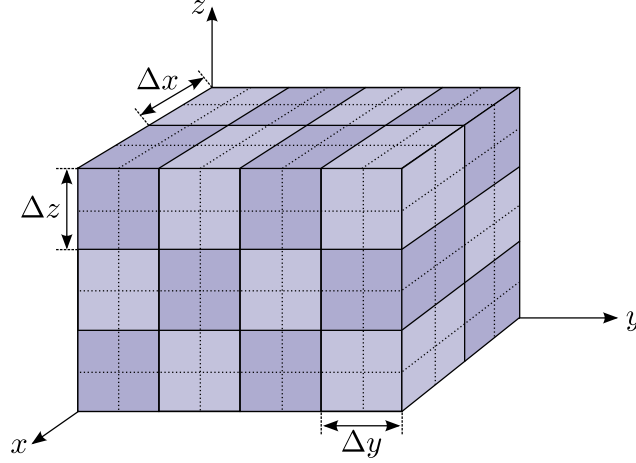


Figure A.1: Primary cells.

- *Secondary faces* (Figure A.6):
 - *Internal secondary faces*: secondary cell faces that do not lie on the boundary, but may touch the boundary. The number of internal secondary faces is the same as the number of the conventional primary edges in the region. These faces are denoted using a double prime.
 - *Boundary secondary faces*: secondary cell faces that coincide with the boundary of the region. The number of boundary secondary faces is the same as the number of normal primary edges. These faces are denoted using a double prime and a dagger †.
- *Secondary edges* (Figure A.7):
 - *Conventional secondary edges*: secondary cell edges that are either inside the region or are normal to the boundary. Their amount is the same as the number of primary faces. These edges are denoted with a double prime.
 - *Tangential secondary edges*: edges of the secondary cells that are tangential to the boundary of the region. On the west boundary of the region, there are $(n_y + 1)n_z$ edges oriented in the $+y$ direction and $n_y(n_z + 1)$ edges oriented in the $+z$ direction. The notation for these edges is a double prime and the symbol $\parallel$.

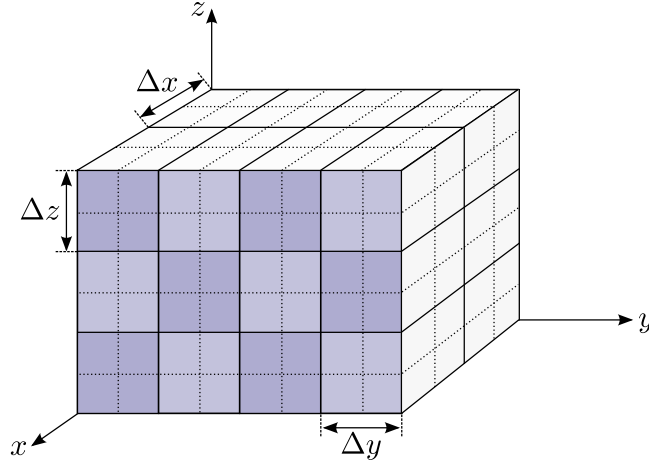
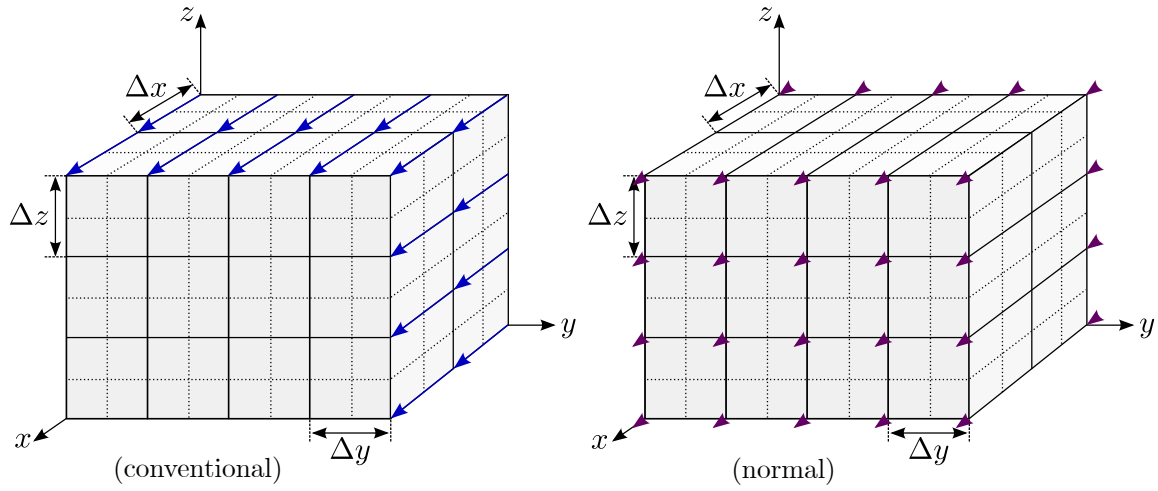
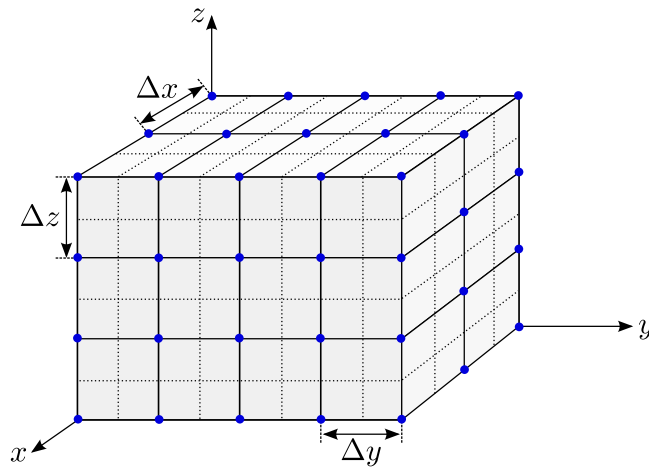
Figure A.2: Primary faces (highlighting those with the normal vectors in the $+x$ direction).Figure A.3: Primary edges (those in the $+x$ direction are shown).

Figure A.4: Primary nodes.

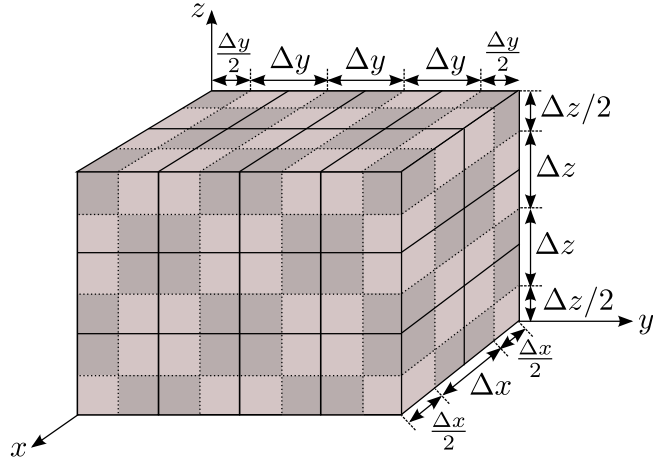


Figure A.5: Secondary cells.

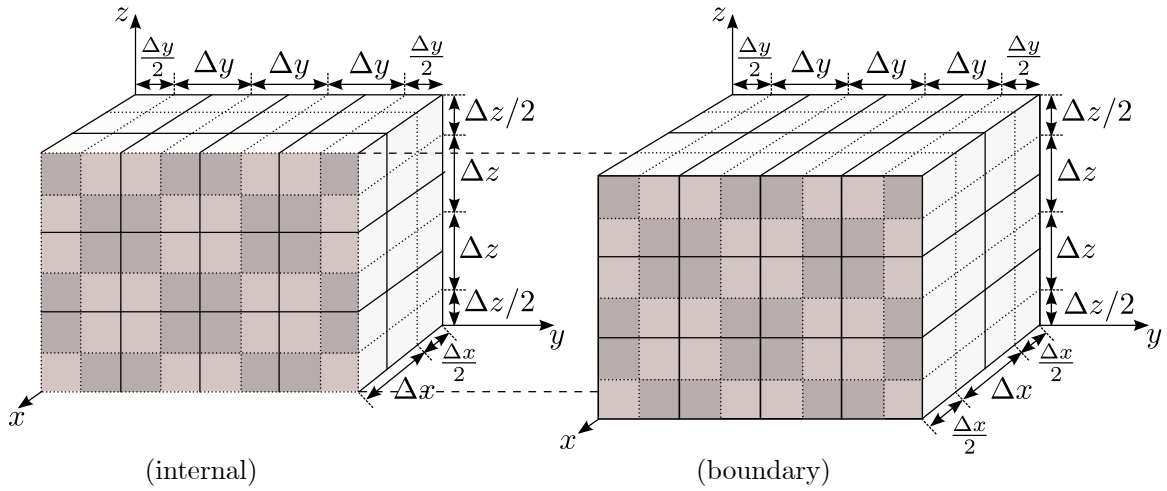


Figure A.6: Secondary faces (those with normal vectors in the $+x$ direction are highlighted).

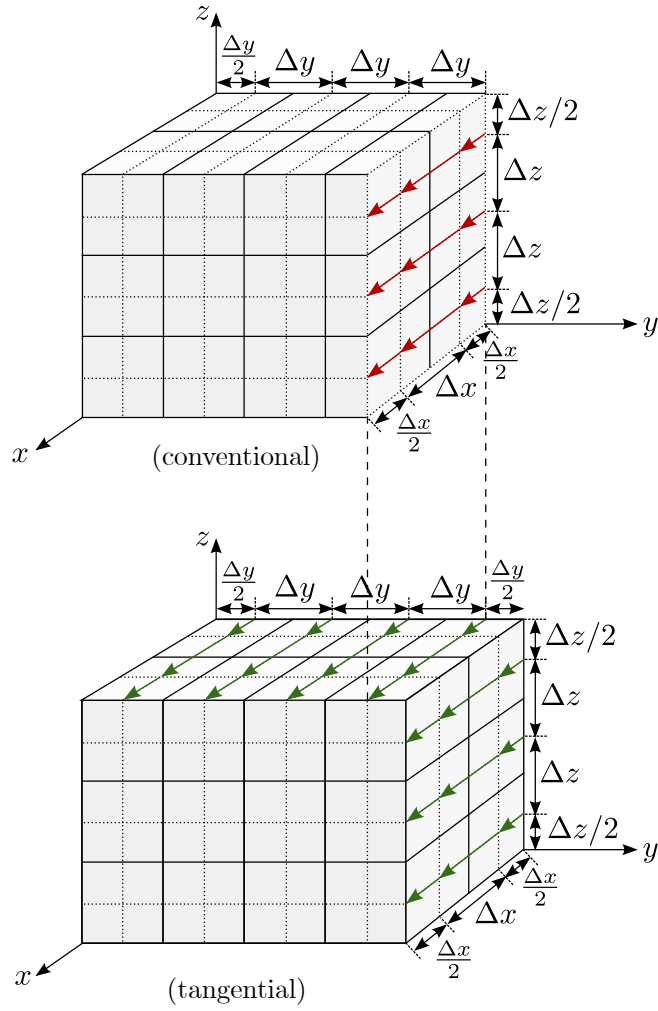


Figure A.7: Secondary edges (showing those in the $+x$ direction).

Appendix B

Expressions of Matrices in Compact Matrix Equations in Chapters 2–4

This appendix provides expressions for the different matrices in Sections 2.1.2, 3.2.3, and 4.1.2. The expressions assume the following numbering convention. For primary nodes, which are indexed as (i, j, k) , the ordering starts with those nodes that have $j = 1$ and $k = 1$, starting with $(1, 1, 1)$ and ending with $(n_x + 1, 1, 1)$. Those are followed by the nodes with $j = 2, k = 1$, followed by those with $j = 3, k = 1$, and so on. The nodes with $k = 1$ are followed by the nodes with $k = 2, k = 3$, and so on, until $k = n_z + 1$. In other words, a node with indices (i, j, k) is numbered as

$$i + (j - 1)(n_x + 1) + (k - 1)(n_x + 1)(n_y + 1).$$

For quantities sampled at the primary nodes, such as $[\partial_t \phi]$ in Chapter 3 or ψ_R in Chapter 4, this numbering dictates the positions of samples in the corresponding vectors, such as $[\partial_t \Phi]$, or Ψ_R . The numbering for the conventional and normal primary edges and the conventional and tangential secondary edges is shown in Table B.1. In vectors collecting samples at the x , y , and z -directed primary edges, the x directed edges are followed by the y -directed edges, which are followed by the z -directed edges. The same reasoning is used for the conventional secondary edges. For the normal primary edges, the ordering is given by

1. x -directed edges on the west face of the boundary,
2. x -directed edges on the east face of the boundary,
3. y -directed edges on the south face of the boundary,
4. y -directed edges on the north face of the boundary,
5. z -directed edges on the bottom face of the boundary,
6. z -directed edges on the top face of the boundary.

For the tangential secondary edges, we define the ordering as

1. x -directed edges on the south face of the boundary,

2. x -directed edges on the north face of the boundary,
3. x -directed edges on the bottom face of the boundary,
4. x -directed edges on the top face of the boundary,
5. y -directed edges on the west face of the boundary,
6. y -directed edges on the east face of the boundary,
7. y -directed edges on the bottom face of the boundary,
8. y -directed edges on the top face of the boundary,
9. z -directed edges on the west face of the boundary,
10. z -directed edges on the east face of the boundary,
11. z -directed edges on the south face of the boundary,
12. z -directed edges on the north face of the boundary.

The secondary cells are numbered in the same way as the corresponding primary nodes. The primary faces are numbered in the same way as the corresponding conventional secondary edges. The internal secondary faces are numbered in the same way as the conventional primary edges. Finally, the boundary secondary faces have the same numbering as the normal primary edges. It should be emphasized that the numbering convention does not enforce any particular implementation and does not affect the validity of the theory. The purpose of stating this convention is to provide concrete definitions for the matrices used in this thesis.

B.1 Diagonal Matrices Describing Grid Entities

Matrix $\mathbf{\Lambda}'_l$ in (2.14) has blocks on the diagonal equal to

$$\mathbf{\Lambda}'_{l,x} = \Delta x \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.1})$$

$$\mathbf{\Lambda}'_{l,y} = \Delta y \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.2})$$

$$\mathbf{\Lambda}'_{l,z} = \Delta z \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x+1}. \quad (\text{B.3})$$

For $\mathbf{\Lambda}''_l$ in (2.15), the corresponding blocks are equal to

$$\mathbf{\Lambda}''_{l,x} = \Delta x \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y} \otimes \tilde{\mathbf{I}}_{n_x+1}, \quad (\text{B.4})$$

$$\mathbf{\Lambda}''_{l,y} = \Delta y \mathbf{I}_{n_z} \otimes \tilde{\mathbf{I}}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.5})$$

$$\mathbf{\Lambda}''_{l,z} = \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \mathbf{I}_{n_y} \otimes \tilde{\mathbf{I}}_{n_x}, \quad (\text{B.6})$$

where $\tilde{\mathbf{I}}_m$ is an $m \times m$ matrix given by

$$\tilde{\mathbf{I}}_m = \text{diag} \left(\frac{1}{2}, 1, 1, \dots, 1, \frac{1}{2} \right). \quad (\text{B.7})$$

For, Λ'_S in (2.16),

$$\Lambda'_{S,x} = \Delta y \Delta z \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.8})$$

$$\Lambda'_{S,y} = \Delta x \Delta z \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.9})$$

$$\Lambda'_{S,z} = \Delta x \Delta y \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x}. \quad (\text{B.10})$$

Matrix Λ''_S in (2.21) has blocks on the diagonal given by

$$\Lambda''_{S,x} = \Delta y \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \tilde{\mathbf{I}}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.11})$$

$$\Lambda''_{S,y} = \Delta x \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \mathbf{I}_{n_y} \otimes \tilde{\mathbf{I}}_{n_x+1}, \quad (\text{B.12})$$

$$\Lambda''_{S,z} = \Delta x \Delta y \mathbf{I}_{n_z} \otimes \tilde{\mathbf{I}}_{n_y+1} \otimes \tilde{\mathbf{I}}_{n_x+1}. \quad (\text{B.13})$$

Submatrices of $\Lambda''_{l,\parallel}$ in (2.26) are given in (2.27)–(2.29), which are in turn comprised of

$$\Lambda''_{l,x,s} = \Delta x \mathbf{I}_{n_z} \otimes \tilde{\mathbf{I}}_{n_x+1}, \quad \Lambda''_{l,x,n} = \Delta x \mathbf{I}_{n_z} \otimes \tilde{\mathbf{I}}_{n_x+1}, \quad (\text{B.14})$$

$$\Lambda''_{l,x,b} = \Delta x \mathbf{I}_{n_y} \otimes \tilde{\mathbf{I}}_{n_x+1}, \quad \Lambda''_{l,x,t} = \Delta x \mathbf{I}_{n_y} \otimes \tilde{\mathbf{I}}_{n_x+1}, \quad (\text{B.15})$$

$$\Lambda''_{l,y,w} = \Delta y \mathbf{I}_{n_z} \otimes \tilde{\mathbf{I}}_{n_y+1}, \quad \Lambda''_{l,y,e} = \Delta y \mathbf{I}_{n_z} \otimes \tilde{\mathbf{I}}_{n_y+1}, \quad (\text{B.16})$$

$$\Lambda''_{l,y,b} = \Delta y \tilde{\mathbf{I}}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad \Lambda''_{l,y,t} = \Delta y \tilde{\mathbf{I}}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.17})$$

$$\Lambda''_{l,z,w} = \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \mathbf{I}_{n_y}, \quad \Lambda''_{l,z,e} = \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \mathbf{I}_{n_y}, \quad (\text{B.18})$$

$$\Lambda''_{l,z,s} = \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \mathbf{I}_{n_x}, \quad \Lambda''_{l,z,n} = \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \mathbf{I}_{n_x}. \quad (\text{B.19})$$

Submatrices of matrix $\Lambda'_{l,\parallel}$ in (2.50) are given by

$$\Lambda'_{l,z,s} = \Delta z \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_x+1}, \quad \Lambda'_{l,z,n} = \Delta z \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.20})$$

$$\Lambda'_{l,y,b} = \Delta y \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad \Lambda'_{l,y,t} = \Delta y \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.21})$$

$$\Lambda'_{l,z,w} = \Delta z \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1}, \quad \Lambda'_{l,z,e} = \Delta z \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1}, \quad (\text{B.22})$$

$$\Lambda'_{l,x,b} = \Delta x \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad \Lambda'_{l,x,t} = \Delta x \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.23})$$

$$\Lambda'_{l,y,w} = \Delta y \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y}, \quad \Lambda'_{l,y,e} = \Delta y \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y}, \quad (\text{B.24})$$

$$\Lambda'_{l,x,s} = \Delta x \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_x}, \quad \Lambda'_{l,x,n} = \Delta x \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_x}. \quad (\text{B.25})$$

Matrices Λ''_V and $\Lambda''_{S,\dagger}$, which appear in (3.31b), are given by

$$\Lambda''_V = \Delta x \Delta y \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \tilde{\mathbf{I}}_{n_y+1} \otimes \tilde{\mathbf{I}}_{n_x+1}, \quad (\text{B.26})$$

$$\Lambda''_{S,\dagger} = \text{diag} \left(\Lambda''_{S,\dagger,x}, \Lambda''_{S,\dagger,y}, \Lambda''_{S,\dagger,z} \right), \quad (\text{B.27})$$

where

$$\Lambda''_{S,\dagger,x} = \text{diag} \left(\Lambda''_{S,\dagger,w}, \Lambda''_{S,\dagger,e} \right), \quad (\text{B.28})$$

$$\Lambda''_{S,\dagger,y} = \text{diag} \left(\Lambda''_{S,\dagger,s}, \Lambda''_{S,\dagger,n} \right), \quad (\text{B.29})$$

$$\Lambda''_{S,\dagger,z} = \text{diag} \left(\Lambda''_{S,\dagger,b}, \Lambda''_{S,\dagger,t} \right), \quad (\text{B.30})$$

and

$$\begin{aligned} \Lambda''_{S,\dagger,w} &= \Delta y \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \tilde{\mathbf{I}}_{n_y+1}, & \Lambda''_{S,\dagger,e} &= \Delta y \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \tilde{\mathbf{I}}_{n_y+1}, \\ \Lambda''_{S,\dagger,s} &= \Delta x \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \tilde{\mathbf{I}}_{n_x+1}, & \Lambda''_{S,\dagger,n} &= \Delta x \Delta z \tilde{\mathbf{I}}_{n_z+1} \otimes \tilde{\mathbf{I}}_{n_x+1}, \\ \Lambda''_{S,\dagger,b} &= \Delta x \Delta y \tilde{\mathbf{I}}_{n_y+1} \otimes \tilde{\mathbf{I}}_{n_x+1}, & \Lambda''_{S,\dagger,t} &= \Delta x \Delta y \tilde{\mathbf{I}}_{n_y+1} \otimes \tilde{\mathbf{I}}_{n_x+1}. \end{aligned} \quad (\text{B.31})$$

B.2 Discrete Integration/Differentiation Operators

The blocks comprising $\mathbf{C}$ in (2.18) are given by

$$\mathbf{C}_{xy} = -\mathbf{W}_{n_z} \otimes \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad \mathbf{C}_{xz} = +\mathbf{I}_{n_z} \otimes \mathbf{W}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.32})$$

$$\mathbf{C}_{yx} = +\mathbf{W}_{n_z} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad \mathbf{C}_{yz} = -\mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{W}_{n_x}, \quad (\text{B.33})$$

$$\mathbf{C}_{zx} = -\mathbf{I}_{n_z+1} \otimes \mathbf{W}_{n_y} \otimes \mathbf{I}_{n_x}, \quad \mathbf{C}_{zy} = +\mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y} \otimes \mathbf{W}_{n_x}. \quad (\text{B.34})$$

where $\mathbf{W}_m$ is an $m \times (m+1)$ matrix given by

$$\mathbf{W}_m = \begin{bmatrix} \mathbf{0}_{m \times 1} & \mathbf{I}_m \end{bmatrix} - \begin{bmatrix} \mathbf{I}_m & \mathbf{0}_{m \times 1} \end{bmatrix}, \quad (\text{B.35})$$

where $\mathbf{0}_{kp}$ is a matrix of zeros with k rows and p columns.

Matrix $\mathbf{D}$ is given by

$$\mathbf{D} = \begin{bmatrix} \mathbf{D}_x & \mathbf{D}_y & \mathbf{D}_z \end{bmatrix}, \quad (\text{B.36})$$

where

$$\mathbf{D}_x = -\mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{W}_{n_x}^T, \quad (\text{B.37})$$

$$\mathbf{D}_y = -\mathbf{I}_{n_z+1} \otimes \mathbf{W}_{n_y}^T \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.38})$$

$$\mathbf{D}_z = -\mathbf{W}_{n_z}^T \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x+1}. \quad (\text{B.39})$$

B.3 Selection Matrices

Submatrices of $\mathbf{L}_{\parallel}$ in (2.31)–(2.33) are, in turn, composed of

$$\mathbf{L}_{xy,b} = \text{elem}\{1, n_z + 1\} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.40})$$

$$\mathbf{L}_{xy,t} = \text{elem}\{n_z + 1, n_z + 1\} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.41})$$

$$\mathbf{L}_{xz,s} = \mathbf{I}_{n_z+1} \otimes \text{elem}\{1, n_y + 1\} \otimes \mathbf{I}_{n_x}, \quad (\text{B.42})$$

$$\mathbf{L}_{xz,n} = \mathbf{I}_{n_z+1} \otimes \text{elem}\{n_y + 1, n_y + 1\} \otimes \mathbf{I}_{n_x}, \quad (\text{B.43})$$

$$\mathbf{L}_{yx,b} = \text{elem}\{1, n_z + 1\} \otimes \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.44})$$

$$\mathbf{L}_{yx,t} = \text{elem}\{n_z + 1, n_z + 1\} \otimes \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.45})$$

$$\mathbf{L}_{yz,w} = \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y} \otimes \text{elem}\{1, n_x + 1\}, \quad (\text{B.46})$$

$$\mathbf{L}_{yz,e} = \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y} \otimes \text{elem}\{n_x + 1, n_x + 1\}, \quad (\text{B.47})$$

$$\mathbf{L}_{zx,s} = \mathbf{I}_{n_z} \otimes \text{elem}\{1, n_y + 1\} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.48})$$

$$\mathbf{L}_{zx,n} = \mathbf{I}_{n_z} \otimes \text{elem}\{n_y + 1, n_y + 1\} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.49})$$

$$\mathbf{L}_{zy,w} = \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1} \otimes \text{elem}\{1, n_x + 1\}, \quad (\text{B.50})$$

$$\mathbf{L}_{zy,e} = \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1} \otimes \text{elem}\{n_x + 1, n_x + 1\}. \quad (\text{B.51})$$

where $\text{elem}\{k, m\}$ is an $m \times 1$ column vector with a 1 in row k and zeros in all other rows.

Matrix $\mathbf{L}_{\perp}$ in (3.31b) is given by

$$\mathbf{L}_{\perp} = \begin{bmatrix} \mathbf{L}_{\perp,x} & \mathbf{L}_{\perp,y} & \mathbf{L}_{\perp,z} \end{bmatrix}, \quad (\text{B.52})$$

where

$$\mathbf{L}_{\perp,x} = \begin{bmatrix} \mathbf{L}_{x,w} & \mathbf{L}_{x,e} \end{bmatrix}, \quad (\text{B.53})$$

$$\mathbf{L}_{\perp,y} = \begin{bmatrix} \mathbf{L}_{y,s} & \mathbf{L}_{y,n} \end{bmatrix}, \quad (\text{B.54})$$

$$\mathbf{L}_{\perp,z} = \begin{bmatrix} \mathbf{L}_{z,b} & \mathbf{L}_{z,t} \end{bmatrix}, \quad (\text{B.55})$$

and

$$\mathbf{L}_{x,w} = \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y+1} \otimes \text{elem}\{1, n_x + 1\}, \quad (\text{B.56})$$

$$\mathbf{L}_{x,e} = \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y+1} \otimes \text{elem}\{n_x + 1, n_x + 1\}, \quad (\text{B.57})$$

$$\mathbf{L}_{y,s} = \mathbf{I}_{n_z+1} \otimes \text{elem}\{1, n_y + 1\} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.58})$$

$$\mathbf{L}_{y,n} = \mathbf{I}_{n_z+1} \otimes \text{elem}\{n_y + 1, n_y + 1\} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.59})$$

$$\mathbf{L}_{z,b} = \text{elem}\{1, n_z + 1\} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.60})$$

$$\mathbf{L}_{z,t} = \text{elem}\{n_z + 1, n_z + 1\} \otimes \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x+1}. \quad (\text{B.61})$$

$$(\text{B.62})$$

B.4 Diagonal Matrices Determining Signs on the Boundary

Matrix $\Lambda_{[\cdot\hat{n}\times]}$ has submatrices given by (2.35)–(2.37), where

$$\Lambda_{[\hat{z}\cdot\hat{n}_s\times\hat{x}]} = (\hat{z} \cdot \hat{n}_s \times \hat{x}) \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_x+1}, \quad \Lambda_{[\hat{z}\cdot\hat{n}_n\times\hat{x}]} = (\hat{z} \cdot \hat{n}_n \times \hat{x}) \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.63})$$

$$\Lambda_{[\hat{y}\cdot\hat{n}_b\times\hat{x}]} = (\hat{y} \cdot \hat{n}_b \times \hat{x}) \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad \Lambda_{[\hat{y}\cdot\hat{n}_t\times\hat{x}]} = (\hat{y} \cdot \hat{n}_t \times \hat{x}) \mathbf{I}_{n_y} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.64})$$

$$\Lambda_{[\hat{z}\cdot\hat{n}_w\times\hat{y}]} = (\hat{z} \cdot \hat{n}_w \times \hat{y}) \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1}, \quad \Lambda_{[\hat{z}\cdot\hat{n}_e\times\hat{y}]} = (\hat{z} \cdot \hat{n}_e \times \hat{y}) \mathbf{I}_{n_z} \otimes \mathbf{I}_{n_y+1}, \quad (\text{B.65})$$

$$\Lambda_{[\hat{x}\cdot\hat{n}_b\times\hat{y}]} = (\hat{x} \cdot \hat{n}_b \times \hat{y}) \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad \Lambda_{[\hat{x}\cdot\hat{n}_t\times\hat{y}]} = (\hat{x} \cdot \hat{n}_t \times \hat{y}) \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x}, \quad (\text{B.66})$$

$$\Lambda_{[\hat{y}\cdot\hat{n}_w\times\hat{z}]} = (\hat{y} \cdot \hat{n}_w \times \hat{z}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y}, \quad \Lambda_{[\hat{y}\cdot\hat{n}_e\times\hat{z}]} = (\hat{y} \cdot \hat{n}_e \times \hat{z}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y}, \quad (\text{B.67})$$

$$\Lambda_{[\hat{x}\cdot\hat{n}_s\times\hat{z}]} = (\hat{x} \cdot \hat{n}_s \times \hat{z}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_x}, \quad \Lambda_{[\hat{x}\cdot\hat{n}_n\times\hat{z}]} = (\hat{x} \cdot \hat{n}_n \times \hat{z}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_x}. \quad (\text{B.68})$$

where

$$\hat{n}_w = -\hat{x}, \quad \hat{n}_e = +\hat{x}, \quad \hat{n}_s = -\hat{y}, \quad \hat{n}_w = +\hat{y}, \quad \hat{n}_b = -\hat{z}, \quad \hat{n}_t = +\hat{z}. \quad (\text{B.69})$$

Matrix $\Lambda_{[\hat{n}\cdot]}$ in (3.31b) is given by

$$\Lambda_{[\hat{n}\cdot]} = \text{diag}(\Lambda_{[\hat{n}\cdot\hat{x}]}, \Lambda_{[\hat{n}\cdot\hat{y}]}, \Lambda_{[\hat{n}\cdot\hat{z}]}) , \quad (\text{B.70})$$

where

$$\Lambda_{[\hat{n}\cdot\hat{x}]} = \text{diag}(\Lambda_{[\hat{n}_w\cdot\hat{x}]}, \Lambda_{[\hat{n}_e\cdot\hat{x}]}) , \quad (\text{B.71})$$

$$\Lambda_{[\hat{n}\cdot\hat{y}]} = \text{diag}(\Lambda_{[\hat{n}_s\cdot\hat{y}]}, \Lambda_{[\hat{n}_n\cdot\hat{y}]}) , \quad (\text{B.72})$$

$$\Lambda_{[\hat{n}\cdot\hat{z}]} = \text{diag}(\Lambda_{[\hat{n}_b\cdot\hat{z}]}, \Lambda_{[\hat{n}_t\cdot\hat{z}]}) , \quad (\text{B.73})$$

and

$$\Lambda_{[\hat{n}_w\cdot\hat{x}]} = (\hat{n}_w \cdot \hat{x}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y+1}, \quad \Lambda_{[\hat{n}_e\cdot\hat{x}]} = (\hat{n}_e \cdot \hat{x}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_y+1}, \quad (\text{B.74})$$

$$\Lambda_{[\hat{n}_s\cdot\hat{y}]} = (\hat{n}_s \cdot \hat{y}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_x+1}, \quad \Lambda_{[\hat{n}_n\cdot\hat{y}]} = (\hat{n}_n \cdot \hat{y}) \mathbf{I}_{n_z+1} \otimes \mathbf{I}_{n_x+1}, \quad (\text{B.75})$$

$$\Lambda_{[\hat{n}_b\cdot\hat{z}]} = (\hat{n}_b \cdot \hat{z}) \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x+1}, \quad \Lambda_{[\hat{n}_t\cdot\hat{z}]} = (\hat{n}_t \cdot \hat{z}) \mathbf{I}_{n_y+1} \otimes \mathbf{I}_{n_x+1}. \quad (\text{B.76})$$

Table B.1: Numbering of samples in vectors appearing in the matrix equations in Chapters 2–4.

Grid entity		Indices	Element in vectors	Examples of vectors		
				Ch 2	Ch 3	Ch 4
Primary nodes		(i, j, k)	$i + (j - 1)(n_x + 1) + (k - 1)(n_x + 1)(n_y + 1)$	N/A	$[\partial_t \Phi]$	Ψ_R
Conventional primary edges ^a	x -directed	$(i + \frac{1}{2}, j, k)$	$i + (j - 1)n_x + (k - 1)n_x(n_y + 1)$	$\mathbf{e}_x$	$[\nabla \Phi]$	$[\nabla \Psi_R]$
	y -directed	$(i, j + \frac{1}{2}, k)$	$i + (j - 1)(n_x + 1) + (k - 1)(n_x + 1)n_y$	$\mathbf{e}_y$		
	z -directed	$(i, j, k + \frac{1}{2})$	$i + (j - 1)(n_x + 1) + (k - 1)(n_x + 1)(n_y + 1)$	$\mathbf{e}_z$		
Normal primary edges ^b	x -directed, w	$(1, j, k)$	$j + (k - 1)(n_y + 1)$	N/A	$[\nabla \Phi]_\perp$	$[\nabla \Psi_R]_\perp$
	x -directed, e	$(n_x + 1, j, k)$	$j + (k - 1)(n_y + 1)$			
	y -directed, s	$(i, 1, k)$	$i + (k - 1)(n_x + 1)$			
	y -directed, n	$(i, n_y + 1, k)$	$i + (k - 1)(n_x + 1)$			
	z -directed, b	$(i, j, 1)$	$i + (j - 1)(n_x + 1)$			
	z -directed, t	$(i, j, n_z + 1)$	$i + (j - 1)(n_x + 1)$			
Conventional secondary edges ^c	x -directed	$(i, j + \frac{1}{2}, k + \frac{1}{2})$	$i + (j - 1)(n_x + 1) + (k - 1)(n_x + 1)n_y$	$\mathbf{h}_x$	$\mathbf{b}$	N/A
	y -directed	$(i + \frac{1}{2}, j, k + \frac{1}{2})$	$i + (j - 1)n_x + (k - 1)n_x(n_y + 1)$	$\mathbf{h}_y$		
	z -directed	$(i + \frac{1}{2}, j + \frac{1}{2}, k)$	$i + (j - 1)n_x + (k - 1)n_x n_y$	$\mathbf{h}_z$		
Tangential secondary edges ^d	x -directed, s	$(i, 1, k + \frac{1}{2})$	$i + (k - 1)(n_x + 1)$	$\mathbf{h}_{x,s}$	$\mathbf{b}_\parallel$	N/A
	x -directed, n	$(i, n_y + 1, k + \frac{1}{2})$	$i + (k - 1)(n_x + 1)$	$\mathbf{h}_{x,n}$		
	x -directed, b	$(i, j + \frac{1}{2}, 1)$	$i + (j - 1)(n_x + 1)$	$\mathbf{h}_{x,b}$		
	x -directed, t	$(i, j + \frac{1}{2}, n_z + 1)$	$i + (j - 1)(n_x + 1)$	$\mathbf{h}_{x,t}$		
	y -directed, w	$(1, j, k + \frac{1}{2})$	$j + (k - 1)(n_y + 1)$	$\mathbf{h}_{y,w}$		
	y -directed, e	$(n_x + 1, j, k + \frac{1}{2})$	$j + (k - 1)(n_y + 1)$	$\mathbf{h}_{y,e}$		
	y -directed, b	$(i + \frac{1}{2}, j, 1)$	$i + (j - 1)n_x$	$\mathbf{h}_{y,b}$		
	y -directed, t	$(i + \frac{1}{2}, j, n_z + 1)$	$i + (j - 1)n_x$	$\mathbf{h}_{y,t}$		
	z -directed, w	$(1, j + \frac{1}{2}, k)$	$j + (k - 1)n_y$	$\mathbf{h}_{z,w}$		
	z -directed, e	$(n_x + 1, j + \frac{1}{2}, k)$	$j + (k - 1)n_y$	$\mathbf{h}_{z,e}$		
	z -directed, s	$(i + \frac{1}{2}, 1, k)$	$i + (k - 1)n_x$	$\mathbf{h}_{z,s}$		
	z -directed, n	$(i + \frac{1}{2}, n_y + 1, k)$	$i + (k - 1)n_x$	$\mathbf{h}_{z,n}$		

^a In vectors collecting quantities sampled on all of the conventional primary edges, the x -directed samples are followed by the y -directed samples, which are followed by the z -directed samples. For example, $[\nabla \Phi]$ consists of three blocks, $[\partial_x \Phi]$, $[\partial_y \Phi]$, and $[\partial_z \Phi]$, with samples numbered as shown in this table.

^b In vectors collecting quantities sampled on all of the normal primary edges, the blocks corresponding to each face of the boundary are arranged consistently with matrices such as (B.27).

^c The arrangement of blocks within the vectors is analogous to that in $\mathbf{h}$, given in (2.13).

^d The arrangement of blocks within the vectors is analogous to that in $\mathbf{h}_\parallel$, given in (2.24).

Treatment of the Fine Region Edges in Subgridding of Section 2.4

This appendix handles the case where the fine grid is surrounded by the coarse grid on all six sides. In this scenario, the edges of the region's boundary require a special treatment. As an example, we consider the samples near the south-west edge of the subgridding interface, shown in Figure C.1. The grid ratios r_x , r_y , and r_z are assumed to be odd integers greater or equal to 3. The samples considered are:

- the coarse sample $E_z|_{j,k,k+0.5}$, denoted as E_z ,
- r_z fine samples $\hat{E}_z|_{1,1,\hat{k}+0.5}$, which are collected into vector $\hat{\mathbf{e}}_{z,sw}$,
- $r_z(r_y - 1)/2$ samples strictly on the west face of the fine grid boundary, grouped into $\hat{\mathbf{e}}_{z,w}$,
- $r_z(r_x - 1)/2$ samples strictly on the south face of the fine grid boundary, collected into $\hat{\mathbf{e}}_{z,s}$,
- the coarse hanging variable $H_x|_{j,k,k+0.5}$, denoted as $H_{\parallel,x}$,
- the coarse hanging variable $H_y|_{j,k,k+0.5}$, denoted as $H_{\parallel,y}$,
- r_z fine hanging variables $\hat{H}_x|_{1,1,\hat{k}+0.5}$, collected into vector $\hat{\mathbf{h}}_{x,sw}$,
- $r_z(r_x - 1)/2$ fine hanging variables $\hat{H}_x|_{\hat{i},1,\hat{k}+0.5}$, with $\hat{i} = 2, \dots, (r_x + 1)/2$, grouped into $\hat{\mathbf{h}}_{x,s}$,
- r_z fine hanging variables $\hat{H}_x|_{1,1,\hat{k}+0.5}$, collected into vector $\hat{\mathbf{h}}_{y,sw}$,
- $r_z(r_y - 1)/2$ fine hanging variables $\hat{H}_y|_{1,\hat{j},\hat{k}+0.5}$, with $\hat{j} = 2, \dots, (r_y + 1)/2$, grouped into $\hat{\mathbf{h}}_{y,w}$.

The indexing convention for vectors $\hat{\mathbf{e}}_{z,sw}$, $\hat{\mathbf{e}}_{z,w}$, $\hat{\mathbf{e}}_{z,s}$, $\hat{\mathbf{h}}_{x,sw}$, $\hat{\mathbf{h}}_{x,s}$, $\hat{\mathbf{h}}_{y,sw}$, $\hat{\mathbf{h}}_{y,w}$ is analogous to that in Appendix B, with the $\hat{i}$ indices iterated first, $\hat{j}$ indices iterated next, and $\hat{k}$ indices iterated after the $\hat{i}$ and $\hat{j}$ indices. The goal of this appendix is to derive the update procedure for the electric field variables at the interface, namely E_z^n , $\hat{\mathbf{e}}_{z,sw}^n$, $\hat{\mathbf{e}}_{z,w}^n$, $\hat{\mathbf{e}}_{z,s}^n$, in a way that guarantees stability.

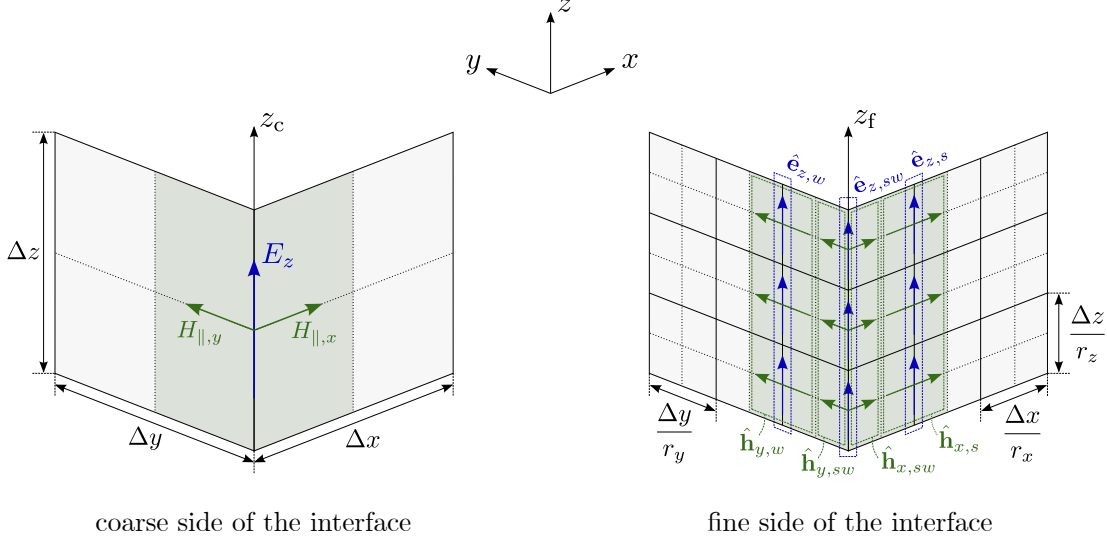


Figure C.1: Coarse and fine sides of the subgridding interface in Appendix C.

C.1 Interpolation Conditions

Similarly to Section 2.4, we impose equality of the fine electric field samples in the z direction and define auxiliary variables $\bar{E}_{z,sw}$, $\bar{\mathbf{e}}_{z,w}$, and $\bar{\mathbf{e}}_{z,s}$, representing the distinct electric field samples of the fine grid¹. With the help of these variables, we write the interpolation conditions between electric field samples and hanging magnetic field variables on the coarse and fine sides of the interface. The first two conditions are analogous to (2.76a) and are given by

$$\frac{\Delta x}{2} E_z^n = \frac{\Delta x}{2r_x} \bar{E}_{z,sw}^n + \frac{\Delta x}{r_x} \mathbf{T}_{\frac{r_x-1}{2}}^T \bar{\mathbf{e}}_{z,s}^n, \quad (\text{C.1a})$$

$$\frac{\Delta y}{2} E_z^n = \frac{\Delta y}{2r_y} \bar{E}_{z,sw}^n + \frac{\Delta y}{r_y} \mathbf{T}_{\frac{r_y-1}{2}}^T \bar{\mathbf{e}}_{z,w}^n. \quad (\text{C.1b})$$

Condition (C.1a) ensures that the weighted average of the fine electric fields over the south portion of the shaded area in Figure C.1 is equal to the value of the coarse field sample E_z . Likewise, (C.1b) ensures that the weighted average of the fine samples over the west portion of the shaded area are equal to the same value E_z .

Next, we write equations that are analogous to (2.76b). However, instead of relating the hanging variables to each other, we describe the relationship between their contributions to the right hand sides of the update equations in (2.20). This approach gives

$$\begin{aligned} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{2r_x} H_{\parallel,x}^{n+\frac{1}{2}} + \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{2r_y} H_{\parallel,y}^{n+\frac{1}{2}} \\ + \mathbf{T}_{\frac{r_z-1}{2}}^T \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{2r_x} \hat{\mathbf{h}}_{x,sw}^{n+\frac{1}{2}} + \mathbf{T}_{\frac{r_z-1}{2}}^T \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{2r_y} \hat{\mathbf{h}}_{y,sw}^{n+\frac{1}{2}} = 0, \end{aligned} \quad (\text{C.1c})$$

$$\mathbf{T}_{\frac{r_y-1}{2}} \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{r_y} H_{\parallel,y}^{n+\frac{1}{2}} + \left(\mathbf{T}_{\frac{r_z-1}{2}}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{r_y} \hat{\mathbf{h}}_{y,w}^{n+\frac{1}{2}} = 0, \quad (\text{C.1d})$$

¹In the particular example shown in Figure C.1 for $r_x = r_y = 3$, vectors $\bar{\mathbf{e}}_{z,w}$ and $\bar{\mathbf{e}}_{z,s}$ reduce to scalar quantities.

$$\mathbf{T}_{\frac{r_x-1}{2}} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{r_x} H_{\parallel,x}^{n+\frac{1}{2}} + \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{h}}_{x,s}^{n+\frac{1}{2}} = 0. \quad (\text{C.1e})$$

For a flat portion of the interface discussed in Section 2.4, the approach used to write (C.1c)–(C.1e) would give equivalent interpolation conditions to (2.76b). For the case of the edge of the subgridding interface discussed in this appendix, this approach ensures that the interpolation conditions are precisely those needed for showing the energy conservation. Equations (C.1c)–(C.1e) will also facilitate elimination of some of the unknowns, specifically the hanging variables of the fine grid.

The third group of conditions is analogous to (2.76c), enforcing the equality of the fine electric field samples in the longitudinal direction,

$$\hat{\mathbf{e}}_{z,sw}^n = \mathbf{T}_{r_z} \bar{E}_{z,sw}^n, \quad (\text{C.1f})$$

$$\hat{\mathbf{e}}_{z,w}^n = \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \bar{\mathbf{e}}_{z,w}^n, \quad (\text{C.1g})$$

$$\hat{\mathbf{e}}_{z,s}^n = \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \bar{\mathbf{e}}_{z,s}^n. \quad (\text{C.1h})$$

C.1.1 Stability Proof

In this section, we prove that the interpolation conditions (C.1a)–(C.1h) are lossless. Similarly to Section 2.4.2, we do this by showing that the contribution of the variables shown in Figure C.1 to the supply rate (2.48) of the fine grid and the supply rate of the coarse grid adds to zero.

From (2.54), the contribution of the coarse samples to the supply rate of the coarse grid is given by

$$\delta s^{n+\frac{1}{2}} = \Delta t \frac{E_z^n + E_z^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta z}{2} H_{\parallel,x}^{n+\frac{1}{2}} + \Delta t \frac{E_z^n + E_z^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{2} H_{\parallel,y}^{n+\frac{1}{2}}. \quad (\text{C.2})$$

The contribution of the fine samples to the supply rate of the fine grid is

$$\begin{aligned} \delta \hat{s}^{n+\frac{1}{2}} &= \Delta t \frac{(\hat{\mathbf{e}}_{z,sw}^n + \hat{\mathbf{e}}_{z,sw}^{n+1})^T}{2} \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{2r_x} \hat{\mathbf{h}}_{x,sw}^{n+\frac{1}{2}} \\ &\quad + \Delta t \frac{(\hat{\mathbf{e}}_{z,sw}^n + \hat{\mathbf{e}}_{z,sw}^{n+1})^T}{2} \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{2r_y} \hat{\mathbf{h}}_{y,sw}^{n+\frac{1}{2}} \\ &\quad + \Delta t \frac{(\hat{\mathbf{e}}_{z,w}^n + \hat{\mathbf{e}}_{z,w}^{n+1})^T}{2} \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{r_y} \hat{\mathbf{h}}_{y,w}^{n+\frac{1}{2}} \\ &\quad + \Delta t \frac{(\hat{\mathbf{e}}_{z,s}^n + \hat{\mathbf{e}}_{z,s}^{n+1})^T}{2} \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{h}}_{x,s}^{n+\frac{1}{2}}. \quad (\text{C.3}) \end{aligned}$$

Substituting (C.1f)–(C.1h) into (C.3),

$$\begin{aligned}\delta\hat{s}^{n+\frac{1}{2}} = & \Delta t \frac{\bar{E}_{z,sw}^n + \bar{E}_{z,sw}^{n+1}}{2} \mathbf{T}_{r_z}^T \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{2r_x} \hat{\mathbf{h}}_{x,sw}^{n+\frac{1}{2}} \\ & + \Delta t \frac{\bar{E}_{z,sw}^n + \bar{E}_{z,sw}^{n+1}}{2} \mathbf{T}_{r_z}^T \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{2r_y} \hat{\mathbf{h}}_{y,sw}^{n+\frac{1}{2}} \\ & + \Delta t \frac{(\bar{\mathbf{e}}_{z,w}^n + \bar{\mathbf{e}}_{z,w}^{n+1})^T}{2} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{r_y} \hat{\mathbf{h}}_{y,w}^{n+\frac{1}{2}} \\ & + \Delta t \frac{(\bar{\mathbf{e}}_{z,s}^n + \bar{\mathbf{e}}_{z,s}^{n+1})^T}{2} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{h}}_{x,s}^{n+\frac{1}{2}}. \quad (\text{C.4})\end{aligned}$$

From (C.1c)–(C.1e),

$$\begin{aligned}\delta\hat{s}^{n+\frac{1}{2}} = & -\Delta t \frac{\bar{E}_{z,sw}^n + \bar{E}_{z,sw}^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{2r_x} H_{\parallel,x}^{n+\frac{1}{2}} \\ & - \Delta t \frac{\bar{E}_{z,sw}^n + \bar{E}_{z,sw}^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{2r_y} H_{\parallel,y}^{n+\frac{1}{2}} \\ & - \Delta t \frac{(\bar{\mathbf{e}}_{z,w}^n + \bar{\mathbf{e}}_{z,w}^{n+1})^T}{2} \mathbf{T}_{\frac{r_y-1}{2}} \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{r_y} H_{\parallel,y}^{n+\frac{1}{2}} \\ & - \Delta t \frac{(\bar{\mathbf{e}}_{z,s}^n + \bar{\mathbf{e}}_{z,s}^{n+1})^T}{2} \mathbf{T}_{\frac{r_x-1}{2}} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{r_x} H_{\parallel,x}^{n+\frac{1}{2}}. \quad (\text{C.5})\end{aligned}$$

Finally, using (C.1a)–(C.1b),

$$\delta\hat{s}^{n+\frac{1}{2}} = -\Delta t \frac{E_z^n + E_z^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{2} H_{\parallel,x}^{n+\frac{1}{2}} - \Delta t \frac{E_z^n + E_z^{n+1}}{2} \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{2} H_{\parallel,y}^{n+\frac{1}{2}}. \quad (\text{C.6})$$

Comparing (C.6) to (C.2),

$$\delta\hat{s}^{n+\frac{1}{2}} + \delta s^{n+\frac{1}{2}} = 0, \quad (\text{C.7})$$

which proves that the interpolation conditions (C.1a)–(C.1h) conserve energy. Hence, the scheme in Section 2.4 remains stable after generalizing it to the case of embedded fine grid in the way described in this appendix.

C.2 Update Equations

Next, we derive the update equations that implicitly enforce (C.1a)–(C.1h). Similarly to Section 2.4.3, we write (2.20) for the coarse electric field sample in Figure C.1 as

$$\begin{aligned}\Delta z \, 3 \frac{\Delta x}{2} \frac{\Delta y}{2} \left(\frac{\varepsilon_z}{\Delta t} + \frac{\sigma_z}{2} \right) E_z^{n+1} = & \Delta z \, 3 \frac{\Delta x}{2} \frac{\Delta y}{2} \left(\frac{\varepsilon_z}{\Delta t} - \frac{\sigma_z}{2} \right) E_z^n \\ & + \Delta z \mathbf{C}^T \mathbf{\Lambda}_l'' \mathbf{h}^{n+\frac{1}{2}} + \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{2} H_{\parallel,x}^{n+\frac{1}{2}} + \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{2} H_{\parallel,y}^{n+\frac{1}{2}}, \quad (\text{C.8})\end{aligned}$$

Equation (C.8) is written for a volume resulting from extrusion in the z direction of the L-shaped internal secondary face of area $3(\Delta x/2)(\Delta y/2)$, which is adjacent to the edge of the coarse region in Figure C.1. Quantities ε_z , σ_z , $\mathbf{h}$, $\mathbf{\Lambda}_l''$, and $\mathbf{C}$ are defined accordingly. Likewise, we can write the

corresponding equations for the fine grid as,

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{2r_y} \left(\frac{\hat{\Lambda}_{\varepsilon,z,sw}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,sw}}{2} \right) \hat{\mathbf{e}}_{z,sw}^{n+1} &= \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{2r_y} \left(\frac{\hat{\Lambda}_{\varepsilon,z,sw}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,sw}}{2} \right) \hat{\mathbf{e}}_{z,sw}^n \\ &+ \frac{\Delta z}{r_z} \hat{\mathbf{C}}_{sw}^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} + \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{2r_x} \hat{\mathbf{h}}_{x,sw}^{n+\frac{1}{2}} + \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{2r_y} \hat{\mathbf{h}}_{y,sw}^{n+\frac{1}{2}}, \quad (\text{C.9}) \end{aligned}$$

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{r_y} \left(\frac{\hat{\Lambda}_{\varepsilon,z,w}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,w}}{2} \right) \hat{\mathbf{e}}_{z,w}^{n+1} &= \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{r_y} \left(\frac{\hat{\Lambda}_{\varepsilon,z,w}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,w}}{2} \right) \hat{\mathbf{e}}_{z,w}^n \\ &+ \frac{\Delta z}{r_z} \hat{\mathbf{C}}_w^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} + \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{r_y} \hat{\mathbf{h}}_{y,w}^{n+\frac{1}{2}}, \quad (\text{C.10}) \end{aligned}$$

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\frac{\hat{\Lambda}_{\varepsilon,z,s}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,s}}{2} \right) \hat{\mathbf{e}}_{z,s}^{n+1} &= \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\frac{\hat{\Lambda}_{\varepsilon,z,s}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,s}}{2} \right) \hat{\mathbf{e}}_{z,s}^n \\ &+ \frac{\Delta z}{r_z} \hat{\mathbf{C}}_s^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} + \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{h}}_{x,s}^{n+\frac{1}{2}}. \quad (\text{C.11}) \end{aligned}$$

Rewriting (C.9) using (C.1f) and left-multiplying both sides of the equation by $\mathbf{T}_{r_z}^T$, we obtain

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{2r_y} \mathbf{T}_{r_z}^T \left(\frac{\hat{\Lambda}_{\varepsilon,z,sw}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,sw}}{2} \right) \mathbf{T}_{r_z} \bar{E}_{z,sw}^{n+1} &= \\ \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{2r_y} \mathbf{T}_{r_z}^T \left(\frac{\hat{\Lambda}_{\varepsilon,z,sw}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,sw}}{2} \right) \mathbf{T}_{r_z} \bar{E}_{z,sw}^n &+ \mathbf{T}_{r_z}^T \frac{\Delta z}{r_z} \hat{\mathbf{C}}_{sw}^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} \\ &+ \mathbf{T}_{r_z}^T \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{2r_x} \hat{\mathbf{h}}_{x,sw}^{n+\frac{1}{2}} + \mathbf{T}_{r_z}^T \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{2r_y} \hat{\mathbf{h}}_{y,sw}^{n+\frac{1}{2}}, \quad (\text{C.12}) \end{aligned}$$

Similarly, rewriting (C.10) using (C.1g) and left-multiplying the result by $\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{(r_y-1)/2}$,

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,w}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,w}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \bar{\mathbf{e}}_{z,w}^{n+1} &= \\ \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,w}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,w}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \bar{\mathbf{e}}_{z,w}^n & \\ + \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \frac{\Delta z}{r_z} \hat{\mathbf{C}}_w^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} &+ \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_w \times \hat{y}) \frac{\Delta y}{r_y} \hat{\mathbf{h}}_{y,w}^{n+\frac{1}{2}}, \quad (\text{C.13}) \end{aligned}$$

Finally, rewriting (C.11) using (C.1h) and left-multiplying the resulting equation by $\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{(r_x-1)/2}$, we obtain

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,s}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,s}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \bar{\mathbf{e}}_{z,s}^{n+1} = \\ \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,s}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,s}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \bar{\mathbf{e}}_{z,s}^n \\ + \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \frac{\Delta z}{r_z} \hat{\mathbf{C}}_s^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} + \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \frac{\Delta z}{r_z} (\hat{z} \cdot \hat{n}_s \times \hat{x}) \frac{\Delta x}{r_x} \hat{\mathbf{h}}_{x,s}^{n+\frac{1}{2}}. \quad (\text{C.14}) \end{aligned}$$

Next, we eliminate the fine hanging variables from (C.12)–(C.14) with the use of (C.1c)–(C.1e). Equation (C.12) becomes

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{2r_y} \mathbf{T}_{r_z}^T \left(\frac{\hat{\Lambda}_{\varepsilon,z,sw}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,sw}}{2} \right) \mathbf{T}_{r_z} \bar{E}_{z,sw}^{n+1} = \\ \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{2r_y} \mathbf{T}_{r_z}^T \left(\frac{\hat{\Lambda}_{\varepsilon,z,sw}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,sw}}{2} \right) \mathbf{T}_{r_z} \bar{E}_{z,sw}^n + \mathbf{T}_{r_z}^T \frac{\Delta z}{r_z} \hat{\mathbf{C}}_{sw}^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} \\ - \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{2r_x} H_{\parallel,x}^{n+\frac{1}{2}} - \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{2r_y} H_{\parallel,y}^{n+\frac{1}{2}}. \quad (\text{C.15}) \end{aligned}$$

Equation (C.13) becomes

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,w}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,w}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \bar{\mathbf{e}}_{z,w}^{n+1} = \\ \frac{\Delta z}{r_z} \frac{\Delta x}{2r_x} \frac{\Delta y}{r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,w}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,w}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \bar{\mathbf{e}}_{z,w}^n \\ + \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_y-1}{2}} \right) \frac{\Delta z}{r_z} \hat{\mathbf{C}}_w^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} - \mathbf{T}_{\frac{r_y-1}{2}} \Delta z (\hat{z} \cdot \hat{n}_e \times \hat{y}) \frac{\Delta y}{r_y} H_{\parallel,y}^{n+\frac{1}{2}}. \quad (\text{C.16}) \end{aligned}$$

Lastly, (C.14) becomes

$$\begin{aligned} \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,s}}{\Delta t} + \frac{\hat{\Lambda}_{\sigma,z,s}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \bar{\mathbf{e}}_{z,s}^{n+1} = \\ \frac{\Delta z}{r_z} \frac{\Delta x}{r_x} \frac{\Delta y}{2r_y} \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \left(\frac{\hat{\Lambda}_{\varepsilon,z,s}}{\Delta t} - \frac{\hat{\Lambda}_{\sigma,z,s}}{2} \right) \left(\mathbf{T}_{r_z} \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \bar{\mathbf{e}}_{z,s}^n \\ + \left(\mathbf{T}_{r_z}^T \otimes \mathbf{I}_{\frac{r_x-1}{2}} \right) \frac{\Delta z}{r_z} \hat{\mathbf{C}}_s^T \hat{\Lambda}_l'' \hat{\mathbf{h}}^{n+\frac{1}{2}} - \mathbf{T}_{\frac{r_x-1}{2}} \Delta z (\hat{z} \cdot \hat{n}_n \times \hat{x}) \frac{\Delta x}{r_x} H_{\parallel,x}^{n+\frac{1}{2}}. \quad (\text{C.17}) \end{aligned}$$

We now collect the update equations (C.8) for the coarse electric sample, (C.15)–(C.17) for the unique fine electric samples, and the interpolation rules (C.1a)–(C.1b), into a linear system of the form

$$\mathbf{A} \mathbf{z}^{n+1} = \mathbf{b}^n, \quad (\text{C.18})$$

with an $(r_x/2 + r_y/2 + 3) \times 1$ vector of unknowns

$$\mathbf{z}^{n+1} = \begin{bmatrix} E_z^{n+1} \\ \bar{E}_{z,sw}^{n+1} \\ \bar{\mathbf{e}}_{z,w}^{n+1} \\ \bar{\mathbf{e}}_{z,s}^{n+1} \\ H_{\parallel,x}^{n+\frac{1}{2}} \\ H_{\parallel,y}^{n+\frac{1}{2}} \end{bmatrix}. \quad (\text{C.19})$$

One can show that $\mathbf{A}$ in (C.18) is invertible. Solving (C.18) for $\mathbf{z}^{n+1}$ gives values of E_z^{n+1} , $\bar{E}_{z,sw}^{n+1}$, $\bar{\mathbf{e}}_{z,w}^{n+1}$, and $\bar{\mathbf{e}}_{z,s}^{n+1}$. The fine grid samples can then be obtained by duplicating the samples in $\bar{E}_{z,sw}^{n+1}$, $\bar{\mathbf{e}}_{z,w}^{n+1}$, and $\bar{\mathbf{e}}_{z,s}^{n+1}$ as dictated by (C.1f)–(C.1h). The size of system (C.18) is only $(r_x/2 + r_y/2 + 3) \times (r_x/2 + r_y/2 + 3)$ and the resulting overhead is typically small. So far in this appendix, we have only shown that (C.1a)–(C.1h) and (C.8)–(C.11) imply (C.18). One can also prove the converse: that the update procedure described above guarantees that (C.1a)–(C.1h) and (C.8)–(C.11) hold.

Proofs of Theorems Relating the Generalized and Conventional CFL Limits

D.1 Proof of Theorem 3

The proof is similar to [197], where an analogous result was shown for the case of an isolated region.

Lemma 4. *Consider the dynamical system (2.38a)–(2.38b) for an $n_x \times n_y \times n_z$ region. Let $\mathbf{x}$ be an arbitrary real vector with the number of elements equal to the total number of the conventional primary edges and the conventional secondary edges. Then, $\mathbf{x}^T \mathbf{R} \mathbf{x}$ can be broken down into the contributions from individual primary cells as*

$$\mathbf{x}^T \mathbf{R} \mathbf{x} = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \sum_{k=1}^{n_z} \mathbf{x}_{ijk}^T \mathbf{R}_{ijk} \mathbf{x}_{ijk}, \quad (\text{D.1})$$

where the subscript ijk denotes quantities pertaining to a single-cell region with the bottom-south-west corner at the node (i, j, k) . Vector $\mathbf{x}_{ijk}$ for each single-cell region is constructed as follows.

1. Vector $\mathbf{x}$ is broken down into two blocks $\mathbf{e}$ and $\mathbf{h}$, as per (2.39).
2. Each element of $\mathbf{e}$ and $\mathbf{h}$ is associated with a conventional primary edge or a conventional secondary edge, respectively, using the convention in Appendix B.
3. The elements of $\mathbf{e}$ and $\mathbf{h}$ associated with the edges belonging to the single-cell region are collected into $\mathbf{x}_{ijk}$ using the same convention for ordering the elements.

The edge permittivity and permeability values used to define $\mathbf{R}_{ijk}$ are uniform and equal to the primary cell values $\varepsilon|_{i+0.5, j+0.5, k+0.5}$ and $\mu|_{i+0.5, j+0.5, k+0.5}$, respectively. The approach taken in the proof of Theorem 3 to first separate $\mathbf{x}^T \mathbf{R} \mathbf{x}$ into the contributions from individual cells is similar to [197].

Proof. The left hand side of (D.1) can be written as

$$\mathbf{x}^T \mathbf{R} \mathbf{x} = \mathbf{e}^T \Lambda'_l \Lambda''_S \frac{\Lambda_\varepsilon}{\Delta t} \mathbf{e} - \mathbf{h}^T \Lambda'_l \mathbf{C} \Lambda'_l \mathbf{e} + \mathbf{h}^T \Lambda''_l \Lambda'_S \frac{\Lambda_\mu}{\Delta t} \mathbf{h}. \quad (\text{D.2})$$

The first term in (D.2) is a summation over primary edges of terms such as

$$\Delta x \Delta y \Delta z \frac{\varepsilon_x|_{i+\frac{1}{2},j,k}}{\Delta t} (E_x|_{i+\frac{1}{2},j,k})^2, \quad (\text{D.3})$$

where the latter is the term associated with an internal primary edge between (i, j, k) and $(i+1, j, k)$. Using the fact that $\varepsilon_x|_{i+0.5,j,k}$ is averaged from the primary cell values as in (2.1), we can break (D.3) into contributions from the four adjacent primary cells

$$\begin{aligned} & \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_x|_{i+\frac{1}{2},j-\frac{1}{2},k-\frac{1}{2}}}{\Delta t} (E_x|_{i+\frac{1}{2},j,k})^2 + \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_x|_{i+\frac{1}{2},j+\frac{1}{2},k-\frac{1}{2}}}{\Delta t} (E_x|_{i+\frac{1}{2},j,k})^2 \\ & + \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_x|_{i+\frac{1}{2},j-\frac{1}{2},k+\frac{1}{2}}}{\Delta t} (E_x|_{i+\frac{1}{2},j,k})^2 + \Delta x \frac{\Delta y}{2} \frac{\Delta z}{2} \frac{\varepsilon_x|_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}}}{\Delta t} (E_x|_{i+\frac{1}{2},j,k})^2. \end{aligned} \quad (\text{D.4})$$

For example, in the first term in (D.4), the factor $\Delta y/2 \times \Delta z/2$ can be recognized as the area of the internal secondary face adjacent to the top-north edge of the boundary of the corresponding single-cell region. Continuing the argument for the remaining two terms in (D.2), we can split all of the terms into the contributions that can be attributed to individual single-cell regions. Grouping the terms corresponding to each such region, we arrive at (D.1). $\square$

Lemma 5. *Let $\Delta t_{CFL,gen}$ and $\Delta t_{CFL,gen,ijk}$ be the generalized CFL limits (2.64) corresponding, respectively, to the entire region and to a single-cell region formed by the primary cell with the bottom-south-west corner at the node (i, j, k) . Then,*

$$\min_{i,j,k} \Delta t_{CFL,gen,ijk} \leq \Delta t_{CFL,gen}. \quad (\text{D.5})$$

Proof. Consider matrices $\mathbf{R}$ and $\mathbf{R}_{ijk}$ and vectors $\mathbf{x}$ and $\mathbf{x}_{ijk}$ as described in Lemma 4. Consider a time step Δt such that

$$\Delta t < \min_{i,j,k} \Delta t_{CFL,gen,ijk}. \quad (\text{D.6})$$

With this time step, by Theorem 2 applied to the single-cell regions,

$$\mathbf{R}_{ijk} \succ 0, \quad \forall i \in \{1, 2, \dots, n_x\}, \forall j \in \{1, 2, \dots, n_y\}, \forall k \in \{1, 2, \dots, n_z\}, \quad (\text{D.7})$$

and for each i, j , and k , we have

$$\begin{cases} \mathbf{x}_{ijk}^T \mathbf{R}_{ijk} \mathbf{x}_{ijk} > 0, & \forall \mathbf{x}_{ijk} \in \mathbb{R}^{(N'_{1,1,1} + N''_{1,1,1}) \times 1} \text{ where } \mathbf{x}_{ijk} \neq \mathbf{0}, \\ \mathbf{x}_{ijk}^T \mathbf{R}_{ijk} \mathbf{x}_{ijk} = 0, & \text{if } \mathbf{x}_{ijk} = \mathbf{0}, \end{cases} \quad (\text{D.8})$$

where N'_{n_x, n_y, n_z} and N''_{n_x, n_y, n_z} are, respectively, the number of conventional primary edges and the number of conventional secondary edges in an $n_x \times n_y \times n_z$ region. By Lemma 4,

$$\mathbf{x}^T \mathbf{R} \mathbf{x} > 0, \quad \forall \mathbf{x} \in \mathbb{R}^{(N'_{n_x, n_y, n_z} + N''_{n_x, n_y, n_z}) \times 1} \text{ with } \mathbf{x} \neq \mathbf{0}, \quad (\text{D.9})$$

which means that $\mathbf{R}$ is positive definite. Hence, by Theorem 2 applied to the entire $n_x \times n_y \times n_z$ region, $\Delta t < \Delta t_{CFL,gen}$.

In conclusion, the following implication holds

$$\Delta t < \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk} \implies \Delta t < \Delta t_{\text{CFL,gen}}. \quad (\text{D.10})$$

This is only possible if (D.5) is true. $\square$

Lemma 6. *Let $\Delta t_{\text{CFL},ijk}$ and $\Delta t_{\text{CFL,gen},ijk}$ be, respectively, the CFL limit (1.36) and the generalized CFL limit (2.64) for the single-cell region composed of a primary cell with the bottom-south-west corner at the node (i, j, k) . Then,*

$$\Delta t_{\text{CFL},ijk} \leq \Delta t_{\text{CFL,gen},ijk}. \quad (\text{D.11})$$

Proof. Consider a time step $\Delta t < \Delta t_{\text{CFL},ijk}$. The goal is to show that this time step is also below $\Delta t_{\text{CFL,gen},ijk}$. This proof follows the same steps as [197]. We consider a vector $\mathbf{x}_{ijk} \in \mathbb{R}^{18 \times 1}$, where the size of the vector is the total number of the conventional primary and conventional secondary edges in the single-cell region. Each element in $\mathbf{x}_{ijk}$ is associated with a sample in the region, as per (2.39) and the numbering convention in Appendix B. By analyzing the expansion (D.2), written for the single-cell region, we can split $\mathbf{x}_{ijk}^T \mathbf{R}_{ijk} \mathbf{x}_{ijk}$ as

$$\mathbf{x}_{ijk}^T \mathbf{R}_{ijk} \mathbf{x}_{ijk} = (\mathbf{x}^{(a)})^T \mathbf{R}^{(a)} \mathbf{x}^{(a)} + (\mathbf{x}^{(b)})^T \mathbf{R}^{(b)} \mathbf{x}^{(b)} + (\mathbf{x}^{(c)})^T \mathbf{R}^{(c)} \mathbf{x}^{(c)} + (\mathbf{x}^{(d)})^T \mathbf{R}^{(d)} \mathbf{x}^{(d)}, \quad (\text{D.12})$$

where

$$\mathbf{x}^{(a)} = \begin{bmatrix} E_x^{(a)} \\ E_y^{(a)} \\ E_z^{(a)} \\ H_x^{(ad)} \\ H_y^{(ac)} \\ H_z^{(ab)} \end{bmatrix}, \quad \mathbf{x}^{(b)} = \begin{bmatrix} E_x^{(b)} \\ E_y^{(b)} \\ E_z^{(b)} \\ H_x^{(bc)} \\ H_y^{(bd)} \\ H_z^{(ab)} \end{bmatrix}, \quad \mathbf{x}^{(c)} = \begin{bmatrix} E_x^{(c)} \\ E_y^{(c)} \\ E_z^{(c)} \\ H_x^{(bc)} \\ H_y^{(ac)} \\ H_z^{(cd)} \end{bmatrix}, \quad \mathbf{x}^{(d)} = \begin{bmatrix} E_x^{(d)} \\ E_y^{(d)} \\ E_z^{(d)} \\ H_x^{(ad)} \\ H_y^{(bd)} \\ H_z^{(cd)} \end{bmatrix}, \quad (\text{D.13a})$$

$$\mathbf{R}^{(a)} = \begin{bmatrix} \frac{\Delta x \Delta y \Delta z}{4} \frac{\varepsilon}{\Delta t} \mathbf{I}_3 & -\frac{1}{2} \mathbf{\Lambda}_l'^{(a)} (\mathbf{C}^{(a)})^T \mathbf{\Lambda}_l''^{(a)} \\ -\frac{1}{2} \mathbf{\Lambda}_l''^{(a)} \mathbf{C}^{(a)} \mathbf{\Lambda}_l'^{(a)} & \frac{\Delta x \Delta y \Delta z}{4} \frac{\mu}{\Delta t} \mathbf{I}_3 \end{bmatrix}, \quad (\text{D.14})$$

and similarly for $\mathbf{R}^{(b)}$, $\mathbf{R}^{(c)}$, and $\mathbf{R}^{(d)}$. In (D.14),

$$\varepsilon = \varepsilon|_{i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}, \quad \mu = \mu|_{i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}, \quad (\text{D.15})$$

which are the permittivity and permeability of the primary cell with the bottom-south-west corner at (i, j, k) ,

$$\mathbf{\Lambda}_l'^{(a)} = \text{diag}(\Delta x, \Delta y, \Delta z), \quad \mathbf{\Lambda}_l''^{(a)} = \text{diag}\left(\frac{\Delta x}{2}, \frac{\Delta y}{2}, \frac{\Delta z}{2}\right), \quad (\text{D.16})$$

and

$$\mathbf{C}^{(a)} = \begin{bmatrix} 0 & +1 & -1 \\ -1 & 0 & -1 \\ +1 & +1 & 0 \end{bmatrix}. \quad (\text{D.17})$$

In the definitions of $\mathbf{R}^{(b)}$, $\mathbf{R}^{(c)}$, and $\mathbf{R}^{(d)}$,

$$\mathbf{C}^{(b)} = \begin{bmatrix} 0 & +1 & +1 \\ -1 & 0 & +1 \\ -1 & -1 & 0 \end{bmatrix}, \quad \mathbf{C}^{(c)} = \begin{bmatrix} 0 & -1 & -1 \\ +1 & 0 & +1 \\ +1 & -1 & 0 \end{bmatrix}, \quad \mathbf{C}^{(d)} = \begin{bmatrix} 0 & -1 & +1 \\ +1 & 0 & -1 \\ -1 & +1 & 0 \end{bmatrix}. \quad (\text{D.18})$$

Using the Schur complement [213] and the assumption that ε is positive, $\mathbf{R}^{(a)} \succ 0$ if and only if

$$\frac{\Delta x \Delta y \Delta z}{4} \frac{\mu}{\Delta t} \mathbf{I}_3 - \frac{\Delta t}{4} \mathbf{\Lambda}_l''^{(a)} \mathbf{C}^{(a)} \mathbf{\Lambda}_l'^{(a)} \left(\frac{\Delta x \Delta y \Delta z}{4} \varepsilon \right)^{-1} \mathbf{\Lambda}_l'^{(a)} (\mathbf{C}^{(a)})^T \mathbf{\Lambda}_l''^{(a)} \succ 0. \quad (\text{D.19})$$

Multiplying the left hand side of (D.19) by the positive scalar $4\varepsilon/(\Delta t \Delta x \Delta y \Delta z)$, (D.19) simplifies to

$$\begin{bmatrix} \frac{\mu\varepsilon}{\Delta t^2} - \frac{1}{\Delta y^2} - \frac{1}{\Delta z^2} & -\frac{C_{xz}^{(a)}C_{yz}^{(a)}}{\Delta x \Delta y} & -\frac{C_{xy}^{(a)}C_{zy}^{(a)}}{\Delta x \Delta z} \\ -\frac{C_{yz}^{(a)}C_{xz}^{(a)}}{\Delta x \Delta y} & \frac{\mu\varepsilon}{\Delta t^2} - \frac{1}{\Delta x^2} - \frac{1}{\Delta z^2} & -\frac{C_{yx}^{(a)}C_{zx}^{(a)}}{\Delta y \Delta z} \\ -\frac{C_{zy}^{(a)}C_{xy}^{(a)}}{\Delta x \Delta z} & -\frac{C_{zx}^{(a)}C_{yx}^{(a)}}{\Delta y \Delta z} & \frac{\mu\varepsilon}{\Delta t^2} - \frac{1}{\Delta x^2} - \frac{1}{\Delta y^2} \end{bmatrix} \succ 0, \quad (\text{D.20})$$

Using the assumption that Δt is below the conventional CFL limit (1.36), applying the Schur complement twice, one can show that condition (D.20) holds and hence $\mathbf{R}^{(a)} \succ 0$. Likewise, one can show that $\mathbf{R}^{(b)}$, $\mathbf{R}^{(c)}$, and $\mathbf{R}^{(d)}$ are positive definite. From (D.12) this implies that $\mathbf{R}$ is positive definite as well. Hence, from Theorem 2, $\Delta t < \Delta t_{\text{CFL,gen},ijk}$.

In conclusion, we have shown that $\Delta t < \Delta t_{\text{CFL},ijk}$ implies $\Delta t < \Delta t_{\text{CFL,gen},ijk}$. This means that $\Delta t_{\text{CFL},ijk} < \Delta t_{\text{CFL,gen},ijk}$. □

Corollary 1. *Let Δt_{CFL} be the conventional CFL limit (1.36) for the entire region, with $\varepsilon_{\min}$ and $\mu_{\min}$ being the smallest values of permeability and permittivity across all of the primary cells in the region. Let $\Delta t_{\text{CFL,gen},ijk}$ be the generalized CFL limit (2.64) for a single-cell region formed by a primary cell with the bottom-south-west corner at the node (i, j, k) . Then,*

$$\Delta t_{\text{CFL}} \leq \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk}. \quad (\text{D.21})$$

Proof. Since Δt_{CFL} is evaluated using the smallest values of permittivity and permeability across all primary cells,

$$\Delta t_{\text{CFL}} \leq \Delta t_{\text{CFL},ijk}, \quad \forall i \in 1, 2, \dots, n_x, \quad \forall j \in 1, 2, \dots, n_y, \quad \forall k \in 1, 2, \dots, n_z, \quad (\text{D.22})$$

where $\Delta t_{\text{CFL},ijk}$ is the conventional CFL limit (1.36), computed for the corresponding single-cell region. Moreover, using Lemma 6, (D.22) implies

$$\Delta t_{\text{CFL}} \leq \Delta t_{\text{CFL,gen},ijk} \quad \forall i \in 1, 2, \dots, n_x, \quad \forall j \in 1, 2, \dots, n_y, \quad \forall k \in 1, 2, \dots, n_z. \quad (\text{D.23})$$

The statement of the corollary follows directly from (D.23). □

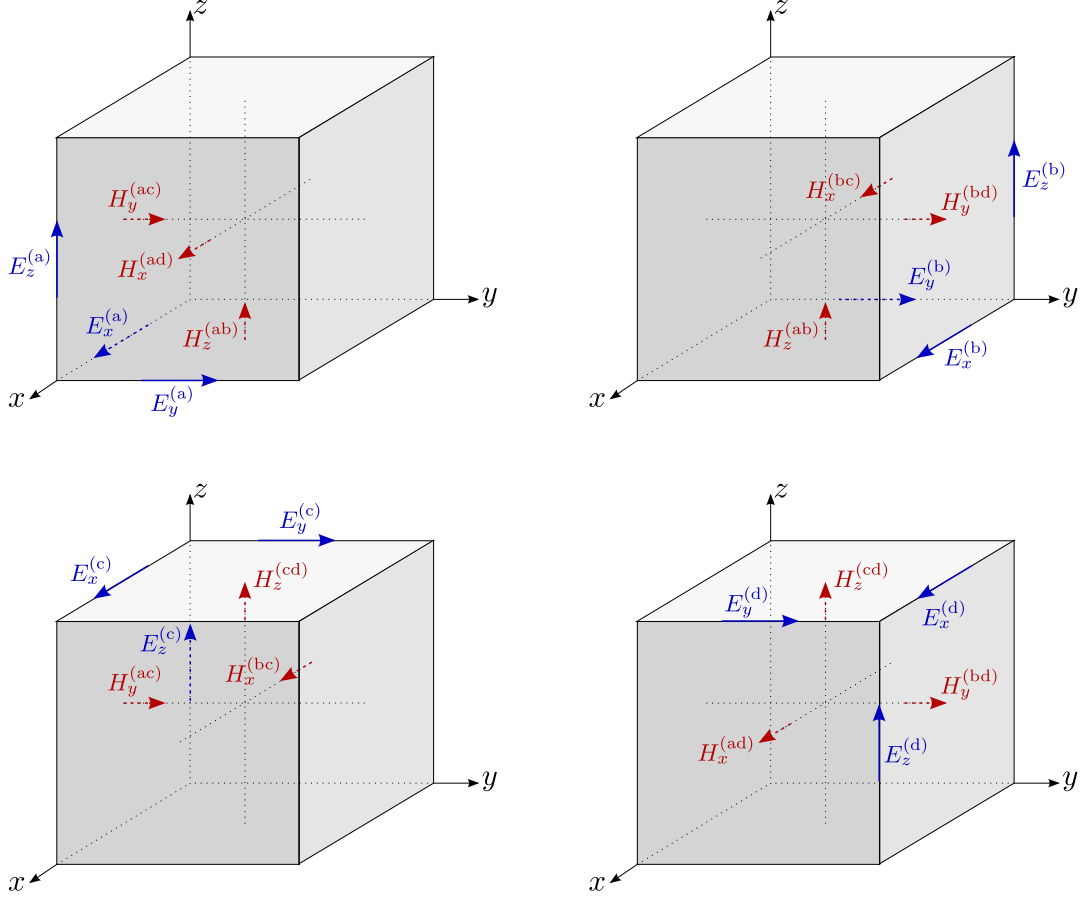


Figure D.1: Splitting of samples into groups in the proof of Lemma 6. The groups are the same as those in [197].

Proof of Theorem 3. Combining the statements of Lemma 5 and Corollary 1,

$$\Delta t_{\text{CFL}} \leq \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk} \leq \Delta t_{\text{CFL,gen}}, \quad (\text{D.24})$$

which proves the theorem. $\square$

D.2 Proof of Theorem 6

Lemma 7. Consider the dynamical systems (3.32a)–(3.32b) and (3.43a)–(3.43b) describing, respectively, the evolution of the scalar potential and the vector potential for a region composed of $n_x \times n_y \times n_z$ primary cells.

1. Let $\mathbf{x}_\phi$ be an arbitrary real vector with the number of elements equal to the total number of conventional primary edges and primary nodes in the region. Then, $\mathbf{x}_\phi^T \mathbf{R}_\phi \mathbf{x}_\phi$ can be written in terms of the contributions from individual primary cells as

$$\mathbf{x}_\phi^T \mathbf{R}_\phi \mathbf{x}_\phi = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \sum_{k=1}^{n_z} \mathbf{x}_{\phi,ijk}^T \mathbf{R}_{\phi,ijk} \mathbf{x}_{\phi,ijk}, \quad (\text{D.25})$$

2. Let $\mathbf{x}_A$ be an arbitrary real vector with the number of elements equal to the total number of conventional secondary edges, conventional primary edges, and the primary nodes. Then, $\mathbf{x}_A^T \mathbf{R}_A \mathbf{x}_A$ can be broken down as

$$\mathbf{x}_A^T \mathbf{R}_A \mathbf{x}_A = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \sum_{k=1}^{n_z} \mathbf{x}_{A,ijk}^T \mathbf{R}_{A,ijk} \mathbf{x}_{A,ijk}. \quad (\text{D.26})$$

In (D.25) and (D.26), the subscript ijk denotes quantities pertaining to a single-cell region with the bottom-south-west corner at the node (i, j, k) . Vectors $\mathbf{x}_{\phi,ijk}$ and $\mathbf{x}_{A,ijk}$ for each single-cell region is constructed analogously to Lemma 4. Matrices $\mathbf{R}_{\phi,ijk}$ and $\mathbf{R}_{A,ijk}$ are defined using the constant permittivity and permeability values of the corresponding primary cell.

Lemma 8. Consider the dynamical systems (3.32a)–(3.32b) and (3.43a)–(3.43b) with the corresponding generalized CFL limits (3.82) and (3.89). The following relation holds between the generalized CFL limits for the entire $n_x \times n_y \times n_z$ region and for the regions composed of single primary cells.

1. Let $\Delta t_{CFL,gen}^{(\phi)}$ and $\Delta t_{CFL,gen,ijk}^{(\phi)}$ be the generalized CFL limits (3.82) corresponding, respectively, to the entire region and to a single-cell region with the bottom-south-west corner at (i, j, k) . Then,

$$\min_{i,j,k} \Delta t_{CFL,gen,ijk}^{(\phi)} \leq \Delta t_{CFL,gen}^{(\phi)}. \quad (\text{D.27})$$

2. Let $\Delta t_{CFL,gen}^{(A)}$ be the generalized CFL limit (3.89) for the entire region and let $\Delta t_{CFL,gen,ijk}^{(A)}$ be the generalized CFL limit for a single-cell region with the bottom-south-west corner at (i, j, k) . Then,

$$\min_{i,j,k} \Delta t_{CFL,gen,ijk}^{(A)} \leq \Delta t_{CFL,gen}^{(A)}. \quad (\text{D.28})$$

The proofs of Lemmas 7 and 8 are similar to those of Lemmas 4 and 5, respectively.

Lemma 9. Let $\Delta t_{CFL,ijk}$ be the CFL limit (1.36) of the single-cell region with the bottom-south-west corner at node (i, j, k) , where $\varepsilon_{\min}$ and $\mu_{\min}$ are equal to the values of permittivity and permeability, respectively, in the single primary cell of that region.

1. For the dynamical system (3.32a)–(3.32b) governing the scalar potential,

$$\Delta t_{CFL,ijk} = \Delta t_{CFL,gen,ijk}^{(\phi)}. \quad (\text{D.29})$$

where $\Delta t_{CFL,gen,ijk}^{(\phi)}$ is the generalized CFL limit (3.82) for the single-cell region.

2. For (3.43a)–(3.43b) governing the vector potential,

$$\Delta t_{CFL,ijk} = \Delta t_{CFL,gen,ijk}^{(A)}. \quad (\text{D.30})$$

where $\Delta t_{CFL,gen,ijk}^{(A)}$ is the generalized CFL limit (3.89) for the single-cell region.

Proof. Consider the expression of the generalized CFL limit (3.82). From Appendix B, for the single-cell region,

$$\mathbf{\Lambda}'_{l,x} = \Delta x \mathbf{I}_4, \quad \mathbf{\Lambda}'_{l,y} = \Delta y \mathbf{I}_4, \quad \mathbf{\Lambda}'_{l,z} = \Delta z \mathbf{I}_4, \quad (\text{D.31})$$

$$\mathbf{\Lambda}_{S,x}'' = \frac{\Delta y}{2} \frac{\Delta z}{2} \mathbf{I}_4, \quad \mathbf{\Lambda}_{S,y}'' = \frac{\Delta x}{2} \frac{\Delta z}{2} \mathbf{I}_4, \quad \mathbf{\Lambda}_{S,z}'' = \frac{\Delta x}{2} \frac{\Delta y}{2} \mathbf{I}_4, \quad (\text{D.32})$$

$$\mathbf{D}_x = -\mathbf{I}_2 \otimes \mathbf{I}_2 \otimes \mathbf{W}_1^T, \quad \mathbf{D}_y = -\mathbf{I}_2 \otimes \mathbf{W}_1^T \otimes \mathbf{I}_2, \quad \mathbf{D}_z = -\mathbf{W}_1^T \otimes \mathbf{I}_2 \otimes \mathbf{I}_2, \quad (\text{D.33})$$

$$\mathbf{\Lambda}_V'' = \frac{\Delta x}{2} \frac{\Delta y}{2} \frac{\Delta z}{2} \mathbf{I}_8. \quad (\text{D.34})$$

Letting ε and μ denote the permittivity and permeability of the primary cell and letting $\chi = \mu\varepsilon^2$,

$$\mathbf{\Lambda}_{\varepsilon,x} = \varepsilon \mathbf{I}_4, \quad \mathbf{\Lambda}_{\varepsilon,y} = \varepsilon \mathbf{I}_4, \quad \mathbf{\Lambda}_{\varepsilon,z} = \varepsilon \mathbf{I}_4, \quad (\text{D.35})$$

$$\mathbf{\Lambda}_\chi = \chi \mathbf{I}_8. \quad (\text{D.36})$$

Using (D.31), (D.32), (D.34), (D.35), (D.36) and (B.36), $\mathbf{\Sigma}_\phi$ in (3.83) simplifies to

$$\mathbf{\Sigma}_\phi = \sqrt{2\frac{\varepsilon}{\chi}} \begin{bmatrix} \frac{1}{\Delta x} \mathbf{D}_x & \frac{1}{\Delta y} \mathbf{D}_y & \frac{1}{\Delta z} \mathbf{D}_z \end{bmatrix}^T. \quad (\text{D.37})$$

Hence,

$$\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi = 2\frac{\varepsilon}{\chi} \left(\frac{\mathbf{D}_x \mathbf{D}_x^T}{(\Delta x)^2} + \frac{\mathbf{D}_y \mathbf{D}_y^T}{(\Delta y)^2} + \frac{\mathbf{D}_z \mathbf{D}_z^T}{(\Delta z)^2} \right). \quad (\text{D.38})$$

Using symbolic computations in MATLAB [215], (D.38) was found to have the following eigenvalues:

$$\lambda_1(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 0, \quad (\text{D.39})$$

$$\lambda_2(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \frac{1}{(\Delta x)^2}, \quad (\text{D.40})$$

$$\lambda_3(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \frac{1}{(\Delta y)^2}, \quad (\text{D.41})$$

$$\lambda_4(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \frac{1}{(\Delta z)^2}, \quad (\text{D.42})$$

$$\lambda_5(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} \right), \quad (\text{D.43})$$

$$\lambda_6(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.44})$$

$$\lambda_7(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \left(\frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.45})$$

$$\lambda_8(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right). \quad (\text{D.46})$$

Taking the largest one of these eigenvalues,

$$\rho(\mathbf{\Sigma}_\phi^T \mathbf{\Sigma}_\phi) = 4\frac{\varepsilon}{\chi} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right). \quad (\text{D.47})$$

Hence, the generalized CFL limit (3.82) for the single-cell region reduces to

$$\Delta t_{\text{CFL,gen},ijk}^{(\phi)} = \frac{1}{\sqrt{\frac{\varepsilon}{\chi} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right)}} = \frac{\sqrt{\mu\varepsilon}}{\sqrt{\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2}}}, \quad (\text{D.48})$$

which proves (D.29).

Next, consider (3.89). For the single-cell region, Σ_A in (3.89) reduces to

$$\Sigma_A = \sqrt{\frac{2}{\mu\varepsilon}} \left[\frac{\text{diag}(\Delta x, \Delta y, \Delta z) \mathbf{C} \text{diag}(\Delta x, \Delta y, \Delta z)}{\Delta x \Delta y \Delta z} \right. \\ \left. \mathbf{D} \text{diag}\left(\frac{1}{\Delta x}, \frac{1}{\Delta y}, \frac{1}{\Delta z}\right) \right]. \quad (\text{D.49})$$

Hence, $\Sigma_A^T \Sigma_A$ is given by

$$\Sigma_A^T \Sigma_A = \frac{2}{\mu\varepsilon} \times \begin{bmatrix} \frac{\mathbf{D}_x^T \mathbf{D}_x}{(\Delta x)^2} + \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zx}}{(\Delta y)^2} + \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yx}}{(\Delta z)^2} & \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zy}}{\Delta x \Delta y} + \frac{\mathbf{D}_x^T \mathbf{D}_y}{\Delta x \Delta y} & \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yz}}{\Delta x \Delta z} + \frac{\mathbf{D}_x^T \mathbf{D}_z}{\Delta x \Delta z} \\ \frac{\mathbf{C}_{zy}^T \mathbf{C}_{zx}}{\Delta x \Delta y} + \frac{\mathbf{D}_y^T \mathbf{D}_x}{\Delta x \Delta y} & \frac{\mathbf{C}_{zy}^T \mathbf{C}_{zy}}{(\Delta x)^2} + \frac{\mathbf{D}_y^T \mathbf{D}_y}{(\Delta y)^2} + \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xy}}{(\Delta z)^2} & \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xz}}{\Delta y \Delta z} + \frac{\mathbf{D}_y^T \mathbf{D}_z}{\Delta y \Delta z} \\ \frac{\mathbf{C}_{yz}^T \mathbf{C}_{yx}}{\Delta x \Delta z} + \frac{\mathbf{D}_z^T \mathbf{D}_x}{\Delta x \Delta z} & \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xy}}{\Delta y \Delta z} + \frac{\mathbf{D}_z^T \mathbf{D}_y}{\Delta y \Delta z} & \frac{\mathbf{C}_{yz}^T \mathbf{C}_{yz}}{(\Delta x)^2} + \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xz}}{(\Delta y)^2} + \frac{\mathbf{D}_z^T \mathbf{D}_z}{(\Delta z)^2} \end{bmatrix}, \quad (\text{D.50})$$

where the off-diagonal blocks can be shown to cancel out, giving

$$\Sigma_A^T \Sigma_A = \frac{2}{\mu\varepsilon} \text{diag} \left(\frac{\mathbf{D}_x^T \mathbf{D}_x}{(\Delta x)^2} + \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zx}}{(\Delta y)^2} + \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yx}}{(\Delta z)^2}, \right. \\ \left. \frac{\mathbf{C}_{zy}^T \mathbf{C}_{zy}}{(\Delta x)^2} + \frac{\mathbf{D}_y^T \mathbf{D}_y}{(\Delta y)^2} + \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xy}}{(\Delta z)^2}, \frac{\mathbf{C}_{yz}^T \mathbf{C}_{yz}}{(\Delta x)^2} + \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xz}}{(\Delta y)^2} + \frac{\mathbf{D}_z^T \mathbf{D}_z}{(\Delta z)^2} \right). \quad (\text{D.51})$$

Hence, the spectral radius of $\Sigma_A^T \Sigma_A$ is given by

$$\rho(\Sigma_A^T \Sigma_A) = \frac{2}{\mu\varepsilon} \max \left\{ \rho \left(\frac{\mathbf{D}_x^T \mathbf{D}_x}{(\Delta x)^2} + \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zx}}{(\Delta y)^2} + \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yx}}{(\Delta z)^2} \right), \right. \\ \left. \rho \left(\frac{\mathbf{C}_{zy}^T \mathbf{C}_{zy}}{(\Delta x)^2} + \frac{\mathbf{D}_y^T \mathbf{D}_y}{(\Delta y)^2} + \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xy}}{(\Delta z)^2} \right), \rho \left(\frac{\mathbf{C}_{yz}^T \mathbf{C}_{yz}}{(\Delta x)^2} + \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xz}}{(\Delta y)^2} + \frac{\mathbf{D}_z^T \mathbf{D}_z}{(\Delta z)^2} \right) \right\}. \quad (\text{D.52})$$

Using MATLAB symbolic calculations, the eigenvalues of the three matrices in (D.52) were found to be

$$\lambda_1 \left(\frac{\mathbf{D}_x^T \mathbf{D}_x}{(\Delta x)^2} + \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zx}}{(\Delta y)^2} + \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yx}}{(\Delta z)^2} \right) = 2 \frac{1}{(\Delta x)^2}, \quad (\text{D.53})$$

$$\lambda_2 \left(\frac{\mathbf{D}_x^T \mathbf{D}_x}{(\Delta x)^2} + \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zx}}{(\Delta y)^2} + \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yx}}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} \right), \quad (\text{D.54})$$

$$\lambda_3 \left(\frac{\mathbf{D}_x^T \mathbf{D}_x}{(\Delta x)^2} + \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zx}}{(\Delta y)^2} + \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yx}}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.55})$$

$$\lambda_4 \left(\frac{\mathbf{D}_x^T \mathbf{D}_x}{(\Delta x)^2} + \frac{\mathbf{C}_{zx}^T \mathbf{C}_{zx}}{(\Delta y)^2} + \frac{\mathbf{C}_{yx}^T \mathbf{C}_{yx}}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.56})$$

$$\lambda_1 \left(\frac{\mathbf{C}_{zy}^T \mathbf{C}_{zy}}{(\Delta x)^2} + \frac{\mathbf{D}_y^T \mathbf{D}_y}{(\Delta y)^2} + \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xy}}{(\Delta z)^2} \right) = 2 \frac{1}{(\Delta y)^2}, \quad (\text{D.57})$$

$$\lambda_2 \left(\frac{\mathbf{C}_{zy}^T \mathbf{C}_{zy}}{(\Delta x)^2} + \frac{\mathbf{D}_y^T \mathbf{D}_y}{(\Delta y)^2} + \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xy}}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} \right), \quad (\text{D.58})$$

$$\lambda_3 \left(\frac{\mathbf{C}_{zy}^T \mathbf{C}_{zy}}{(\Delta x)^2} + \frac{\mathbf{D}_y^T \mathbf{D}_y}{(\Delta y)^2} + \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xy}}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.59})$$

$$\lambda_4 \left(\frac{\mathbf{C}_{zy}^T \mathbf{C}_{zy}}{(\Delta x)^2} + \frac{\mathbf{D}_y^T \mathbf{D}_y}{(\Delta y)^2} + \frac{\mathbf{C}_{xy}^T \mathbf{C}_{xy}}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.60})$$

$$\lambda_1 \left(\frac{\mathbf{C}_{yz}^T \mathbf{C}_{yz}}{(\Delta x)^2} + \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xz}}{(\Delta y)^2} + \frac{\mathbf{D}_z^T \mathbf{D}_z}{(\Delta z)^2} \right) = 2 \frac{1}{(\Delta z)^2}, \quad (\text{D.61})$$

$$\lambda_2 \left(\frac{\mathbf{C}_{yz}^T \mathbf{C}_{yz}}{(\Delta x)^2} + \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xz}}{(\Delta y)^2} + \frac{\mathbf{D}_z^T \mathbf{D}_z}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.62})$$

$$\lambda_3 \left(\frac{\mathbf{C}_{yz}^T \mathbf{C}_{yz}}{(\Delta x)^2} + \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xz}}{(\Delta y)^2} + \frac{\mathbf{D}_z^T \mathbf{D}_z}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.63})$$

$$\lambda_4 \left(\frac{\mathbf{C}_{yz}^T \mathbf{C}_{yz}}{(\Delta x)^2} + \frac{\mathbf{C}_{xz}^T \mathbf{C}_{xz}}{(\Delta y)^2} + \frac{\mathbf{D}_z^T \mathbf{D}_z}{(\Delta z)^2} \right) = 2 \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right). \quad (\text{D.64})$$

From (D.52),

$$\rho(\Sigma_A^T \Sigma_A) = \frac{4}{\mu \varepsilon} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right). \quad (\text{D.65})$$

Hence, the generalized CFL limit (3.89) for the single-cell region is given by

$$\Delta t_{\text{CFL,gen},ijk}^{(A)} = \frac{\sqrt{\mu \varepsilon}}{\sqrt{\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2}}}, \quad (\text{D.66})$$

which proves (D.30). □

Corollary 2. Let Δt_{CFL} be the conventional CFL limit (1.36) for the entire region. Let $\Delta t_{\text{CFL,gen},ijk}^{(\phi)}$ be the generalized CFL limit (3.82) for a single-cell region formed by a primary cell with the bottom-south-west corner at the node (i, j, k) . Let $\Delta t_{\text{CFL,gen},ijk}^{(A)}$ be the generalized CFL limit (3.89) for that single-cell region. Then,

$$\Delta t_{\text{CFL}} \leq \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk}^{(\phi)}. \quad (\text{D.67})$$

$$\Delta t_{\text{CFL}} \leq \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk}^{(A)}. \quad (\text{D.68})$$

The proof is similar to that of Corollary 1 and follows from Lemma 9.

Proof of Theorem 6. From Lemma 8 and Corollary 2,

$$\Delta t_{\text{CFL}} \leq \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk}^{(\phi)} \leq \Delta t_{\text{CFL,gen}}^{(\phi)}, \quad (\text{D.69})$$

$$\Delta t_{\text{CFL}} \leq \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk}^{(A)} \leq \Delta t_{\text{CFL,gen}}^{(A)}, \quad (\text{D.70})$$

proving the theorem. $\square$

D.3 Proof of Theorem 8

Lemma 10. *Consider a region with $n_x \times n_y \times n_z$ primary cells, governed by (4.9a)–(4.9b) or, equivalently, (4.12). Let ψ be an arbitrary real vector with the number of elements equal to double the number of primary nodes in the region. Then, $\psi^T \mathcal{P} \psi$ can be written as*

$$\psi^T \mathcal{P} \psi = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \sum_{k=1}^{n_z} \psi_{ijk}^T \mathcal{P}_{ijk} \psi_{ijk}. \quad (\text{D.71})$$

where $\mathcal{P}_{ijk}$ and ψ_{ijk} are defined similarly to Lemmas 4 and 7. In the definition of $\mathcal{P}_{ijk}$, the value of the potential U at each of the eight primary nodes in the single-cell region equals to the value at the corresponding node in the $n_x \times n_y \times n_z$ region.

The proof of Lemma 10 is similar to the proofs of Lemma 4 and Lemma 7.

Lemma 11. *Let $\Delta t_{\text{CFL,gen}}$ and $\Delta t_{\text{CFL,gen},ijk}$ be the generalized CFL limits (4.31) corresponding, respectively, to the entire region and to a region formed by a single primary cell with the bottom-south-west corner at the node (i, j, k) . Then,*

$$\min_{i,j,k} \Delta t_{\text{CFL,gen},ijk} \leq \Delta t_{\text{CFL,gen}}. \quad (\text{D.72})$$

Lemma 11 can be proved using a similar reasoning to the proof of Lemma 5.

Lemma 12. *Let $\Delta t_{\text{CFL},ijk}$ be the CFL limit (1.39) for a single-cell region composed of a primary cell with the bottom-south-west corner at the node (i, j, k) . Let $\Delta t_{\text{CFL,gen},ijk}$ be the generalized CFL limit (4.31) for that region. Then,*

$$\Delta t_{\text{CFL},ijk} \leq \Delta t_{\text{CFL,gen},ijk}. \quad (\text{D.73})$$

Proof. With reference to Appendix B, for the single-cell region,

$$\Lambda_V'' = \frac{\Delta x \Delta y \Delta z}{8} \mathbf{I}_8, \quad (\text{D.74})$$

$$\Lambda_S'' = \text{diag} \left(\frac{\Delta y \Delta z}{4} \mathbf{I}_4, \frac{\Delta x \Delta z}{4} \mathbf{I}_4, \frac{\Delta x \Delta y}{4} \mathbf{I}_4 \right), \quad (\text{D.75})$$

$$\Lambda_l' = \text{diag} (\Delta x \mathbf{I}_4, \Delta y \mathbf{I}_4, \Delta z \mathbf{I}_4), \quad (\text{D.76})$$

$$\mathbf{D} = - \begin{bmatrix} \mathbf{I}_2 \otimes \mathbf{I}_2 \otimes \mathbf{W}_1^T & \mathbf{I}_2 \otimes \mathbf{W}_1^T \otimes \mathbf{I}_2 & \mathbf{W}_1^T \otimes \mathbf{I}_2 \otimes \mathbf{I}_2 \end{bmatrix}. \quad (\text{D.77})$$

Hence, using (4.10), matrix Σ in (4.32) for that region can be written as

$$\Sigma = \frac{\hbar}{2m} (\Lambda_V'')^{-\frac{1}{2}} \mathbf{D} \Lambda_S'' (\Lambda_l')^{-1} \mathbf{D}^T (\Lambda_V'')^{-\frac{1}{2}} + \frac{1}{\hbar} \Lambda_U = \Sigma_0 + \frac{1}{\hbar} \Lambda_U, \quad (\text{D.78})$$

where

$$\mathbf{\Sigma}_0 = \frac{\hbar}{m} \left(\frac{\mathbf{I}_2 \otimes \mathbf{I}_2 \otimes \mathbf{W}_1^T \mathbf{W}_1}{(\Delta x)^2} + \frac{\mathbf{I}_2 \otimes \mathbf{W}_1^T \mathbf{W}_1 \otimes \mathbf{I}_2}{(\Delta y)^2} + \frac{\mathbf{W}_1^T \mathbf{W}_1 \otimes \mathbf{I}_2 \otimes \mathbf{I}_2}{(\Delta z)^2} \right). \quad (\text{D.79})$$

Matrix $\mathbf{W}_1^T \mathbf{W}_1$ can be diagonalized as

$$\mathbf{W}_1^T \mathbf{W}_1 = \mathbf{V}_2 \text{diag}(0, 2) \mathbf{V}_2^T, \quad (\text{D.80})$$

where $\mathbf{V}_2$ is an orthogonal matrix given by

$$\mathbf{V}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} |\mathbf{W}_1^T| & \mathbf{W}_1^T \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}. \quad (\text{D.81})$$

Hence, $\mathbf{\Sigma}_0$ in (D.79) can be diagonalized as

$$\mathbf{\Sigma}_0 = \frac{\hbar}{m} (\mathbf{V}_2 \otimes \mathbf{V}_2 \otimes \mathbf{V}_2) \left(\frac{\mathbf{I}_2 \otimes \mathbf{I}_2 \otimes \text{diag}(0, 2)}{(\Delta x)^2} + \frac{\mathbf{I}_2 \otimes \text{diag}(0, 2) \otimes \mathbf{I}_2}{(\Delta y)^2} + \frac{\text{diag}(0, 2) \otimes \mathbf{I}_2 \otimes \mathbf{I}_2}{(\Delta z)^2} \right) (\mathbf{V}_2 \otimes \mathbf{V}_2 \otimes \mathbf{V}_2)^T. \quad (\text{D.82})$$

From (D.82), the eigenvalues of $\mathbf{\Sigma}_0$ are given by

$$\lambda_1(\mathbf{\Sigma}_0) = 0, \quad (\text{D.83})$$

$$\lambda_2(\mathbf{\Sigma}_0) = \frac{2\hbar}{m} \frac{1}{(\Delta x)^2}, \quad (\text{D.84})$$

$$\lambda_3(\mathbf{\Sigma}_0) = \frac{2\hbar}{m} \frac{1}{(\Delta y)^2}, \quad (\text{D.85})$$

$$\lambda_4(\mathbf{\Sigma}_0) = \frac{2\hbar}{m} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} \right), \quad (\text{D.86})$$

$$\lambda_5(\mathbf{\Sigma}_0) = \frac{2\hbar}{m} \frac{1}{(\Delta z)^2}, \quad (\text{D.87})$$

$$\lambda_6(\mathbf{\Sigma}_0) = \frac{2\hbar}{m} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.88})$$

$$\lambda_7(\mathbf{\Sigma}_0) = \frac{2\hbar}{m} \left(\frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.89})$$

$$\lambda_8(\mathbf{\Sigma}_0) = \frac{2\hbar}{m} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right), \quad (\text{D.90})$$

$$(\text{D.91})$$

where the eigenvalues are listed in the order corresponding to the diagonal elements of the matrix in the second brackets in (D.82). From (D.78), we have [96]

$$\rho(\mathbf{\Sigma}) = \|\mathbf{\Sigma}\|_2 \leq \|\mathbf{\Sigma}_0\|_2 + \frac{1}{\hbar} \|\mathbf{\Lambda}_U\|_2 = \frac{2\hbar}{m} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right) + \frac{|U|_{\max}}{\hbar}, \quad (\text{D.92})$$

Table D.1: Relative difference of the CFL limit (1.39) and the generalized CFL limit (4.31) in the scenarios considered in Appendix D.3.

Potential	$n_x = n_y = n_z = 1$	$n_x = n_y = n_z = 10$
$U _{i,j,k} = 0$	2.49×10^{-16}	0
$U _{i,j,k} = 0.1 \text{ eV}$	-1.84×10^{-16}	1.47×10^{-15}
$U _{i,j,k} = 1 \text{ eV}$	-1.81×10^{-16}	-1.81×10^{-16}
$U _{i,j,k} = 10 \text{ eV}$	1.91×10^{-16}	9.56×10^{-16}
$U _{i,j,k} = -0.1 \text{ eV}$	-6.51×10^{-1}	-6.51×10^{-1}
$U _{i,j,k} = -1 \text{ eV}$	-1.72×10^{-1}	-1.72×10^{-1}
$U _{i,j,k} = -10 \text{ eV}$	-2.03×10^{-2}	-2.03×10^{-2}
$0 < U _{i,j,k} < 0.1 \text{ eV}$ (randomized)	-5.13×10^{-2}	-9.18×10^{-2}
$0 < U _{i,j,k} < 1 \text{ eV}$ (randomized)	-8.05×10^{-2}	-4.96×10^{-2}
$0 < U _{i,j,k} < 10 \text{ eV}$ (randomized)	-1.15×10^{-2}	-1.01×10^{-2}
$-0.1 \text{ eV} < U _{i,j,k} < 0$ (randomized)	-3.69×10^{-1}	-4.24×10^{-1}
$-1 \text{ eV} < U _{i,j,k} < 0$ (randomized)	-2.55×10^{-1}	-2.28×10^{-1}
$-10 \text{ eV} < U _{i,j,k} < 0$ (randomized)	-3.52×10^{-2}	-3.04×10^{-2}

where

$$|U|_{\max} = \max(|U|_{i,j,k}, |U|_{i+1,j,k}, |U|_{i,j+1,k}, |U|_{i+1,j+1,k}, |U|_{i,j,k+1}, |U|_{i+1,j,k+1}, |U|_{i,j+1,k+1}, |U|_{i+1,j+1,k+1}). \quad (\text{D.93})$$

Rearranging (D.92) and multiplying both sides of the inequality by 2, we obtain

$$\frac{2\hbar}{m} \left(\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2} \right) + \frac{|U|_{\max}}{\hbar} \leq \frac{2}{\rho(\Sigma)}, \quad (\text{D.94})$$

which proves (D.73). $\square$

Lemma 12 leads to the following corollary.

Corollary 3. *Let Δt_{CFL} be the CFL limit (1.39) for the $n_x \times n_y \times n_z$ region. Let $\Delta t_{\text{CFL,gen},ijk}$ be the generalized CFL limit (4.31) for the region composed of a single cell with the bottom-west-south corner at the primary node (i, j, k) . Then,*

$$\Delta t_{\text{CFL}} \leq \min_{i,j,k} \Delta t_{\text{CFL,gen},ijk}. \quad (\text{D.95})$$

The proof of Corollary 3 is similar to that of Corollary 1.

Proof of Theorem 8. The statement of the theorem follows directly from Lemma 11 and Corollary 3. $\square$

Table D.1 shows the relative difference between the two time steps in (1.39) and (4.31), computed as $(\Delta t_{\text{CFL}} - \Delta t_{\text{CFL,gen}})/\Delta t_{\text{CFL,gen}}$. The cell dimensions were taken as $\Delta x = 1 \text{ nm}$, $\Delta y = 2 \text{ nm}$, and $\Delta z = 3 \text{ nm}$. The mass of the particle was that of an electron. As expected, for the regions consisting of a single cell with constant non-negative potential U at each of the eight nodes, the two time steps Δt_{CFL} and $\Delta t_{\text{CFL,gen}}$ were the same, with small discrepancy due to finite machine

precision. Interestingly, the two time steps were also equal for the multi-cell regions in Table D.1 with constant non-negative potential. However, when the potential either had a spacial variation or was negative, the two time steps deviated, although the difference tended to be small. In all regions where the CFL limit (1.39) and the generalized CFL limit (4.31) deviated, the CFL limit (1.39) had a lower value, confirming the statement of Theorem 8.

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